Asset Prices, Risk, and Information

ECON 450

Scott Condie

2026

Table of contents

Chapter 1

Asset Prices, Risk, and Information

Welcome to the course notes for ECON 450: Financial Economics, at Brigham Young University. These notes cover the core topics of a first calculus-based course in financial economics, progressing from the mechanics of financial markets and the principles of portfolio theory through the Capital Asset Pricing Model and its extensions, to the analysis of fixed income securities and the pricing of derivatives including forwards, futures, and options. The goal is to give students a rigorous yet accessible foundation for understanding how financial markets work, how assets are priced, and how the tools developed here can be used to make sound financial decisions.

This edition was compiled on August 18, 2026.

Chapter 2

Introduction

2.1 Why Financial Markets Matter

At some point in your life — perhaps soon, perhaps already — you will face a set of decisions that no amount of goodwill or hard work alone can resolve without knowledge: how to save for retirement, whether to buy or rent a home, how to evaluate a job offer that includes stock options, whether to borrow to fund a degree, and how to protect your family against risks you cannot fully control. These are not abstract puzzles. They are the decisions that separate people who accumulate wealth and security from people who, despite working equally hard, find themselves financially fragile. Financial economics is the discipline that provides the tools to navigate them well.

But the stakes extend far beyond personal finance. Financial markets are the mechanism by which a modern economy allocates capital — directing the savings of millions of households toward the firms, governments, and projects that can put those resources to their most productive use. When markets function well, they allow a young entrepreneur with a good idea but little wealth to attract funding from investors who have capital but lack opportunities, and they allow a retiree who has saved steadily to earn a return on her accumulated resources by funding those ventures. When they malfunction — because of mispricing, opacity, poorly designed incentives, or failures of information — the consequences ripple through the entire economy, as the events of 2007–2009 demonstrated with devastating force. Understanding how financial markets work is therefore not only personally valuable; it is a prerequisite for informed citizenship in an economy where the health of financial institutions affects the prosperity of everyone.

This text develops the conceptual foundations of financial economics through an opening chapter on market institutions followed by five thematic parts that build on one another. Each chapter poses the questions that its material is designed to answer. By the end, you will have a coherent, analytically grounded view of how portfolios are constructed, how asset prices are determined in equilibrium, how information is reflected in prices, how bonds and interest rates are analyzed, how derivatives are priced and used, and how more advanced market models extend these foundations to practical problems in portfolio management, credit risk, and the mechanics of trading itself. More practically, you will have the tools to make better decisions — for yourself and, if you go into finance, for others.

> **In the news**
>
> Financial economics is not an abstraction — it shows up in the balance of every retirement account. In early 2026, Fidelity reported that the average 401(k) balance *fell about 4%* in a single quarter as markets turned volatile, even though balances had climbed for three straight years on the back of strong stock returns. The same risk–return trade-off that drives those swings, and the tools for deciding how much of it to bear, are the subject of this entire book. Read it at CNBC.

2.2 The Architecture of the Course

The course opens with a standalone chapter on financial markets as institutions and then proceeds through five thematic parts. Understanding how the parts connect will help you see each chapter not as an isolated topic but as one step in a cumulative argument.

Financial markets as institutions. Before any theory of prices or portfolios can be built, it is necessary to understand the institutional machinery through which financial transactions actually take place. The opening chapter asks how buyers and sellers find each other, complete transactions reliably, and protect themselves against the failure of their counterparties. The answers — exchanges, over-the-counter dealer networks, clearinghouses, margin accounts, and broker-dealer relationships — are the infrastructure on which everything else in the course depends. The chapter also introduces the main asset classes — equity, fixed income, currencies, and derivatives — that the subsequent parts analyze in depth.

Part I: Individual portfolio selection and market equilibrium. The first part addresses the most fundamental quantitative question in finance: given a universe of risky assets, how should a rational investor allocate her wealth? Mean-variance portfolio theory provides the answer. Investors are modeled as caring about the expected return and variance of their portfolio, and the central insight is that combining assets whose returns are imperfectly correlated reduces total risk — the principle of diversification. Solving the variance-minimization problem traces out the efficient frontier, and when a risk-free asset is available the frontier collapses to a single ray — the capital market line — along which every investor, regardless of risk aversion, holds the same risky portfolio. The Capital Asset Pricing Model then aggregates individual optimization across all investors to produce an equilibrium theory of expected returns: when all investors hold efficient portfolios and markets clear, an asset's required return is determined not by its standalone volatility but by its beta — its covariance with the aggregate market portfolio, which is the only component of risk that cannot be diversified away. A final chapter turns this equilibrium theory to a practical use, asking how to evaluate whether a portfolio manager has genuine skill: the Sharpe ratio, Treynor measure, Jensen's alpha, tracking error, and information ratio each strip the reward for bearing risk out of a raw return so that what remains can be attributed to skill rather than luck.

Part II: Information, trading and market efficiency. The second part asks how information is reflected in market prices, and what that implies for the value of active investing. The efficient markets hypothesis examines the conditions under which prices aggregate the dispersed private information of all market participants — distinguishing the weak, semi-strong, and strong forms of efficiency and surveying the evidence from behavioral finance on systematic departures from rationality. It confronts the question of whether biases documented at the level of individual investors need translate into distorted prices, and it turns the demanding assumptions of the CAPM into a practical checklist for judging any investment that is pitched as offering high returns with little risk.

Part III: Fixed income securities and pricing. The third part covers the world's largest asset class: bonds and other fixed income instruments. Four chapters build sequentially from first principles. The first establishes the time value of money, deriving present value from the arbitrage principle and showing how the various compounding conventions relate to one another. The second introduces the yield to maturity, exposes its limitations as a measure of realized return, and shows how coupon income and capital gains combine to determine the total return over any holding period. The third studies the yield curve — the relationship between yield and maturity across the spectrum of maturities — developing forward rates, the expectations hypothesis, and the liquidity preference theory, and providing the tools needed to read the curve as a signal about future interest rates. The fourth chapter introduces duration and convexity as measures of interest rate sensitivity, and develops immunization — the strategy of matching the duration of assets and liabilities to protect a portfolio against adverse rate movements.

Part IV: Derivatives. The fourth part turns to instruments whose payoffs are defined by reference to an underlying asset. Forwards and futures allow market participants to lock in today a price for a future transaction; their pricing follows from a clean cost-of-carry arbitrage argument, and the chapter works through both the institutional mechanics of futures markets — margin, marking to market, basis risk — and the three classical theories of the relationship between futures prices and expected future spot prices. Options add the crucial asymmetry of the right without the obligation to transact, and pricing them requires either a replication argument in a discrete binomial framework or the Black-Scholes formula in continuous time. Understanding options is essential not only for hedging and speculation but because option-like payoffs are embedded throughout corporate finance and increasingly in retail financial products.

Part V: Advanced market models. The final part develops more advanced models of markets and risk, extending the core framework in three directions of growing practical importance. The Black-Litterman model addresses a well-known failure mode of mean-variance optimization: unconstrained optimizers produce extreme, unstable portfolios because they treat noisy return estimates as known with certainty. The Black-Litterman remedy is to treat expected returns as random variables with a prior anchored to market equilibrium weights, and then to update that prior using the investor's own subjective views through Bayesian methods. The result is a principled framework for translating investment conviction into portfolio tilts in proportion to the confidence placed in each view. The credit models chapter then turns to default risk — the defining feature of corporate bonds and structured products — developing the Gaussian copula framework for modeling default times and exploring how correlation in defaults across borrowers determines the risk of individual tranches in a securitized pool. The 2007–2009 financial crisis, whose severity was amplified by a collective failure to model default correlation correctly, provides both the motivation and the cautionary lesson. A final chapter goes inside the trading process itself to ask why a bid-ask spread exists and how privately informed traders interact with competitive dealers; the three canonical models studied there — Glosten-Milgrom, Kyle, and Grossman-Stiglitz — show that spreads arise from the rational response of dealers to the risk of trading against someone who knows more than they do, and that prices can only be partially revealing when information acquisition is costly.

2.3 How This Course Will Serve You

The most immediate application of this material is to your own financial life. The mean-variance framework gives you the theoretical justification for index investing: if the CAPM is approximately right and you cannot consistently identify undervalued securities, holding the market portfolio at the lowest possible cost is the rational choice. The efficient markets hypothesis tells you what level of evidence you would need to see before concluding that an active manager is genuinely skilled rather than merely lucky. The fixed income chapters give you the tools to evaluate bond investments, understand the interest rate risk embedded in your mortgage, and interpret the economic signal in the slope of the yield curve. The chapters on derivatives explain the instruments that are increasingly embedded in everything from employee compensation to retail investment products, enabling you to evaluate their value and risk with a clear head rather than in confusion.

At a broader level, this course is an education in rigorous reasoning under uncertainty. The skills you develop here — the habit of specifying a model precisely, deriving its implications carefully, and asking whether those implications are consistent with observed data — are transferable far beyond financial economics. Whether you go into portfolio management, corporate finance, policy, consulting, or any field where decisions must be made under uncertainty with incomplete information, the analytical habits cultivated here will serve you. Financial markets are a remarkably clean laboratory for learning these habits, because they provide immediate feedback: if your model is wrong, prices will tell you. That discipline is part of what makes financial economics, at its best, one of the most intellectually honest and practically powerful branches of applied economics.

Finally, it is worth saying plainly what this course is not. It is not a guide to getting rich quickly, and it is not a set of trading strategies that will guarantee outsized returns. The efficient markets hypothesis — whose content you will understand rigorously by the end of the semester — implies that reliable excess returns require either superior information, superior analysis, or a willingness to bear risks that others are not. This course will improve your analysis and deepen your understanding of risk, but it cannot manufacture informational advantages from thin air. What it can do is help you avoid the costly mistakes — the failure to diversify, the misunderstanding of leverage, the confusion between risk and volatility, the inability to recognize when a financial product is not what it appears — that are the primary source of financial distress for households and institutions alike. A thorough understanding of financial economics will not make you wealthy by itself, but ignorance of it can make it very hard to stay that way.

2.4 Application: A new graduate's first portfolio

Consider Maya, a twenty-two-year-old who has just started her first salaried job and been handed a retirement account with a single decision to make: how to divide each paycheck between a stock fund and a bond fund. Every idea in this book bears on that choice. The trade-off between risk and expected return tells her why stocks have historically paid more and why that extra return is compensation for bearing volatility she will feel in bad years; the principle of diversification explains why she should hold a broad fund rather than a handful of names; and the time value of money shows how decades of compounding magnify even small differences in her allocation. What looks like a simple slider between "stocks" and "bonds" is in fact the practical face of the entire theory of financial economics, and the chapters that follow give Maya the tools to set it deliberately rather than by guesswork.

> Further listening
>
> *Planet Money* (NPR) — "Summer School 1: The Stock Market": an accessible tour of what a stock is, how prices are set, and the risk–reward trade-off that runs through this whole book.

Chapter 3

Markets

3.1 Introduction

Three questions animate the survey of financial markets taken up in this chapter. First, what are the broad classes of financial assets, and how much of the world's wealth does each one represent? Second, what has each class actually paid its investors historically, and how much did investors have to endure to earn it? Third, do the different classes rise and fall together, or do their bad months arrive at different times? These questions matter because their answers are the raw material for everything that follows: the pricing theory of the coming chapters exists to explain the patterns that this chapter documents.

The first question is important because the terrain has to be mapped before it can be modeled. Financial claims come in a small number of economically distinct forms — *equity*, which represents ownership of a firm; *fixed income*, which represents a promise to repay borrowed money; *currency*, the claims used to settle transactions across national monies; and *derivatives*, whose payoffs are defined in terms of some other, underlying asset — and their relative sizes are frequently misunderstood. Debt markets are larger than equity markets despite attracting a fraction of the attention, and derivatives are routinely quoted at a *notional* value forty times the economically meaningful one. Getting these magnitudes right is a prerequisite for judging which markets matter and why.

The second question is where measurement begins. Reporting what an asset "returned" requires first defining a return precisely, and then confronting the fact that a single average conceals almost everything of interest. The chapter therefore looks at whole *distributions* of monthly returns rather than at averages alone — for U.S. equities, for short and long Treasury debt, and for investment-grade and high-yield corporate credit. Read side by side, those distributions reveal a ladder in which average return and volatility rise together, along with two facts that no single summary number captures: that the left tails of risky assets are fatter than a normal distribution allows, and that two assets with identical standard deviations can carry entirely different risks.

The third question is the seed of the theory of portfolio choice. An asset's risk in isolation is not the same as the risk it contributes to a portfolio, and the difference turns on correlation. The chapter closes by reporting how the five assets co-move — Treasuries with each other, high-yield credit with equities, and equities with Treasuries — and by noting that it is precisely the *low* correlations that make a mixed portfolio steadier than any of its components. That observation is the point at which description gives way to analysis, and it is taken up formally in the mean-variance chapter that follows.

The chapter is organized as a survey. It opens by sizing the four broad asset classes at market value, and by tabulating the average return and volatility of five representative assets in the annual units investors actually quote. It then treats U.S. equities in detail — what a share of common stock actually is, how many such companies there are to invest in and why that number has shrunk by half since 1996, and what the historical record of the S&P 500 reveals about the distribution of market returns. A parallel treatment of government and corporate debt follows, and the chapter ends with the summary moments and the correlation matrix that the rest of these notes will draw on.

> **In the news**
>
> When a private company "goes public," it taps the *primary market* — selling shares to investors for the first time — and thereafter its stock trades in the *secondary market* whose returns this chapter measures. In June 2026, SpaceX completed the largest IPO in history: it priced at \$135 a share, raised roughly \$75 billion at a valuation near \$1.8 trillion, and closed its first full day up about 20%. Two even more anticipated offerings are lined up behind it — Anthropic has filed for a listing that bankers expect could top \$1 trillion, and OpenAI is reportedly preparing one of its own. Each of these listings adds a new name to the equity asset class surveyed below — and, from its first day of trading onward, a new stream of the monthly returns whose distribution this chapter examines.

3.2 The assets we will study

Financial markets exist to move a handful of broad classes of asset between the people who own them and the people who want them. The remainder of these notes studies those classes in turn: *equity*, which represents ownership of a firm; *fixed income*, which represents a promise to repay borrowed money; *currency*, the claims used to settle transactions across national monies; and *derivatives*, whose payoffs are defined in terms of some other, underlying asset. We begin with the class most investors meet first, U.S. equities, and take up the others in the chapters that follow.

How much of the world's financial wealth does each class account for? Figure 3.1 gives the answer at market value. At the end of 2024, global equity markets were worth about \$127 trillion and global debt securities about \$145 trillion — so debt, taken as a whole, is the larger market, even though equities dominate the financial news. The debt total divides into two roughly equal halves. Governments account for slightly more than half of all bonds outstanding — about $75 trillion — a share that has grown steadily since the financial crisis of 2008 and the pandemic of 2020, both of which were financed with heavy sovereign issuance. The remainder, about $70 trillion, is owed by corporations — banks and other financial firms, non-financial companies — and by the securitization vehicles that package mortgages and other loans into bonds.

The derivatives slice requires a word of caution, because derivatives are usually described with a much larger and much more misleading number. The *notional* value of outstanding derivative contracts — the face amount on which payoffs are computed — stood at roughly $700 trillion at the end of 2024, more than double everything else in the figure combined. But notional value measures the scale of the promises written, not the value of the claims they create: an interest-rate swap on $100 million of notional obligates the parties to exchange only the *difference* between two interest payments on that amount, which is worth a tiny fraction of $100 million. The economically comparable number is the *gross market value* — the cost of replacing every outstanding contract at current market prices — and by that yardstick derivatives were worth about $17.6 trillion, only around 6% of the total in Figure 3.1. The gap between the two numbers is itself a lesson in how derivatives work, and we will return to it in the chapters on forwards, futures, and options.

Before the details, the headline facts. Table 3.1 reports, for the five assets this section studies, the average return and the volatility of returns in the annual units investors actually quote. These are the orders of magnitude worth committing to memory: short Treasury debt has returned about 5% per year with very little volatility; long Treasuries and investment-grade corporate bonds sit in the 5–7% range with moderate volatility; high-yield bonds have earned close to 8% and equities close to 9% per year, with volatility that rises step for step. Everything in this section — the markets, the figures, and the distributions — is an unpacking of where this table comes from.

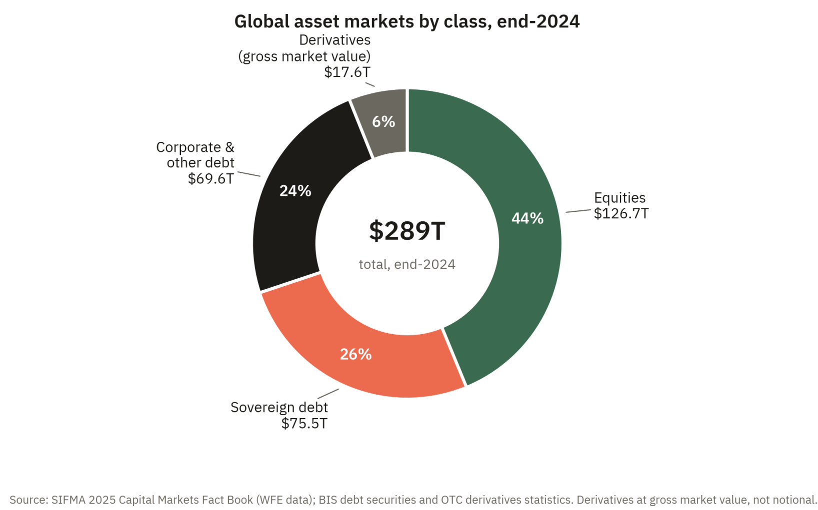

Figure 3.1: Global asset markets by class at the end of 2024, at market value. Derivatives are measured at gross market value — the replacement cost of outstanding contracts — rather than notional value. Sovereign and corporate debt split the $145 trillion in global debt securities using the BIS estimate that governments account for about 52% of the total. Source: SIFMA 2025 Capital Markets Fact Book (WFE data); BIS debt securities and OTC derivatives statistics.

Table 3.1: Average annual returns and volatilities of the five assets studied in this section, annualized from the monthly returns underlying Table 3.2 (mean multiplied by 12, standard deviation by $\sqrt{12}$).

Asset	Sample	Mean return (% per year)	Std. dev. (% per year)
2-year Treasury	1976–2026	5.0	2.6
10-year Treasury	1953–2026	5.4	6.4
Investment-grade corporate	1976–2026	7.1	6.7
High-yield corporate	1986–2026	7.7	7.9
U.S. equities (S&P 500)	1950–2026	8.7	12.0

3.2.1 U.S. equities

A share of *common stock* is a unit of ownership in a corporation. The shareholder is not a lender to the firm but a part-owner of it, and this distinction organizes everything about how equity behaves.

Common stock. A security representing fractional ownership of a corporation. Its holder is a *residual claimant* on the firm's assets and earnings — entitled to what remains only after employees, suppliers, and debtholders have been paid — and ordinarily carries the right to vote on corporate matters in proportion to shares held.

Three features follow from ownership. First, the shareholder's claim is a *residual* one: bondholders and other creditors are paid on fixed schedules, and the equityholder receives whatever is left, whether that is a large distribution or nothing at all. This is why equity is riskier than the firm's debt and why, in compensation, it has historically earned a higher average return — a trade-off that is the subject of much of the rest of these notes. Second, ownership usually confers *control* through the right to vote for directors and on major decisions. Third, ownership is protected by *limited liability*: a shareholder can lose the amount she paid for her shares but no more, so her downside is bounded while her upside is not. Firms may return cash to shareholders directly, as *dividends*, but a shareholder's return comes just as much from *capital gains* — changes in the price at which the share itself trades in the secondary market.

How many such companies are there to invest in? Fewer than there used to be. The number of U.S.-listed domestic companies rose through the 1980s and early 1990s to a peak of **8,090 in 1996**, then fell by more than half over the following three decades, to roughly **3,900 by 2025** (Figure 3.2). The decline reflects a combination of forces: waves of mergers and acquisitions that fold public firms into one another, the growth of private capital that lets companies raise large sums without listing, a slower pace of new initial public offerings, and the fixed costs and disclosure burdens of being public. The result is a market that is narrower in the number of names it contains even as, in aggregate, it has grown enormously in value.

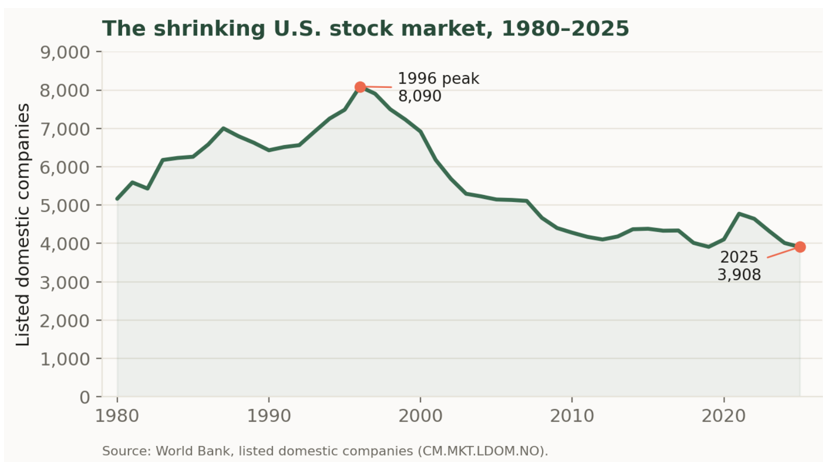

Figure 3.2: The number of U.S.-listed domestic companies peaked in 1996 and has fallen by roughly half since. Figures before 1980 use a narrower definition and are omitted. Source: World Bank.

That growth in aggregate value is easiest to see through a stock market *index*. An index summarizes the prices of many stocks in a single number, so that its movement over time tracks the fortunes of the market as a whole rather than any one firm. The best known is the **S&P 500**, a capitalization-weighted index of about five hundred of the largest U.S. companies; because those firms account for the bulk of the market's total value, the S&P 500 serves throughout these notes as a practical stand-in for "the market" whose risk and return we will study. Its long climb — from roughly 17 in 1950 to more than 7,000 in 2026 — is shown in Figure 3.3, and the contrast with the previous figure is instructive: the population of listed firms has shrunk, but the value of the survivors, and of the market they compose, has multiplied many times over.

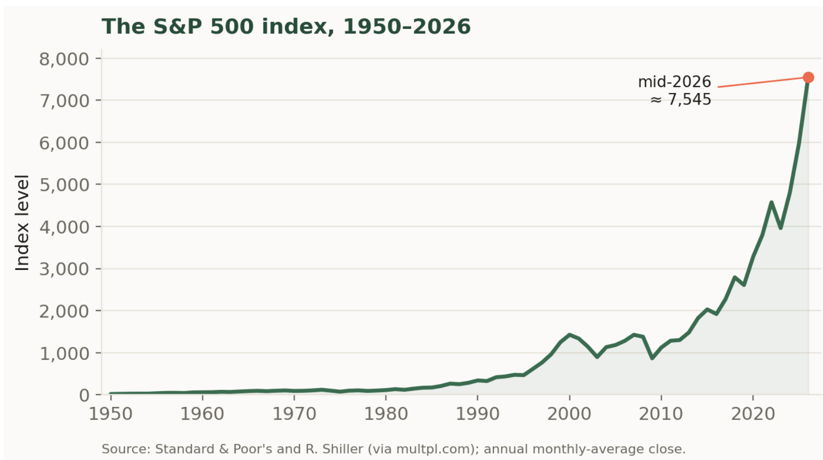

Figure 3.3: The S&P 500 index, a capitalization-weighted measure of about five hundred large U.S. companies, from 1950 to 2026. Source: Standard & Poor's and R. Shiller (via multpl.com).

An index level is useful for seeing the long sweep of the market, but what an investor actually earns from one period to the next is a *return*. The simple return on the index over a month is the percentage change in its level,

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} - 1,$$

where P_t is the index level at the end of month t. Returns, rather than price levels, are the raw material of everything that follows: risk, diversification, and the pricing models of later chapters are all stated in terms of the distribution of returns. It is worth looking at that distribution directly. Figure 3.4 is a histogram of the roughly nine hundred monthly returns to the S&P 500 from 1950 to 2026.

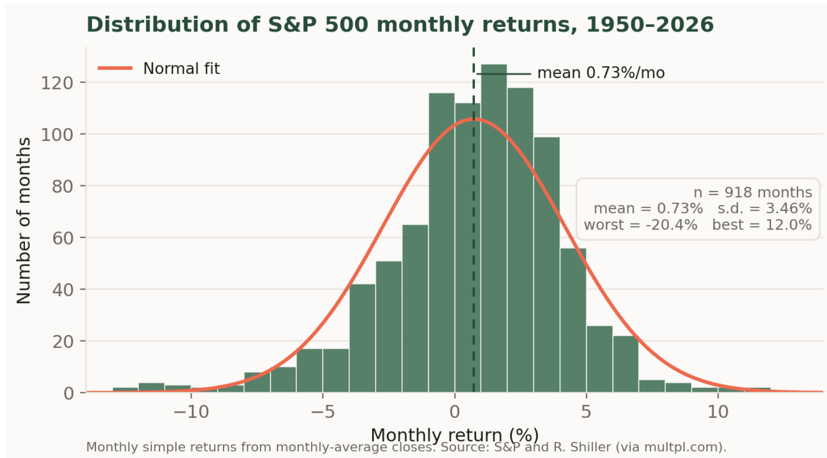

Figure 3.4: Histogram of monthly returns to the S&P 500, 1950–2026, with a fitted normal density (coral). Returns are computed from monthly-average index closes. Source: S&P and R. Shiller (via multpl.com).

Several features of the picture recur throughout these notes. The distribution is centered at a small *positive* number — the market rose in a typical month, averaging about 0.7%, so that a bit under two-thirds of all months were up months — which is the compensation investors earn for bearing risk. It is also *spread out*, with a standard deviation of roughly 3.5% per month; this dispersion is precisely the risk that the return is compensation for, and measuring it is the business of the next several chapters. The bell-shaped normal curve overlaid in coral fits the center of the data reasonably well, which is why the normal distribution is such a convenient modeling assumption. But the fit is imperfect in an important way: the far *left tail* is fatter than the normal distribution allows, because occasional crashes — the worst month here is about –20% — occur more often, and are more severe, than a normal distribution would predict. This combination of a modest positive average, substantial month-to-month volatility, and rare but serious losses is the empirical backdrop against which the rest of the theory is built.

3.2.2 Other asset classes

Equity is only one of the broad asset classes an investor can hold, and the S&P 500 histogram becomes far more informative when compared to those of other asset classes. This section introduces the other markets that will occupy us in later chapters — government debt and corporate debt of high and low quality — and draws for each the same pair of pictures we drew for the S&P 500: the long climb of an index and the histogram of its monthly returns. Reading these histograms side by side is the fastest way to see what makes each asset class distinctive, because the differences between them show up in the four features we highlighted above: where the distribution is centered, how spread out it is, whether it leans left or right, and how heavy its tails are.

Fixed income. A security representing a promise to repay borrowed money on a fixed schedule. Its holder is a *creditor* of the issuer, not an owner: she is entitled to specified coupon and principal payments, paid ahead of any distribution to shareholders, but does not share in the issuer's success beyond them.

Treasury securities. The market for U.S. Treasury debt is the largest and most liquid bond market in the world, and Treasury yields are the benchmark against which nearly every other dollar-denominated asset is priced. Because the securities are backed by the taxing power of the U.S. government, investors treat them as free of default risk — yet Figure 3.5 and Figure 3.6 show that *default-free* is not *risk-free*. A Treasury note held for one month must be sold at whatever yield then prevails, so its return varies with interest rates; this is **interest-rate risk**, and it grows with the length of the promise. The two figures are constructed identically — a par note is bought at the prevailing constant-maturity yield and repriced one month later — and differ only in maturity, yet the 10-year note's monthly returns (standard deviation about 1.8%) are two and a half times as volatile as the 2-year's (about 0.7%). The sensitivity of a bond's price to its yield is called *duration*, and we will measure it carefully in the fixed-income chapters; the comparison here is the phenomenon that concept exists to describe. Note also what neither histogram shows: the violent *left* tail of the equity distribution. The worst month for the 2-year note in fifty years was a loss of about 2%, and for the 10-year about 8% — losses an order of magnitude milder than equities'. What asymmetry these series do display runs the other way: the 2-year note's skewness of +2.3 in Table 3.2 reflects a long *right* tail, the occasional very large gain that arrives when interest rates fall sharply. A default-free bond's bad outcomes are bounded by how far rates can rise; its good outcomes are not correspondingly bounded, which is the reverse of the credit-risky pattern taken up next.

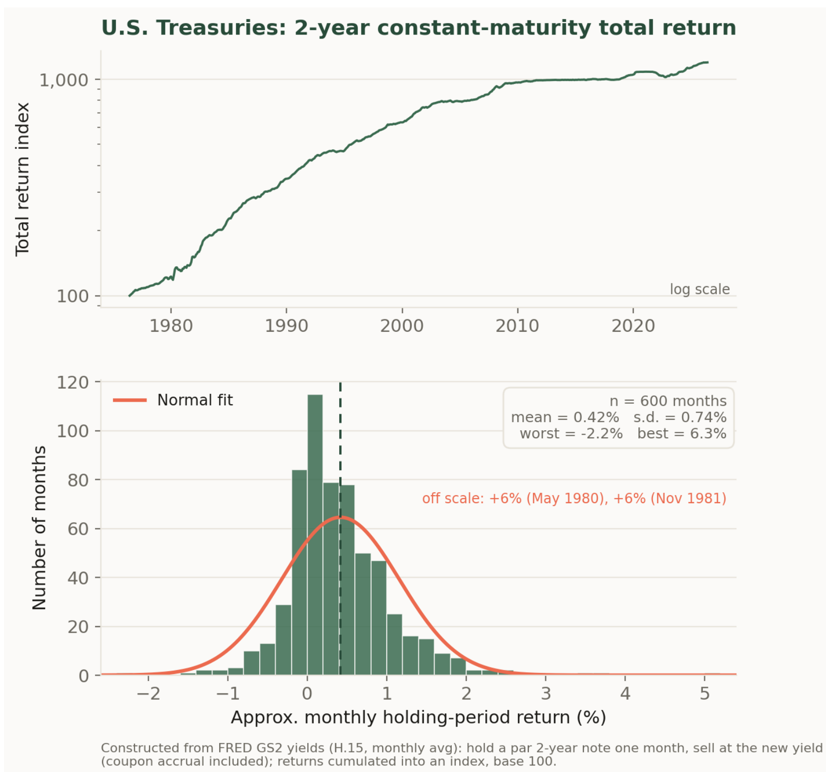

Figure 3.5: The 2-year Treasury note: total return index (top) and monthly holding-period returns (bottom), 1976–2026. Returns approximate buying a par note at the constant-maturity yield and selling one month later at the new yield. Source: Federal Reserve H.15 (FRED series GS2).

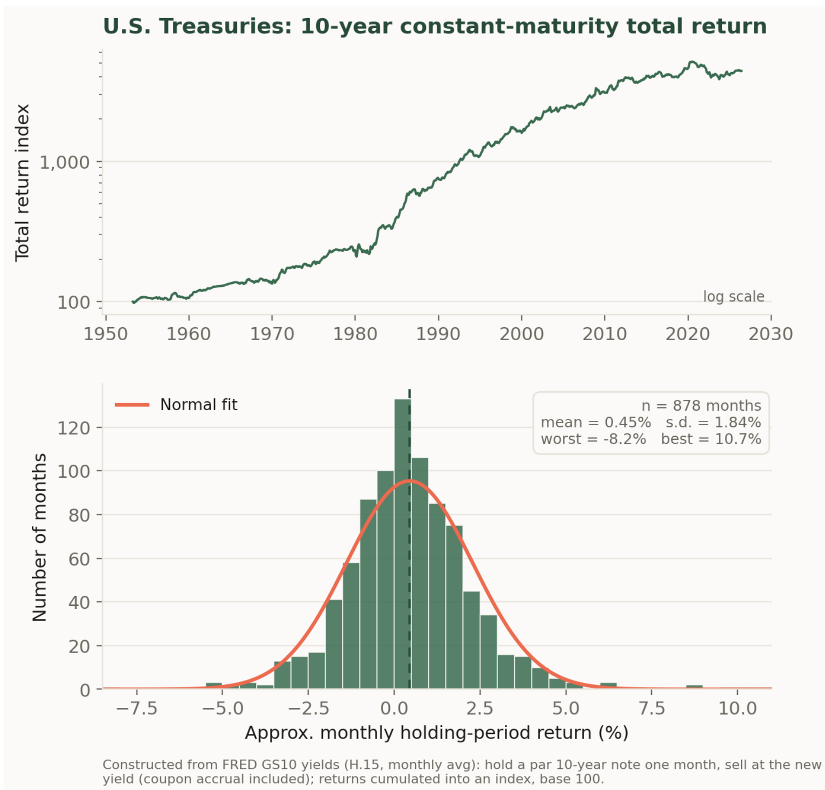

Figure 3.6: The 10-year Treasury note: total return index (top) and monthly holding-period returns (bottom), 1953–2026, constructed as in Figure 3.5. The longer maturity makes the same movements in interest rates far more consequential. Source: Federal Reserve H.15 (FRED series GS10).

Corporate credit. Corporations borrow too, but a corporate promise may not be kept, so corporate bonds carry **default risk** on top of interest-rate risk. Rating agencies grade that risk, and the market splits at a bright line: bonds rated BBB−/Baa3 or better are **investment grade**, while those below are **high yield** (less politely, *junk bonds*). Figure 3.7 and Figure 3.8 show total return indices and monthly returns for the two halves of the market. Investment-grade credit looks like a slightly riskier cousin of the 10-year Treasury — similar volatility, a modestly higher average return as compensation for bearing default risk, and a nearly symmetric distribution. High yield is a different animal. Its overall standard deviation (about 2.3%) is only modestly higher than investment grade's, but the *shape* of its distribution is sharply asymmetric: the skewness is strongly negative and the left tail is long, with the worst month — October 2008 — a loss of more than 16%, roughly seven standard deviations below the mean if the normal fit were taken literally. The economics behind the shape is worth internalizing: a bond's upside is capped, since the best that can happen is that it is repaid in full, while its downside runs all the way to default. High-yield bonds are therefore sometimes described as bond-like in good times and equity-like in bad times — a character their histogram wears openly.

Comparing the five distributions. Table 3.2 collects the summary statistics, and reading down its columns turns the five histograms into a single picture of how risk and return are organized across markets. Two of its columns need a word of definition, since they describe features of shape that the mean and standard deviation cannot capture.

Skewness. A measure of a distribution's asymmetry about its mean. A *negative* value indicates a long left tail — losses that are rarer than gains but much deeper when they arrive. A *positive* value indicates the mirror image, a long right tail. A symmetric distribution, such as the normal, has skewness zero.

Excess kurtosis. A measure of tail thickness relative to the normal distribution, which is the benchmark and has excess kurtosis zero. A *positive* value means extreme outcomes in either direction occur more often

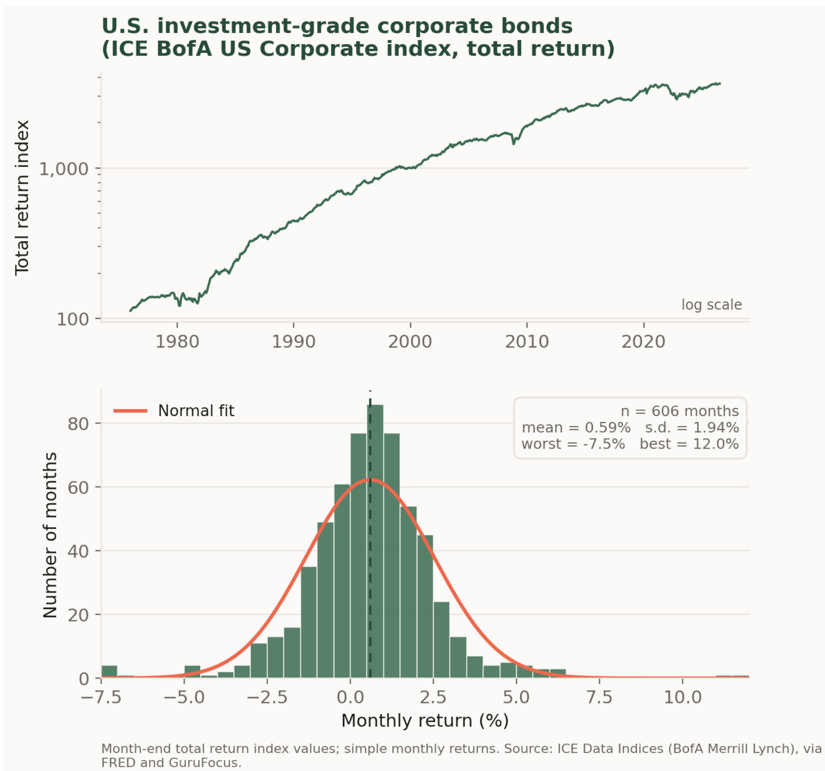

Figure 3.7: U.S. investment-grade corporate bonds: the ICE BofA US Corporate index, total return, 1976–2026. Source: ICE Data Indices (BofA Merrill Lynch), via FRED and GuruFocus.

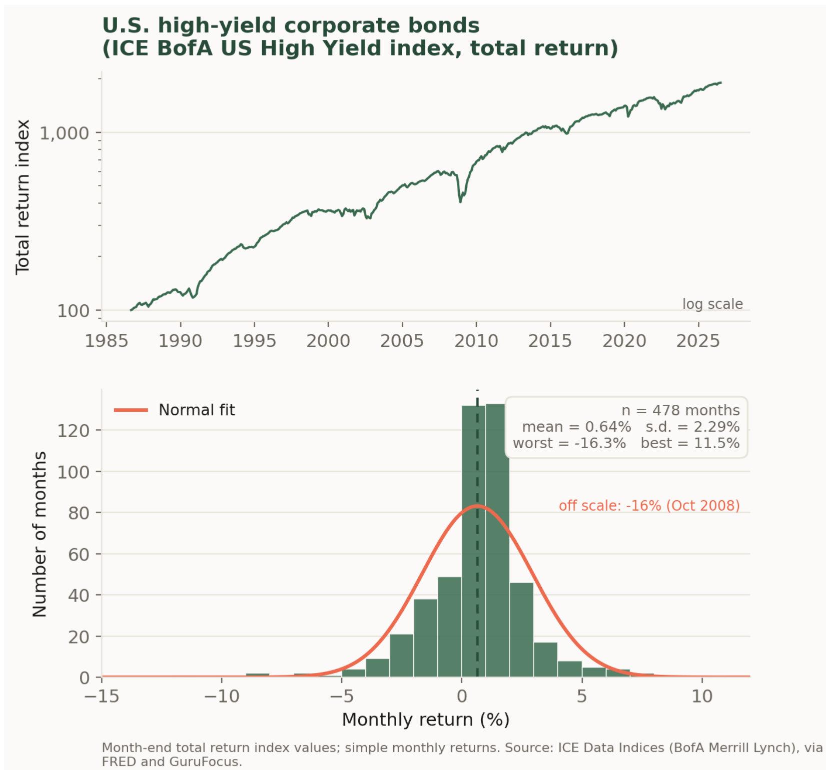

Figure 3.8: U.S. high-yield corporate bonds: the ICE BofA US High Yield index, total return, 1986–2026. Note the long left tail: the worst month, October 2008, lies off the edge of the chart. Source: ICE Data Indices (BofA Merrill Lynch), via FRED and GuruFocus.

than the normal predicts — the "fat tails" that make the normal approximation dangerous precisely where accuracy matters most.

The samples differ in length, and the equity series is built from monthly averages, which smooths it slightly, so the comparisons are broad-brush — but the patterns are robust. First, *dispersion is a ladder*: volatility rises from the 2-year Treasury, to the 10-year Treasury and investment-grade credit, to high yield, to equities, and average returns rise broadly along the same ladder. That co-movement of mean and standard deviation — more compensation where there is more risk to compensate — is the central empirical regularity that the pricing theory of the coming chapters is built to explain. Second, the ladder is not a law: the 2-year Treasury's ratio of mean to standard deviation is the highest in the table, flattered by a sample that begins near the 1981 peak in interest rates and rides yields down for four decades. Averages computed over particular histories are evidence, not guarantees. Third, and most important for what follows, *risk is not one number*. The 10-year Treasury and investment-grade credit have nearly identical standard deviations, but they are exposed to different underlying forces — interest rates alone for one, interest rates plus default for the other. High yield and the 10-year Treasury also have similar standard deviations, yet one is roughly symmetric while the other loses 16% in its worst month. Standard deviation is where our study of risk begins in the next chapter, but the tails and asymmetries visible in these histograms — and the question of *when* each asset's bad months arrive, and whether they arrive together — are where it will lead.

Table 3.2: Summary statistics of monthly returns, in percent per month. Treasury returns are approximate holding-period returns constructed from constant-maturity yields; equity returns are computed from monthly-average prices; corporate bond returns are from month-end total return index values.

Asset	Sample	n	Mean	Std. dev.	Skewness	Excess kurtosis	Worst	Best
2-year Treasury	1976–2026	600	0.42	0.74	2.3	13.8	−2.2	6.3
10-year Treasury	1953–2026	878	0.45	1.84	0.6	3.9	−8.2	10.7
Investment-grade corporate	1976–2026	606	0.59	1.94	0.1	5.2	−7.5	12.0
High-yield corporate	1986–2026	478	0.64	2.29	−1.2	10.0	−16.3	11.5
U.S. equities (S&P 500)	1950–2026	917	0.73	3.47	−0.9	3.5	−20.4	12.0

One more set of numbers belongs in this first look at the data, because everything in the chapters on portfolio choice depends on it: how these assets move *together*.

Correlation. A standardized measure of the tendency of two assets' returns to move together, ranging from –1 to +1. A correlation of +1 means the two move in lockstep; –1 means they move in exactly opposite directions; and 0 means knowing one asset's return this month tells you nothing about the other's. The formal definition, in terms of the covariance between the two return series, is given in the next chapter.

Diversification. The reduction in portfolio risk obtained by combining assets that do not move together. Because their fluctuations partially offset one another within the portfolio, the portfolio's volatility falls below the weighted average of its components' volatilities — and it does so without any corresponding sacrifice of expected return, which is what makes diversification the rare free lunch in finance.

Table 3.3 reports the correlations between the five assets' monthly returns over the forty years for which all five series overlap. Three of its entries deserve to be remembered. The two Treasuries are strongly correlated (0.76) — no surprise, since a single force, the level of interest rates, drives them both. High-yield bonds co-move far more with equities (0.64) than with the 10-year Treasury (0.11): their histogram looked equity-like in bad times, and their correlation says the same thing — the default risk that dominates high yield is the same business-cycle risk that drives the stock market. And equities have been roughly *uncorrelated* with Treasuries over this sample (about –0.1), which is precisely why the classic stock-and-bond portfolio works: when assets do not move together, the bad months of one are, on average, ordinary months for the other, and a mix of the two is steadier than either alone. Low correlation is the raw material of diversification, and measuring its consequences precisely is where the mean-variance chapter begins.

Table 3.3: Correlations of monthly returns, September 1986–June 2026 — the longest period over which all five series overlap (n = 478 months).

	2-yr Treasury	10-yr Treasury	IG corporate	High yield	Equities
2-year Treasury	1.00	0.76	0.39	0.03	−0.14
10-year Treasury	0.76	1.00	0.52	0.11	−0.09
Investment-grade corporate	0.39	0.52	1.00	0.59	0.28
High-yield corporate	0.03	0.11	0.59	1.00	0.64
U.S. equities (S&P 500)	−0.14	−0.09	0.28	0.64	1.00

Two classes of asset remain. **Currencies** — claims used to settle transactions across national monies — and **derivatives** — securities such as forwards, futures, and options whose payoffs are defined in terms of some other, underlying asset — have return properties that depend so directly on their contractual structure that they are best studied alongside that structure, and we take them up in the chapters on forwards, futures, and options later in these notes.

3.3 Application: A founder deciding how to go public

Imagine the chief financial officer of a fast-growing software company weighing how to raise its next round of capital and give early investors a way to cash out. The choices on the table are a traditional initial public offering, a direct listing, or remaining private and selling a stake to a large fund. Framed in this chapter's terms, the decision is about which asset class the firm's claims will belong to and who will bear their risk. Listing converts a concentrated private stake into publicly traded equity, and equity — as Table 3.1 records — is the class investors have historically demanded roughly 9% a year to hold, precisely because its returns carry the volatility and the fat left tail visible in Figure 3.4. That required return *is* the firm's cost of equity capital. It is also why the CFO's alternative of raising debt is cheaper in coupon terms and more dangerous in obligation terms: bondholders are paid first and their claim is fixed, which is the same asymmetry that gives corporate credit its own distinctive return distribution. Deciding how to go public is, at bottom, deciding what the firm will pay for capital, and the historical record surveyed in this chapter is where that price comes from.

Further listening

The Indicator from Planet Money (NPR) — "The SpaceX IPO drama explained": why a company goes public, and how the primary market has changed now that giants like SpaceX can stay private for so long.

3.4 Homework problems

3.4.1 Conceptual

MKT-C1. Distinguish *equity* from *fixed income* in terms of the claim each represents. In your answer, state who is an owner and who is a creditor, which claim is paid first when the issuing firm distributes cash, and why the label *residual claimant* applies to only one of them. Then explain why this ordering — not any difference in the firms involved — is the fundamental reason equity has historically earned a higher average return than the same firm's debt.

MKT-C2. The chapter reports that outstanding derivative contracts had a *notional* value of roughly \$700 trillion at the end of 2024 but a *gross market value* of only about \$17.6 trillion. Explain what each of these two numbers measures, and use the chapter's interest-rate-swap example to explain why the notional figure vastly overstates the economic size of the derivatives market. Which of the two belongs in a chart comparing derivatives to stocks and bonds, and why?

MKT-C3. Global debt securities (about \$145 trillion) are worth more than global equities (about \$127 trillion), yet equities dominate financial news coverage. Offer an explanation for this mismatch in attention. In your answer, note what fraction of the debt total is sovereign rather than corporate, and explain why the sovereign share has grown since 2008 and again since 2020.

MKT-C4. A share of common stock makes its holder a *residual claimant* on the firm. Explain what this means for the shareholder's cash flows in a very good year and in a year in which the firm barely covers its obligations. Then explain why residual status is the source of both equity's higher average return and the fat left tail visible in Figure 3.4.

MKT-C5. Explain the roles of *control* and *limited liability* in the definition of common stock. Why does limited liability make a shareholder's downside bounded but her upside unbounded, and how does this asymmetry differ from the payoff to a bondholder of the same firm? Relate your answer to why *high-yield* corporate bond returns are strongly negatively skewed (skewness −1.2 in Table 3.2) while equity returns, though also negatively skewed, retain a long right tail as well.

MKT-C6. The number of U.S.-listed domestic companies peaked at 8,090 in 1996 and had fallen to roughly 3,900 by 2025 (Figure 3.2), even as the aggregate value of the U.S. stock market multiplied many times over (Figure 3.3). Explain how both facts can be true simultaneously, and give three distinct forces that contributed to the decline in the number of listings.

MKT-C7. Explain what a stock market *index* is and what it means for the S&P 500 to be *capitalization-weighted*. Why does a capitalization-weighted index of roughly five hundred firms serve as a reasonable stand-in for "the market" as a whole, and what kind of company's fortunes would such an index systematically underweight?

MKT-C8. The chapter defines the simple return $r_t = P_t/P_{t-1} - 1$ and asserts that returns, not price levels, are "the raw material of everything that follows." Explain why the theory of the following chapters is stated in returns rather than prices. In your answer, address comparability across assets of different price levels and why the *level* of an index is close to meaningless on its own.

MKT-C9. Describe the four features of the S&P 500 monthly return histogram (Figure 3.4) that the chapter highlights: where the distribution is centered, how spread out it is, how well the fitted normal density describes the middle, and how it fails in the left tail. Explain what economic meaning attaches to each of the first two, and why the failure in the left tail is the one that matters most to an investor.

MKT-C10. U.S. Treasury securities are regarded as free of default risk, yet Figure 3.5 and Figure 3.6 show that their monthly returns are far from constant. Explain the source of this variation, why the 10-year note's returns are roughly two and a half times as volatile as the 2-year's when the two figures are constructed identically, and what the term *duration* names. Explain the slogan "default-free is not risk-free."

MKT-C11. High-yield corporate bonds have a standard deviation only modestly above investment-grade credit's, yet their return distribution is sharply negatively skewed with a worst month of roughly –16%. Explain the economics that produces this asymmetry, using the observation that a bond's upside is capped at repayment in full while its downside runs all the way to default. Explain what is meant by the description that high-yield bonds are "bond-like in good times and equity-like in bad times," and how Table 3.3 corroborates it.

MKT-C12. The chapter concludes that "risk is not one number." Support this claim using two comparisons from Table 3.2: the 10-year Treasury versus investment-grade credit (nearly identical standard deviations), and the 10-year Treasury versus high yield (similar standard deviations, very different shapes). Then explain, using Table 3.3, why an asset's standard deviation in isolation is not the same as the risk it contributes to a portfolio, and why the near-zero equity–Treasury correlation is what makes the classic stock-and-bond portfolio work.

3.4.2 Quantitative

MKT-Q1. A stock index stands at $P_0 = 4{,}820$ at the end of December, $P_1 = 4{,}675$ at the end of January, and $P_2 = 4{,}953$ at the end of February. (a) Using $r_t = P_t/P_{t-1} - 1$, compute the January and February simple returns. (b) Compute the two-month return directly from P_0 and P_2. (c) Show that the two-month return is *not* the sum of the two monthly returns, and state the correct relationship between them.

MKT-Q2. The S&P 500 stood at roughly 17 in 1950 and above 7,000 in 2026 (Figure 3.3), a span of 76 years. (a) Compute the total growth factor over the period. (b) Compute the average annual compound growth rate g satisfying $17(1+g)^{76} = 7000$. (c) Table 3.1 reports an average annual return of 8.7% for the S&P 500. Explain why your answer to (b) is *smaller* than that figure, and why it would still be smaller even if the two were measured over identical samples.

MKT-Q3. Table 3.2 reports S&P 500 monthly returns with a mean of 0.73% and a standard deviation of 3.47%. Annualize both using the chapter's convention — multiply the mean by 12 and the standard deviation by $\sqrt{12}$ — and confirm that your answers reproduce the equity row of Table 3.1. Then explain why the standard deviation scales with $\sqrt{12}$ rather than with 12.

MKT-Q4. Repeat the annualization of MKT-Q3 for the 2-year Treasury (monthly mean 0.42%, standard deviation 0.74%) and the high-yield corporate index (mean 0.64%, standard deviation 2.29%). Compute the ratio of annualized mean return to annualized standard deviation for each, and for the S&P 500 from MKT-Q3. Which asset has the highest ratio, and what caution does the chapter attach to reading that result as a statement about the future?

MKT-Q5. Treat S&P 500 monthly returns as normally distributed with mean

0.73% and standard deviation 3.47%. (a) How many standard deviations below the mean is a return of –10%? Using a normal table, what probability does the normal assign to such a month, and how many such months would you expect in the $n = 917$ months of the sample? (b) The actual worst month in the sample was –20.4%. How many standard deviations below the mean is that, and roughly how often would a normal distribution produce it? (c) What do your answers demonstrate about the adequacy of the normal approximation?

MKT-Q6. High-yield corporate bonds have a monthly mean of 0.64% and standard deviation of 2.29% over $n = 478$ months, with a worst month (October 2008) of –16.3%. (a) Express the worst month as a number of standard deviations below the mean and verify the chapter's claim that it is "roughly seven standard deviations." (b) Compare the frequency a normal distribution would assign to such an event against the once-in-478-months frequency actually observed. (c) Explain why an investor who sized her positions using standard deviation alone would have badly misjudged this market.

MKT-Q7. Figure 3.1 is built from the following end-2024 market values: equities \$126.7 trillion, sovereign debt \$75.5 trillion, corporate debt \$69.6 trillion, and derivatives at a gross market value of \$17.6 trillion. (a) Compute the total and each class's percentage share. (b) Now rebuild the pie using the \$700 trillion *notional* value of derivatives in place of gross market value, and recompute the derivatives share. (c) Comment on the difference between your two answers and on what the second version would lead a reader to believe.

MKT-Q8. Consider a portfolio holding equities and the 10-year Treasury, with monthly standard deviations $\sigma_E = 3.47\%$ and $\sigma_B = 1.84\%$ and correlation $\rho = -0.09$ from Table 3.3. The standard deviation of a portfolio holding weight w in equities is

$$\sigma_P = \sqrt{w^2\sigma_E^2 + (1-w)^2\sigma_B^2 + 2w(1-w)\rho\,\sigma_E\sigma_B}.$$

(a) Compute σ_P for $w = 0.5$. (b) Compute the weighted average $0.5\sigma_E + 0.5\sigma_B$

and explain why σ_P falls short of it. (c) Recompute σ_P at $w = 0.5$ assuming instead that $\rho = 0.9$, and state what happens to the diversification benefit as correlation rises toward 1.

Part I

Individual portfolio selection and market equilibrium

Chapter 4

Mean-Variance Portfolios

4.1 Introduction

Three questions motivate the study of mean-variance portfolio theory. First, given a risky asset and a risk-free asset, how should a rational investor divide her wealth between them, and how does the answer depend on her degree of risk aversion? Second, when multiple risky assets are available, how does the correlation structure among their returns shape the risk of the combined portfolio, and what role does diversification play? Third, across all possible ways of combining risky assets, which portfolios minimize risk for each level of expected return — and what does the answer imply about which risky portfolio every investor should hold regardless of her individual risk tolerance?

These questions are central to finance for both practical and theoretical reasons. Investors do not seek high returns in isolation; they seek high returns relative to the risk they must bear. Without a formal way to describe the feasible set of return-risk combinations and to find the combinations that are not dominated by any other, portfolio construction would rely on intuition rather than optimization. Moreover, the efficient frontier and the separation property derived in this chapter are the direct inputs to the Capital Asset Pricing Model, which translates the logic of optimal diversification into a theory of equilibrium asset prices.

The first question is addressed through the capital allocation line. When an investor can combine a single risky portfolio with a risk-free asset, the achievable mean-standard deviation combinations form a straight line whose slope — the Sharpe ratio — measures the reward per unit of risk. A mean-variance investor with risk aversion coefficient A optimally allocates a fraction of wealth to the risky asset equal to the risk premium divided by the product of risk aversion and portfolio variance. This formula makes precise the intuition that more risk-averse investors hold less of the risky asset and that assets with higher risk premia or lower variance attract greater demand.

The second question is answered by working out the variance of a portfolio of two or more risky assets. Portfolio variance depends not only on the individual asset variances and portfolio weights, but crucially on the covariances — or equivalently the correlations — between asset returns. When assets are not perfectly correlated, combining them reduces total portfolio variance below the weighted average of individual variances, a result known as diversification. An asset with negative correlation with the rest of the portfolio — a hedge asset — is especially valuable because it reduces portfolio risk while contributing positively to expected return in proportion to its own expected return. The degree of risk reduction available through diversification is therefore a function of the entire covariance matrix of asset returns.

The third question leads to the efficient frontier and the separation property. The chapter develops this by solving the variance-minimization problem formally with Lagrangian methods for the two-asset case; the general case of many assets, which uses matrix algebra to characterize the entire set of minimum-variance portfolios, is worked out in an appendix at the end of the chapter. The first-order conditions reveal that at any point on the efficient frontier, the ratio of excess expected return to covariance with the portfolio must be equal across all assets. When a risk-free asset is available, the efficient frontier collapses to a single ray: the capital allocation line constructed from the risk-free rate to the tangency portfolio on the risky-asset frontier. Because this tangency portfolio is the same for every investor regardless of risk aversion, all investors hold the same risky portfolio composition, differing only in how much of their wealth they allocate to it — a result known as the separation property.

> **In the news**
>
> Diversification is only as real as the covariances beneath it. By early 2026, the "Magnificent Seven" technology stocks had grown to roughly a third of the S&P 500 — so concentrated that a broad index fund, the textbook diversified portfolio, had quietly become a large bet on a handful of correlated names. Fortune argued there was "nowhere to hide." This is the mean–variance problem in the headlines: how much true diversification does a portfolio actually provide once its holdings move together? Read it at Fortune.

4.2 Preferences

Consider an investor that has preferences that can be represented by the utility function $U = Er - \frac{1}{2}A\sigma^2$. Indifference curves for this investor can be plotted with the expected return on the y axis and the *standard deviation* on the x axis. They slope upward and are convex.

4.3 Assets

The investor has available to her one risky and one risk-free asset. Looking forward we view the risky asset as a portfolio of assets that has been selected optimally. We will learn how to select this portfolio later. The (random) rate of return for the risky portfolio is r_P. Its standard deviation is σ_P. The rate of return on the risk free asset is r_f. If the investor puts proportion y of her total available capital in the risky asset, after uncertainty has been resolved the investor will have $W_0(1 + r_C) = W_0 y(1 + r_P) + W_0(1 - y)(1 + r_f)$.

In terms of the state-preference model, if the risky asset costs p and the risk free asset costs p_f, then total expenditures initially are $x_P p + x_f p_f$. The value of the portfolio next period is $x_P v_P + x_f v_f$. Algebra implies that $r_C = w_P r_p + w_f r_f = y r_P + (1-y) r_f$. The expected value of this return is $yEr_P + (1-y)r_f$. The variance of this portfolio is $y^2\sigma_P^2$ and the standard deviation is $y\sigma_P$.

Suppose $y = 0$. What is the expected return of the portfolio? Standard deviation? Now suppose that $y = 1$. What are expected return and standard deviation? What about $y = \frac{1}{2}$?

One can think about increasing the expected return of the portfolio from r_f to Er_P by moving y from 0 to 1. As one does so, one must trade off increases in expected return with increases in the standard deviation of the complete portfolio.

Notice that the expected return of the complete portfolio as a function of its standard deviation is a straight line with intercept r_f and slope of

$$\frac{Er_P - r_f}{\sigma_P}$$

Why? The standard deviation of the portfolio is given by $\sigma_C = y\sigma_P$. This implies that $y = \frac{\sigma_C}{\sigma_P}$. Plugging this into the equation $Er_C = yEr_P + (1-y)r_f = r_f + y(Er_P - r_f)$ yields

$$Er_C = r_f + \frac{\sigma_C}{\sigma_P}(Er_P - r_f)$$

showing that the set of possible complete portfolios implies a trade off between risk and return that has slope $\frac{Er_P - r_f}{\sigma_P}$. This can be drawn in the σ, Er plane as an upward sloping line. The slope of this line represents the trade off that the market presents this investor between risk and return. This slope is also known as the *Sharpe ratio*. The line is called the *capital allocation line* (CAL).

4.4 The decision problem

From this the decision problem of the investor can be seen to be

$$\max Er_C - \frac{1}{2}A\sigma_C^2 \text{ s.t. } Er_C = r_f + \frac{\sigma_C}{\sigma_P}(Er_P - r_f)$$

This can be solved by setting up the Lagrangian, or simply by substitution. Since $Er_C = r_f + y(Er_P - r_f)$ and $\sigma_C = y\sigma_P$ we can rewrite utility as

$$U = r_f + y(Er_P - r_f) - \frac{1}{2}Ay^2\sigma_P^2$$

The first order conditions with respect to y are

$$(Er_P - r_f) - yA\sigma_P^2 = 0$$

which implies that the optimal amount to put into the risky asset is

$$y = \frac{Er_P - r_f}{A\sigma_P^2}$$

Notice that this increases *ceteris paribus* as the risk premium increases and as risk aversion or the variance of the portfolio decreases. So, investors that are more risk averse, will choose to be further to the left on the CAL whereas less risk averse investors will be more to the right.

4.5 Diversification

Where does the optimal risky portfolio come from? To understand this, we must understand the return/risk tradeoff for general portfolios. Consider a portfolio of two risky assets labelled D and E. We have already seen that if the weight put on asset i in the portfolio is w_i then, the return to the whole portfolio will be $r_P = w_D r_D + w_E r_E$. What is the variance of the portfolio?

$$\sigma_P^2 = w_D^2\sigma_D^2 + w_E^2\sigma_E^2 + 2w_D w_E Cov(r_D, r_E)$$

Why? Write the variance as the expected squared deviation from the mean, $\mu_P = w_D\mu_D + w_E\mu_E$, and group the deviation asset by asset:

$$\begin{aligned}
\sigma_P^2 &= E\left[(w_D r_D + w_E r_E - w_D\mu_D - w_E\mu_E)^2\right] \\
&= E\left[\left(w_D(r_D - \mu_D) + w_E(r_E - \mu_E)\right)^2\right] \\
&= E\left[w_D^2(r_D - \mu_D)^2 + w_E^2(r_E - \mu_E)^2 + 2w_D w_E(r_D - \mu_D)(r_E - \mu_E)\right] \\
&= w_D^2\sigma_D^2 + w_E^2\sigma_E^2 + 2w_D w_E Cov(r_D, r_E),
\end{aligned}$$

where the last line uses the definitions $\sigma_i^2 = E(r_i - \mu_i)^2$ and $Cov(r_D, r_E) = E(r_D - \mu_D)(r_E - \mu_E)$. Recall also that if ρ_{DE} is the correlation coefficient of r_D and r_E then

$$\rho_{DE} = \frac{Cov(r_D, r_E)}{\sigma_D\sigma_E}.$$

This implies that the variance of the portfolio can be rewritten

$$\sigma_P^2 = w_D^2\sigma_D^2 + w_E^2\sigma_E^2 + 2w_D w_E\sigma_D\sigma_E\rho_{DE}$$

Think about the case when $\rho_{DE} = 1$. In that case the rhs of the above equation is a perfect square and simplifies to

$$\sigma_P^2 = (w_D\sigma_D + w_E\sigma_E)^2$$

so the portfolio standard deviation is the weighted average of the two standard deviations, $\sigma_P = w_D\sigma_D + w_E\sigma_E$ — diversification buys nothing. It can be seen that the portfolio variance is strictly decreasing in the correlation between assets. An asset is called a *hedge* if it has negative correlation with the other assets in the portfolio. The lowest possible portfolio variance is given by $\rho_{DE} = -1$. In this case

$$\sigma_P^2 = (w_D\sigma_D - w_E\sigma_E)^2$$

So one can eliminate all risk by taking w_D and w_E to make the above zero.

All of this algebra becomes a single picture when drawn with real assets. Figure 4.1 traces the mean and standard deviation of every portfolio of the U.S. stock market and a 10-year Treasury, using the last fifteen years of monthly returns. Over this sample the two assets are essentially uncorrelated ($\rho = -0.03$), and the dashed line shows the counterfactual just derived: if ρ were +1, the frontier would collapse to the straight chord and portfolio risk would be the weighted average of the two standard deviations. The bow of the solid curve to the left of that chord *is* the diversification benefit, in units of standard deviation avoided. Note the point marked on the curve: the minimum-variance mix holds only about a fifth of the portfolio in stocks, yet carries less risk than a portfolio of bonds alone — an impossibility if risks simply added.

One interesting statistic to calculate is the covariance of any particular asset's returns to the return of the whole portfolio. We are interested in finding $Cov(r_D, w_Dr_D + w_Er_E)$ for example.

$$\begin{aligned}
Cov(r_D, r_P) &= E(r_D - \mu_D)(w_Dr_D + w_Er_E - w_D\mu_D - w_E\mu_E) \\
&= E(r_D - \mu_D)\left(w_D(r_D - \mu_D) + w_E(r_E - \mu_E)\right) \\
&= w_DE(r_D - \mu_D)^2 + w_EE(r_D - \mu_D)(r_E - \mu_E) \\
&= w_DVar(r_D) + w_ECov(r_D, r_E).
\end{aligned}$$

The second line simply collects the portfolio's mean-deviation asset by asset, and the third distributes the factor $(r_D - \mu_D)$ across the two terms and takes expectations of each.

In general, for a portfolio with weights $w_1, \ldots, w_n$ and return $r_P = \sum_{i=1}^n w_ir_i$, the covariance of the return to asset j with the whole portfolio follows from the linearity of covariance:

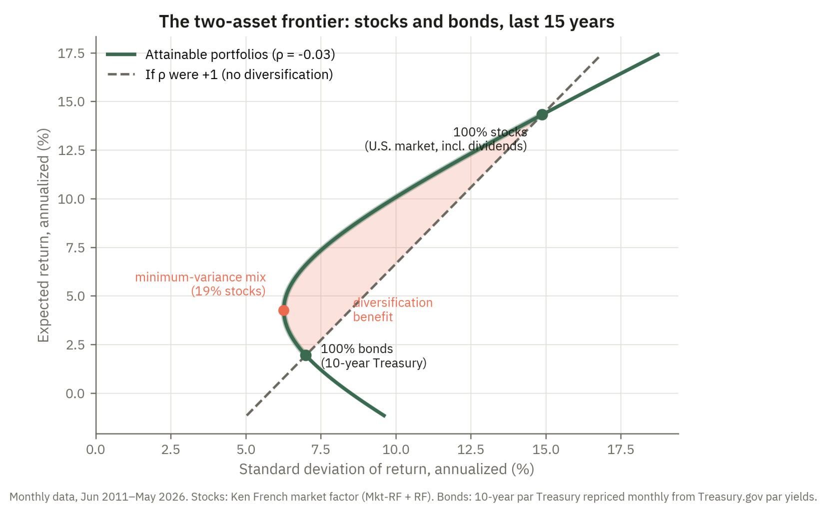

Figure 4.1: Every attainable mix of the U.S. stock market (Ken French market factor, including dividends) and a 10-year Treasury, from the last fifteen years of monthly data. The dashed chord is what the frontier would be if the correlation were +1; the shaded gap is the diversification benefit. Sources: Ken French Data Library; U.S. Department of the Treasury.

$$Cov(r_j, r_P) = Cov\left(r_j, \sum_{i=1}^{n} w_i r_i\right) = \sum_{i=1}^{n} w_i \, Cov(r_i, r_j).$$

This is interesting because it means that the variance of the portfolio can be broken up into the covariance of each of the components of the portfolio with the whole portfolio: multiplying by w_j and summing over j gives $\sum_j w_j Cov(r_j, r_P) = Cov(r_P, r_P) = \sigma_P^2$.

4.6 Minimum-variance portfolios

By changing the portfolio weights, one changes the variance and standard deviation of the portfolio. Since preferences depend only on the mean and standard deviation of returns, a natural question is: for each expected return, what is the lowest-variance portfolio one can obtain? This is the problem

$$\min Var(r_P) \text{ s.t. } Er_P \geq \bar{r}.$$

The set of solutions — one minimum-variance portfolio for each target expected return — traces out the *efficient frontier*. We solve this problem here for two risky assets; the general case of many assets, which uses matrix algebra, is developed in the appendix at the end of the chapter.

4.7 The two asset case

Consider the problem

$$\min \frac{1}{2}\left(w_D(w_D\sigma_D^2 + w_E\sigma_{DE}) + w_E(w_E\sigma_E^2 + w_D\sigma_{DE})\right) \text{ s.t. } w'\mu \geq \bar{r} \text{ and } w'\mathbf{1} = 1$$

The first order conditions are

$$w_D\sigma_D^2 + w_E\sigma_{DE} + \lambda\mu_D - \gamma = 0$$
$$w_E\sigma_E^2 + w_D\sigma_{DE} + \lambda\mu_E - \gamma = 0$$

Recall that $w_D\sigma_D^2 + w_E\sigma_{DE}$ is the covariance of the asset D with the rest of the portfolio, which we have denoted σ_{DP}. To interpret these first order conditions, divide the first one by σ_{DP} and the second one by σ_{EP} to obtain

$$1 + \frac{\lambda\mu_D - \gamma}{\sigma_{DP}} = 0$$
$$1 + \frac{\lambda\mu_E - \gamma}{\sigma_{EP}} = 0$$

Now define $\theta = \gamma/\lambda$. Solving each first-order condition for the ratio $(\mu_i - \theta)/\sigma_{iP}$ gives $-1/\lambda$ in both cases, so the two equations imply that

$$\frac{\mu_D - \theta}{Cov(r_D, r_P)} = \frac{\mu_E - \theta}{Cov(r_E, r_P)}.$$

Because this common ratio is the same for asset D and asset E, it is also unchanged if we replace a single asset by any portfolio of the two. In particular, apply it to the risky portfolio P itself. Taking a weighted average of the numerators and denominators with weights w_D, w_E leaves the ratio equal to $(Er_P - \theta)/Cov(r_P, r_P)$, and since $Cov(r_P, r_P) = \sigma_P^2$,

$$\frac{Er_i - \theta}{Cov(r_i, r_P)} = \frac{Er_P - \theta}{\sigma_P^2} \text{ for all } i.$$

Finally, identify θ. The quantity θ is the expected return of an asset that has zero covariance with P (set $Cov(r_i, r_P) = 0$ above and the numerator must vanish, giving $Er_i = \theta$). When a risk-free asset is available it has exactly this property, so $\theta = r_f$, yielding

$$\frac{Er_i - r_f}{Cov(r_i, r_P)} = \frac{Er_P - r_f}{\sigma_P^2} \text{ for all } i.$$

The same minimization can be carried out for an arbitrary number of risky assets using matrix algebra. That derivation traces out the full minimum-variance frontier, identifies the global minimum-variance portfolio, and establishes the two-fund separation result. Readers who are interested can find it in the appendix at the end of this chapter.

4.8 The capital allocation line (CAL)

Notice that for a given risk free rate r_f, and a given efficient frontier, one may draw a capital allocation line as the highest line that passes through the risk free rate r_f and the efficient frontier. This will be the same for all investors as long as everyone agrees on the efficient frontier. This implies that the composition of the risky portfolio is independent of the risk aversion of the individual. This property is called *separation*.

4.9 Application: An advisor confronting concentration

A financial advisor sits across from a client who is delighted that her low-cost index fund has risen sharply, but who does not realize that a handful of correlated technology names now drive most of that fund's movements. The advisor's job is to translate that unease into numbers, and mean-variance analysis is the instrument. By estimating the covariances among the client's holdings, the advisor can show how much of the portfolio's risk is genuinely diversified away and how much is concentrated systematic exposure that a few disappointing earnings reports could expose. The efficient frontier then frames the conversation about whether to trim the position: it makes precise the gain in expected return per unit of risk the client is giving up — or buying — by tilting toward or away from the crowded trade.

Further listening

Odd Lots (Bloomberg) — "Meb Faber on the Big Bear Market in Diversification and Tactical Allocation": a practitioner's take on why diversification works and how to spread risk across assets — the mean–variance problem in action.

4.10 Appendix: The general N-asset case

This appendix solves the minimum-variance problem for an arbitrary number of risky assets using matrix algebra.

Suppose that the assets are labelled $1, \ldots, k$, all of which have different expected returns and variances. Let w be a vector of portfolio weights for each asset. Let the expected return *vector* for these assets be μ and let Σ be the variance-covariance matrix of the assets. The expected portfolio return will be given by $w'\mu$ and the variance will be $w'\Sigma w$. The problem of interest is

$$\min w'\Sigma w \text{ s.t. } w'\mu \geq \bar{r} \text{ and } w'\mathbf{1} = 1$$

It is convenient to minimize $\frac{1}{2}w'\Sigma w$ because it has the same solution, but more convenient expressions. The Lagrangian for this problem is

$$\mathcal{L} = \frac{1}{2}w'\Sigma w + \lambda(\bar{r} - w'\mu) + \gamma(1 - w'\mathbf{1})$$

You know from 388 that the first order conditions to this problem are

$$\begin{aligned} \frac{1}{2}(\Sigma + \Sigma')w - \lambda\mu - \gamma\mathbf{1} &= 0 \\ \Sigma w &= \lambda\mu + \gamma\mathbf{1} \end{aligned}$$

Premultiplying both sides by Σ^{-1} gives

$$w = \lambda \Sigma^{-1} \mu + \gamma \Sigma^{-1} \mathbf{1}$$

To eliminate the multipliers, premultiply the previous equation by μ' and $\mathbf{1}$ to get

$$\begin{aligned} \mu' w &= \lambda \mu' \Sigma^{-1} \mu + \gamma \mu' \Sigma^{-1} \mathbf{1} \\ \mathbf{1}' w &= \lambda \mathbf{1}' \Sigma^{-1} \mu + \gamma \mathbf{1}' \Sigma^{-1} \mathbf{1} \end{aligned}$$

The left-hand side of the first of these equations is the expected return to the portfolio and the left-hand side of the second is 1. Using the constraints $\mu' w = \bar{r}$ and $\mathbf{1}' w = 1$, the system becomes

$$\begin{aligned} \bar{r} &= \lambda \mu' \Sigma^{-1} \mu + \gamma \mu' \Sigma^{-1} \mathbf{1} \\ 1 &= \lambda \mathbf{1}' \Sigma^{-1} \mu + \gamma \mathbf{1}' \Sigma^{-1} \mathbf{1} \end{aligned}$$

It is convenient to introduce three scalar constants that depend only on the assets (not on the target return $\bar{r}$):

$$A = \mu' \Sigma^{-1} \mathbf{1} = \mathbf{1}' \Sigma^{-1} \mu, \qquad B = \mu' \Sigma^{-1} \mu, \qquad C = \mathbf{1}' \Sigma^{-1} \mathbf{1}, \qquad D = BC - A^2.$$

Note that Σ is a positive definite matrix (since it is a covariance matrix of non-redundant assets), which implies $B > 0$, $C > 0$, and by the Cauchy-Schwarz inequality $D = BC - A^2 > 0$. The two-equation system can now be written compactly as

$$\begin{pmatrix} B & A \\ A & C \end{pmatrix} \begin{pmatrix} \lambda \\ \gamma \end{pmatrix} = \begin{pmatrix} \bar{r} \\ 1 \end{pmatrix}.$$

The determinant of the coefficient matrix is $BC - A^2 = D > 0$, so the system has a unique solution. Applying Cramer's rule gives

$$\lambda = \frac{C\bar{r} - A}{D}, \qquad \gamma = \frac{B - A\bar{r}}{D}.$$

Substituting these back into $w = \lambda\Sigma^{-1}\mu + \gamma\Sigma^{-1}\mathbf{1}$ yields the optimal portfolio weights as an explicit function of the target return:

$$\boxed{w(\bar{r}) = \frac{C\bar{r} - A}{D}\,\Sigma^{-1}\mu + \frac{B - A\bar{r}}{D}\,\Sigma^{-1}\mathbf{1}.}$$

It is instructive to factor this as $w(\bar{r}) = g + \bar{r}\,h$, where

$$g = \frac{B\,\Sigma^{-1}\mathbf{1} - A\,\Sigma^{-1}\mu}{D}, \qquad h = \frac{C\,\Sigma^{-1}\mu - A\,\Sigma^{-1}\mathbf{1}}{D}.$$

Both g and h are fixed vectors (independent of $\bar{r}$). This shows that every minimum-variance portfolio is a linear combination of the two fixed portfolios g and $g + h$, which is the **two-fund separation** result: any point on the efficient frontier can be constructed by mixing two particular portfolios.

4.10.1 Portfolio variance on the frontier

From the first-order condition $\Sigma w = \lambda\mu + \gamma\mathbf{1}$, premultiplying by w' gives

$$\sigma_P^2 = w'\Sigma w = \lambda\,w'\mu + \gamma\,w'\mathbf{1} = \lambda\bar{r} + \gamma.$$

Substituting the expressions for λ and γ:

$$\sigma_P^2 = \frac{C\bar{r} - A}{D}\,\bar{r} + \frac{B - A\bar{r}}{D} = \frac{C\bar{r}^2 - 2A\bar{r} + B}{D}.$$

This is a **parabola** in the $(\bar{r},\,\sigma_P^2)$ plane, and equivalently a **hyperbola** in the $(\sigma_P,\,\bar{r})$ plane that is conventionally plotted with σ_P on the horizontal axis. The entire minimum-variance frontier is traced out as $\bar{r}$ varies.

4.10.2 The global minimum-variance portfolio

The minimum of σ_P^2 over all target returns is found by differentiating with respect to $\bar{r}$ and setting the result to zero:

$$\frac{d\sigma_P^2}{d\bar{r}} = \frac{2C\bar{r} - 2A}{D} = 0 \implies \bar{r}^* = \frac{A}{C}.$$

Substituting back gives the minimum achievable variance:

$$\sigma_{min}^2 = \frac{C(A/C)^2 - 2A(A/C) + B}{D} = \frac{B - A^2/C}{D} = \frac{BC - A^2}{CD} = \frac{1}{C}.$$

The weights of the global minimum-variance portfolio are obtained by setting $\bar{r} = A/C$ in $w(\bar{r})$:

$$w^* = \frac{C(A/C) - A}{D}\,\Sigma^{-1}\mu + \frac{B - A(A/C)}{D}\,\Sigma^{-1}\mathbf{1} = \frac{B - A^2/C}{D}\,\Sigma^{-1}\mathbf{1} = \frac{1}{C}\,\Sigma^{-1}\mathbf{1}.$$

Since $C = \mathbf{1}'\Sigma^{-1}\mathbf{1}$, this simplifies to

$$\boxed{w^* = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}'\Sigma^{-1}\mathbf{1}}.}$$

The global minimum-variance portfolio weights are proportional to $\Sigma^{-1}\mathbf{1}$, normalized so that they sum to one. The portfolio allocates more weight to assets that have low variance and low covariance with other assets, and is independent of expected returns entirely — it is determined solely by the covariance structure.

The **efficient frontier** is the upper half of this hyperbola: the set of minimum-variance portfolios with $\bar{r} \geq A/C$. Portfolios on the lower half offer the same variance as a portfolio on the upper half but with a lower expected return and are therefore dominated.

4.11 Homework problems

4.11.1 Conceptual

MV-C1. Two investors, Ana and Ben, face the same risky portfolio (with return r_P, standard deviation σ_P) and the same risk-free rate r_f. Ana chooses to hold a complete portfolio with $\sigma_C = 0.05$, while Ben holds one with $\sigma_C = 0.15$. Using the geometry of the capital allocation line, explain who is more risk averse and why. Would the *composition* of the risky portion of their portfolios differ? Explain what this reveals about the CAL.

MV-C2. A new brokerage advertises that it can offer clients a "steeper" capital allocation line than its competitors while using the very same underlying risky portfolio. Given that the slope of the CAL is $\frac{Er_P - r_f}{\sigma_P}$, explain what would have to be true for the brokerage's claim to hold, and why an ordinary broker cannot simply make the CAL steeper at will. What does the slope represent economically?

MV-C3. An investor currently holds all her wealth in the risky portfolio ($y = 1$). The risk premium $Er_P - r_f$ then doubles because the risk-free rate falls, while σ_P and her risk aversion A stay the same. Using the optimal allocation rule $y = \frac{Er_P - r_f}{A\sigma_P^2}$, explain the direction in which she should adjust y, and describe qualitatively how her position on the CAL moves. Would a very risk-averse investor and a nearly risk-neutral investor respond in the same direction?

MV-C4. Explain, in words, why the optimal risky fraction $y = \frac{Er_P - r_f}{A\sigma_P^2}$ falls when the coefficient of risk aversion A rises but rises when the variance σ_P^2 falls. Two investors face identical Er_P, σ_P, and r_f but one has twice the risk aversion of the other. Where does each end up on the CAL, and does the difference in A change which risky portfolio they hold?

MV-C5. Suppose two risky assets are being considered for a portfolio. Asset D has a slightly higher variance than asset E, but D has strongly *negative* correlation with E, whereas a third candidate asset F has the same variance as E but is *positively* correlated with it. Explain, using the role of covariance in portfolio variance, why the investor might prefer to add the higher-variance asset D rather than F. What is the term for an asset like D?

MV-C6. Starting from $\sigma_P^2 = w_D^2\sigma_D^2 + w_E^2\sigma_E^2 + 2w_D w_E \sigma_D \sigma_E \rho_{DE}$, explain what happens to portfolio variance as ρ_{DE} moves from +1 down to −1 with the weights and individual standard deviations held fixed. Why does $\rho_{DE} = 1$ give a portfolio standard deviation equal to the weighted average of the individual standard deviations, while $\rho_{DE} < 1$ gives strictly less? What does this reveal about the source of the diversification benefit?

MV-C7. A client protests that her index fund "is already diversified across 500 companies," so correlation should not matter. Referring to the chapter's decomposition $\sigma_P^2 = \sum_j w_j\, Cov(r_j, r_P)$ of portfolio variance into the covariance of each asset with the whole portfolio, explain why the number of holdings is not what determines diversification. Connect your answer to the chapter's "In the news" observation about the Magnificent Seven.

MV-C8. Explain what the statistic $Cov(r_j, r_P)$ measures and why it, rather than an asset's own variance σ_j^2, is the right notion of the "risk that asset j contributes" to a portfolio. Use the identity $\sum_j w_j\, Cov(r_j, r_P) = \sigma_P^2$ to make the argument precise, and describe how an asset with low or negative $Cov(r_j, r_P)$ affects total portfolio risk.

MV-C9. Consider the minimum-variance frontier traced by the problem $\min Var(r_P)$ s.t. $Er_P \geq \bar{r}$. An analyst identifies a portfolio that lies strictly *inside* the frontier (i.e., not on it). Explain in what sense this portfolio is "dominated," and describe the two distinct improvements an investor could make by moving to the frontier. Why does no rational mean-variance investor hold an interior portfolio?

MV-C10. Two portfolios on the minimum-variance frontier have the *same* variance σ_P^2 but different expected returns. Explain how this is possible given that the frontier is a parabola in the $(\bar{r}, \sigma_P^2)$ plane with vertex at $\bar{r}^* = A/C$, and state which of the two a mean-variance investor would ever choose. What is the name for the acceptable (upper) half of the frontier?

MV-C11. The separation property says every investor holds the same risky portfolio regardless of risk aversion. A skeptic argues, “That can’t be right — surely a cautious retiree and an aggressive 25-year-old hold different things.” Reconcile the skeptic’s intuition with the separation result. Precisely where does the difference between the two investors show up, and where does it *not*?

MV-C12. In the appendix, every frontier portfolio is written as $w(\bar{r}) = g + \bar{r}\,h$ for two fixed vectors g and h. Explain how this algebraic fact establishes the two-fund separation result — that any frontier portfolio can be built by mixing just two fixed portfolios. When a risk-free asset is added, why does the relevant risky portfolio in the separation result become a single one (the tangency portfolio) common to all investors?

MV-C13. The global minimum-variance portfolio weights $w^* = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}'\Sigma^{-1}\mathbf{1}}$ depend only on the covariance matrix Σ and not at all on expected returns μ. Explain intuitively why this is the case, and describe a practical situation in which an investor might deliberately choose to hold this portfolio despite giving up expected return.

MV-C14. Explain, using the structure of $w^* = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}'\Sigma^{-1}\mathbf{1}}$, why the global minimum-variance portfolio tends to place more weight on assets with low variance and low covariance with the others. In practice, estimates of expected returns are far noisier than estimates of covariances; explain why this makes w^* attractive to some practitioners relative to a portfolio that also targets a particular expected return.

4.11.2 Quantitative

MV-Q1. A risky portfolio has $Er_P = 0.11$ and $\sigma_P = 0.20$, and the risk-free rate is $r_f = 0.03$. (a) Write the equation of the capital allocation line, Er_C as a function of σ_C. (b) What is the Sharpe ratio? (c) What expected return corresponds to a complete-portfolio standard deviation of $\sigma_C = 0.08$?

MV-Q2. Using the CAL $Er_C = r_f + \frac{\sigma_C}{\sigma_P}(Er_P - r_f)$ with $r_f = 0.02$, $Er_P = 0.10$, and $\sigma_P = 0.25$: (a) An investor targets an expected complete-portfolio return of $Er_C = 0.06$. What standard deviation σ_C must she accept? (b) What fraction

y of wealth is she holding in the risky portfolio? (Recall $\sigma_C = y\sigma_P$.)

MV-Q3. An investor has risk aversion $A = 4$, faces $Er_P = 0.12$, $\sigma_P = 0.20$, and $r_f = 0.04$. (a) Compute the optimal fraction y in the risky portfolio using $y = \frac{Er_P - r_f}{A\sigma_P^2}$. (b) What are the expected return and standard deviation of her complete portfolio? (c) If instead $A = 2$, recompute y and comment on the change.

MV-Q4. Using $y = \frac{Er_P - r_f}{A\sigma_P^2}$: an investor with $A = 3$ currently sets $y = 0.5$ when $\sigma_P = 0.20$ and $r_f = 0.03$. (a) What risk premium $Er_P - r_f$ is implied? (b) Now suppose σ_P rises to 0.25 with the same risk premium and A. Recompute y and explain the direction of the change.

MV-Q5. Two risky assets D and E have $\sigma_D = 0.30$, $\sigma_E = 0.20$, and correlation $\rho_{DE} = 0.25$. The portfolio weights are $w_D = 0.4$ and $w_E = 0.6$. Using $\sigma_P^2 = w_D^2\sigma_D^2 + w_E^2\sigma_E^2 + 2w_Dw_E\sigma_D\sigma_E\rho_{DE}$, compute the portfolio variance and standard deviation. Compare the result to the weighted average of the two standard deviations and comment on the diversification benefit.

MV-Q6. Take assets D and E with $\sigma_D = 0.40$, $\sigma_E = 0.10$, and $\rho_{DE} = -1$. Using the perfect-negative-correlation form $\sigma_P^2 = (w_D\sigma_D - w_E\sigma_E)^2$ together with $w_D + w_E = 1$, find the weights w_D, w_E that make the portfolio risk-free ($\sigma_P = 0$). Verify your answer.

MV-Q7. For a three-asset problem, the frontier constants are $A = \mu'\Sigma^{-1}\mathbf{1} = 2.0$, $B = \mu'\Sigma^{-1}\mu = 0.30$, and $C = \mathbf{1}'\Sigma^{-1}\mathbf{1} = 20$. (a) Compute $D = BC - A^2$. (b) Using $\sigma_P^2 = \frac{C\bar{r}^2 - 2A\bar{r} + B}{D}$, find the minimum-variance portfolio's variance at a target return of $\bar{r} = 0.15$. (c) What is the expected return $\bar{r}^* = A/C$ of the global minimum-variance portfolio, and what is its variance $\sigma_{min}^2 = 1/C$?

MV-Q8. For a set of risky assets the frontier constants are $A = 1.2$, $B = 0.20$, and $C = 8$. (a) Compute D. (b) Using $\sigma_P^2 = \frac{C\bar{r}^2 - 2A\bar{r} + B}{D}$, compute the minimum-variance-frontier variance at target returns $\bar{r} = 0.10$ and $\bar{r} = 0.20$. (c) Verify that the smaller variance occurs at the return closer to $\bar{r}^* = A/C$, and confirm that σ_P^2 evaluated at $\bar{r}^*$ equals $1/C$.

MV-Q9. Consider two assets with covariance matrix $\Sigma = \begin{pmatrix} 0.04 & 0.01 \\ 0.01 & 0.09 \end{pmatrix}$. Using the global minimum-variance weights $w^* = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}'\Sigma^{-1}\mathbf{1}}$, compute the weights on the two assets and the resulting minimum variance $1/C$. Comment on which asset receives more weight and why.

MV-Q10. Two uncorrelated assets have variances $\sigma_1^2 = 0.01$ and $\sigma_2^2 = 0.04$, so $\Sigma = \begin{pmatrix} 0.01 & 0 \\ 0 & 0.04 \end{pmatrix}$. Using $w^* = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}'\Sigma^{-1}\mathbf{1}}$, find the global minimum-variance weights and the minimum variance $1/C$. Show that the weights are inversely proportional to the individual variances and explain why that makes sense.

4.11.3 Data exercises

These problems use the live-data figure in this chapter.

MV-D1. Figure 4.1 reports, for the last fifteen years of monthly data: stocks $\mu_D = 14.3\%$, $\sigma_D = 14.9\%$; 10-year Treasury $\mu_E = 1.95\%$, $\sigma_E = 7.0\%$; correlation $\rho_{DE} = -0.03$ (all annualized). (a) Compute the mean and standard deviation of the 50/50 portfolio. (b) Compute what the 50/50 standard deviation would be if ρ_{DE} were +1, and interpret the difference as the diversification benefit shown by the shaded region in the figure.

MV-D2. Using the same inputs as MV-D1 and the two-asset minimum-variance weight $w_D^* = \dfrac{\sigma_E^2 - \sigma_D\sigma_E\rho_{DE}}{\sigma_D^2 + \sigma_E^2 - 2\sigma_D\sigma_E\rho_{DE}}$, compute the minimum-variance share in stocks and verify that it matches the mix marked in Figure 4.1. Then explain why the minimum-variance portfolio holds *any* stocks at all, given that stocks are twice as volatile as the bond.

Chapter 5

The Capital Asset Pricing Model

5.1 Introduction

Two questions lie at the heart of the Capital Asset Pricing Model. First, if all investors optimize in the mean-variance sense and share identical beliefs about expected returns and covariances, what must the market portfolio look like in equilibrium? Second, given that equilibrium, how should the expected return on any individual asset be determined by its contribution to the risk of a well-diversified investor's portfolio? These questions matter profoundly because they address the most basic practical problem in finance: what return should a rational investor demand as compensation for holding an asset, given the risk it carries? A credible answer to this question is needed to estimate the cost of capital, to assess whether a security is fairly priced, and to evaluate the performance of a portfolio manager.

The first question has a striking answer. Because all investors face the same inputs — identical beliefs about means, variances, and covariances — they all solve the same mean-variance optimization problem and therefore all select the same risky portfolio. In equilibrium, since every investor holds the same risky portfolio and all assets must be held, that portfolio must be the value-weighted portfolio of all traded assets: the market portfolio. This result, which follows directly from the separation property established in the preceding chapter, implies that the Capital Market Line — the capital allocation line through the market portfolio — is the single efficient frontier that every investor faces, regardless of her risk aversion. A more risk-averse investor simply holds more of the risk-free asset and less of the market; she does not change the composition of the risky portion of her portfolio.

The second question yields the CAPM's central quantitative prediction: the Security Market Line. In equilibrium, the expected excess return on any asset is proportional to its beta, where beta is defined as the covariance of the asset's return with the market portfolio divided by the variance of the market. Beta is the right measure of risk here because it captures exactly the contribution of an asset to the variance of the market portfolio — the only portfolio that any investor holds. An asset's standalone variance is irrelevant; what matters is how much it adds to aggregate risk when held as part of the market. High-beta assets amplify market swings, demand a higher risk premium to be held voluntarily, and have higher expected returns in equilibrium. Low-beta and negative-beta assets reduce portfolio risk and can therefore be held at lower expected returns. To see why standalone variance does not matter, the chapter decomposes each asset's risk into a systematic component tied to its beta and an idiosyncratic component specific to the asset, and shows that the idiosyncratic part can be diversified away at no cost — so only systematic risk is compensated in equilibrium.

The chapter derives both implications rigorously from six clearly stated assumptions: investors are price takers; all plan over a single identical holding period; all assets are publicly traded; there are no transaction costs or taxes; all investors are mean-variance optimizers; and expectations are homogeneous. The derivation shows that equilibrium market clearing, combined with the first-order conditions from mean-variance optimization, forces the market portfolio to be the tangency portfolio and pins down individual asset risk premia through the ratio of covariance with the market to market variance. The quantitative relationship between the market risk premium and average risk aversion also emerges directly from the clearing condition.

> **In the news**
>
> The CAPM's required return is the rate at which investors discount a company's future cash flows — so when that rate rises, prices fall, and they fall most for firms whose payoffs lie furthest in the future. NBC News reported a sharp sell-off in technology shares on fears of higher interest rates, precisely because high-growth, high-beta stocks are the most sensitive to a rising discount rate. The link between an asset's risk, its required return, and its price is the heart of this chapter. Read it at NBC News.

5.2 Assumptions

Consider a world where:

1. All investors are price takers
2. All investors plan for one identical holding period. This behavior may not be optimal for those planning for a longer horizon.
3. All investments are publicly traded (this rules out education, labor income, private capital and others).
4. There are no transactions costs and no taxes to be paid on returns.
5. All investors are mean-variance optimizers.
6. Investors agree on the means and variances of all of the securities that they trade. This is called *homogeneous expectations*.

5.3 Implications

1. All investors will hold the risky portfolio (although in perhaps different amounts).

2. The market portfolio will be the tangency portfolio to the CAL. This

will be called the *Capital Market Line* (CML). This line is the optimal CAL for all investors (homogeneous expectations).

3. The risk premium on the market portfolio will be given by $Er_M - r_f = \bar{A}\sigma_M^2$ where σ_M^2 is the variance on the market portfolio and $\bar{A}$ is the average degree of risk aversion. Notice that σ_M^2 is the systematic risk in the environment.

4. For asset i, let

$$\beta_i = \frac{Cov(r_i, r_M)}{\sigma_M^2}$$

Then it will be true that

$$Er_i - r_f = \beta_i(Er_M - r_f) = \frac{Cov(r_i, r_M)}{\sigma_M^2}(Er_M - r_f)$$

Why is this true?

5.4 Derivation

By assumptions 2,3,5 and 6 each of the investors in this economy will want to hold the same risky portfolio. Call this portfolio *the market portfolio* M. So the weights given to each asset in the risky portion of every investor's portfolio will be the same. What this means is that this weight will be the same weight that is given throughout the market. To see this, imagine that there are 2 risk assets and that the current value of both assets added together is $1 M. Suppose that there are two investors in the market. Together they own the whole market. Investor 1 and investor 2 have 20% of their wealth in asset 1. Since the total of these two investor's wealths is $1 M and both have exactly 20% in asset 1, it must be that asset 1 comprises 20% of the market value.

Claim 2 follows from Claim 1 together with the separation property. By separation, the single risky portfolio that every investor holds is the tangency portfolio — the portfolio at which a ray from r_f touches the risky-asset efficient frontier. Since every investor holds this same tangency portfolio and their combined risky holdings sum to the market, the market portfolio *is* the tangency portfolio. The CAL running from r_f through it — the Capital Market Line — is therefore the optimal CAL for every investor. (A fully rigorous statement is given in the appendix.)

Claim three follows from the demand of an individual MV optimizer who puts some fraction y of wealth into the risky asset and the fraction $1 - y$ into the risk free asset. Recall that the demand for this individual is given by

$$y = \frac{Er_M - r_f}{A\sigma_M^2}$$

Index investors by i, so investor i with wealth W_i and risk aversion A_i demands

$$y_i = \frac{Er_M - r_f}{A_i\sigma_M^2}.$$

We now aggregate across investors. In this model any borrowing done by one investor must be offset by lending from another, so the risk-free asset is in zero net supply and total wealth is held in the market portfolio. The dollars each investor places in risky assets, $W_i y_i$, must therefore sum to aggregate wealth $W = \sum_i W_i$:

$$\sum_i W_i y_i = W \quad \iff \quad \sum_i \frac{W_i}{W} y_i = 1,$$

so the *wealth-weighted* average fraction placed in the risky asset is 1. Substituting the demands and pulling the common factor out of the sum,

$$\sum_i \frac{W_i}{W} \cdot \frac{Er_M - r_f}{A_i\sigma_M^2} = \frac{Er_M - r_f}{\sigma_M^2} \sum_i \frac{W_i}{W} \frac{1}{A_i} = 1.$$

Solving for the risk premium gives

$$\bar{A}\sigma_M^2 = Er_M - r_f, \qquad \bar{A} \equiv \left(\sum_i \frac{W_i}{W}\frac{1}{A_i}\right)^{-1}.$$

The "average" risk aversion $\bar{A}$ that appears is thus the *wealth-weighted harmonic mean* of the individual coefficients — the same object derived formally in the appendix — because it is the reciprocals $1/A_i$, not the A_i themselves, that average linearly across investors.

The most often cited implication of the CAPM model is claim 4. Recall from the previous lecture that the expected reward to risk ratio (the market price of risk) must be the same for all assets and the market portfolio. Thus

$$\frac{Er_i - r_f}{Cov(r_i, r_M)} = \frac{Er_j - r_f}{Cov(r_j, r_M)} = \frac{Er_M - r_f}{Cov(r_M, r_M)} \text{ for all } i, j$$

This implies (by rearranging) that

$$Er_i - r_f = \frac{Cov(r_i, r_M)}{\sigma_M^2}(Er_M - r_f)$$

5.5 Systematic and idiosyncratic risk

The CAPM says that only an asset's beta — its covariance with the market — determines its expected return, and that an asset's standalone variance is irrelevant. To see why, it helps to split the risk of any asset into two pieces: a part that moves with the market and a part that is specific to the asset itself.

Write the return on asset i as its projection on the market return plus a residual:

$$r_i = \alpha_i + \beta_i r_M + \epsilon_i,$$

where $\beta_i = Cov(r_i, r_M)/\sigma_M^2$ as before, and ϵ_i is the firm-specific surprise, constructed so that it is uncorrelated with the market: $Cov(\epsilon_i, r_M) = 0$ and $E\epsilon_i = 0$. This is simply the regression of the asset's return on the market's: $\beta_i r_M$ is the fitted part and ϵ_i is the residual.

Taking the variance of both sides and using $Cov(\epsilon_i, r_M) = 0$ gives the **variance decomposition**

$$\sigma_i^2 = \underbrace{\beta_i^2 \sigma_M^2}_{\text{systematic}} + \underbrace{\sigma_{\epsilon_i}^2}_{\text{idiosyncratic}} .$$

The first term, $\beta_i^2 \sigma_M^2$, is the **systematic** (or market) risk of the asset: it comes from the asset's exposure to economy-wide movements and is governed entirely by beta. The second term, $\sigma_{\epsilon_i}^2$, is the **idiosyncratic** (or firm-specific) risk: it reflects events that affect this asset alone — a product recall, a lawsuit, a management shake-up — and is uncorrelated with what happens to the rest of the market.

The decomposition is something you can estimate in one regression, and Figure 5.1 shows it for two familiar stocks over the last five years of monthly data. NVIDIA's fitted line is steep ($\hat{\beta} = 2.15$): when the market moves a percentage point, NVIDIA moves more than two on average, the signature of a high-beta stock. Coca-Cola's line is nearly flat ($\hat{\beta} = 0.37$). The vertical scatter of points around each line is the residual ϵ_i made visible, and the R^2 of each regression is exactly the systematic share of variance from the decomposition above, $\beta_i^2 \sigma_M^2 / \sigma_i^2$ — about 45% for NVIDIA and only 12% for Coca-Cola. Both stocks, in other words, carry most of their risk as idiosyncratic risk, which is precisely the part the next subsection shows a diversified investor can shed for free.

5.5.1 Diversification eliminates idiosyncratic risk

Idiosyncratic risk is special because it can be removed at essentially no cost simply by holding many assets. Consider an equally weighted portfolio of n assets, $r_P = \frac{1}{n} \sum_{i=1}^{n} r_i$. Substituting the decomposition,

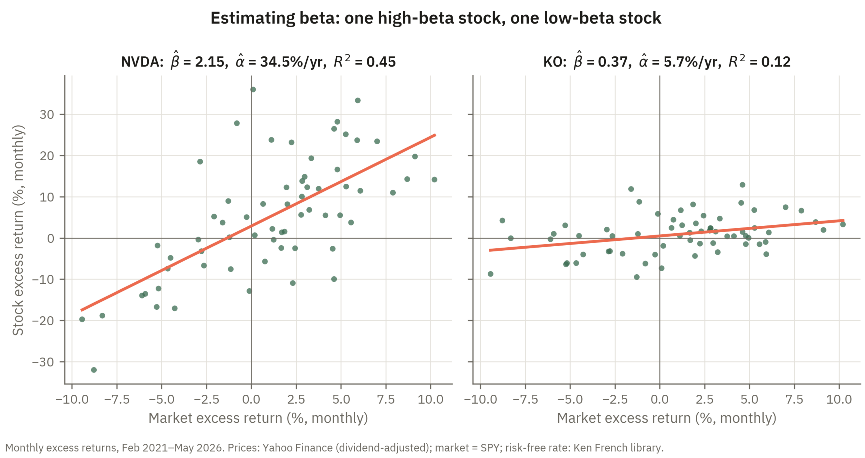

Figure 5.1: Monthly excess returns of NVIDIA and Coca-Cola against the market's excess return, with fitted regression lines, over the most recent five years. The slope of each line is the stock's beta; the scatter around the line is its idiosyncratic risk. Sources: Yahoo Finance; Ken French Data Library.

$$r_P = \bar{\alpha} + \left(\frac{1}{n}\sum_{i=1}^{n}\beta_i\right) r_M + \frac{1}{n}\sum_{i=1}^{n}\epsilon_i.$$

The systematic part has coefficient equal to the average beta $\bar{\beta} = \frac{1}{n}\sum_i \beta_i$, which does not shrink as n grows. The idiosyncratic part is the average of the residuals. If the firm-specific shocks are uncorrelated across assets, each with variance no larger than some bound $\bar{\sigma}_\epsilon^2$, then

$$Var\left(\frac{1}{n}\sum_{i=1}^{n}\epsilon_i\right) = \frac{1}{n^2}\sum_{i=1}^{n}\sigma_{\epsilon_i}^2 \le \frac{\bar{\sigma}_\epsilon^2}{n} \xrightarrow{n\to\infty} 0.$$

As the number of assets grows, the portfolio's idiosyncratic variance vanishes while its systematic variance, $\bar{\beta}^2\sigma_M^2$, remains. A well-diversified portfolio therefore carries essentially only systematic risk:

$$\sigma_P^2 \longrightarrow \bar{\beta}^2\,\sigma_M^2.$$

Market Insight

Because idiosyncratic risk can be diversified away for free, an investor who holds the market portfolio bears none of it. Risk that everyone can costlessly eliminate cannot command a reward in equilibrium: if an asset offered a premium for its firm-specific risk, investors would buy it, diversify that risk away, and bid the premium to zero. Only systematic risk — the part that survives in every diversified portfolio — must be compensated with higher expected return.

This is precisely why the Security Market Line prices assets by beta rather than by total variance σ_i^2. Two assets with the same standalone standard deviation can have very different expected returns if one is mostly systematic risk and the other mostly idiosyncratic; the CAPM rewards the first and not the second. The practical lesson is twofold. An investor who holds only a handful of stocks is bearing firm-specific risk that the market does not pay her to hold — she could earn the same expected return at lower risk by diversifying. And when judging how an asset contributes to a portfolio that is already diversified, its standalone volatility is the wrong measure; what matters is its beta, the systematic risk it adds.

5.6 Application: A CFO setting a hurdle rate

When a manufacturing company considers building a new plant, its chief financial officer must decide what return the project has to clear to be worth undertaking — and that hurdle rate is, in practice, the cost of equity the CAPM is designed to produce. The CFO estimates the project's beta from comparable firms, combines it with the prevailing risk-free rate and an estimate of the market risk premium, and reads off the Security Market Line the return that shareholders require for bearing the project's systematic risk. That single number flows directly into the discounted-cash-flow valuation that determines whether the plant is approved. The CAPM is therefore not merely a description of equilibrium prices; it is the everyday machinery by which corporate investment decisions, securities valuations, and the evaluation of portfolio managers are carried out.

5.7 Appendix: Formal Derivation of the CAPM

This appendix derives the four claims of the CAPM rigorously from the six assumptions stated above. The arguments proceed in three steps: individual optimization, aggregation via market clearing, and the resulting asset pricing equation.

5.7.1 Setup and notation

There are N risky assets with random returns $r_1, \ldots, r_N$. Let $\mu = (\mu_1, \ldots, \mu_N)'$ denote the vector of expected returns and Σ the $N \times N$ positive-definite variance-covariance matrix. A risk-free asset with return r_f is also available. There are I investors; investor i has initial wealth W_i and risk-aversion coefficient $A_i > 0$. Let $W = \sum_i W_i$ denote aggregate wealth.

By assumptions 5 and 6 every investor solves the same mean-variance problem, taking μ, Σ, and r_f as given.

5.7.2 Step 1: Individual optimization and the tangency portfolio

Investor i chooses portfolio weights $w \in \mathbb{R}^N$ on the risky assets (the remainder $1 - \mathbf{1}'w$ is placed in the risk-free asset) to maximize

$$U_i = w'(\mu - r_f \mathbf{1}) + r_f - \frac{A_i}{2} w'\Sigma w.$$

The first-order condition is

$$(\mu - r_f \mathbf{1}) - A_i \Sigma w_i^* = 0,$$

which gives the optimal risky-asset weights for investor i:

$$w_i^* = \frac{1}{A_i}\,\Sigma^{-1}(\mu - r_f\mathbf{1}).$$

Every investor holds the same vector $\Sigma^{-1}(\mu - r_f\mathbf{1})$, scaled by $1/A_i$. The composition of the risky portion of the portfolio is therefore identical across investors; only the overall scale differs. Normalising to unit weight gives the **tangency portfolio**

$$w_T = \frac{\Sigma^{-1}(\mu - r_f\mathbf{1})}{\mathbf{1}'\Sigma^{-1}(\mu - r_f\mathbf{1})},$$

and we can write $w_i^* = y_i\,w_T$ where the scalar

$$y_i = \frac{\mathbf{1}'\Sigma^{-1}(\mu - r_f\mathbf{1})}{A_i}$$

is the fraction of wealth investor i places in the risky tangency portfolio. This confirms **Claim 1** and **Claim 2**: every investor holds the same risky portfolio w_T, and the capital market line passes through w_T.

5.7.3 Step 2: Market clearing and the market risk premium

Market clearing requires that the aggregate demand for every risky asset equals its supply. The aggregate dollar demand for the risky-asset vector is

$$\sum_i W_i\,w_i^* = \left(\sum_i \frac{W_i}{A_i}\right)\Sigma^{-1}(\mu - r_f\mathbf{1}).$$

The supply of risky assets equals their total market capitalisation, which as a share of aggregate wealth is precisely w_M, the vector of **market-portfolio weights**. Equilibrium therefore requires

$$W\,w_M = \left(\sum_i \frac{W_i}{A_i}\right)\Sigma^{-1}(\mu - r_f\mathbf{1}).$$

This confirms that $w_M \propto \Sigma^{-1}(\mu - r_f\mathbf{1})$, so the tangency portfolio and the market portfolio coincide: $w_T = w_M$.

To obtain the market risk premium, note that the net supply of the risk-free asset is zero in the aggregate, so the wealth-weighted mean fraction invested in risky assets must equal one:

$$\frac{\sum_i y_i W_i}{W} = 1.$$

Substituting $y_i = (\mu_M - r_f)/(A_i\sigma_M^2)$, where $\mu_M = w_M'\mu$ and $\sigma_M^2 = w_M'\Sigma w_M$:

$$\frac{\mu_M - r_f}{\sigma_M^2} \cdot \frac{\sum_i W_i/A_i}{W} = 1.$$

Define the **wealth-weighted harmonic mean of risk aversion**

$$\bar{A} = \left(\frac{\sum_i W_i/A_i}{W}\right)^{-1}.$$

Then the market-clearing condition becomes

$$\boxed{\mu_M - r_f = \bar{A}\,\sigma_M^2,}$$

which is **Claim 3**. The market risk premium equals the product of average risk aversion and market variance. When investors are on average more risk-averse, or when the market is more volatile, the equilibrium risk premium is larger.

5.7.4 Step 3: The Security Market Line

Returning to the first-order condition $\Sigma w_M = \frac{1}{\bar{A}}(\mu - r_f\mathbf{1})$, the n-th row reads

$$\sum_{j=1}^{N} \sigma_{nj}\, w_{M,j} = \frac{\mu_n - r_f}{\bar{A}}.$$

The left-hand side is exactly $Cov(r_n,\ r_M) = Cov\Big(r_n,\ \sum_j w_{M,j} r_j\Big)$, so

$$Cov(r_n, r_M) = \frac{\mu_n - r_f}{\bar{A}}.$$

Applying the same equation to the market portfolio itself ($n = M$) gives

$$\sigma_M^2 = Cov(r_M, r_M) = \frac{\mu_M - r_f}{\bar{A}}.$$

Dividing the first equation by the second eliminates $\bar{A}$:

$$\frac{\mu_n - r_f}{\mu_M - r_f} = \frac{Cov(r_n, r_M)}{\sigma_M^2} \equiv \beta_n.$$

Rearranging yields **Claim 4**, the **Security Market Line**:

$$\boxed{\mu_n - r_f = \beta_n\,(\mu_M - r_f), \qquad \beta_n = \frac{Cov(r_n, r_M)}{\sigma_M^2}.}$$

Every asset's expected excess return is proportional to its beta. Assets with $\beta_n > 1$ amplify market risk and must offer a premium above the market; assets with $\beta_n < 1$ contribute less than proportionally to aggregate variance and therefore command a smaller premium; assets with $\beta_n = 0$ are uncorrelated with the market and, in equilibrium, earn only the risk-free rate as their expected return.

5.7.5 Summary

The four claims of the CAPM follow in sequence. Homogeneous expectations and mean-variance optimization (Assumptions 5–6) imply that every investor holds the same tangency portfolio. The public-markets assumption (Assumption 3) and price-taking (Assumption 1) ensure that aggregate demand must equal aggregate supply at every asset, which forces the tangency portfolio to be the market portfolio and pins down the market risk premium as $\bar{A}\sigma_M^2$. The first-order conditions then deliver the Security Market Line as an immediate algebraic consequence, with beta as the sole determinant of an asset's required excess return.

> **Further listening**
>
> *Odd Lots* (Bloomberg) — "Cliff Asness on How Markets Got Dumber in the Last 10 Years": a student of Eugene Fama on risk, return, and the factor models that grew out of the CAPM.

5.8 Homework problems

5.8.1 Conceptual

CAPM-C1. Two investors, a cautious retiree and an aggressive young professional, share identical beliefs about the means, variances, and covariances of all traded assets and both are mean-variance optimizers. Explain why, despite their very different attitudes toward risk, the *composition* of the risky portion of their portfolios must be identical. In your answer, describe how each investor expresses her risk preference and what role the risk-free asset plays.

CAPM-C2. Explain why, in equilibrium, the single risky portfolio that every investor holds must be the *market* portfolio — the value-weighted portfolio of all traded assets. Use the fact that all assets must be held by someone and that every investor gives the same weights to the risky assets. What would go wrong if the common risky portfolio placed zero weight on some traded asset?

CAPM-C3. Suppose a new class of investors enters the market who, unlike everyone else, believe technology stocks are far riskier than the rest of the market believes. Explain how this violation of homogeneous expectations could cause different investors to hold different risky portfolios, and why the market portfolio would then no longer necessarily be the tangency portfolio for every investor.

CAPM-C4. Assumptions 5 and 6 (mean-variance optimization and homogeneous expectations) are what force every investor to solve the *same* optimization problem. Explain how each of these two assumptions contributes to the conclusion that the market portfolio equals the tangency portfolio, and describe qualitatively what happens to that conclusion if investors instead disagree about expected returns.

CAPM-C5. Two assets have exactly the same standalone standard deviation σ_i, but asset A has $\beta_A = 1.4$ and asset B has $\beta_B = 0.3$. Which asset should have the higher expected return in equilibrium, and why? In your answer, explain why the CAPM prices assets by beta rather than by total variance σ_i^2.

CAPM-C6. A financial advisor tells a client that a stock with a very high standalone volatility is "too risky to hold" and should command a huge risk premium. Using the definition $\beta_i = Cov(r_i, r_M)/\sigma_M^2$ and the idea that beta measures an asset's contribution to the variance of the market portfolio, explain the flaw in this reasoning. What if the stock had a beta of zero despite its high volatility?

CAPM-C7. Explain why, according to the equilibrium condition $Er_M - r_f = \bar{A}\sigma_M^2$, the market risk premium would be expected to rise during a period in which investors collectively become more risk-averse (for example, during a financial panic), even if the underlying variance of the market σ_M^2 did not change.

CAPM-C8. In the derivation of the market risk premium, the wealth-weighted average of the fraction y placed in the risky asset is set equal to one. Explain the market-clearing logic behind this step: why must this average equal exactly one, and what does this have to do with borrowing and lending across investors and the risk-free asset being in zero net supply?

CAPM-C9. Using the decomposition $\sigma_i^2 = \beta_i^2\sigma_M^2 + \sigma_{\epsilon_i}^2$, explain in words the difference between the systematic and idiosyncratic components of an asset's risk. Give a concrete example of a real-world event that would show up in ϵ_i but not in the market return r_M.

CAPM-C10. Consider the single-index representation $r_i = \alpha_i + \beta_i r_M + \epsilon_i$.

Explain why ϵ_i is constructed to be uncorrelated with r_M, and how that requirement is exactly what allows the variance of r_i to split cleanly into a systematic term and an idiosyncratic term with no cross term. Which term captures economy-wide movements and which captures firm-specific events?

CAPM-C11. An investor holds a portfolio of only three stocks and complains that it swings wildly. Using the result that the idiosyncratic variance of an equally weighted portfolio behaves like $\bar{\sigma}_\epsilon^2/n$, explain why adding many more uncorrelated stocks would reduce her risk at essentially no cost, and explain what portion of the portfolio's variance would remain no matter how many stocks she added.

CAPM-C12. Explain why the systematic risk of a well-diversified equally weighted portfolio, $\bar{\beta}^2\sigma_M^2$, does *not* shrink as the number of assets n grows, while its idiosyncratic risk does. Relate this to the statement that "there is no free lunch" for market risk: what is the fundamental difference between the two components that makes one diversifiable and the other not?

CAPM-C13. Explain why, in CAPM equilibrium, an asset earns no premium for bearing idiosyncratic risk. Frame your answer as an arbitrage-style argument: what would investors do if an asset offered a premium for its firm-specific risk, and how would that drive the premium to zero?

CAPM-C14. A stock is found to have $\beta_i = 0$ even though its standalone standard deviation is large. According to the CAPM, what expected return should it earn, and why does its large idiosyncratic risk earn it nothing? Contrast this with the expectation of a naive investor who prices the stock by its total volatility.

5.8.2 Quantitative

CAPM-Q1. An asset has $\beta_i = 1.3$. The risk-free rate is $r_f = 0.02$ and the expected return on the market is $Er_M = 0.09$. Use the Security Market Line $Er_i - r_f = \beta_i(Er_M - r_f)$ to compute the asset's equilibrium expected return Er_i.

CAPM-Q2. The market risk premium is $Er_M - r_f = 0.06$ and the risk-free rate is $r_f = 0.03$. Two assets have betas $\beta_A = 0.5$ and $\beta_B = 1.6$. Use the Security Market Line to compute the equilibrium expected return of each asset, and find the difference $Er_B - Er_A$.

CAPM-Q3. An analyst estimates that a stock's covariance with the market is $Cov(r_i, r_M) = 0.018$ and the variance of the market is $\sigma_M^2 = 0.036$. Compute the stock's beta $\beta_i = Cov(r_i, r_M)/\sigma_M^2$. Then, using the market risk premium $Er_M - r_f = 0.055$ and $r_f = 0.025$, find the stock's expected return Er_i from the Security Market Line.

CAPM-Q4. A stock has standard deviation $\sigma_i = 0.30$ and correlation $\rho_{iM} = 0.4$ with the market, and the market has standard deviation $\sigma_M = 0.20$. Using $Cov(r_i, r_M) = \rho_{iM}\sigma_i\sigma_M$ and $\beta_i = Cov(r_i, r_M)/\sigma_M^2$, compute the stock's beta.

CAPM-Q5. The average degree of risk aversion in the market is $\bar{A} = 2.5$ and the variance of the market portfolio is $\sigma_M^2 = 0.04$. Using $Er_M - r_f = \bar{A}\sigma_M^2$, compute the equilibrium market risk premium. If the risk-free rate is $r_f = 0.03$, what is the expected return on the market portfolio Er_M?

CAPM-Q6. In equilibrium the market risk premium is observed to be $Er_M - r_f = 0.06$ and the variance of the market is $\sigma_M^2 = 0.05$. Using the relation $Er_M - r_f = \bar{A}\sigma_M^2$, solve for the implied average degree of risk aversion $\bar{A}$ in the market.

CAPM-Q7. An investor with risk-aversion coefficient $A = 3$ faces a market with risk premium $Er_M - r_f = 0.07$ and variance $\sigma_M^2 = 0.04$. Using the individual demand rule $y = (Er_M - r_f)/(A\sigma_M^2)$, compute the fraction y of wealth this investor places in the risky market portfolio. Is she a net borrower or lender at the risk-free rate?

CAPM-Q8. Using the demand rule $y = (Er_M - r_f)/(A\sigma_M^2)$ with a market risk premium of $Er_M - r_f = 0.05$ and market variance $\sigma_M^2 = 0.025$, find the risk-aversion coefficient A of an investor who chooses to hold exactly $y = 1$ (all wealth in the market and nothing in the risk-free asset). Interpret how this value relates to the average risk aversion $\bar{A}$ in equilibrium.

CAPM-Q9. An asset has $\beta_i = 1.2$, and the market has variance $\sigma_M^2 = 0.05$. The idiosyncratic variance of the asset is $\sigma_{\epsilon_i}^2 = 0.02$. Using the decomposition $\sigma_i^2 = \beta_i^2 \sigma_M^2 + \sigma_{\epsilon_i}^2$, compute the asset's total variance σ_i^2 and its total standard deviation σ_i. What fraction of the total variance is systematic?

CAPM-Q10. A stock has total variance $\sigma_i^2 = 0.10$ and beta $\beta_i = 0.8$. The market variance is $\sigma_M^2 = 0.045$. Using the variance decomposition $\sigma_i^2 = \beta_i^2 \sigma_M^2 + \sigma_{\epsilon_i}^2$, solve for the stock's idiosyncratic variance $\sigma_{\epsilon_i}^2$, and compute the systematic share of total variance.

5.8.3 Data exercises

These problems use the live-data figure in this chapter.

CAPM-D1. In Figure 5.1, NVIDIA's regression has $\hat{\beta} = 2.15$ and $R^2 = 0.45$; the market's monthly standard deviation over the sample was about 4.4%. (a) Using the variance decomposition, compute NVIDIA's systematic monthly variance $\hat{\beta}^2 \sigma_M^2$. (b) Using $R^2 = \beta^2 \sigma_M^2 / \sigma_i^2$, compute NVIDIA's total monthly standard deviation and its idiosyncratic monthly standard deviation. (c) What fraction of NVIDIA's risk would a diversified investor actually be compensated for bearing?

CAPM-D2. Take $r_f = 4\%$ and an expected market premium of 6% per year. (a) Using the SML and the betas in Figure 5.1, compute the CAPM-required expected returns for NVIDIA ($\beta = 2.15$) and Coca-Cola ($\beta = 0.37$). (b) The figure reports NVIDIA's estimated $\hat{\alpha} \approx 34.5\%$ per year over 2021–2026. Give two reasons this is *not* decisive evidence against the CAPM, drawing on the chapter's discussion of estimation and the next chapter's joint-hypothesis problem.

Chapter 6

Portfolio performance evaluation

6.1 Introduction

Having just derived the Capital Asset Pricing Model, we turn to a question the model itself appears to rule out: how do we tell a skilled portfolio manager from a lucky one? The tension is worth stating at the outset. The CAPM says that expected return is *completely* determined by systematic risk, so that every asset and every portfolio plots exactly on the Security Market Line and no manager can earn more than her beta entitles her to. In that world the abnormal return we will call alpha is identically zero, and there is simply nothing for performance evaluation to measure. The enterprise of this chapter therefore presupposes that the CAPM does not hold exactly. Alpha is what appears when one of the model's assumptions fails — above all the assumption that markets are *informationally efficient*, so that prices already embed every scrap of available information and no one can systematically profit from gathering or interpreting it better than the crowd. A manager who does process information better than the market can earn a return the model cannot explain, and it is precisely that residual return, invisible in a frictionless efficient market, that the measures below are built to detect. Each performance characteristic we develop — from Jensen's alpha to the information ratio — earns its keep only in a world where markets are informationally inefficient enough for genuine skill to leave a footprint; informational efficiency itself is the subject we take up in the chapters that follow.

Three questions organize the evaluation of portfolio performance. First, when a portfolio earns a high return, how much of that return is compensation for bearing risk rather than evidence of skill, and how can we strip the one from the other? Second, *which* notion of risk is the right one to adjust for, given that a portfolio held in isolation exposes an investor to its total volatility, while the very same portfolio held as one sleeve of a diversified whole contributes only its systematic risk? Third, when a manager is hired to beat a specific benchmark, how do we measure the return she adds relative to the risk she takes *away from* that benchmark? These questions matter because the raw return on a portfolio — the number that catches an investor's eye first — is almost useless on its own: it confounds skill with luck, and reward with risk.

The first question is the problem of risk adjustment, and the answer depends on the role the portfolio plays. If the portfolio is the entirety of an investor's risky wealth, then the relevant risk is the portfolio's *total* standard deviation, because that is the volatility the investor actually experiences. The **Sharpe ratio** measures reward per unit of total risk and is the natural yardstick in this case. If instead the portfolio is one component of a larger, well-diversified holding, then its idiosyncratic risk has already been diversified away, and only its systematic risk — its beta — contributes to the risk the investor bears. The **Treynor measure** and **Jensen's alpha** adjust for systematic risk and are the appropriate yardsticks in this case. Both build directly on the Capital Asset Pricing Model of the previous chapter: the CAPM supplies the "fair" expected return that a portfolio of given beta ought to earn, and abnormal performance is measured as the gap between the return realized and the return the CAPM predicts.

The third question arises because most professional portfolios are not managed in a vacuum but against a stated benchmark — an index the manager is paid to outperform. Here the natural object of study is the *active return*, the difference between the portfolio's return and the benchmark's, and the natural measure of risk is the volatility of that difference, called **tracking error**. The **information ratio**, the active return per unit of tracking error, is the reward-to-risk ratio of active management itself, and it is the measure most directly tied to the statistical question of whether a manager has genuine skill or has merely been lucky.

The chapter takes the five measures in turn — the Sharpe ratio, the Treynor measure, Jensen's alpha, tracking error, and the information ratio — developing the intuition and the algebra behind each, and is careful throughout to say which measure is appropriate in which situation. It then works a single numerical example through all five, and closes by drawing out the relationships among them: in particular, that the Sharpe and Treynor measures rank fully diversified portfolios identically but diverge precisely to the extent that a portfolio carries diversifiable risk, and that the information ratio is, up to a factor of $\sqrt{T}$, the t-statistic on a manager's alpha.

> **In the news**
>
> The whole point of these measures is to separate reward from risk — and once you do, the case for active management gets harder to make. S&P Dow Jones Indices' SPIVA scorecard found that even on a *risk-adjusted* basis, roughly 97% of U.S. domestic active funds underperformed the broad market over the 20 years through 2024 — an even worse showing than on raw returns. Measuring performance correctly, rather than being dazzled by a single year's headline number, is exactly what this chapter is about. Read it at S&P Dow Jones Indices.

Market Insight

A natural hope for active management is that raw return comparisons are unfair — perhaps active funds earn their keep by delivering smoother rides, so that adjusting for risk would flatter them. The data say the opposite. Figure 6.1 compares, for large-cap funds and the horizons S&P reports, the share of funds underperforming on raw returns with the share underperforming after the returns of every fund and the benchmark are put on an equal-risk footing using Sharpe ratios — the first measure this chapter develops. At every horizon the risk-adjusted bar is *higher*: over fifteen years, 90% of large-cap funds trail the S&P 500 on raw returns but 99% trail it after risk adjustment. The economic reading is that active funds have tended to take *more* total risk than their benchmarks, not less, so adjusting for risk removes what looked like a partial offset to their fee drag. Keep this figure in mind as the chapter proceeds: it is a preview of why every serious performance evaluation begins by asking not "what did it return?" but "what did it return per unit of risk?"

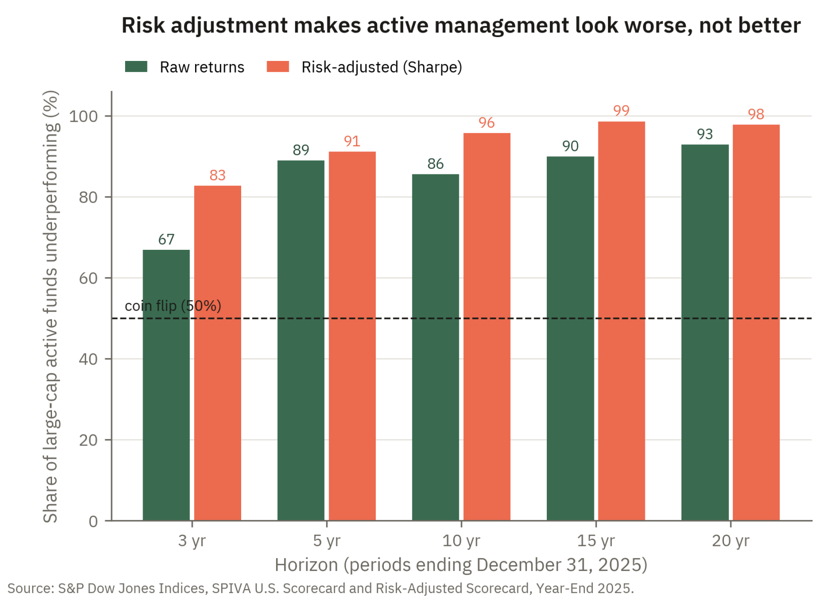

Figure 6.1: Share of actively managed U.S. large-cap funds underperforming the S&P 500 on raw returns and on a risk-adjusted (Sharpe ratio) basis, periods ending December 31, 2025. Source: S&P Dow Jones Indices, SPIVA U.S. Scorecard and Risk-Adjusted Scorecard, Year-End 2025.

6.2 Why returns alone mislead

Let r_P denote the (random) return on a portfolio P over an evaluation period, r_f the risk-free rate, r_M the return on the market portfolio, and r_b the return on a benchmark against which P is judged. The *excess return* of the portfolio is $r_P - r_f$, and its expected value $Er_P - r_f$ is the risk premium the portfolio is expected to deliver. We will measure total risk by the standard deviation $\sigma_P = \sqrt{Var(r_P)}$ and systematic risk by the beta

$$\beta_P = \frac{Cov(r_P, r_M)}{\sigma_M^2},$$

exactly as in the preceding chapter. In practice none of these population quantities is observed; they are estimated from a sample of historical returns, so each measure below has an *ex ante* form written in terms of expectations and an *ex post* form computed from sample averages and sample standard deviations. We write sample averages with a bar, as in $\bar{r}_P$.

The reason raw return is an inadequate measure of performance is that it answers the wrong question. An investor does not want the portfolio that earned the most last year; she wants the portfolio that delivers the most expected return for the risk she is willing to bear. Two portfolios that earned the same average return are not equally good if one did so with twice the volatility, and a portfolio that earned a spectacular return by taking a spectacular amount of risk may have been a poor choice that happened to pay off. Every measure in this chapter is therefore a ratio or a difference that places return in the numerator and some notion of risk in the denominator or the baseline. What distinguishes the measures from one another is *which* notion of risk they use, and that choice is dictated by how the portfolio is held.

6.3 The Sharpe ratio

Sharpe ratio. The ratio of a portfolio's expected excess return to the standard deviation of its return,

$$S_P = \frac{Er_P - r_f}{\sigma_P}.$$

The Sharpe ratio, introduced by William Sharpe in 1966 as the "reward-to-variability" ratio, measures the expected excess return earned per unit of total risk. Its *ex post* counterpart replaces the population moments with their sample estimates,

$$\hat{S}_P = \frac{\bar{r}_P - \bar{r}_f}{s_P},$$

where $\bar{r}_P - \bar{r}_f$ is the average realized excess return and s_P is the sample standard deviation of the portfolio's return.

The intuition is geometric. Recall that combining a risky portfolio P with the risk-free asset traces out the *capital allocation line* through P: an investor who places a fraction y of wealth in P and $1 - y$ in the risk-free asset earns expected return $r_f + y(Er_P - r_f)$ with standard deviation $y\sigma_P$. Eliminating y shows that the attainable combinations lie on a straight line in (σ, Er) space that begins at r_f and passes through P, and the slope of that line is precisely

$$\text{slope of the CAL} = \frac{Er_P - r_f}{\sigma_P} = S_P.$$

A higher Sharpe ratio is a steeper capital allocation line, which means more expected return at *every* level of total risk the investor might choose. This is why, among all portfolios that an investor might hold as her entire risky position, the one with the highest Sharpe ratio is unambiguously best: it offers the most favorable trade-off, and any desired risk level is then reached by mixing it with the risk-free asset. It is also why the tangency portfolio of the previous chapters is exactly the maximum-Sharpe-ratio portfolio.

Because it uses *total* risk in the denominator, the Sharpe ratio is the right measure when the portfolio under evaluation represents all of the investor's risky wealth. It charges the portfolio for every source of volatility it carries, diversifiable or not — which is appropriate, because an investor holding nothing else is exposed to all of that volatility.

6.4 The Treynor measure

Treynor measure. The ratio of a portfolio's expected excess return to its beta,

$$T_P = \frac{Er_P - r_f}{\beta_P}.$$

The Treynor measure, sometimes called the reward-to-volatility ratio, replaces total risk with systematic risk in the denominator. It answers the question: how much excess return did the portfolio earn for each unit of market risk it took on?

The reason one might prefer beta to standard deviation in the denominator is the central lesson of the CAPM. If the portfolio being evaluated is not held in isolation but as one component of a large, well-diversified portfolio, then its idiosyncratic risk is diversified away by the other holdings and does not contribute to the risk the investor ultimately bears. Only the portfolio's covariance with the rest of the world — summarized by its beta — survives diversification. For such a component portfolio, reward should be measured per unit of the risk it actually adds, namely its beta, and that is what the Treynor measure does.

Like the Sharpe ratio, the Treynor measure has a geometric reading: it is the slope of the line in (β, Er) space that runs from the risk-free asset, plotted at $\beta = 0$, through the portfolio. The market portfolio has $\beta_M = 1$, so its Treynor measure is

$$T_M = \frac{Er_M - r_f}{1} = Er_M - r_f,$$

which is just the slope of the Security Market Line. A portfolio beats the market on a Treynor basis precisely when $T_P > T_M$, which is the same as saying that the portfolio plots *above* the Security Market Line. Note that the Treynor measure is denominated in excess return per unit of beta, while the Sharpe ratio is denominated in excess return per unit of standard deviation; the two numbers are not directly comparable, and as we will see they can rank portfolios differently when those portfolios are not fully diversified.

6.5 Jensen's alpha

Jensen's alpha. The difference between a portfolio's expected return and the return predicted for it by the CAPM,

$$\alpha_P = Er_P - \left[r_f + \beta_P (Er_M - r_f) \right].$$

Where the Treynor measure rescales excess return by beta, Jensen's alpha subtracts the CAPM benchmark return outright. The bracketed term is exactly the expected return the Security Market Line assigns to an asset of beta β_P; alpha is the vertical distance by which the portfolio sits above (if positive) or below (if negative) that line. Under the CAPM, every asset and every portfolio has $\alpha = 0$, because the CAPM says that expected return is *completely* determined by beta. A nonzero alpha is therefore return that the single-factor model cannot account for, and as the introduction stressed, it can arise only when one of the model's assumptions fails. The economically interesting failure is a breakdown of informational efficiency: if prices do not already embed all available information, a manager who collects or interprets that information better than the market can earn a return beta cannot explain, and alpha is the visible trace of that advantage. A positive alpha is thus putative evidence of skill — though it may equally reflect exposure to a priced risk that the one-factor model omits rather than any informational edge.

In practice alpha is estimated as the intercept of the time-series regression of the portfolio's excess return on the market's excess return,

$$r_{P,t} - r_{f,t} = \alpha_P + \beta_P\,(r_{M,t} - r_{f,t}) + \varepsilon_{P,t},$$

where the slope β_P is the estimated systematic-risk exposure and the residual $\varepsilon_{P,t}$ captures the portion of the portfolio's return that is uncorrelated with the market. The fitted intercept $\hat{\alpha}_P$ is the average return the manager delivered over and above what her market exposure alone would have produced.

Two cautions accompany alpha. First, it is an *absolute* measure, expressed in units of return, and so it does not by itself reveal how much risk was taken to earn it: a manager who produces 2% of alpha while subjecting investors to wild idiosyncratic swings is not obviously better than one who produces 1% of alpha with a steady hand. Second, an estimated alpha is a noisy statistic, and a positive point estimate is not proof of skill. Whether a measured alpha is distinguishable from luck is a question about its t-statistic, and that question is answered most cleanly by the information ratio developed below.

6.6 Tracking error

Tracking error. The standard deviation of the difference between a portfolio's return and its benchmark's return,

$$TE_P = \sigma(r_P - r_b) = \sqrt{Var(r_P - r_b)}.$$

The quantity $r_P - r_b$ is the portfolio's *active return*: the amount by which it beats or trails its benchmark over the period. Tracking error is the volatility of that active return, and it measures how tightly the portfolio hugs the benchmark it is managed against. A portfolio that is constructed to replicate an index — a pure index fund — has a tracking error close to zero, because its return moves nearly in lockstep with the benchmark. A high-conviction active fund that takes large positions away from the index has a large tracking error.

Expanding the variance of the difference shows what drives tracking error,

$$TE_P^2 = \sigma_P^2 + \sigma_b^2 - 2\,\rho_{Pb}\,\sigma_P\,\sigma_b,$$

where ρ_{Pb} is the correlation between the portfolio and its benchmark. Tracking error is small when the portfolio and benchmark have similar volatilities and are highly correlated, and it grows as the portfolio's holdings depart from the benchmark's. It is important to recognize that tracking error is a measure of *risk*, not of performance: it tells you how far a portfolio is willing to stray from its benchmark, not whether straying paid off. A manager can run a large tracking error and still underperform. What tracking error provides is the denominator for the one measure that does combine active return with active risk.

6.7 The information ratio

Information ratio. The ratio of a portfolio's expected active return to its tracking error,

$$IR_P = \frac{Er_P - Er_b}{\sigma(r_P - r_b)} = \frac{\overline{r_P - r_b}}{TE_P}.$$

The information ratio is to active management what the Sharpe ratio is to total investing. The Sharpe ratio measures excess return over the risk-free rate per unit of total risk; the information ratio measures excess return over the *benchmark* per unit of *active* risk. It answers the question that matters when judging a manager paid to beat an index: for each unit of benchmark-relative risk she chose to take, how much did she add?

There is a closely related definition that uses the regression of the previous section rather than a benchmark difference. Writing the active return as the manager's alpha plus residual noise, the information ratio can be expressed as alpha divided by the standard deviation of the residual,

$$IR_P = \frac{\alpha_P}{\sigma(\varepsilon_P)},$$

where $\sigma(\varepsilon_P)$ is the *residual* (non-systematic) risk left over after accounting for market exposure. In this form the information ratio is the reward for skill per unit of the diversifiable risk the manager imposes. The two definitions coincide when the benchmark is the market portfolio and the manager's only systematic exposure is to the market; in general they differ slightly, and it is worth stating which one is in use.

The information ratio is the measure most tightly linked to the statistical question of skill versus luck. To see why, recall that $\hat{\alpha}_P$ is the intercept of the regression above, estimated from T observations of the residual $\varepsilon_{P,t}$. The standard error of that intercept is approximately the residual volatility divided by the square root of the sample size,

$$se(\hat{\alpha}_P) \approx \frac{\sigma(\varepsilon_P)}{\sqrt{T}}.$$

The t-statistic for testing whether alpha is zero is the estimate divided by its standard error, and substituting $IR_P = \alpha_P/\sigma(\varepsilon_P)$ gives

$$t(\hat{\alpha}_P) = \frac{\hat{\alpha}_P}{se(\hat{\alpha}_P)} \approx \frac{\hat{\alpha}_P}{\sigma(\varepsilon_P)/\sqrt{T}} = \frac{\hat{\alpha}_P}{\sigma(\varepsilon_P)}\sqrt{T} = IR_P \cdot \sqrt{T}.$$

So, over an evaluation period of T years, the t-statistic on a manager's estimated alpha is approximately

$$t(\hat{\alpha}_P) \approx IR_P \cdot \sqrt{T}.$$

An information ratio of 0.5 sustained over sixteen years yields a t-statistic of about 2, the conventional threshold for statistical significance — which is a sobering way of seeing how much data it takes to establish skill with any confidence. The information ratio also sits at the center of Grinold's *fundamental law of active management*, $IR \approx IC \times \sqrt{\text{breadth}}$, which decomposes a manager's skill into the quality of her individual forecasts (the information coefficient IC) and the number of independent bets she makes (the breadth). Both relationships make the same point: a respectable information ratio, consistently achieved, is a far more demanding standard than a single year's outperformance.

6.8 Which measure when?

The five measures are not competitors to be ranked from best to worst; each is the correct measure in a particular situation, and using the wrong one gives a misleading answer.

- The **Sharpe ratio** is appropriate when the portfolio under evaluation is the investor's entire risky position, because then the investor is exposed to the portfolio's total risk.
- The **Treynor measure** and **Jensen's alpha** are appropriate when the portfolio is one component of a larger, well-diversified portfolio, because then only its systematic risk matters. Treynor gives a reward-to-risk ratio suited to ranking such components; Jensen's alpha gives the absolute abnormal return.
- **Tracking error** measures how much active risk a benchmarked portfolio takes, and the **information ratio** measures the active return earned per unit of that active risk — the right standard for a manager judged against a benchmark.

A useful relationship ties the first two together. If a portfolio carries no residual risk — that is, if it is perfectly diversified so that $\varepsilon_P \equiv 0$ — then its total and systematic risk are linked by $\sigma_P = \beta_P \, \sigma_M$, and the Sharpe and Treynor measures become

$$S_P = \frac{Er_P - r_f}{\sigma_P} = \frac{Er_P - r_f}{\beta_P \sigma_M} = \frac{T_P}{\sigma_M}.$$

Since σ_M is the same for every portfolio, the Sharpe and Treynor measures rank fully diversified portfolios *identically*. They diverge only to the extent that a portfolio carries diversifiable risk, and the size of the gap between a portfolio's Sharpe ranking and its Treynor ranking is itself a signal of how much idiosyncratic risk the manager has left on the table. This is the precise sense in which the choice of denominator encodes an assumption about how the portfolio is held.

6.9 A worked example

Suppose that over an evaluation period the risk-free rate was $r_f = 3\%$ and the market portfolio earned $Er_M = 9\%$ with a standard deviation of $\sigma_M = 16\%$. An active equity fund P earned $Er_P = 12\%$ with a standard deviation of $\sigma_P = 20\%$ and a beta of $\beta_P = 1.2$. The fund is managed against a benchmark index with expected return $Er_b = 9.5\%$, and the standard deviation of the fund's active return relative to that benchmark was $TE_P = 5\%$. We compute all five measures.

The **Sharpe ratio** of the fund and of the market are

$$S_P = \frac{12 - 3}{20} = 0.45, \qquad S_M = \frac{9 - 3}{16} = 0.375,$$

so the fund delivered more expected return per unit of total risk than the market did. The **Treynor measure** of the fund and of the market are

$$T_P = \frac{12 - 3}{1.2} = 7.5\%, \qquad T_M = \frac{9 - 3}{1} = 6.0\%,$$

so the fund also beat the market per unit of systematic risk, which is the same as saying it plots above the Security Market Line. The size of that gap is **Jensen's alpha**,

$$\alpha_P = 12 - \left[\, 3 + 1.2\,(9 - 3)\,\right] = 12 - 10.2 = 1.8\%.$$

The fund earned 1.8% more than its market exposure alone would justify. Relative to its benchmark, the fund's active return was $Er_P - Er_b = 12 - 9.5 = 2.5\%$, carried at a **tracking error** of 5%, giving an **information ratio** of

$$IR_P = \frac{2.5}{5.0} = 0.50.$$

Finally, the fund's residual risk can be recovered from the identity $\sigma_P^2 = \beta_P^2\,\sigma_M^2 + \sigma^2(\varepsilon_P)$:

$$\sigma^2(\varepsilon_P) = 20^2 - 1.2^2 \cdot 16^2 = 400 - 368.64 = 31.36, \qquad \sigma(\varepsilon_P) = 5.6\%,$$

so the residual-risk form of the information ratio is $\alpha_P/\sigma(\varepsilon_P) = 1.8/5.6 \approx 0.32$, somewhat below the benchmark-relative value of 0.50 because the two definitions measure active risk differently. Collecting the results:

Measure	Fund P	Market / benchmark
Sharpe ratio	0.45	0.375
Treynor measure	7.5%	6.0%
Jensen's alpha	1.8%	0
Tracking error	5.0%	—
Information ratio	0.50	—

By every measure the fund outperformed, which is reassuring but not guaranteed: had the fund earned its 12% with a beta of, say, 1.6 instead of 1.2, its Jensen's alpha would have been negative even though its raw return was unchanged, because a beta of 1.6 would entitle it to a CAPM return above 12%. It is exactly this kind of reversal — a high raw return masking poor risk-adjusted performance — that the measures of this chapter are built to expose.

6.10 Summary

The raw return on a portfolio confounds reward with risk and skill with luck, and the measures of this chapter are devices for separating them. The

Sharpe ratio divides expected excess return by total risk and is the slope of the capital allocation line; it is the right measure when the portfolio is all an investor holds. The Treynor measure divides expected excess return by beta and is appropriate when the portfolio is a diversified component, in which case only systematic risk matters. Jensen's alpha measures the same systematic-risk-adjusted performance as an absolute return — the distance above the Security Market Line — and is estimated as the intercept of a market-model regression. Tracking error measures the volatility of a portfolio's active return relative to a benchmark, and the information ratio divides active return by tracking error to give the reward-to-risk ratio of active management, a quantity that is essentially the t-statistic of manager skill divided by $\sqrt{T}$. Fully diversified portfolios are ranked identically by the Sharpe and Treynor measures; the two diverge precisely to the degree that a portfolio carries diversifiable risk. Underlying the whole apparatus is a single premise: because the CAPM assigns zero alpha to every portfolio, any abnormal performance these measures uncover is a symptom of the model's assumptions failing — most importantly, of markets that are not perfectly informationally efficient. That premise is what makes the question of market efficiency, taken up in the chapters that follow, the natural sequel to performance evaluation.

6.11 Application: An investment consultant evaluating managers

A consultant advising a pension board must recommend whether to keep or replace an equity manager who beat her benchmark by three points last year. Raw outperformance is exactly the number the consultant must look past. She first asks how the excess return was earned: a Jensen's alpha regression separates the part attributable to simply running a higher beta from the part the manager actually added, and the t-statistic on that alpha — roughly the information ratio times the square root of the years of data — tells her whether the result is distinguishable from luck. She computes the information ratio to see whether the active return justified the active risk the manager took relative to the mandate, and the Sharpe ratio to confirm that adding the manager improves, rather than merely inflates, the risk-adjusted return of the plan as a whole. Only a manager whose outperformance survives all of these adjustments — not the one with the gaudiest single-year number — is worth the fees. The measures of this chapter are the instruments that turn a board's hunch about a manager into a defensible decision.

> Further listening
>
> *Planet Money* (NPR) — "Episode 688: Brilliant vs. Boring": a famous $1 million, ten-year bet on whether hedge funds can beat a simple index fund frames the chapter's central question — how do you tell genuine skill from luck?

6.12 Homework problems

6.12.1 Conceptual

PERF-C1. An investor is deciding how to allocate *all* of her risky wealth between two mutual funds. Fund A has a higher average return than Fund B, but also a higher standard deviation, and Fund A's Sharpe ratio is lower than Fund B's. Explain which fund she should choose and why, appealing to the capital allocation line and the fact that the Sharpe ratio is its slope. In your answer, explain how she could nonetheless end up with a higher expected return than Fund B offers on its own.

PERF-C2. Two analysts each compute a Sharpe ratio for the same equity fund over the same period but get different numbers. One used the fund's *total* standard deviation σ_P in the denominator; the other, believing the fund would be held inside a diversified portfolio, used only its systematic risk. Explain which analyst has applied the Sharpe ratio correctly, and explain what measure the second analyst has effectively (mis)computed and when that alternative measure would actually be the appropriate one.

PERF-C3. A pension fund holds forty distinct equity sleeves, each managed by a different manager, and is considering adding a forty-first. A trustee argues the new sleeve should be judged by its Sharpe ratio. Explain why the Treynor measure is the more appropriate yardstick here, and what feature of the CAPM justifies charging the sleeve only for its beta rather than its total volatility.

PERF-C4. Two component portfolios have identical Treynor measures, but portfolio X has a much larger standard deviation than portfolio Y. Explain what this tells you about the two portfolios' idiosyncratic risk, and explain why, despite the difference in total volatility, an investor holding either as one small piece of a well-diversified whole might be nearly indifferent between them.

PERF-C5. A manager reports a positive Jensen's alpha of 2%. Explain precisely what "abnormal" means here in terms of the Security Market Line, and give two distinct reasons why a positive estimated alpha is *not* by itself convincing evidence that the manager has genuine skill.

PERF-C6. Fund G and Fund H earned the same raw return of 11% over the same period, but Fund G had a beta of 0.8 and Fund H a beta of 1.4. Without doing arithmetic, explain which fund almost certainly has the larger Jensen's alpha and why, and explain the general lesson this illustrates about why raw return can mask poor risk-adjusted performance.

PERF-C7. A pure index fund and a high-conviction active fund are both benchmarked to the S&P 500. Explain which one has the larger tracking error and why, using the decomposition $TE_P^2 = \sigma_P^2 + \sigma_b^2 - 2\rho_{Pb}\sigma_P\sigma_b$. Then explain why a large tracking error is *not* by itself either good or bad news about the manager.

PERF-C8. A consultant says, "Two funds have the same tracking error, so they are equally good." Explain why this statement is wrong by contrasting tracking error as a measure of *risk* with the information ratio as a measure of *performance*, and describe what additional information about active return would be needed to rank the two funds.

PERF-C9. The information ratio is often described as "the Sharpe ratio of active management." Explain the parallel precisely: identify what plays the role of the excess return in the numerator and what plays the role of total risk in the denominator, and explain why the information ratio, rather than Jensen's alpha, is the natural reward-to-risk measure for a manager judged against a benchmark.

PERF-C10. The chapter gives two definitions of the information ratio: active return over tracking error, $\overline{r_P - r_b}/TE_P$, and alpha over residual risk, $\alpha_P/\sigma(\varepsilon_P)$. Explain the sense in which these two coincide, the circumstances under which they diverge, and why it matters to state which one is being reported.

PERF-C11. A manager has sustained an information ratio of 0.5. Using the relationship $t(\hat{\alpha}_P) \approx IR_P \cdot \sqrt{T}$, explain what this implies about how many years of data are needed before her alpha would clear the conventional significance threshold, and explain why this makes the information ratio "a far more demanding standard than a single year's outperformance."

PERF-C12. Explain why the relationship $t(\hat{\alpha}_P) \approx IR_P \cdot \sqrt{T}$ holds, walking through the two ingredients: that the t-statistic is $\hat{\alpha}_P/se(\hat{\alpha}_P)$ and that $se(\hat{\alpha}_P) \approx \sigma(\varepsilon_P)/\sqrt{T}$. Then explain what the $\sqrt{T}$ factor implies: why two managers with the *same* information ratio can have very different t-statistics, and what a manager can and cannot do to raise her t-statistic.

6.12.2 Quantitative

PERF-Q1. Over a five-year evaluation period a balanced fund earned an average return of $\bar{r}_P = 10\%$ while the risk-free rate averaged $\bar{r}_f = 2.5\%$, and the fund's sample standard deviation was $s_P = 15\%$. The market earned $\bar{r}_M = 8\%$ with a standard deviation of 16%. Compute the fund's *ex post* Sharpe ratio $\hat{S}_P$ and the market's Sharpe ratio S_M, and state which delivered more reward per unit of total risk.

PERF-Q2. An investor will hold Fund Z as her entire risky position and mix it with the risk-free asset. Fund Z has $Er_P = 14\%$, $\sigma_P = 25\%$, and $r_f = 4\%$.

She wants an expected return of 11%. Using the Sharpe ratio and the capital allocation line, find (a) the Sharpe ratio S_P, (b) the fraction y of wealth she must place in Fund Z, and (c) the standard deviation of her resulting position.

PERF-Q3. A portfolio has $Er_P = 13\%$, a beta of $\beta_P = 1.5$, and the risk-free rate is $r_f = 3\%$; the market earned $Er_M = 9\%$. Compute the Treynor measure of the portfolio T_P and of the market T_M, and state whether the portfolio plots above or below the Security Market Line.

PERF-Q4. Two diversified component portfolios are to be compared on a Treynor basis. Portfolio D has $Er_D = 10\%$ and $\beta_D = 0.9$; portfolio E has $Er_E = 15\%$ and $\beta_E = 1.6$. The risk-free rate is $r_f = 3\%$. Compute T_D and T_E and determine which component is the better performer per unit of systematic risk.

PERF-Q5. A fund earned $Er_P = 11\%$ with a beta of $\beta_P = 0.7$. The risk-free rate is $r_f = 2\%$ and the market earned $Er_M = 10\%$. Compute the CAPM-predicted return for the fund and its Jensen's alpha α_P, and interpret the sign.

PERF-Q6. A fund earned a raw return of $Er_P = 12\%$. The risk-free rate is $r_f = 3\%$ and the market earned $Er_M = 8\%$. Find the beta β_P at which the fund's Jensen's alpha α_P would be exactly zero, and state whether a beta *above* that value makes the fund's alpha positive or negative.

PERF-Q7. A fund's active return relative to its benchmark averaged $\overline{r_P - r_b} = 3\%$ over the period, with a tracking error of $TE_P = 6\%$. Compute the information ratio IR_P. Then explain what the resulting number means as a reward-to-active-risk ratio.

PERF-Q8. A benchmarked fund earned an average return of $\bar{r}_P = 10.5\%$ while its benchmark earned $\bar{r}_b = 8\%$. Over the same period the portfolio had a standard deviation of $\sigma_P = 18\%$, the benchmark had $\sigma_b = 15\%$, and the two had a correlation of $\rho_{Pb} = 0.9$. Using $TE_P^2 = \sigma_P^2 + \sigma_b^2 - 2\rho_{Pb}\sigma_P\sigma_b$, compute the tracking error, and then compute the information ratio $\overline{r_P - r_b}/TE_P$.

PERF-Q9. A fund's active return relative to its benchmark averaged 3% with a tracking error of 6%, giving an information ratio of 0.5. Using $t(\hat{\alpha}_P) \approx$

$IR_P \cdot \sqrt{T}$, find the number of years T of data needed for the manager's alpha to reach a t-statistic of 2.

PERF-Q10. A manager's estimated alpha over $T = 25$ years of data is $\hat{\alpha}_P = 1.5\%$, with a residual standard deviation of $\sigma(\varepsilon_P) = 6\%$. Compute her information ratio $IR_P = \alpha_P/\sigma(\varepsilon_P)$ and the approximate t-statistic $t(\hat{\alpha}_P) \approx IR_P \cdot \sqrt{T}$, and state whether her alpha is statistically distinguishable from zero at the conventional threshold.

PERF-Q11. A manager's market-model regression yields an estimated alpha of $\alpha_P = 2.4\%$ and a residual standard deviation of $\sigma(\varepsilon_P) = 8\%$. Separately, her active return relative to a benchmark averaged $\overline{r_P - r_b} = 2.0\%$ with tracking error $TE_P = 5\%$. Compute the information ratio under both the residual-risk definition and the benchmark-difference definition, and explain in one sentence why they differ.

PERF-Q12. A fund has a total standard deviation of $\sigma_P = 22\%$ and a beta of $\beta_P = 1.1$; the market has a standard deviation of $\sigma_M = 15\%$. Using the identity $\sigma_P^2 = \beta_P^2\sigma_M^2 + \sigma^2(\varepsilon_P)$, compute the fund's residual (idiosyncratic) risk $\sigma(\varepsilon_P)$. If the fund's Jensen's alpha is $\alpha_P = 1.9\%$, compute the residual-risk form of its information ratio $\alpha_P/\sigma(\varepsilon_P)$.

6.12.3 Data exercises

These problems use the live-data figure in this chapter.

PERF-D1. A fund earned a mean excess return of 6.8% per year with volatility 22%; the index earned a mean excess return of 6.5% with volatility 17%. (a) On raw returns, which performed better? (b) Compute both Sharpe ratios. Which performed better per unit of risk? (c) Explain how this example illustrates why the risk-adjusted bars in Figure 6.1 are *higher* than the raw-return bars at every horizon.

PERF-D2. Figure 6.1 reports that over fifteen years 90% of large-cap active funds underperform the S&P 500 on raw returns but 99% underperform after Sharpe-ratio adjustment. What does the direction of this gap reveal about the *total risk* active funds have taken relative to their benchmark?

Connect your answer to the chapter's claim that every performance measure places return relative to a notion of risk.

Part II

Information, trading and market efficiency

Chapter 7

The efficient markets hypothesis

7.1 Introduction

Three questions give this chapter its organizing logic. First, what does it mean to say a market is "efficient," and how do the three standard versions of that claim — weak, semi-strong, and strong — differ in what they assume prices already reflect? Second, what are the precise empirical implications of each level of efficiency, and what kinds of trading strategies would each render unprofitable? Third, even if individual investors display systematic psychological biases, does that necessarily mean market prices will be distorted, or can arbitrageurs correct for those biases? Together these questions trace a path from the information contained in prices all the way to the aggregate efficiency — or inefficiency — of the market.

The first question matters because "efficiency" is not a single thing. Weak-form efficiency asserts only that past price history contains no exploitable information. Semi-strong efficiency is more demanding: it asserts that all publicly available information — earnings, filings, news — is already impounded in prices. Strong-form efficiency is the most demanding of all: it asserts that even private, insider information is reflected in prices. The distinction is crucial because it determines how much information one is willing to assume the market has already digested before drawing conclusions about what analysis can accomplish.

The second question translates these levels into testable empirical claims. Weak-form efficiency rules out profitable technical analysis, since it holds that patterns in past prices cannot be exploited. Semi-strong efficiency rules out returns to fundamental analysis and factor-based strategies, since all public information is already priced. Strong-form efficiency rules out profits even from insider trading. Each form has direct implications for the value of different investment strategies, and the chapter is candid about the formidable empirical difficulties in testing any of them.

The third question — whether individual behavioral biases translate into market-level inefficiency — grounds the chapter's discussion of behavioral finance. Psychologists have documented robust and systematic departures from expected-utility maximization: overconfidence, representativeness bias, loss aversion, and others. But the existence of such biases at the individual level does not automatically imply that prices will be distorted, because rational arbitrageurs can in principle trade against biased investors and eliminate the resulting mispricings. Whether they actually do so depends on whether arbitrage is limited by transaction costs, financing constraints, or the risk that mispricings worsen before they correct — a set of frictions that the chapter examines as the boundary conditions for behavioral influences on asset prices.

The chapter then draws out the close kinship between market efficiency and the CAPM. Because the CAPM's sharp "no free lunch" conclusion rests on six assumptions, the chapter examines each in turn and asks how strong it really is, and it uses that examination to make precise what *alpha* means: a return in excess of what a pricing model warrants, which can arise either because a CAPM assumption fails or because the efficient markets hypothesis does — the joint-hypothesis problem that makes any measured alpha ambiguous.

Turning this logic to personal use, the chapter closes by developing the CAPM as a practical checklist for evaluating any investment pitched as offering high returns with little risk: walking through the assumptions one at a time reveals which risk, if any, a purportedly extraordinary opportunity is really paying you to bear — and when the honest answer is none, that the promised returns are almost certainly illusory.

> **In the news**
>
> The efficient-markets hypothesis makes a testable prediction: most professional managers should not be able to beat the market after fees. CNBC reported that *fewer* active managers beat their index benchmarks again last year — part of a long pattern that has pushed trillions of dollars from active funds into low-cost index funds. Whether that pattern reflects truly efficient prices, and what the exceptions tell us, is the question this chapter takes up. Read it at CNBC.

Market Insight

The news item above has a remarkably stable statistical shadow. Twice a year, S&P's SPIVA scorecard counts the fraction of actively managed U.S. funds that failed to beat their benchmark, and Figure 7.1 shows the latest tally by horizon. Two features deserve attention. First, even over a single year, active large-cap funds underperform far more often than a coin flip would suggest — 79% in the most recent year. Second, and more telling, the failure rate *rises* with the horizon: 89% of large-cap funds trail the S&P 500 over five years and 93% over twenty. That pattern is exactly what the efficient markets hypothesis predicts. If prices already impound public information, a manager's gross return is approximately the market's plus noise, and fees subtract from it every single year; over long horizons the noise washes out while the fee drag compounds, so ever fewer funds stay ahead. Note the one wrinkle: small-cap managers fare noticeably better at short horizons — 41% underperforming over one year — which is just where one would expect prices to be furthest from fully efficient. Even there, however, nine in ten trail their benchmark once the horizon stretches to fifteen years.

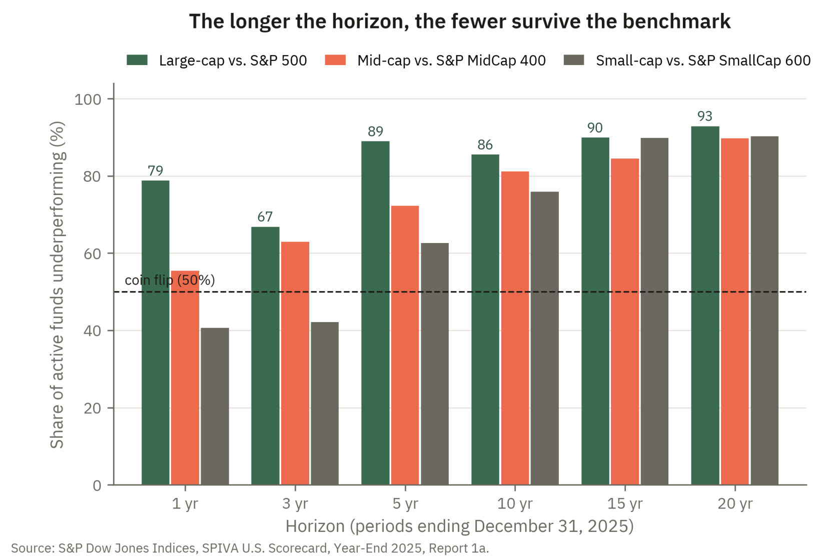

Figure 7.1: Share of actively managed U.S. equity funds underperforming their benchmark, by fund category and horizon, for periods ending December 31, 2025. Source: S&P Dow Jones Indices, SPIVA U.S. Scorecard, Year-End 2025.

7.2 The efficient markets hypothesis

The efficient markets hypothesis is a term that has many definitions. All of the definitions center around the idea that participants in the market will use all of the available information in a market as soon as it becomes available and hence that trading on information will likely have no value. Some imprecise definitions of the efficient market hypothesis are

Weak-form efficient markets hypothesis. The assumption that all data that is available from observing the history of market prices has already been incorporated into current market prices. Thus, investment strategies that rely on past market data should not yield returns to investors beyond what the investors can expect from holding the market requisite amount of risk.

Semistrong-form efficient market hypothesis. The assumption that all information available from observing *any* publicly available data has already been incorporated into the market price. Thus, investment strategies that rely on earnings or other balance sheet data should not yield returns to investors beyond what they can expect from holding the market requisite amount of risk.

Strong-form efficient market hypothesis. The assumption that asset prices reflect *all* information that is relevant to the payoffs of that asset. This includes all the information that a CEO or other insider might know.

Note that these hypotheses imply that certain trading strategies will not be successful. For example the weak-form assumption implies that Technical Analysis is unlikely to be successful. Likewise, the semistrong-hypothesis implies that both fundamental analysis and quantitative (factor) models are likely to not be successful. The strong-form hypothesis implies that no analysis, including insider trading, will help in picking stocks. If the strong form hypothesis holds then there is no reason for anyone to ever hold anything but the market portfolio (i.e. everyone should have a passive strategy).

Of course these are only assumptions. One might want to test whether these hypotheses are true. There are several reasons why testing these hypotheses is difficult. The first is that if you truly had a very profitable trading strategy you would never tell anyone about it. Instead you would make millions on your own and once it stopped working you would tell people about it. Also, unless markets are highly inefficient, the volatility of the market would make it hard to notice a strategy that increased returns by 5%. An increase of 20% would be observable, but the ability to have such an increase would mean the market was really inefficient. A third reason is that we can only observe finite time periods of data. Being successful over 5 years (or 10 or 20) does not necessarily imply that you have a successful trading strategy. You may be a monkey who picked stocks by throwing darts, but happened to throw darts in the right direction.

7.3 Behavioral Finance

There is some indication that the expected utility model with no other constraints does not generate predictions that reflect the data that is observed about asset prices. The field of behavioral finance (and behavioral economics in general) has sought to improve the predictions of traditional economic models by changing the set of assumptions that underly the preferences of the participating agents. Often these changes are based on research performed by psychologists about the nature of individual decision makers. These psychologists speak of behavioral "biases" that are evidenced by behavior that is inconsistent with expected utility. Some of these biases have to do with the ability of investors to process information. These include (one should read the definitions of these in the book)

Overconfidence. Some individuals overestimate the precision of their beliefs or forecasts. This may cause them to take larger positions in particular equities than they would otherwise take.

Representativeness bias. This is sometimes called "the law of small numbers." It is the bias that at times individuals believe that small samples provide a better estimate than they actually do about the future. This leads investors to invest based on extrapolated trends that are not believed to be as noisy as they really are. If all investors do this, it has been shown that mutual funds with otherwise irrational managers may survive.

Others have to do with the nature of investors' vN-M utility functions. These are discussed in the book and include prospect theory, regret avoidance, framing and others.

Note that simply because *individuals* display these types of biases does not imply that market prices will display biases consistent with these individual behaviors. If arbitrageurs are able to profit from the existence of these traders, then they may drive the resulting mispricings out of the market.

Thus, in order for these behaviors to have an effect on market prices, it must be true that there is some limit to the ability of these arbitrageurs to buy and sell in the market. Can you think of some of these limits? (Transactions costs, etc.)

7.4 How strong are the CAPM assumptions?

The efficient markets hypothesis and the CAPM are close relatives. The CAPM makes a very sharp version of the "no free lunch" claim: in equilibrium every asset lies on the security market line, so its expected return is exactly $r_f + \beta_i(Er_M - r_f)$ — compensation for its systematic risk and nothing more. There is no room for a positive "alpha," no room for an investment whose expected return exceeds what its risk warrants. That is a powerful conclusion, but it is only as strong as the assumptions it rests on. Recall the six assumptions of the CAPM, and notice that on closer inspection each of them can seem very strong.

All investors are price takers. This treats every participant as too small to move prices. But some players are clearly large enough to affect the prices at which they trade — a large institution unwinding a position, an activist investor, or a single large order in a thin or illiquid market. When a trader's own actions move prices, the tidy competitive equilibrium behind the CAPM is disturbed.

One identical holding period. Investors plainly do not share a single horizon. A twenty-five-year-old saving for retirement, a retiree drawing income, and a trading desk judged on quarterly performance are optimizing over very different lengths of time. The single-period framing also ignores that the investment opportunity set itself shifts over time, so that a long-horizon investor may rationally want to hedge against future changes in interest rates or expected returns — considerations the one-period model cannot see.

All investments are publicly traded. This is perhaps the strongest assumption of all. Enormous fractions of real wealth are not tradable:

your human capital and future labor income, privately held businesses, closely held real estate, and the returns to education. These assets are correlated with the market and cannot be freely diversified or hedged, so the "market portfolio" the theory prices is not the true portfolio of aggregate wealth. An investor whose largest asset is her own earning power faces a risk-return problem the CAPM does not fully describe.

No transactions costs and no taxes. Real trading incurs commissions and bid-ask spreads, and returns are taxed. Taxes in particular differ across investors — capital gains rates, holding-period rules, and the distinction between taxable and tax-advantaged accounts all drive a wedge between what different investors will pay for the very same stream of cash flows. Once such wedges exist, investors need not agree on prices even if they agree on everything else.

All investors are mean-variance optimizers. Mean-variance optimization is exactly correct only under quadratic utility or (roughly) normally distributed returns. But investors demonstrably care about more than the mean and the variance. They care about downside risk, about skewness and lottery-like payoffs, and about the possibility of ruin. The behavioral biases discussed in the previous section are themselves direct departures from this assumption.

Homogeneous expectations. Investors visibly disagree about the means and variances of asset payoffs. Analyst price targets diverge, and the sheer volume of trade is hard to reconcile with universal agreement — if everyone truly held identical beliefs, there would be far less reason to trade with one another at all. Disagreement is the normal condition of financial markets, not the exception.

7.5 What does alpha mean?

The discussion above referred informally to an investment that offers "more return than its risk warrants." That idea has a name. *Alpha* is the return an asset or portfolio earns in excess of what a benchmark asset-pricing model says it should earn given its risk. When the benchmark is the CAPM, the relevant risk is systematic risk and the benchmark return is the security market line, so the alpha of asset *i* is

$$\alpha_i = Er_i - \left[r_f + \beta_i(Er_M - r_f) \right].$$

Geometrically, α_i is the vertical distance between the asset's expected return and the security market line: a positive alpha means the asset plots *above* the line, delivering more than its beta would entitle it to, while a negative alpha means it plots below. If both the CAPM and the efficient markets hypothesis hold, then every asset lies exactly on the line and $\alpha_i = 0$ for everything. A nonzero alpha is therefore not a free-standing fact; it is a statement that something in that joint prediction has given way. There are two distinct ways this can happen.

The first is that the CAPM is simply the wrong benchmark — one of its assumptions fails. If trading is not frictionless, if investors hold large non-traded assets such as human capital, if they care about more than mean and variance, or if they do not share the same beliefs, then the security market line is not the correct description of equilibrium expected returns. In that case an asset can show a positive alpha *relative to the CAPM* while earning exactly fair compensation for some risk the CAPM leaves out. Here the alpha is an artifact of measuring against the wrong model — a "bad benchmark" — rather than a genuine reward in excess of risk.

The second is that the CAPM correctly describes the trade-off between risk and return, but the efficient markets hypothesis fails, so prices depart from the values that available information implies. Then a nonzero alpha reflects a true mispricing: a return available, at least for a time, to an investor who identifies it before prices adjust.

Because either channel — a broken pricing-model assumption or a failure of market efficiency — can produce the same number, a measured alpha is never self-interpreting. This is the *joint hypothesis problem*: any test of alpha is simultaneously a test of the pricing model used to define it and of market efficiency, and a nonzero estimate on its own cannot say which is responsible. A third possibility further complicates matters, namely that an estimated alpha is nonzero purely by chance in a finite sample of returns, an issue the performance-measurement material takes up when it distinguishes skill from luck.

This section takes no position on how large alpha typically is, or on whether investors can reliably identify it in advance. Those are empirical questions, and reasonable people disagree about the answers. The narrower point here is definitional: alpha is a return that a given asset-pricing model cannot account for, and its existence in any particular case depends on a violation of that model's assumptions or of the efficient markets hypothesis.

7.6 The CAPM as a personal investment checklist

None of this means the CAPM is useless. On the contrary, once you appreciate how demanding its assumptions are, the model becomes a remarkably practical guide to personal investing — especially when you are being pitched an opportunity you did not go looking for. Sooner or later a family member, a friend, or a friend of a friend will describe an investment that supposedly offers high returns with little corresponding risk. The CAPM gives you a disciplined way to think about that claim.

The logic is simple. The CAPM result says that unless one of its six assumptions is violated, there cannot be a project or investment opportunity that delivers returns higher than its risk would predict. Every legitimate opportunity sits on the security market line. So when someone asserts that an investment beats that line — high return, modest risk — you are left with only two possibilities. Either the claim is simply false, or one of the assumptions is genuinely being violated in a way that creates the opportunity. The value of the CAPM is that it hands you a checklist for telling these two cases apart. Walk through the assumptions one at a time and ask which one the pitch relies on:

1. **Is the return really compensation for illiquidity or non-tradability?**

A private business, a startup, or an interest in real estate is not publicly traded (assumption 3). Its extra return may be payment for the fact that you cannot easily sell it or diversify it away — a real risk you would be bearing, not a free lunch.

2. **Is there a genuine tax advantage?** If the opportunity exploits a feature of the tax code that applies to you (assumption 4), the higher after-tax return may be real — but it depends on your specific tax situation, not on any mispricing.

3. **Do you actually possess information others don't?** An edge that depends on private information touches both the strong-form EMH and the homogeneous-expectations assumption (assumption 6). Be honest about whether you truly know something the market does not — and whether acting on it is even legal.

4. **Is the "high return" hiding a fat downside?** Many high-return pitches are really lottery tickets or strategies that pay off steadily until they collapse (assumption 5). Mean-variance intuition understates that danger; the return is compensation for a tail risk you may not want.

5. **Could someone be moving the price?** If the opportunity depends on a large player pushing prices around (assumption 1), you need to ask whether you are the one benefiting from that or the one being taken advantage of.

If you go through this list and cannot identify any assumption that is plausibly violated, then the CAPM's verdict is blunt: the promised returns are almost certainly illusory, mismeasured because the risk is being ignored, or outright fraudulent. And when an opportunity *is* real, naming the specific assumption it violates tells you exactly what risk you are being paid to bear — and whether you are the right person to bear it.

❗ Market insight: affinity fraud and the trusted messenger

The checklist above assumes you can weigh a pitch dispassionately. But the most dangerous “high return” offers arrive through exactly the channel that tends to switch off scrutiny: a trusted member of your own community — the family member or friend who introduced the opportunity to you in the first place. This is *affinity fraud*, fraud that spreads through tight-knit groups in which shared identity substitutes for due diligence.

The empirical record bears this out. Analyzing data on more than half a million participants in the collapsed U.S. pyramid scheme Fortune Hi-Tech Marketing, Bosley and Knorr (2018) found that recruitment flourished in counties with identifiable affinity groups, with religious communities prominent among them. Reviewing why people join such schemes, Hock and Button (2023) point on the participants’ side to the vision of high reward for little work, and on the organizers’ side to the deliberate exploitation of specific groups combined with high-pressure sales — with victims frequently enlisted to become recruiters themselves.

The connection to this chapter is direct. When a relative or fellow congregant offers you an opportunity with unusually high returns, the trust you place in the *messenger* does the work that the CAPM says the *numbers* cannot: it makes an off-the-security-market-line promise feel safe. The checklist is precisely the antidote. Run through the assumptions regardless of who is pitching — and treat a trusted-community source not as a reason to skip the checklist but as a reason to work through it more carefully.

- Bosley, S. and Knorr, M. (2018), “Pyramids, Ponzis and fraud prevention: lessons from a case study,” *Journal of Financial Crime*, 25(1), 81–94. https://doi.org/10.1108/JFC-10-2016-0062
- Hock, B. and Button, M. (2023), “Why do people join pyramid schemes?,” *Journal of Financial Crime*, 30(5), 1130–1139. https://doi.org/10.1108/JFC-09-2022-0225

7.7 Application: A saver choosing index or active

A university employee reviewing the menu of funds in her retirement plan faces a deceptively simple question: should she pay for an actively managed fund that promises to beat the market, or accept the market's return through a low-cost index fund? The efficient-markets hypothesis is the framework that answers her. If prices already impound available information, then consistently identifying mispriced securities is extraordinarily difficult, and the fees charged by active managers become a near-certain drag on her returns — which is precisely what the long-run evidence on active underperformance shows. Understanding the different forms of market efficiency, and the anomalies that genuinely challenge them, lets her judge how much, if anything, she should be willing to pay in pursuit of returns above the index.

> Further listening
>
> *Planet Money* (NPR) — "Summer School 2: Index Funds & The Bet": Warren Buffett's million-dollar wager that a simple index fund would beat the experts — a real-world test of market efficiency.

7.8 Homework problems

7.8.1 Conceptual

EMH-C1. A hedge fund advertises a proprietary system that studies charts of past prices and trading volume to time entries and exits. State which form of the EMH, if true, would render this strategy unprofitable in expectation, and explain why. Then contrast this with a fund that trades on newly released accounting ratios: which form is required to rule out that second strategy, and why is it a stronger assumption?

EMH-C2. Carefully distinguish the three forms of the efficient markets hypothesis by the information set each assumes is already impounded in prices. Explain why the three forms are nested — that is, why the strong form implies the semi-strong form, which in turn implies the weak form — and give one concrete example of information that is included in the semi-strong information set but not the weak-form information set.

EMH-C3. A journalist claims: "A corporate insider who traded ahead of a merger announcement made a large profit, therefore markets are not efficient." Explain which form of the EMH this observation bears on, whether it is evidence against the weak or semi-strong forms, and why the strong-form EMH is generally regarded as the hardest to defend empirically.

EMH-C4. For each of the three standard investment approaches — technical analysis, fundamental (and factor) analysis, and insider trading — state which form of the EMH must hold to render it unprofitable in expectation, and explain the reasoning. Then explain the claim in the chapter that if the strong-form hypothesis holds, no one should ever hold anything but the market portfolio.

EMH-C5. Suppose a researcher documents that a real trading strategy earned 5% per year above a risk-matched benchmark over the past ten years. The chapter gives several reasons why this is weak evidence against the EMH. Explain at least two of these reasons (for example, market volatility masking the signal and the finite-sample "dart-throwing monkey" problem) and describe what additional evidence would make the case against efficiency more convincing.

EMH-C6. The chapter argues that testing the EMH is genuinely hard. One reason it gives is that a trader with a truly profitable strategy has an incentive to keep it secret. Explain how this incentive complicates empirical tests of efficiency, and explain separately why an increase in returns of, say, 5% is much harder to detect in the data than an increase of 20% — and why the very detectability of a 20% edge would itself signal a highly inefficient market.

EMH-C7. Psychological studies find that many individual investors are overconfident and exhibit representativeness bias. A colleague concludes that market prices must therefore be systematically distorted. Explain why this conclusion does not follow automatically, and describe the conditions on arbitrage under which individual behavioral biases can nonetheless move equilibrium prices. Give two concrete limits to arbitrage that would allow biased traders to affect prices.

EMH-C8. Define overconfidence and representativeness bias as the chapter uses them, and for each describe a specific way it could lead an investor to make a decision inconsistent with expected-utility maximization. The chapter notes that representativeness bias ("the law of small numbers") can allow mutual funds run by otherwise irrational managers to survive. Explain the intuition for how a bias in investors' beliefs could sustain such funds rather than driving them out.

EMH-C9. The chapter presents the requirement that arbitrage be limited as the bridge between individual behavioral biases and market-level mispricing. Explain why, in a world of frictionless arbitrage, biased traders would have no lasting effect on prices. Then discuss how the risk that a mispricing worsens before it corrects — as opposed to simple transaction costs — can deter an arbitrageur even when the arbitrageur correctly identifies the mispricing.

EMH-C10. Choose three of the six CAPM assumptions and, for each, explain why it may be "very strong" in the sense the chapter means, giving a concrete real-world situation in which it fails. Then explain why the failure of any one of these assumptions undermines the sharp CAPM conclusion that every asset must lie exactly on the security market line.

EMH-C11. The chapter singles out "all investments are publicly traded" as "perhaps the strongest assumption of all." Explain what kinds of wealth are not publicly traded, and why the existence of large non-traded assets such as human capital means the CAPM's "market portfolio" is not the true portfolio of aggregate wealth. Discuss why this matters for an investor whose largest asset is her own future labor income.

EMH-C12. Define alpha relative to the CAPM benchmark in words, and explain geometrically what a positive versus a negative alpha means in relation to the security market line. Explain why, if both the CAPM and the

EMH hold, every asset must have an alpha of exactly zero.

EMH-C13. State the joint-hypothesis problem in your own words. A researcher estimates a statistically significant positive alpha for a portfolio. Enumerate the distinct explanations the chapter offers for a nonzero measured alpha — a broken CAPM assumption ("bad benchmark"), a genuine failure of market efficiency, and finite-sample chance — and explain why a single nonzero estimate cannot by itself tell you which is responsible.

EMH-C14. A friend describes an investment that supposedly offers high returns with little corresponding risk. Using the chapter's "personal investment checklist," walk through how you would evaluate the pitch by asking which CAPM assumption, if any, the opportunity relies on. Explain what conclusion you should draw if you cannot identify any assumption that is plausibly violated.

EMH-C15. Explain what "affinity fraud" is and why the chapter argues that a trusted-community source is a reason to work through the CAPM checklist *more* carefully rather than to skip it. Connect this to the chapter's point that trust in the *messenger* can do the work that the CAPM says the *numbers* cannot.

EMH-C16. The "In the news" note reports that fewer active managers beat their benchmarks and that trillions of dollars have flowed from active into low-cost index funds. Explain how the efficient markets hypothesis predicts this pattern, and then explain why persistent active underperformance is not, by itself, decisive proof that markets are efficient — identify what alternative explanations the chapter's framework leaves open.

7.8.2 Quantitative

EMH-Q1. A stock has a beta of 1.2. The risk-free rate is 3% and the expected return on the market is 9%. An analyst forecasts that the stock will actually earn an expected return of 12%.

(a) Using the security market line, compute the return the CAPM says the stock should earn given its risk.

(b) Compute the stock's alpha, $\alpha_i = Er_i - [\, r_f + \beta_i(Er_M - r_f)\,]$.

(c) State whether the stock plots above or below the security market line, and explain what the joint-hypothesis problem implies about how you should interpret this nonzero alpha.

EMH-Q2. Two portfolios are evaluated against the CAPM. The risk-free rate is 2% and the market risk premium $Er_M - r_f$ is 5%. Portfolio A has a beta of 0.8 and an expected return of 7%. Portfolio B has a beta of 1.5 and an expected return of 9%.

(a) Compute the CAPM-implied expected return and the alpha for each portfolio.

(b) Which portfolio has the larger alpha? Does the portfolio with the higher *raw* expected return also have the higher alpha? Explain what this shows about comparing investments on raw return versus risk-adjusted return.

(c) If both alphas are in fact zero in the true population and your estimates are nonzero, name the reason from the chapter that would explain the discrepancy.

7.8.3 Data exercises

These problems use the live-data figure in this chapter.

EMH-D1. Suppose the index earns 8% per year and an active fund holds essentially the market portfolio but charges a 1% annual fee, netting 7%. (a) Compute the ratio of the active investor's wealth to the index investor's wealth after 1 year, 5 years, and 20 years. (b) Explain how this simple fee arithmetic accounts for the pattern in Figure 7.1, where the share of large-cap funds underperforming rises from 79% at one year to 93% at twenty years.

EMH-D2. In Figure 7.1, only 41% of active small-cap funds underperformed their benchmark over one year — better than a coin flip — yet roughly 90% underperform over fifteen years. Does the one-year number refute the efficient markets hypothesis? Discuss with reference to (i) where prices are most plausibly far from fully efficient, (ii) the role of luck in short samples, and (iii) what the long-horizon number implies about whether any short-horizon edge survives fees.

Part III

Fixed income securities and pricing

Chapter 8

Interest

8.1 Introduction

Two questions structure this chapter. First, how does the passage of time affect the value of money, and what mathematical framework allows us to place cash flows occurring at different dates on a common footing? Second, how do the various conventions for quoting and compounding interest rates relate to one another, and which measure of interest — simple, compound, or continuously compounded — is the right one to use when pricing a financial security? These questions form the conceptual bedrock of fixed-income analysis and are prerequisites for virtually every quantitative application in finance.

The first question matters because almost every significant financial decision involves comparing cash flows that do not occur at the same time. A firm evaluating a capital investment must weigh an immediate outlay against revenues that arrive over years or decades. An individual choosing between a lump-sum retirement payout and a life annuity must compare a certain payment today against a stream of future payments that depends on longevity. A government issuing a bond must price the security so that the proceeds it receives today equal what the market regards as the present value of future coupon and principal payments. Without a rigorous discounting framework these comparisons collapse into apples-and-oranges arithmetic, and mispricing is inevitable. The chapter establishes this framework from first principles using the arbitrage principle: the present value of a future cash flow is precisely the amount one would need to invest today, at the prevailing interest rate, to replicate that payment. The resulting present-value formula extends naturally from a single payment to a finite stream of cash flows, and then — via the mathematics of geometric series — to infinite streams and fixed-horizon annuities.

The second question matters because the same nominal interest rate implies very different rates of wealth accumulation depending on how frequently interest is compounded. A stated annual rate of 12% compounded monthly leaves the investor meaningfully richer after one year than the same rate compounded annually, and continuous compounding — the limiting case as the compounding frequency tends to infinity — produces the largest accumulation of all. Financial contracts, regulatory disclosures, and security prospectuses employ all three conventions, and the ability to convert among them accurately is an essential practical skill. The limit of continuous compounding also connects interest-rate arithmetic to the exponential function via a calculus argument using L'Hôpital's rule, illuminating why the natural exponential appears so pervasively in mathematical finance. The chapter closes by applying the full discounting framework to the pricing of canonical fixed-income instruments: zero-coupon Treasury bills, perpetuities (including the historically important British consol bond), and fixed-term annuities.

The goal of this lecture is to understand the pricing of risk free securities. All of the relationships presented rely on an understanding of the *time value of money*. The phrase *time value of money* refers to the fact that the preferences of most individuals are such that they are not indifferent between receving \$1000 today and receiving \$1000 in one year.

In the news

Interest rates are set in the headlines as well as the textbook. In June 2026 the Federal Reserve held its policy rate at 3.50–3.75% even as inflation climbed toward 3.6%, leaving the *real* return on cash slim or negative. The distinction between nominal and real rates, and the compounding that turns a quoted rate into actual growth, is the machinery this chapter builds. Read it at Fox Business.

8.2 Present value

Unless noted otherwise, suppose that an investor faces a risk-free interest rate of r every period for the foreseeable future. This implies that the present value of x dollars received one period from now is

$$PV_1 = \frac{1}{1+r}x.$$

What is the present value of x dollars received in two periods? To answer this we consider exactly how much would have to be saved today, to obtain a payment of x in two periods. By the arbitrage principle the present value of a risk free payment in two periods must be the same as the amount you would be required to save today in order to receive that payment in two periods. Let PV_2 be the required amount of saving today. If you save PV_2, then at the end of one period your total wealth from this investment will be $PV_2(1+r)$. Since you want to save just enough to receive the payment x in *two* periods, you reinvest your principal and interest ($PV_2(1+r)$) at the market interest rate of r and at the end of two periods you will have $PV_2(1+r)(1+r) = PV_2(1+r)^2$. Recall that you want PV_2 to have been just enough to yield x at the end of two periods so you want PV_2 to solve

$$PV_2(1+r)^2 = x.$$

Solving this implies that $PV_2 = (1+r)^{-2}x$. In other words, the present value (or present discounted value) of the payment x received in two periods is $(1+r)^{-2}x$.

Suppose that one were to receive the payment x in T years. The required amount of investment capital to generate the payment x in T years satisfies $PV_T(1+r)^T = x$ and so the present value of the payment x to be received in T years is $PV_T = (1+r)^{-T}x$.

Notice that PV_T falls as the horizon T lengthens and as the interest rate r rises. Both effects follow from the same replication logic: the longer the wait, or the higher the rate, the less you must set aside today to reach a given future sum. (Problem INT-Q11 asks you to put numbers on this.)

Now imagine that you own the right to receive the amount Y_1 one period from now and the amount Y_2 two periods from now? What is the present value of this asset? $PV = Y_1/(1+r) + Y_2/(1+r)^2$. Why? How much would you have to save in order to earn this payoff? You start by saving PV. At the end of the first period you now have $PV(1+r)$. From that, you remove the payment Y_1 that you are to receive and save $PV(1+r) - Y_1$ at interest. At the end of the second period you have $(PV(1+r) - Y_1)(1+r)$. By the arbitrage principle it must be that $(PV(1+r) - Y_1)(1+r) = Y_2$. Solving this equation for PV gives $PV = Y_1/(1+r) + Y_2/(1+r)^2$.

In general, suppose that one owned the right to receive the sequence of payments $Y_1, Y_2, \ldots, Y_T$. In a manner analogous to the above, it can be shown that the present value of this sequence of payments is

$$PV(Y) = \sum_{t=1}^{T} \frac{Y_t}{(1+r)^t}$$

8.3 A digression on geometric series

An infinite geometric series is a function of the form

$$f(x) = \sum_{t=0}^{\infty} x^t$$

Notice that $f(x)$ is not defined for all $x \in \mathbb{R}$. If $x \geq 1$ then there is no real number r such that $r = \sum_{t=0}^{\infty} x^t$. If $|x| < 1$ then it can be shown that there exists a number r such that $r = \sum_{t=0}^{\infty} x^t$.[1]

Infinite (and convergent) geometric series. Let $y = \sum_{t=0}^{\infty} x^t = 1+x+x^2+\cdots$. Therefore, $yx = x + x^2 + x^3 + \cdots$. This implies that $y - yx = 1$ which in turn implies that

$$y = \frac{1}{1-x}$$

Finite geometric series. Let $y = \sum_{t=0}^{T} x^t = 1 + x + \cdots + x^T$. Then $yx = x + x^2 + \cdots + x^{T+1}$. Therefore

$$y = \frac{1-x^{T+1}}{1-x}$$

Notice that as $T \to \infty$ the value of the finite geometric series converges to the value of the infinite geometric series if $|x| < 1$.

Why is this useful? Consider an asset that makes the constant payment of M once a year, forever. Assume that the first payment will be one year from now. What is the present value of such an asset if the interest rate is assumed to be the constant rate r forever?

$$V = \sum_{t=1}^{\infty} \frac{M}{(1+r)^t} = M\left(\frac{1}{1-\frac{1}{1+r}} - 1\right) = M\frac{1}{r}$$

To see the cancellation explicitly, set $x = \frac{1}{1+r}$, so that $|x| < 1$ whenever $r > 0$. The sum starts at $t = 1$ rather than $t = 0$, so we subtract the missing $t = 0$

[1] Satchell, Stephen, and Alan Scowcroft. "A demystification of the Black-Litterman model: Managing quantitative and traditional portfolio construction." In Forecasting expected returns in the financial markets, pp. 39-53. Academic Press, 2007.

term:

$$V = M\sum_{t=1}^{\infty} x^t = M\left(\sum_{t=0}^{\infty} x^t - 1\right) = M\left(\frac{1}{1-x} - 1\right).$$

Now substitute $x = \frac{1}{1+r}$ into $\frac{1}{1-x}$:

$$\frac{1}{1-x} = \frac{1}{1-\frac{1}{1+r}} = \frac{1+r}{(1+r)-1} = \frac{1+r}{r},$$

so that

$$V = M\left(\frac{1+r}{r} - 1\right) = M \cdot \frac{(1+r)-r}{r} = M\frac{1}{r}.$$

The first equality follows from the rule on infinite and convergent geometric series. The quantity $\frac{1}{r}$ might be known as the *perpetuity factor* since a payment of M made *in perpetuity* (i.e. forever) is worth $\frac{M}{r}$.

Now consider the case of an annuity.

Annuity. A constant payment to be made regularly for a fixed period of time.

Example. Consider an annuity that makes annual payments of M for T years. Assume that the first payment is to be made one year from now. If the interest rate over this period is to remain constant at r, what is the value (i.e. present value) of such an asset?

Using the rule on finite geometric series above, we see that

$$V = \sum_{t=1}^{T} \frac{M}{(1+r)^t} = M\frac{1-\left(\frac{1}{1+r}\right)^T}{r} = M\frac{1}{r}\left(1 - \frac{1}{(1+r)^T}\right)$$

Again writing $x = \frac{1}{1+r}$, the sum runs from $t = 1$ to T, so factor out one power of x and apply the finite-series rule to the remaining T terms:

$$V = M \sum_{t=1}^{T} x^t = M\,x \sum_{t=0}^{T-1} x^t = M\,x\,\frac{1-x^T}{1-x}.$$

The prefactor collapses just as in the perpetuity case, since $\dfrac{x}{1-x} = \dfrac{1/(1+r)}{r/(1+r)} = \dfrac{1}{r}$. Hence

$$V = M\,\frac{1}{r}\,(1-x^T) = M\frac{1}{r}\left(1 - \frac{1}{(1+r)^T}\right).$$

The term $\frac{1}{r}(1 - \frac{1}{(1+r)^T})$ is called the *annuity factor* at interest rate r.

Another way of thinking about this formula (using arbitrage) is the following. Consider purchasing a perpetuity today that makes constant payments of M and selling the right to all payments after period T. The value of the perpetuity that you purchase originally is $M\frac{1}{r}$. The value at time T (after the T-period payment) of the payments that you sell is $M\frac{1}{r}$, but the value of these payments today is $M\frac{1}{r(1+r)^T}$. Since the annuity makes exactly the same payments as purchasing a perpetuity today and selling all of the payments after time T, the prices of these assets must be the same. Therefore the value of the annuity is

$$V = M\frac{1}{r} - M\frac{1}{r(1+r)^T} = M\frac{1}{r}\left(1 - \frac{1}{(1+r)^T}\right)$$

as above.

8.4 Simple and compound interest

Simple interest. Interest that is calculated on the original principal only.

> **Example.** If the amount x is deposited for two years in an account paying simple interest at the rate r per year, at the end of the first year the account will have a balance of $B_1 = x + xr$ and at the end

of two years the balance will be $B_2 = x + xr + xr = x(1 + 2r)$.

Compound interest. Interest that is calculated on both the original principal and on previous interest payments.

Example. If the amount x is deposited for two years in an account paying compound interest at the rate r per year, at the end of the first year the account will have a balance of $B_1 = x + xr = x(1 + r)$ and at the end of two years the balance will be $B_2 = x + xr + (x + xr)r = x(1 + r)^2$.

Compound period. The time beginning immediately after interest is calculated on an asset and ending with the next calculation of interest on the asset.

Example. In the U.S. savings accounts often pay compound interest that is calculated monthly. Thus, every month the balance on the account is used to calculate the amount of interest that the bank will pay to the holder of the account. The compound period in this case is 1 month.

Periodic interest rate. The rate of interest paid over one compound period.

Annual interest rate (annual rate). The rate of interest paid on an asset annually without accounting for compound interest. This is calculated as the periodic interest rate multiplied by the number of compound periods in a year.

Example. A savings account in the U.S. that pays a periodic interest rate of 1% pays an annual interest rate of 12%.

Effective annual rate. The rate of interest paid on an asset annually when compound interest is taken into account. The effective annual rate is the percentage increase in funds invested over a 1-year horizon.

Example. Consider depositing x in the bank at a stated annual interest rate of r that is compounded every 6 months. This means that every 6 months interest is calculated on the amount in the account. Thus the compound period is 6 months and the periodic interest rate is $\frac{r}{2}$. Therefore, at the end of six months (after interest has been paid) the balance on the account is

$$B_6 = x + x\left(\frac{r}{2}\right)$$

and at the end of the year the amount in the account is

$$\begin{aligned} B_{12} &= B_6 + B_6\left(\frac{r}{2}\right) \\ &= B_6\left(1 + \frac{r}{2}\right) \\ &= x\left(1 + \frac{r}{2}\right)^2 \end{aligned}$$

If interest is compounded n times during the year then at the end of the first period the balance in the account is $B_1 = x(1 + \frac{r}{n})$. At the end of two periods the balance is $B_2 = x(1 + \frac{r}{n})^2$ and after m periods the balance is $B_m = x(1 + \frac{r}{n})^m$. The *effective annual rate* is then

$$r^* = \left(1 + \frac{r}{n}\right)^n - 1.$$

The effective annual rate always exceeds the stated annual rate whenever $n > 1$, and the gap widens with n. A certificate of deposit quoting an annual rate of 1.99% compounded monthly, for instance, has a compound period of one month, a periodic rate of $0.0199/12 = 0.16583\%$, and an effective annual rate of $(1 + 0.0199/12)^{12} - 1 = 2.008\%$.

If an asset with initial value x is to be held for t years at the annual interest rate r and compounding occurs n times a year, the balance on the account at the end of the t years will be

$$B_t = x\left(1 + \frac{r}{n}\right)^{nt}$$

8.4.1 Continuous compounding

Consider what happens to the effective annual rate when the number of compounding periods in a year goes to ∞.

$$\lim_{n\to\infty}\left(1+\frac{r}{n}\right)^n$$

This is the gross effective annual rate (i.e. 1 plus the effective annual interest rate) of an asset that is re-compounded at every instant of time. This type of compounding is known as *continuous compounding*. While very few assets are actually compounded continuously, mathematically this is a very interesting case.

> **Mathematical note.** There are a few rules from calculus that are needed here.
>
> 1. If f and g are continuous functions, g has a continuous inverse and $\lim_{x\to\infty} f(x)$ exists then
>
> $$g^{-1}\left(\lim_{x\to\infty} g(f(x))\right) = \lim_{x\to\infty} f(x).$$
>
> This follows from the continuity of g and its inverse.
>
> 2. L'Hôpital's rule. It says that if the functions f and g are differentiable and
>
> $$\lim_{x\to\infty}\frac{f'(x)}{g'(x)} = L$$
>
> then
>
> $$\lim_{x\to\infty}\frac{f(x)}{g(x)} = L.$$

By rule 1 above, if

$$\lim_{n\to\infty} \log\left(1+\frac{r}{n}\right)^n = L$$

then

$$\lim_{n\to\infty} \left(1+\frac{r}{n}\right)^n = e^L.$$

We will find this first limit.

$$\begin{aligned}
\lim_{n\to\infty} \log\left(1+\frac{r}{n}\right)^n &= \lim_{n\to\infty} n\log\left(1+\frac{r}{n}\right)\\
&= \lim_{n\to\infty} \frac{\log\left(1+\frac{r}{n}\right)}{\frac{1}{n}}
\end{aligned}$$

Notice that this bottom limit has the form $\frac{0}{0}$, a perfect candidate for L'Hôpital's rule. Finding the derivative of top and bottom

$$\begin{aligned}
\lim_{n\to\infty} \frac{\left(\frac{1}{1+\frac{r}{n}}\right)\frac{-r}{n^2}}{\frac{-1}{n^2}} &= \lim_{n\to\infty}\left(\frac{1}{1+\frac{r}{n}}\right)r\\
&= r
\end{aligned}$$

By L'Hôpital's rule this implies that $\lim_{n\to\infty}\log\left(1+\frac{r}{n}\right)^n = r$ which implies (by rule 1) that

$$\lim_{n\to\infty} \left(1+\frac{r}{n}\right)^n = e^r.$$

Thus the value of assets that are compounded continously is easy to calculate. An initial deposit of x held for t years at the continuously compounded annual rate r grows to

$$B_t = xe^{rt},$$

and the effective annual rate is $r^* = e^r - 1$. Because e^r is the *supremum* of $(1 + r/n)^n$ over n, continuous compounding places an upper bound on how much more frequent compounding can be worth: at $r = 5\%$, annual compounding returns 5.000%, semiannual 5.0625%, monthly 5.1162%, daily 5.1267%, and continuous 5.1271%. The gains shrink rapidly once compounding is more frequent than monthly.

8.5 Pricing Securities

It turns out that for a lot of risk-free securities (like U.S. government issued bills, notes and bonds) one can calculate the correct price of the security using exactly the methods just discussed. Three canonical instruments illustrate the point.

Zero-coupon bonds. The U.S. government issues short-term debt (usually 28, 91, or 182 days to maturity) referred to as T-Bills (Treasury Bills). These generally have a face value of \$1000 and pay no coupons: the issuer agrees only to pay the holder the face value at maturity. A zero-coupon security is therefore a single future cash flow, so its price is just the present value of that one payment. A six-month bill of face value F, when the six-month interest rate is r, must sell for

$$P = \frac{F}{1 + r}.$$

Since $r > 0$ the bill trades at a *discount* to face value, and the discount deepens as r rises. The bill's return is earned entirely through the gap between purchase price and face value rather than through interim payments.

Perpetuities and the British consol. The British government has at times issued what is referred to as a consol bond (for *consolidated fund stock* — these were first issued in the 1750s). These securities pay interest forever and never repay principal.[2] A consol with face value M paying

[2] Actually there is a call provision, but we will ignore that for now.

quarterly at the periodic rate $r/4$ delivers a coupon of $rM/4$ every quarter in perpetuity, so by the perpetuity formula its price immediately *after* a coupon has been paid is

$$P = \sum_{t=1}^{\infty} \frac{rM/4}{\left(1 + \frac{r}{4}\right)^t} = \frac{rM}{4} \cdot \frac{4}{r} = M.$$

A consol whose coupon rate equals the market rate is thus worth exactly its face value — it trades at par. Purchased immediately *before* a coupon payment, the buyer also collects the imminent $rM/4$, so the price is $M + rM/4$. The general lesson is that a perpetual security's price is pinned down entirely by the ratio of its coupon to the market rate, which is why consol prices moved so sharply with interest rates.

Annuities. An annuity pays a constant amount for a fixed number of periods rather than forever, so its price is the perpetuity value net of the value of the payments given up after date T — the annuity factor derived above. Annuities are ubiquitous outside the bond market as well: a mortgage, a car lease, a level pension payout, and a monthly service contract attached to a piece of hardware are all annuities, and the present-value machinery prices each of them the same way. Problem INT-Q14 works through one such case in which two purchase plans with different up-front costs and different monthly fees are compared, and in which the ranking of the two plans reverses once the discount rate is high enough.

8.6 Application: A retiree weighing a CD

A saver approaching retirement is deciding whether to lock a year of living expenses into a certificate of deposit advertising an attractive rate. The tools of this chapter tell her what that rate actually means. Compounding determines how the quoted annual rate translates into the sum she will hold at maturity, and the Fisher relation — real return equals nominal return minus expected inflation — reveals the figure that truly matters: with inflation running above the rate the CD pays, her money is guaranteed to grow in dollars while quietly shrinking in purchasing power. Distinguishing nominal from real returns, and understanding how the time value of money links payments across dates, is what separates a decision that feels safe from one that actually preserves what she has earned.

> Further listening
>
> *The Indicator from Planet Money* (NPR) — "How the Fed's interest rate hikes are like Ratatouille": a short, vivid explanation of the two interest rates that really matter when the Fed sets policy.

8.7 Homework problems

8.7.1 Conceptual

INT-C1. A friend offers you a choice: receive \$1000 today, or receive \$1000 exactly one year from now. Using the idea of the *time value of money*, explain why most individuals strictly prefer the payment today even when they are certain the future payment will actually arrive. In your answer, connect the preference to the risk-free interest rate r and to what the arbitrage principle says about the present value of the delayed payment.

INT-C2. Two risk-free assets, A and B, each promise a single payment of \$5000. Asset A pays in 3 years and asset B pays in 8 years. Without doing any arithmetic, explain which asset has the larger present value and why. Then explain what happens to *each* asset's present value, and to the gap between them, as the interest rate r rises. Relate your reasoning to the present-value formula $PV_T = (1 + r)^{-T} x$.

INT-C3. The chapter values a perpetuity paying M forever as $V = M/r$ using the rule on infinite geometric series. Explain, without algebra, why the value is *decreasing* in r and why a small perpetual payment can nonetheless be worth a large lump sum today. What feature of the arbitrage/replication argument guarantees that M/r is exactly the amount you would need to invest today, at rate r, to reproduce the payment stream M forever?

INT-C4. The chapter derives the annuity value $V = M\frac{1}{r}\left(1 - (1+r)^{-T}\right)$ two ways: from the finite geometric series, and from an *arbitrage* argument in which a perpetuity is bought today and the rights to all payments after period T are sold. Explain in words why these two routes must give the same answer, and identify the role played by the "law of one price" (assets with identical cash flows have identical prices). As $T \to \infty$, what does the annuity value approach, and why should that be obvious from the arbitrage picture?

INT-C5. A student claims: "Whether the bank uses simple or compound interest doesn't really matter — over a single year the balance is the same either way, and over longer horizons the difference is tiny." Evaluate this claim. Explain the precise sense in which the student is right for a one-year horizon with interest paid once at year-end, and the sense in which they are wrong once interest is credited more than once per period or the money is left for many years.

INT-C6. Using the chapter's two-year examples, explain why a balance earning *compound* interest at rate r grows as $x(1+r)^2$ while the same principal earning *simple* interest grows only as $x(1+2r)$. Identify precisely the dollar term that compounding adds over the two years, explain in words what that term represents, and state whether the gap between the two balances widens or narrows as the horizon lengthens.

INT-C7. Two certificates of deposit both advertise an *annual interest rate* of 6%. The first compounds semiannually; the second compounds monthly. Explain why the second leaves the investor richer after one year even though the two quoted annual rates are identical. Define the *effective annual rate* r^* in your answer and explain why r^*, rather than the quoted annual rate r, is the correct number to compare across the two CDs.

INT-C8. Explain, in words, why increasing the number of compounding periods n within a year raises the effective annual rate $r^* = (1 + r/n)^n - 1$, yet the effective rate does not grow without bound as $n \to \infty$. What is the limiting gross return the chapter derives, and why does the natural exponential e^r appear as the ceiling on how much more frequent compounding can do for you?

INT-C9. The chapter obtains continuous compounding by taking $n \to \infty$ in $(1 + r/n)^n$ and, via L'Hôpital's rule applied to $\log(1 + r/n)^n$, finds the limit equals e^r. Explain in words why the logarithm is introduced before the limit is taken, and why the intermediate limit has the indeterminate form $0/0$ that makes L'Hôpital's rule applicable. What does the resulting balance formula $B_t = xe^{rt}$ say about how a continuously compounded deposit grows through time?

INT-C10. In the chapter's "In the news" callout, the Fed's policy rate sits near 3.6% while inflation runs near 3.6%. Using the distinction between *nominal* and *real* returns (the Fisher relation, real return equals nominal return minus expected inflation), explain why a saver holding cash at this nominal rate may see her balance grow in dollars while her purchasing power stays flat or shrinks. Why does the compounding machinery of this chapter operate on the nominal rate, not the real rate?

INT-C11. A retiree can take her pension as a single lump sum today or as a level lifelong monthly payment. Explain how the Fisher relation and the annuity/perpetuity valuation of this chapter together shape the comparison: which quantity should she discount the payment stream at, and why does inflation erode a *fixed* nominal annuity in a way it does not erode a payment that is indexed to prices? Keep the discussion conceptual — no arithmetic required.

INT-C12. A T-Bill is a zero-coupon security: the issuer pays only the face value at maturity, with no intermediate coupons. Explain why the price of such a bill must be *below* its face value in a world with a positive interest rate, and describe how the bill's price would move if the six-month interest rate r suddenly rose. Relate your answer to the single-payment present value formula $PV_T = (1 + r)^{-T}x$.

8.7.2 Quantitative

INT-Q1. An investor is entitled to a stream of risk-free payments: \$400 one year from now, \$600 two years from now, and \$1000 three years from now. The risk-free rate is $r = 0.08$ every period. Using $PV(Y) = \sum_{t=1}^{T} \frac{Y_t}{(1+r)^t}$, compute the present value of this stream.

INT-Q2. You own the right to receive \$2500 exactly $T = 4$ years from now, and the risk-free interest rate is $r = 0.07$. Using $PV_T = (1 + r)^{-T} x$, find the present value. Then find the present value if the horizon is instead $T = 9$ years, holding r fixed.

INT-Q3. A perpetuity pays a constant \$750 at the end of each year, forever, with the first payment one year from today. The interest rate is $r = 0.05$ forever. Using $V = M/r$, compute the present value of the perpetuity. Then recompute it if the interest rate falls to $r = 0.04$, and comment on the direction of the change.

INT-Q4. A British consol has face value $M = \$1000$ and pays quarterly interest at the periodic rate $r/4$ where the annual rate is $r = 0.08$. The quarterly coupon is therefore $rM/4$. Using the perpetuity logic $P = \sum_{t=1}^{\infty} \frac{rM/4}{(1+r/4)^t} = \frac{rM/4}{r/4}$, compute the price immediately *after* a coupon has been paid, and then the price immediately *before* a coupon (when the buyer also collects the imminent payment).

INT-Q5. An annuity pays $M = \$1500$ at the end of each year for $T = 15$ years, with the first payment one year from now. The interest rate is $r = 0.06$. Using the annuity factor $V = M\frac{1}{r}\left(1 - \frac{1}{(1+r)^T}\right)$, compute the present value of the annuity.

INT-Q6. An annuity pays $M = \$800$ at the end of each year for $T = 25$ years, with the first payment one year from now, at interest rate $r = 0.09$. Using $V = M\frac{1}{r}\left(1 - \frac{1}{(1+r)^T}\right)$, compute the present value. As a check, confirm it is smaller than the value of the corresponding perpetuity M/r.

INT-Q7. You deposit $x = \$5000$ at an annual interest rate of $r = 0.10$ compounded $n = 4$ times per year, and leave it for $t = 8$ years. Using $B_t = x\left(1 + \frac{r}{n}\right)^{nt}$, compute the ending balance.

INT-Q8. A credit card quotes an annual rate of $r = 0.18$ compounded monthly ($n = 12$). Using $r^* = \left(1 + \frac{r}{n}\right)^n - 1$, compute the effective annual rate. Then compute the effective annual rate if the same 18% annual rate were compounded daily ($n = 365$).

INT-Q9. You deposit $3000 at an annual rate of $r = 0.05$ compounded *continuously*. Using $B_t = xe^{rt}$ (the continuous-compounding limit $\lim_{n\to\infty}(1 + r/n)^n = e^r$), compute the balance after $t = 4$ years, and compute the effective annual rate $r^* = e^r - 1$ for this account.

INT-Q10. You deposit $10{,}000 at an annual rate of $r = 0.03$ compounded *continuously* and leave it for $t = 10$ years. Using $B_t = xe^{rt}$, compute the ending balance, and compute the effective annual rate $r^* = e^r - 1$.

INT-Q11. The interest rate is $r = 0.10$ in every period. Using $PV_T = (1+r)^{-T}x$, compute the present discounted value of $1000 received in one year, in two years, and in seven years. By what percentage does the present value fall between year 1 and year 7, and explain why the decline is not simply seven times the one-year discount.

INT-Q12. The U.S. Treasury issues a six-month T-Bill with face value $F =$ $1000 and no coupon payments. (a) Derive the price of the bill as a function of the six-month interest rate r. (b) Compute the price when $r = 0.021$ (2.1% over six months). (c) What annual effective rate does that six-month rate correspond to, assuming the rate can be rolled over at the same terms? (d) If the six-month rate rises to $r = 0.025$, recompute the price and state the direction and dollar size of the price change.

INT-Q13. A savings account pays an annual interest rate of $r = 0.05$ compounded *continuously*. You deposit $1000 today. (a) How much will you have in one year, and what effective annual rate does that imply? (b) Recompute the one-year balance if interest were instead compounded semiannually, monthly, and daily. (c) Order the four balances and comment on how much of the total gain from more frequent compounding is already captured by moving from semiannual to monthly compounding.

INT-Q14. The original iPhone had an up-front cost of $399 plus a required data plan of \$20 per month; a later model had an up-front cost of \$199 plus a required data plan of \$30 per month. Treat each as a two-year commitment: an immediate payment plus 24 month-end payments. (a) At a monthly interest rate of 0.4074% (roughly 5% annually with monthly compounding), compute the present value of the total cost of each phone using the annuity factor $\frac{1}{r}\left(1-(1+r)^{-T}\right)$, and say which is cheaper. (b) Repeat at monthly rates of 1% and 2%. (c) At which of these rates does the ranking reverse, and explain intuitively why a higher discount rate favors the phone with the larger up-front cost.

Chapter 9

Fixed Income Yields

9.1 Introduction

Three questions animate the study of fixed income yields. First, what interest rate is embedded in an observed bond price — and how is that rate extracted from a security's promised cash flows? Second, does the yield to maturity actually represent the return an investor will earn, and under what conditions do the two coincide? Third, what determines the return to holding a bond over a finite period that ends before maturity? These questions matter because fixed income securities constitute one of the largest asset classes in the global economy. The interest rates implied by bond prices serve as benchmarks for borrowing costs across the entire financial system, transmit the effects of monetary policy to households and firms, and anchor the discount rates used to value virtually every other financial asset. A precise command of these concepts is therefore foundational for anyone who studies or works in financial markets.

The first question leads to the concept of yield to maturity: the single discount rate that equates the present value of a bond's entire stream of promised payments to its observed market price. This is not merely a computational convention. Yield to maturity is the language through which bond markets communicate value and make comparisons across instruments. Understanding how it is derived — and how it depends on whether the bond trades above or below par — provides the analytical core of fixed income valuation. The chapter covers both the bond equivalent yield, which uses simple annualization, and the effective annual yield, which accounts for compounding, and works through the relationship between them.

The second question exposes a critical limitation of the yield to maturity as a measure of investment return: it assumes that every coupon payment can be reinvested at exactly the yield prevailing at the time of purchase. In practice, interest rates change, and coupon income is reinvested at whatever rate the market offers at the time. When those reinvestment rates differ from the original yield, the realized compound return — the actual annualized gain from holding the bond to maturity — diverges from the quoted yield. This distinction is not merely theoretical; it has direct implications for how investors should evaluate and compare bond investments under different interest rate scenarios.

The third question introduces the holding period return, which is relevant whenever an investor sells a bond before it matures. Because bond prices move inversely with yields, a rise in market interest rates causes the price of an existing bond to fall, and a sale at that lower price generates a capital loss that offsets some or all of the coupon income earned during the holding period. Conversely, a decline in rates produces a capital gain. Together, the coupon income and the capital gain or loss determine the total holding period return, which can differ substantially from the yield to maturity depending on how rates evolve and how long the bond is held.

The chapter begins by mapping the landscape of fixed income securities, distinguishing money market instruments from longer-term bonds and surveying the main categories — Treasury securities, corporate bonds, and asset-backed securities — along with the role of credit ratings in explaining yield differences across issuers. It then derives the yield to maturity formally, verifies that a bond priced at par has a yield equal to its coupon rate (and conversely), establishes the intuitive rules linking price to yield, and illustrates the relationship between bond equivalent and effective annual yields through worked examples. The chapter closes by showing how the realized compound return depends on reinvestment rates and how capital gains and coupon income combine to determine the return over any finite holding period.

> **In the news**
>
> A bond's price and its yield move in opposite directions, and nothing illustrates it like a hot inflation report. CNBC reported the 10-year Treasury yield climbing to about 4.46% after consumer prices rose to their highest level in nearly three years — as yields rose, existing bond prices fell, and the cost of mortgages and other loans rose with them. Computing exactly that relationship between price, coupon, and yield to maturity is the work of this chapter. Read it at CNBC.

9.2 Fixed Income Securities

The following definitions for fixed income securities are useful.

Maturity. The date at which a fixed income security's payments expire. For traditional bonds this is the day that the principal (or face value--see below) is repaid.

Coupon payment. A regular interest payment made on a fixed income security. Most securities that have coupon payments make them semi-annually.

Face value (par value). The maturity value of the bond. Par value and face value are generally interchangeable.

Bond indenture. The contract issued between the issuer and the buyer in a bond transaction. It specifies the rights of the debt holder in the case of default and the actions (in terms of further debt issuance) what kind of debt can be issued in the future and so on.

Fixed income securities come in many varieties. Broadly this market can be divided into two groups: the money market and the bond market. Within the bond market there are several main classes. These include:

Treasury securities. Various government securities including: bills (usually maturities of 28, 61 or 182 days--usually no coupons), notes (maturities of up to 10 years) and bonds (maturities greater than 10 years).

Corporate bonds. Issued by individual corporations to finance investment and operations. These usually pay coupon payments and the coupon rate is usually set so that when first sold the bonds sell at par.

Asset backed securities. Fixed income instruments that give the holder the right to receive payments from the asset that backs the security. A prominent example are mortgage backed securities that give to the holder the right to collect payments from a group of mortgage holders.

9.3 Credit Risk

Bonds provide a guaranteed source of fixed income, but this guarantee may not always be perfect. That is, there is some chance that a firm or issuing entity will go bankrupt and be unable to make payments on its outstanding debt. This risk is known as *credit risk*. Ratings agencies attempt to provide the market with information on the likelihood of a group defaulting. The common ratings agencies are Moody's Investor Services, Standard & Poor's Corporation and Fitch Investors Service. The ratings are

1. **Moody's:**
    - Investment grade: Aaa, Aa, Baa
    - High Yield (junk): Ba, B, Ca, C
2. **S & P:**
    - Investment grade: AAA, AA, A, BBB
    - High Yield (junk): BB, B, CCC, CC, C, etc.
3. **Fitch:**
    - Investment grade: AAA, AA, A, BBB

- Speculative: BB, B, CCC, CC, C, etc.

In order for investors to purchase bonds that have a higher probability of default, they must be compensated in the form of higher returns. Therefore, bonds in lower ratings categories generally offer higher yields. In other words, investors are willing to pay less for them.

9.4 Yield to maturity

Suppose that a security pays no coupons, has a face value of M and will mature in exactly 1 year. The observed price of the security is P. The bond's *yield to maturity* is then defined to be the discount rate y that solves the following equation

$$P = \frac{M}{1+y}$$

Yield to maturity. The constant interest rate y such that the present value of a security's future payments is exactly equal to its price. For a bond making T coupon payments b with face value M it is the solution to the equation

$$P = \sum_{t=1}^{T} \frac{b}{(1+y)^t} + \frac{M}{(1+y)^T}$$

For a bond making semiannual payments, the annualized yield to the above would be $2y$.

Example. Consider a bond with face value of \$1000 with an 8% coupon rate paid in two semiannual payments of 4% each that matures in 5 years. What is the yield to maturity if the observed price is \$1000? It is found by looking for the solution to the above equation. Note that there are 10 coupon payments, the first coming in 6 months.

$$1000 = \sum_{t=1}^{10} \frac{40}{(1+y)^t} + \frac{1000}{(1+y)^{10}}$$

Solving this for y yields, $y = 0.04$, or the periodic yield to maturity is the value of the coupon rate, which implies that the annualized yield to maturity is exactly the coupon rate. This is not a coincidence. In general the yield to maturity of a coupon paying bond that is selling for par will be the coupon rate of the bond.

To verify this, write the per-period coupon as $b = cM$, where c is the coupon rate, and evaluate the pricing equation at the candidate yield $y = c$. The coupon stream is an annuity, so applying the annuity factor $\frac{1}{c}\left(1-(1+c)^{-T}\right)$ derived earlier,

$$P = \sum_{t=1}^{T} \frac{cM}{(1+c)^t} + \frac{M}{(1+c)^T} = cM \cdot \frac{1}{c}\left(1 - \frac{1}{(1+c)^T}\right) + \frac{M}{(1+c)^T}.$$

The two occurrences of $M(1+c)^{-T}$ cancel:

$$P = M\left(1 - \frac{1}{(1+c)^T}\right) + \frac{M}{(1+c)^T} = M.$$

So setting y equal to the coupon rate prices the bond exactly at par. Because the pricing equation has a unique yield solution for a given price, the converse also holds: a bond priced at par ($P = M$) must have $y = c$. For this example 8% is the *bond equivalent yield.* This is the annualized yield to maturity when simple interest is used. The effective annual yield for this bond is then $(1.04)^2 - 1 = 0.0816$ or 8.16%.

Bond prices can be found in lots of places, including the Wall Street Journal Online.[1]

[1] Satchell, Stephen, and Alan Scowcroft. "A demystification of the Black-Litterman model: Managing quantitative and traditional portfolio construction." In Forecasting expected returns in the financial markets, pp. 39-53. Academic Press, 2007.

Example. Select a bond from `wsj.com` and calculate its YTM using the Excel spreadsheet `YTM Calculation.xls`. Notice that bond prices are usually expressed as a percent of the face value.

In general this equation cannot be solved algebraically if the bond is not selling at par. Excel and many other programs can be used to calculate it for you. In Excel the function that you want to use is YIELD( ). Some general rules apply however.

1. If the bond is selling above par then the YTM will be less than the coupon rate.
2. If the bond is selling below par then the YTM will be more than the coupon rate.

Yield to call. Some bonds are callable before their maturity date. For such bonds, the *yield to call* can be calculated as the yield to maturity with the call date used instead of the maturity date.

9.5 Realized compound return

The yield to maturity gives the return that will be earned on the bond *if all coupon payments can be reinvested at the current yield*. It does not generally give the return to holding the bond.

Example. Consider a 2 year bond paying coupon rate of 5% that is selling at par (for a face value of \$1000). The bond makes annual coupon payments of \$50 and the yield to maturity is $y = 0.05$ (from above). At the end of two years, the holder receives one coupon payment of \$50 and the face value of the bond which is \$1000. If the coupon payment received after the first year (\$50) is reinvested at 5% then it will now be worth $50 \cdot 1.05 = 52.50$. So the total amount of cash held at the end of the two years is \$1000 + \$50 + \$52.50 = \$1102.50. To find the effective annual yield, we must solve the equation $1000(1 + y)^2 = 1102.50$. Therefore the effective annual yield to holding this security is

$$y = \sqrt{\frac{1102.50}{1000}} - 1 = 0.05$$

Suppose however that in the first year the interest rate increases from 5% to 10%. Therefore the coupon payment that is received at the end of year 1 can be reinvested at 10%. What is the return to holding the security? At the end of the two years the cash in hand will be $1000 + $50 + $55 = $1105. This implies an effective annual yield of

$$r = \sqrt{\frac{1105}{1000}} - 1 = 0.0512$$

9.6 Holding period return

The price of bonds and thus the return to holding them will change over time. These capital gains work together with the coupon payments to give the return to holding the bond.

Example. Consider a bond with maturity of ten years from now paying semiannual coupons of 6% (i.e. the annual coupon rate is 12%). Assume that the current 6-month periodic interest rate (risk adjusted) is 6%. Therefore the price of the bond will be 100% of the face value. Now, consider what happens if the moment after the bond is issued the 6-month periodic interest rate changes from 6% to 5% and you purchase the bond. The price that you pay for it is

$$P = \frac{6}{0.05}\left(1 - \frac{1}{1.05^{20}}\right) + \frac{100}{(1.05^{20})} = \$112.462$$

At the end of the six months after you have received one coupon payment, the price of the bond will be given by

$$P = \frac{6}{0.05}\left(1 - \frac{1}{1.05^{19}}\right) + \frac{100}{(1.05^{19})} = \$112.0853$$

The rate of return to holding the bond for those six months has been

$$R = \frac{112.0853 + 6 - 112.462}{112.462} = 0.050$$

there is a small difference due to roundoff error.

9.7 Application: An investor comparing a bond to cash

An individual investor with a lump sum to deploy is choosing between buying a ten-year Treasury note and leaving the money in a money-market fund. The pricing tools of this chapter are what let her compare the two on equal footing. By computing the bond's yield to maturity she can see the annualized return she would lock in by holding to maturity, weigh it against the floating rate the money-market fund pays, and understand how the bond's price would move if interest rates rose or fell before she sold. The relationship between price, coupon, and yield — abstract when written as a formula — becomes the concrete basis on which she decides where to put her savings and how much interest-rate risk she is willing to carry.

> Further listening
>
> *Planet Money* (NPR) — "Summer School 4: Bonds & Becky With The Good Yield": how a bond's price and yield are linked, and the default, inflation, and interest-rate risks a bondholder bears.

9.8 Homework problems

9.8.1 Conceptual

YLD-C1. A colleague quotes you the yield to maturity on a 20-year corporate bond and says, "This is the return you'll earn if you buy the bond today and hold it to maturity." State precisely what the yield to maturity y is defined to be — the single discount rate that solves $P = \sum_{t=1}^{T} \frac{b}{(1+y)^t} + \frac{M}{(1+y)^T}$ — and explain the exact sense in which the colleague's statement is and is not correct.

YLD-C2. Explain why the yield to maturity is a *single* discount rate applied to every promised payment, rather than a separate rate for each maturity. What does it mean to say that y summarizes the entire price–cash-flow relationship in one number, and why is that number the natural way for bond markets to communicate value and compare instruments of different maturities and coupons?

YLD-C3. The chapter shows that a bond priced at par ($P = M$) has yield to maturity exactly equal to its coupon rate ($y = c$), and conversely. Reproduce the logic of the verification in words: starting from $b = cM$ and evaluating the pricing equation at $y = c$, explain why the coupon annuity and the discounted face value combine to give exactly M, and why uniqueness of the yield delivers the converse.

YLD-C4. State the two rules relating a bond's price to its coupon rate: if the bond sells above par the YTM is below the coupon rate, and if it sells below par the YTM is above the coupon rate. Explain *why* each rule must hold, referring to the definition of y as the discount rate that equates the present value of promised payments to the observed price P.

YLD-C5. The chapter distinguishes the quoted yield to maturity from the realized compound return. Suppose an investor buys a coupon bond at par and, over the life of the bond, market interest rates fall well below the original yield. Explain what happens to the realized compound return relative to the quoted yield to maturity, and identify the specific assumption behind the yield to maturity that is violated when rates change.

YLD-C6. "The yield to maturity assumes coupons are reinvested at the yield prevailing at purchase." A student objects: "But I never reinvest my coupons — I spend them, so the reinvestment assumption is irrelevant to me." Evaluate this objection. Under what circumstances does the reinvestment assumption matter for the return an investor realizes, and what does an investor who spends every coupon actually give up relative to the quoted yield when rates rise after purchase?

YLD-C7. An investor holds a long-maturity bond for six months and then sells it. Decompose the return over that period into its two components using the holding period return $R = \frac{P_1+b-P_0}{P_0}$, and explain how each component behaves when market interest rates rise sharply just after purchase. Which component can turn a positive coupon return into a negative total return?

YLD-C8. Two investors both hold the same ten-year bond. One plans to hold it to maturity; the other plans to sell it after one year. Explain why a sudden increase in market interest rates affects these two investors differently, distinguishing between the effect on the price at which the second investor can sell and the effect on the reinvestment of coupons for the first investor.

YLD-C9. Bond prices and yields move in opposite directions. Explain why this inverse relationship follows directly from the pricing equation $P = \sum_{t=1}^{T} \frac{b}{(1+y)^t} + \frac{M}{(1+y)^T}$, and describe how the news item at the head of the chapter — a rising 10-year Treasury yield after a hot inflation report — illustrates the same mechanism for an investor already holding the bond.

YLD-C10. Two bonds have the same yield to maturity but different maturities: one matures in 2 years, the other in 20 years. Using the inverse price–yield relationship, explain which bond's price changes by more (in percentage terms) for a given rise in the market yield, and why the pricing equation makes the longer bond more sensitive.

YLD-C11. Distinguish the *current yield* (annual coupon income divided by price) from the *yield to maturity*. Explain what the current yield omits, and give a scenario in which a bond with the higher current yield delivers the lower yield to maturity.

YLD-C12. Explain the difference between the bond equivalent yield (simple annualization of a semiannual periodic yield) and the effective annual yield (which accounts for compounding of the semiannual coupon). Why is the effective annual yield always at least as large as the bond equivalent yield, and when are the two equal?

9.8.2 Quantitative

YLD-Q1. A bond has face value $M = \$1000$, makes annual coupon payments of $b = \$70$, and matures in $T = 3$ years. The observed price is $P = \$1000$. Using $P = \sum_{t=1}^{T} \frac{b}{(1+y)^t} + \frac{M}{(1+y)^T}$, find the yield to maturity y, and explain how you can determine it here without solving the equation numerically.

YLD-Q2. A bond with face value $M = \$1000$ makes a single annual coupon of $b = \$60$ and matures in $T = 2$ years. Its observed price is $P = \$981.92$. Verify that the yield to maturity is $y = 0.07$ by substituting into $P = \sum_{t=1}^{2} \frac{b}{(1+y)^t} + \frac{M}{(1+y)^2}$ and confirming the present value equals the price. State whether this bond trades above or below par and check that the sign is consistent with the chapter's rules.

YLD-Q3. Consider a 3-year bond with face value $M = \$1000$ paying an annual coupon of $b = \$80$, selling at par so its yield to maturity is $y = 0.08$. All coupons are reinvested. Suppose immediately after purchase the reinvestment rate rises to 10% and stays there for the life of the bond. Compute the total cash held at maturity and the realized compound return r from $M(1 + r)^T = \text{(final cash)}$.

YLD-Q4. A 2-year bond with face value $M = \$1000$ pays an annual coupon of $b = \$50$ and sells at par ($y = 0.05$). Suppose the first coupon is reinvested at 3% rather than 5%. Compute the final cash held at the end of two years and the realized compound return r from $M(1 + r)^2 = \text{(final cash)}$, and compare r to the quoted yield to maturity.

YLD-Q5. A bond has maturity of ten years, pays semiannual coupons of 6 per 100 of face value (annual coupon rate 12%), and the current 6-month periodic interest rate is 6%, so the bond sells at par ($P = 100$). Immediately after issue the 6-month periodic rate falls to 4%. Compute the purchase price P_0 using $P = \frac{6}{y}\left(1 - \frac{1}{(1+y)^{20}}\right) + \frac{100}{(1+y)^{20}}$, then the price P_1 six months later (19 periods remaining), and finally the holding period return $R = \frac{P_1+b-P_0}{P_0}$ over the first six months.

YLD-Q6. You buy a bond at $P_0 = \$980$, receive a coupon of $b = \$40$ over your holding period, and sell it at $P_1 = \$955$. Compute the holding period return $R = \frac{P_1 + b - P_0}{P_0}$, and separately report the coupon component and the capital-gain component of that return.

YLD-Q7. A bond with face value $M = \$1000$ has a coupon rate of 8% paid in two semiannual payments of \$40 each and matures in 5 years. Its periodic yield to maturity solves $1000 = \sum_{t=1}^{10} \frac{40}{(1+y)^t} + \frac{1000}{(1+y)^{10}}$, giving $y = 0.04$. Report the bond equivalent yield (simple annualization) and compute the effective annual yield using $(1 + y)^2 - 1$.

YLD-Q8. A bond sells at par and has a semiannual periodic yield to maturity of $y = 0.035$. Report its bond equivalent yield (simple annualization) and compute its effective annual yield using $(1 + y)^2 - 1$. By how many basis points does the effective annual yield exceed the bond equivalent yield?

YLD-Q9. A bond pays annual coupons of $b = \$65$ and its observed price is $P = \$928.57$, with face value $M = \$1000$. Compute the current yield (annual coupon income divided by price). Then, using the chapter's rules relating price to par and coupon rate, state whether the yield to maturity is above or below the coupon rate for this bond.

YLD-Q10. A bond with face value $M = \$1000$ pays an annual coupon of $b = \$70$ and trades at a price of $P = \$1050$. Compute its current yield, and compute its coupon rate. State whether the bond trades above or below par, and rank the coupon rate, the current yield, and the yield to maturity from highest to lowest, justifying the ordering with the chapter's rules.

Chapter 10

The Yield Curve

10.1 Introduction

Three questions are at the heart of the study of the yield curve. First, how does the arbitrage principle relate yields of different maturities to one another, and what does this imply about bond pricing when the yield curve is not flat? Second, what information about future interest rates is encoded in the forward rates that can be extracted from the term structure today — and can an investor lock in those rates through a trading strategy? Third, why does the yield curve typically slope upward, and is that slope driven by expectations about rising short rates, by risk premia demanded by investors, or by some combination of both? These questions are central to financial markets because the yield curve is among the most closely watched signals in global finance. Central banks use it to assess the transmission of monetary policy; corporations use it to time debt issuance; mortgage lenders use it to set long-term rates; and investors use it to infer the market's collective view of the macroeconomic outlook. A rigorous understanding of how the curve is constructed, what it implies, and why it takes the shape it does is indispensable for fixed income analysis.

The first question establishes that bonds of different maturities cannot be priced independently of one another. Once zero-coupon spot rates for various maturities are observed, the no-arbitrage principle requires that every coupon-paying bond be priced by discounting each cash flow at the spot rate appropriate for its timing. If this were not the case, the bond could be stripped into its individual zero-coupon components and the pieces reassembled or sold separately for a riskless profit. This stripping and reconstitution argument pins down the entire pricing structure for fixed income instruments once the spot curve is known.

The second question concerns forward rates — the future short-term rates that are mathematically implied by today's spot curve. The chapter shows that under certainty, the forward rate is identical to the future short rate, and that the yield to maturity on any bond is simply the geometric average of the implied short rates over its life. More practically, it demonstrates how an investor or borrower can use existing zero-coupon bonds to lock in a future borrowing or lending rate today, effectively trading at the forward rate. This result has direct application for institutions that need to hedge future financing costs.

The third question requires moving beyond arbitrage to theories of investor behavior. The expectations hypothesis asserts that forward rates are unbiased forecasts of future short rates, so the yield curve slopes upward only when the market expects rates to rise. The liquidity preference theory offers an alternative: most investors have relatively short horizons, and they demand a premium — a liquidity premium — to hold long-term bonds that expose them to interest rate risk. Under this theory, the curve can slope upward even when short rates are not expected to rise. The chapter uses Jensen's inequality to derive precise conditions under which each theory implies a particular relationship between forward rates and expected future short rates, providing a formal foundation for the debate.

The chapter opens by explaining the stripping of coupon bonds into zero-coupon components and deriving spot rates from observed prices, then shows how these spot rates pin down the pricing of coupon bonds. It next develops forward rates from the spot curve and demonstrates the geometric average relationship between yields and short rates. The chapter then develops the theoretical framework for the term structure, contrasting the expectations hypothesis with liquidity preference theory and grounding each in a treatment of investor preferences under uncertainty. It closes by working through a detailed example of using existing zero-coupon bonds to lock in a future borrowing rate at the forward rate.

> **In the news**
>
> The shape of the yield curve is among the most-watched signals in finance. CNBC reported that the Fed's "favorite recession indicator" — the gap between short- and long-term Treasury yields — was flashing a warning as the curve inverted, a configuration that has preceded most U.S. recessions. Why the curve takes the shape it does, and what the forward rates embedded in it reveal about expected future rates, is the subject of this chapter. Read it at CNBC.

10.2 The Yield Curve

The United States Treasury and many other institutions like to separate bonds into their principal and coupon payments. In fact, one may think of bonds as being a conglomeration of lots of different securities. Each coupon payment represents a security that guarantees a payment (the coupon payment) to the holder at the specified price. The law of one price (i.e. the arbitrage principle) ensures that if one were to add up the price of all of these individual securities, one would get exactly the value of the group of securities.

Example. Consider a 2 year T-note paying a 10% coupon annually (for simplicity). Suppose that the price of such a security is 100% of par value. This security can be split into three pieces. One that pays 10% of par value in one year, another that pays 10% of par value in 2 years and a third that pays 100% of par value in 2 years. Suppose the price of the first security is 9% of par value, the price of the second security is 9%

of par value and the price of the third is 81% of par value. What strategy should you implement? You should sell the 2 year T-note and receive 100% of par value. Then purchase the three individual securities for a total cost of 99% of par value and then go to the treasury and ask them to reconstitute the security. Once reconstituted, you give the security back to cover the one that you sold short. Since you could do this as long as the prices differ, this shows that the prices must be the same.

For many reasons, treasury securities are often split into their individual payments. This creates a set of zero-coupon bonds. From the prices of these zero-coupon bonds, one may calculate the yield to maturity for zeros of various maturities.

Example. Suppose that zeros of 1,2 and 3 years maturity with $1000 par value sell for the prices $943.396, $881.659, $816.298. What are the YTMs of these zeros? They are 6%, 6.5% and 7% respectively.

Spot rate. The currently observed yield to maturity for a particular duration. For example, "the 3-year spot rate is 10%" or "the one-year spot rate is 5%."

Short rate. The interest rate for a particular interval of time (often a short interval like 1 year). For example, "the 1 year short rate three years from now will be 5%" or "the current short rate is 10% which by definition is the current 1-year spot rate."

The spot rates just defined are not abstractions: the U.S. Treasury publishes a full curve of them every business day. Figure 10.1 plots that curve on four dates chosen to span the shapes the curve actually takes. In June 2021 the curve was steep — short rates pinned near zero and long rates about two percentage points higher. Two years later the same curve was *inverted*: the three-month rate exceeded the ten-year rate by more than a percentage point, the configuration behind the recession-indicator headlines in the news box above. Today's curve is different again, upward sloping from around 3.7% at the short end to above 5% at twenty years. Every calculation in this chapter — stripping, spot rates, forward rates — can be carried out on the green line in this figure, and the worked examples below invite you to do exactly that.

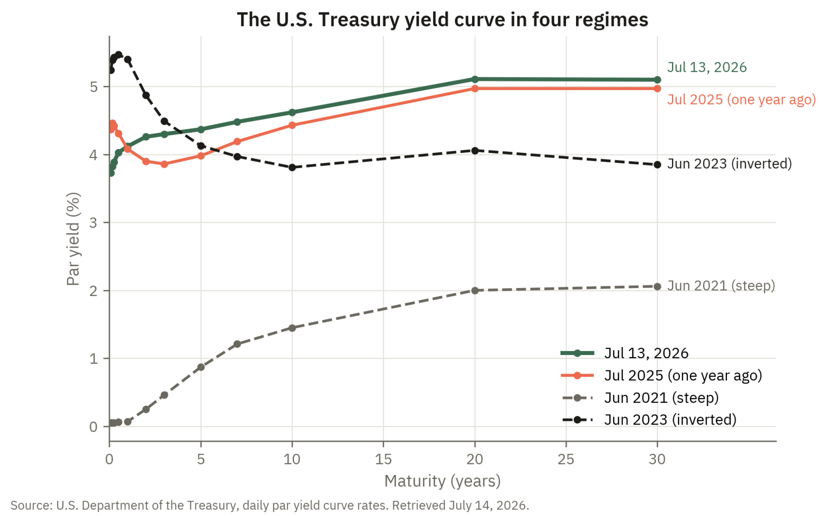

Figure 10.1: The U.S. Treasury par yield curve on four dates: the current curve, the curve one year earlier, the steep curve of June 2021, and the inverted curve of June 2023. Source: U.S. Department of the Treasury, daily par yield curve rates.

10.3 Bond pricing

The existence of differences in yields for zeros of different maturities has implications for the pricing of bonds. In particular, a zero coupon bond to be received at time n should be discounted using the yield for other zeros of the same maturity.

Example. Suppose that the yield curve is given by (y_1,y_2). Then a risk-free bond that matures in 2 years and makes annual coupon payments of b and has face value M would have price

$$P = \frac{b}{1 + y_1} + \frac{b}{(1 + y_2)^2} + \frac{M}{(1 + y_2)^2}$$

10.4 The yield curve without interest rate risk

Under certainty the future short rate can be derived from the current yield curve. This follows from the arbitrage principle. To see this, suppose that the yield curve over one and two years is given by the interest rates y_1 and y_2. Consider saving \$1000 today that you will consume in 2 years. You have several savings options available to you. The first is that you can put the money into a two year zero. At the end of the two years you will be paid $1000(1+y_2)^2$. Another option is that you can purchase a one year zero and at the end of the first year extract your money and put it into another one year zero. Both of these strategies yield exactly the same cash flow, so by the arbitrage principle they must give the same return at the end of the 2 years. Therefore, the short rate (r_2) after 1 year (for the period of the second year) must satisfy

$$1000(1+y_2)^2 = 1000(1+y_1)(1+r_2)$$

which implies that $1+r_2 = \frac{(1+y_2)^2}{1+y_1}$. Thus the future short rate can be calculated from the yield curve. In particular, the yield to maturity will be the *geometric* average of all of the future short rates. This can be seen since $(1+y_n)^n = (1+r_1)(1+r_2)\cdots(1+r_n)$ which implies that $1+y_n = [(1+r_1)(1+r_2)\cdots(1+r_n)]^{1/n}$.

This can be done generally. If n represents the maturity of a risk-free zero-coupon bond in the yield curve then in general we have that $1+r_n = \frac{(1+y_n)^n}{(1+y_{n-1})^{n-1}}$.

What is the rate of return to holding the 1 year zero vs. the 2 year zero for one year and then selling it? Since there is no uncertainty and the two investment strategies have the same payoff, their returns must be equal. To see this, note that at the end of one year, the two year zero is exactly a one year zero. As such it must have price $P = \frac{1000}{1+r_2}$. Therefore, the return to holding the two year zero (which was purchased for $\frac{1000}{(1+y_2)^2}$) will be

$$r = \frac{\frac{1000}{1+r_2} - \frac{1000}{(1+y_2)^2}}{\frac{1000}{(1+y_2)^2}} = \frac{(1+y_2)^2}{1+r_2} - 1.$$

To simplify, substitute the no-arbitrage relation $1 + r_2 = \frac{(1+y_2)^2}{1+y_1}$ from above. Then $\frac{(1+y_2)^2}{1+r_2} = \frac{(1+y_2)^2(1+y_1)}{(1+y_2)^2} = 1 + y_1$, so

$$r = (1 + y_1) - 1 = y_1.$$

(The net one-year holding-period return on the two-year zero is y_1 — not $1 + y_1$; the gross return is $1 + y_1$.) Thus the holding period returns of the two assets are equal.

Example. Suppose that the current yield curve is given by $y_1 = 0.0230$ and $y_2 = 0.0235$. In a world with no uncertainty, what must the 1-year short rate be one year from now? To calculate this, simply match cash flows. Investing \$1 in a two-year zero will yield $1.0235^2 = 1.04755$ at the end of two years. Investing in the 1-year zero and then rolling over will give $1.0230(1 + r_2)$ at the end of two years. Since under certainty these must be the same we have $1.04755 = 1.0230(1 + r_2)$ which implies that $r_2 = 0.02400$.

Of course, future short rates of return are unknown currently. The interest rate could go up or down between now and one, two or 10 years from now.

Forward interest rate. The future short interest rate that is implied by the current yield curve. In particular, if the current yield curve is given by $(y_1, y_2, y_3, \ldots, y_T)$ then the forward rate f_n solves

$$1 + f_n = \frac{(1 + y_n)^n}{(1 + y_{n-1})^{n-1}}$$

Example. The yield curve as of 9/16/08 at approximately 1:30 pm EDT (from Bloomberg[1]) is $(y_1, y_2, y_3) = (0.0154, 0.0174, 0.0198)$. What are

[1] Satchell, Stephen, and Alan Scowcroft. "A demystification of the Black-Litterman model:

the forward rates (f_2, f_3)?

$$f_2 = \frac{(1+y_2)^2}{1+y_1} - 1 = \frac{1.0174^2}{1.0154} = 0.019404$$
$$f_3 = \frac{(1+y_3)^3}{(1+y_2)^2} - 1 = \frac{1.0198^3}{1.0174^2} = 0.024617$$

10.5 Theories of the yield curve

What is the relationship between the future short rate r_n and the forward rate f_n? One difference is that f_n can be calculated directly today while r_n is a random variable. Can we relate Er_n to f_n?

Example. Consider an investor who knows that she wants to invest for a two-year horizon. The investor's preferences can be represented by a utility function $U(x)$ where x represents her consumption at the end of two years. The investor preferences over risky alternatives can be represented by the expected utility functional $EU(x)$. Assume that the investor is risk averse so $U(x)$ is strictly increasing in x and is strictly concave. There are two possible assets that may be used. The first is a two year, zero-coupon bond with current yield to maturity of y_2. The alternative is to invest in a one-year zero coupon bond at current rate y_1 and then roll this investment over at the end of one year. The payoff to this strategy will be $(1+y_1)(1+r_2)$ but r_2 is unknown currently. If the investor were indifferent between the two assets then it must be true that $EU((1+y_1)(1+r_2)) = U((1+y_2)^2)$ (recall that the two-year return to the two-year zero is certain). Recall from Jensen's inequality that for a strictly concave function $f(x)$ we have that $Ef(x) < f(Ex)$. Therefore we have that

$$U(E(1+y_1)(1+r_2)) > U((1+y_2)^2).$$

Managing quantitative and traditional portfolio construction." In Forecasting expected returns in the financial markets, pp. 39-53. Academic Press, 2007.

Recall that the forward rate f_2 is defined as $1 + f_2 = \frac{(1+y_2)^2}{1+y_1}$. From the previous equation we have that $E(1 + y_1)(1 + r_2) > (1 + y_2)^2$ which implies that

$$1 + f_2 = \frac{(1 + y_2)^2}{1 + y_1} < E(1 + r_2).$$

Thus, if all investors are long term investors then the expected future short rate will be larger than the forward rate. Investors in the long-term asset do not have to deal with interest rate risk so they are willing to accept a low yield on that asset. This lower yield is inherent in y_2 since $(1 + y_2)^2 = (1 + y_1)(1 + f_2)$. Investors are willing to accept an implied future short rate that is less than the expected value of the future short rate because they don't have to deal with the interest rate risk.

Example. Now consider an investor who knows that she wants to invest for only one year. She can invest in the one-year zero and earn return y_1 for sure. Or she can invest in the two year security and sell it at the end of the first year. What will be the return to this strategy? It depends on the one-year short rate that is realized at the end of the year. Call this short rate r_2.

$$R = \frac{\frac{1}{1+r_2}}{\frac{1}{(1+y_2)^2}} = \frac{(1 + y_2)^2}{1 + r_2}$$

If the investor is to be indifferent between the two assets it must be that

$$EU\left(\frac{(1 + y_2)^2}{1 + r_2}\right) = U(1 + y_1) = U\left(\frac{(1 + y_2)^2}{1 + f_2}\right)$$

Again by Jensen's inequality we have that

$$U\left(E\frac{(1+y_2)^2}{1+r_2}\right) > U\left(\frac{(1+y_2)^2}{1+f_2}\right)$$

which implies that

$$E\frac{(1+y_2)^2}{1+r_2} > \frac{(1+y_2)^2}{1+f_2}$$

That is, the expected return to holding the 2-year security will have to be larger than the return to holding the one year security because the two year zero bears the risk of the currently unknown forward spot rate r_2. How does this relate to the expected future spot rate Er_2? Suppose that $f_2 = Er_2$. Notice that $\frac{1}{1+r_2}$ is convex in r_2 so $E\frac{1}{1+r_2} < \frac{1}{1+Er_2}$ by Jensen's inequality. Multiplying both sides of this inequality by the positive constant $(1+y_2)^2$ preserves it,

$$E\frac{(1+y_2)^2}{1+r_2} < \frac{(1+y_2)^2}{1+Er_2},$$

and substituting the supposition $Er_2 = f_2$ into the right-hand side gives

$$E\frac{(1+y_2)^2}{1+r_2} < \frac{(1+y_2)^2}{1+f_2}$$

which violates the previous equation. Therefore it must be that $f_2 = Er_2 + LP$ where LP is a positive risk (or liquidity) premium that is sufficient to make it so that

$$E\frac{(1+y_2)^2}{1+r_2} > \frac{(1+y_2)^2}{1+Er_2+RP} = \frac{(1+y_2)^2}{1+f_2}$$

In other words, if all investors are short term investors then they will only be willing to hold the two-year zero if its return incorporates a positive risk premium (i.e. $f_2 > Er_2$).

There are two alternative hypotheses about the yield curve. One is called the *expectations hypothesis* which asserts that forward rates are exactly the expectation of future interest rates. As seen previously, this is equivalent to assuming either that 1) The number of short term and long term investors in the market exactly counteract each other, or 2) there exist investors in the market who are risk-neutral and so care only about expectations. An alternative hypothesis is the *liquidity preference theory* which asserts that most investors in the market are short term investors and so the reason that the yield curve often slopes upward is that investors require a liquidity premium to hold the longer term debt, since doing so requires one to take on interest rate risk.

10.6 Forward rates and future loans

If it is currently known that you will need access to funds in the future, one may lock in the future interest rate by using the current forward rates. Assume that the current yield curve is given by $(y_1, y_2, \ldots, y_T)$. You will need access to the amount x at time n and will repay it at time $n+1$. The general strategy is to sell (i.e. short) enough $n+1$-year zeros to pay for the purchase of x dollars of n-year zeros. The n-year zeros will payoff in period n, but you won't have to purchase the $n+1$-year zeros to cover your short position until year $n+1$. The price of an n-year zero is $P_n = \frac{M}{(1+y_n)^n}$ where M is the face value of the zero. The price of the $n+1$ year zero is $P_{n+1} = \frac{M}{(1+y_{n+1})^{n+1}}$.

To receive the amount x at time n you must purchase x/M units of the n-year zeros. This cost is $\frac{x}{(1+y_n)^n}$. To fund this position with $n+1$-year zeros that have face value M you must sell

$$q = \frac{(1+y_{n+1})^{n+1}}{M(1+y_n)^n}$$

units of the $n+1$-year zero to fund this position. Recall from the definition of f_{n+1} that

$$1 + f_{n+1} = \frac{(1 + y_{n+1})^{n+1}}{(1 + y_n)^n}$$

which implies that

$$q = (1 + f_{n+1})\frac{x}{M}.$$

Thus the amount you must repay in period $n + 1$ is $qM = (1 + f_{n+1})x$. That is, you received x at the beginning of period n and must repay $(1 + f_{n+1})x$ at the beginning of period $n + 1$. You have effectively taken a loan of x at time n at rate f_{n+1}.

Example. Suppose that you want to raise \$1M from the capital markets to be received by you in one year and payed off one year later. Assume that one-year and two-year zeros have face value $M = \$1000$. The current yield curve is $(y_1, y_2) = (0.0154, 0.0174)$. So the price of the one year zero is $1000/1.0154 = 984.834$. The price of the two year zero is $1000/1.0174^2 = 966.0877$. To have \$1M in one year you will need purchase 1000 one year zeros in the market. This will cost you \$984,834. In order to raise that capital you start by selling $q = 984, 834/966.0877 = 1019.404$ two year zeros in the market. This gives you \$984,834 to be used to purchase the 1 year zeros. In 1 year you will have exactly \$1M dollars as a payoff from the one year zeros. In two years you will need to purchase all of the two year zeros that you owe so that you can pay back the person from whom you borrowed them. How much will this cost? The value of the two year zeros in two years will be $1000 \cdot 1019.404 = \$1, 019, 404$. Thus you will have paid \$19,404 in interest on the one year loan. Recall from before that $f_2 = 0.019404$.

10.7 Application: A bank treasurer managing the gap

The treasurer of a regional bank must decide how much to fund long-dated loans with short-term deposits — a choice that determines both the bank's profit and its vulnerability. The term structure of interest rates is the treasurer's map. The shape of the yield curve, and the forward rates embedded in it, reveal what the market expects future short rates to be and how much extra return long maturities currently offer; the expectations hypothesis and the role of term premia explain why the curve slopes as it does. Reading that signal is not academic: a treasurer who borrows short and lends long without accounting for how the curve can move is exposed to exactly the maturity mismatch that destroyed Silicon Valley Bank when rates rose, and the analysis in this chapter is what allows the risk to be measured and managed rather than ignored.

> Further listening
>
> *The Indicator from Planet Money* (NPR) — "The inverted yield curve is screaming RECESSION": why the slope of the yield curve has predicted past recessions, and what an inversion really means.

10.8 Homework problems

10.8.1 Conceptual

YC-C1. A trader claims that a coupon-paying two-year T-note can be priced by discounting all of its cash flows at the single two-year spot rate y_2. Explain, using the stripping-and-reconstitution (no-arbitrage) argument, why this is incorrect when the yield curve is not flat, and state the correct pricing rule for each of the note's cash flows.

YC-C2. A colleague argues that if a coupon bond and a portfolio of zeros replicating its cash flows sold at different prices, "the market would just eventually correct." Explain instead the *immediate* arbitrage trade (which security to buy, which to sell, and the role of the Treasury's stripping and reconstitution facility) that forces the coupon bond's price to equal the sum of the values of its stripped cash flows. Why does the mere possibility of this trade pin down the price, regardless of investor sentiment?

YC-C3. A colleague states, "The three-year spot rate y_3 tells us the market's forecast of the short rate that will prevail three years from now." Explain why this statement confuses a spot rate with a forward (or short) rate. In your answer, describe precisely what y_3 measures today, what the forward rate f_3 measures, and how the two are related through the definition $1 + f_3 = \frac{(1+y_3)^3}{(1+y_2)^2}$.

YC-C4. Carefully distinguish the three rate concepts used in this chapter — the spot rate y_n, the short rate r_n, and the forward rate f_n — by stating which are observable today and which are (in general) random variables. Explain why f_n and r_n coincide in a world of certainty but generally differ under uncertainty, and why y_n can always be written as a geometric average of a sequence of rates.

YC-C5. Explain how an investor who knows she will need to borrow money one year from now, and repay it two years from now, can lock in today's forward rate f_2 without any uncertainty. Describe the trading strategy in words (which zeros to buy, which to short) and explain why the effective borrowing rate is exactly f_2 rather than y_1 or y_2.

YC-C6. A borrower says, "To lock in a future loan I would need a counterparty willing to sign a forward contract with me today." Explain why, given a full set of zero-coupon bonds, no such explicit forward contract is necessary: describe how the long-and-short zero position *manufactures* the forward loan synthetically, and identify precisely at which dates cash flows in and out so that nothing changes hands between today and the start of the loan.

YC-C7. The chapter shows that if all investors are long-term (two-year-horizon) investors, then $f_2 < Er_2$, whereas the expectations hypothesis asserts $f_2 = Er_2$. Explain in words why a market populated entirely by long-horizon investors produces a forward rate *below* the expected future short rate. What feature of the two-year zero makes those investors willing to accept a lower implied rate, and how does Jensen's inequality enter the argument?

YC-C8. Under the expectations hypothesis, an upward-sloping yield curve is interpreted as a signal that the market expects future short rates to rise. Explain the reasoning that connects the slope of the curve to expected future short rates under this hypothesis, using the relationship $f_n = Er_n$. What would a downward-sloping (inverted) curve imply about Er_n relative to today's short rate?

YC-C9. According to the liquidity preference theory, the yield curve can slope upward even when the market expects short rates to remain unchanged. Using the decomposition $f_2 = Er_2 + LP$ with $LP > 0$, explain why this is possible and what economic force LP represents. Contrast this interpretation of an upward slope with the expectations-hypothesis interpretation.

YC-C10. The chapter derives that if all investors are short-term (one-year-horizon) investors, then $f_2 > Er_2$, so a positive liquidity premium $LP = f_2 - Er_2$ is required for anyone to hold the two-year zero. Explain intuitively why short-horizon investors demand this premium, and why it is these investors — not the long-horizon investors — whose behavior the liquidity preference theory takes to dominate the market.

YC-C11. Under the assumption of certainty, the chapter shows that buying a two-year zero and selling it after one year earns exactly the same *net* one-year holding-period return as simply holding a one-year zero, namely y_1. Explain in words why the no-arbitrage principle forces these two strategies to have equal one-year returns. Be careful to state whether it is the *net* return or the *gross* return that equals y_1, and give the value of the other one.

YC-C12. A student writes on an exam that "the one-year holding-period return on a two-year zero equals $1 + y_1$." Explain precisely what is wrong with this statement, distinguishing the gross return (one plus the net return) from the net return. Identify which of y_1 and $1 + y_1$ is the net one-year holding-period return and which is the gross return, and show where in the derivation the "−1" enters.

YC-C13. Suppose that on a given morning the spread between the 10-year and 3-month Treasury yields turns negative (the curve inverts), and financial news outlets warn of a possible recession. Drawing on the chapter's framing of the yield curve as a signal, explain what an inverted curve implies about the market's expectations for future short rates under the expectations hypothesis, and why a bank treasurer who borrows short and lends long should pay attention to this configuration.

YC-C14. The bank-treasurer application describes funding long-dated loans with short-term deposits — "borrowing short and lending long." Using the language of spot rates, forward rates, and term/liquidity premia developed in this chapter, explain both *why* this maturity mismatch is typically profitable when the curve slopes upward and *why* it is risky. Connect the risk explicitly to the possibility that realized future short rates r_n differ from today's forward rates f_n.

10.8.2 Quantitative

YC-Q1. The current yield curve is $(y_1, y_2, y_3) = (0.030, 0.035, 0.038)$. Consider a two-year risk-free note with face value $M = \$1000$ paying an annual coupon of $b = \$60$. Using the no-arbitrage rule $P = \frac{b}{1+y_1} + \frac{b+M}{(1+y_2)^2}$, compute the price of the note. Then show that discounting *both* cash flows at the single rate y_2 gives a different (incorrect) price.

YC-Q2. Zeros of 1-, 2-, and 3-year maturity with \$1000 par sell for \$970.874, \$942.596, and \$915.142 respectively. (a) Extract the spot rates y_1, y_2, y_3. (b) Use them to price a three-year note with face value \$1000 and annual coupon \$50 by discounting each cash flow at the spot rate of matching maturity.

YC-Q3. The current yield curve is $(y_1, y_2, y_3) = (0.030, 0.035, 0.038)$. Compute the forward rates f_2 and f_3 using $1 + f_n = \frac{(1+y_n)^n}{(1+y_{n-1})^{n-1}}$.

YC-Q4. Zero-coupon bonds with \$1000 par sell for the following prices: the 1-year zero for \$952.381 and the 2-year zero for \$898.452. Extract the spot rates y_1 and y_2, then compute the forward rate f_2.

YC-Q5. Suppose the future short rates under certainty are $r_1 = 0.020$, $r_2 = 0.030$, and $r_3 = 0.040$. Using the geometric-average relationship $(1 + y_n)^n = (1 + r_1)(1 + r_2)\cdots(1 + r_n)$, compute the three-year spot rate y_3.

YC-Q6. The two-year spot rate is $y_2 = 0.05$ and the one-year spot rate is $y_1 = 0.04$. Under the assumption of certainty, use the arbitrage relation $(1 + y_2)^2 = (1 + y_1)(1 + r_2)$ to find the one-year short rate r_2 that will prevail one year from now.

YC-Q7. The current yield curve is $(y_1, y_2) = (0.04, 0.05)$ and each zero has face value \$1000. An investor buys the two-year zero today and sells it after one year, once it has become a one-year zero. Under certainty: (a) compute the purchase price today; (b) compute the one-year short rate r_2 and hence the resale price after one year; (c) compute the *net* one-year holding-period return and verify it equals y_1; and (d) state the corresponding *gross* return.

YC-Q8. A two-year zero with \$1000 par is bought today at a one-year spot rate $y_1 = 0.03$ and two-year spot rate $y_2 = 0.045$, and sold after one year under certainty. Compute the resale price and the *gross* one-year holding-period return $\frac{\text{resale price}}{\text{purchase price}}$. Confirm that the *net* return equals $y_1 = 0.03$, and explain in one sentence why reporting $1 + y_1$ as the holding-period return would be an error.

YC-Q9. The current yield curve is $(y_1, y_2) = (0.0154, 0.0174)$, matching the chapter example. Under the expectations hypothesis, the market expects the one-year short rate one year from now to equal the forward rate: $Er_2 = f_2$. Compute f_2 and hence Er_2. Then, if instead the liquidity preference theory holds with a premium $LP = 0.004$ and the forward rate is unchanged, compute the implied expected future short rate Er_2 from $f_2 = Er_2 + LP$.

YC-Q10. Suppose $y_1 = 0.03$, and the market's expected future one-year short rate is $Er_2 = 0.05$. (a) Under the expectations hypothesis ($f_2 = Er_2$), determine the two-year spot rate y_2 consistent with this expectation, using $(1 + y_2)^2 = (1 + y_1)(1 + f_2)$. (b) Now suppose instead that a liquidity premium $LP = 0.01$ raises the forward rate to $f_2 = Er_2 + LP$; recompute y_2 and comment on how the premium steepens the curve.

YC-Q11. You want to raise \$2M from the capital markets to be received in one year and repaid one year later. One-year and two-year zeros have face value $M = \$1000$. The current yield curve is $(y_1, y_2) = (0.020, 0.025)$. Compute (a) the price of each zero, (b) how many one-year zeros you must buy to receive \$2M in one year, (c) how many two-year zeros q you must short to fund the purchase, and (d) the total amount you must repay in two years. Verify that the effective interest paid corresponds to the forward rate f_2.

YC-Q12. Using the general result that a loan of x received at time n and repaid at time $n+1$ carries effective rate f_{n+1}, suppose the yield curve is $(y_1, y_2, y_3) = (0.030, 0.035, 0.040)$ and you wish to lock in a loan of $x = \$500{,}000$ to be received at time 2 and repaid at time 3. Compute the forward rate f_3, the number of zeros involved via $q = (1 + f_3)\frac{x}{M}$ with $M = \$1000$, and the amount $(1 + f_3)x$ you must repay at time 3.

10.8.3 Data exercises

These problems use the live-data figures in this chapter.

YC-D1. Reading from the current (green) curve in Figure 10.1, the one-, two-, and three-year par yields are approximately $(y_1, y_2, y_3) = (0.0412, 0.0426, 0.0430)$. Compute the implied forward rates f_2 and f_3. Then compare your answers with the forward rates computed in the chapter from the September 2008 curve, and explain in one sentence why today's forwards sit so close to today's spots while the 2008 curve implied sharply rising short rates.

YC-D2. In June 2023 (Figure 10.1, dashed black line) the three-month yield was near 5.4% while the ten-year yield was near 3.8%. State what this inversion implies about expected future short rates under the expectations hypothesis. Then explain how the liquidity preference theory changes the reading: is the market's implied expected decline in short rates *larger* or *smaller* than the raw forward rates suggest? Justify using $f_n = Er_n + LP$ with $LP > 0$.

Chapter 11

Fixed Income Portfolio Management

11.1 Introduction

Three questions drive the analysis of fixed income portfolio management. First, how sensitive is a bond's price to a change in interest rates, and how can that sensitivity be measured and summarized in a tractable way? Second, how can a portfolio manager use that measure to construct a portfolio that is protected — immunized — against the losses that arise when yields shift unexpectedly? Third, why does duration-based immunization break down for large yield movements, and what additional tool is needed to achieve more accurate hedging? These questions are of direct practical importance to any institution that holds fixed income assets or issues fixed income liabilities. Pension funds must ensure that the present value of their assets tracks the present value of their obligations; insurance companies must fund future claims; banks must manage the mismatch between the durations of their loan portfolios and their deposit liabilities. A rigorous framework for measuring and managing interest rate risk is therefore not merely an academic exercise but a core competency of institutional finance.

The first question leads to the concept of duration. Macaulay's duration is the time-weighted average of the present values of a bond's cash flows — effectively the center of gravity of the payment stream — and provides an intuitive summary of when, on average, the bondholder receives value. Modified duration translates this directly into a first-order approximation: a bond with modified duration D^* will lose approximately $D^*\Delta y$ percent of its value for each percentage-point increase in yield. This makes precise the familiar intuition that long-maturity, low-coupon bonds are far more sensitive to rate changes than short-maturity, high-coupon bonds, and it provides a single number that can be compared and aggregated across an entire portfolio.

The second question leads to the strategy of immunization. If the duration of a portfolio of assets equals the duration of the corresponding liabilities, then a small parallel shift in yields will change the two sides of the balance sheet by approximately the same amount, leaving the net position intact. The chapter works through this strategy concretely, showing how a company that has issued an annuity-like obligation can select a zero-coupon bond — choosing both its face value and its maturity — so that the two sides of the portfolio are duration-matched and the net value is insensitive to small yield changes.

The third question arises because duration is only a first-order, linear approximation. As the magnitude of yield changes increases, the linear approximation loses accuracy, and a duration-matched portfolio can still show residual losses or gains depending on the curvature of the price-yield relationship. This curvature is captured by convexity — the second derivative of bond price with respect to yield — which is almost always positive for standard bonds. Convexity is generally a desirable property: for a given duration, a more convex bond gains more when yields fall and loses less when yields rise. Matching both duration and convexity across assets and liabilities extends the accuracy of immunization to larger yield movements.

The chapter begins by deriving the price sensitivity of bonds using calculus, introducing modified duration and Macaulay's duration and establishing their key properties — including that a zero-coupon bond's duration equals its maturity and that duration falls as the coupon rate rises. It then develops the immunization strategy through detailed examples and closes with a treatment of convexity, demonstrating how second-order hedging extends the robustness of an immunized portfolio.

In the news

Duration is not a dry statistic — it can sink a bank. NPR explained how Silicon Valley Bank loaded up on long-term government bonds and then watched their market value collapse as interest rates rose in 2022, leaving it with billions in losses when it was forced to sell. A bank or pension that matches the duration of its assets to its liabilities — the immunization strategy in this chapter — is protected from exactly that shock. Read it at NPR.

11.2 The sensitivity of bond prices to yields

Interest rate risk is one of the principle risks faced by the manager of a bond portfolio. Changes in yield affect the value of a bond. This can be seen from definition of yield to maturity. A bond that matures in T periods, pays the coupon b, has face value M and price P will have a YTM defined by

$$P = \sum_{t=1}^{T} \frac{b}{(1+y)^t} + \frac{M}{(1+y)^T}$$

Suppose that you were to ask yourself the question "If yields change by a small amount (say Δy), what would be the approximate change in the value of the bond?" Since you know calculus you know that a good approximation for small Δy is given by

$$\Delta P = \frac{\partial P}{\partial(1+y)} \Delta y.$$

If you wanted to know the percent change in price (i.e. the return) that arises from a small change in yield you would calculate

$$\frac{\Delta P}{P} = \frac{\partial P}{P\partial(1+y)}\Delta y = -D^{*}\Delta y$$

The quantity D^{*} is called a bond's *modified duration*. For what follows, let CF_t represent the cash flows to the bond in question where $CF_t = b$ for all $t < T$ and $CF_T = b + M$.

Modified duration. For a bond with price P, and current YTM y the bond's modified duration is given by

$$D^{*} = -\frac{\partial P}{P\partial(1+y)}$$

The modified duration of a bond with cash flows CF_t is given by

$$D^{*} = \frac{1}{P}\sum_{t=1}^{T}\frac{tCF_t}{(1+y)^{t+1}}$$

To obtain the closed form, differentiate the pricing equation $P = \sum_{t=1}^{T} CF_t(1+y)^{-t}$ term by term with respect to $(1+y)$. Using the power rule, $\frac{\partial}{\partial(1+y)}(1+y)^{-t} = -t(1+y)^{-t-1}$, so

$$\frac{\partial P}{\partial(1+y)} = \sum_{t=1}^{T} CF_t \cdot (-t)\,(1+y)^{-t-1} = -\sum_{t=1}^{T}\frac{t\,CF_t}{(1+y)^{t+1}}.$$

Every term in the sum is positive, so $\partial P/\partial(1+y) < 0$: prices fall when yields rise, as expected. Dividing by $-P$ flips the sign and gives a positive number,

$$D^{*} = -\frac{1}{P}\frac{\partial P}{\partial(1+y)} = \frac{1}{P}\sum_{t=1}^{T}\frac{t\,CF_t}{(1+y)^{t+1}},$$

which is the formula above.

For economists a natural way to think about the sensitivity of this bond's value to changes in yield is the *yield elasticity of bond price*. From your 380/382 experience you know that this can be written as

$$\epsilon = \frac{\partial P}{\partial(1+y)}\frac{1+y}{P}$$

This has an alternative definition.

Duration (Macaulay's duration). The Macaulay duration of a bond with cash flows CF_t is given by

$$D = -\frac{\partial P}{\partial(1+y)}\frac{1+y}{P} = \frac{1}{P}\sum_{t=1}^{T}\frac{tCF_t}{(1+y)^t}$$

An alternative way to interpret Macaulay's duration is to think of it as the time weighted average of present value cash flows. Notice in the above equation that the present value of each cash flow $CF_t/(1+y)^t$ is multiplied by the time at which it is received and divided by the price.

Example. Consider a bond making annual coupon payments at a rate of 10% that will mature in 3 years and is currently selling at par ($1000). What is the bond's duration (Macaulay)? This is calculated as

$$D = \frac{1}{1000}\left(1\cdot\frac{100}{1.10} + 2\cdot\frac{100}{1.10^2} + 3\cdot\frac{1100}{1.10^3}\right) = 2.736$$

This can be interpreted as a "summary statistic of the effective average maturity of the portfolio."[1]

The economic content of modified duration is easiest to see in a picture. The pricing equation makes price a convex, decreasing function of yield, and the duration approximation $\Delta P/P \approx -D^*\Delta y$ is precisely the *tangent line* to that curve at the current yield. Figure 11.1 draws both for a ten-year Treasury

[1] Satchell, Stephen, and Alan Scowcroft. "A demystification of the Black-Litterman model: Managing quantitative and traditional portfolio construction." In Forecasting expected returns in the financial markets, pp. 39-53. Academic Press, 2007.

carrying today's par coupon. Near the current yield the tangent hugs the curve and the approximation is excellent; far from it, the true price curve bends away above the tangent on both sides. That gap is no accident — the price curve is convex — and measuring it is the subject of the convexity section later in this chapter.

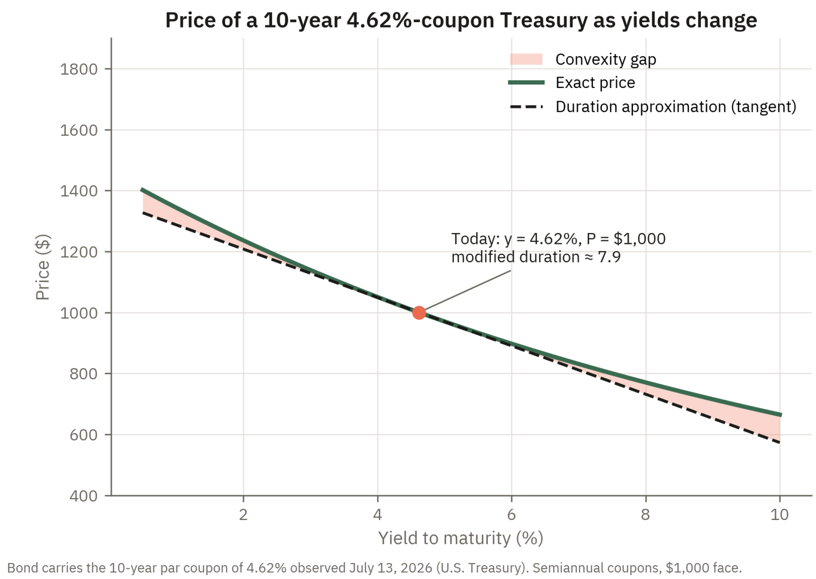

Figure 11.1: The exact price of a 10-year Treasury as a function of its yield (green) and the duration approximation — the tangent line at the current yield (dashed). The bond carries the current 10-year par coupon, so it is priced at par today. The shaded region is the gap the convexity correction will close. Source: U.S. Department of the Treasury.

11.3 Duration's properties

1. *The duration of a zero coupon bond is its time to maturity*. To see this notice that the formula above gives

$$D = \frac{1}{P}\frac{TM}{(1+y)^T} = \frac{(1+y)^T}{M}\frac{TM}{(1+y)^T} = T$$

2. *Duration decreases in the coupon rate*. Increasing the coupon rate moves more of the payments from the bond forward in time. As such it also makes the price of the bond less susceptible to changes in yield.

3. *Duration often increases with the maturity of bonds*. For most bonds, increasing the maturity means spreading payments out over a longer period. However, this may not be true for some bonds selling at a very deep discount.

Market Insight

The duration arithmetic above is exactly what destroyed Silicon Valley Bank, and the numbers are worth working through. In March 2020 the 20-year Treasury yield touched 0.87%. A 20-year bond issued at par on that day has, by the formulas of this chapter, a modified duration of about eighteen. Between then and late 2023 long yields rose by roughly four percentage points, and the approximation $\Delta P/P \approx -D^{*}\Delta y$ predicts a decline in value on the order of $-18 \times 0.04 \approx -70\%$ — an overstatement, because for a move this large the convex price curve bends well above its tangent line, but the exact repricing in Figure 11.2 still shows a collapse of 48% at the trough. That is the mark-to-market hole that sat on SVB's balance sheet in the spring of 2023: the bank held tens of billions of dollars of long-duration bonds bought when yields were near their lows, funded by deposits that could leave overnight. When depositors ran on March 9–10, 2023, the unrealized duration loss became a realized one. Note also what the figure does *not* show: no default, no credit event, no missed coupon. Every one of those Treasury payments will be made in full. Duration risk is a pure price-of-time risk, and it is entirely capable of sinking a bank on its own.

11.4 Portfolio Immunization

Knowledge about the duration of a particular payment stream gives a portfolio manager the ability to *immunize* her portfolio from the riskiness of yields. The strategy employed by such procedures is to purchase the rights to an obligation whose value changes exactly opposite to the change in value

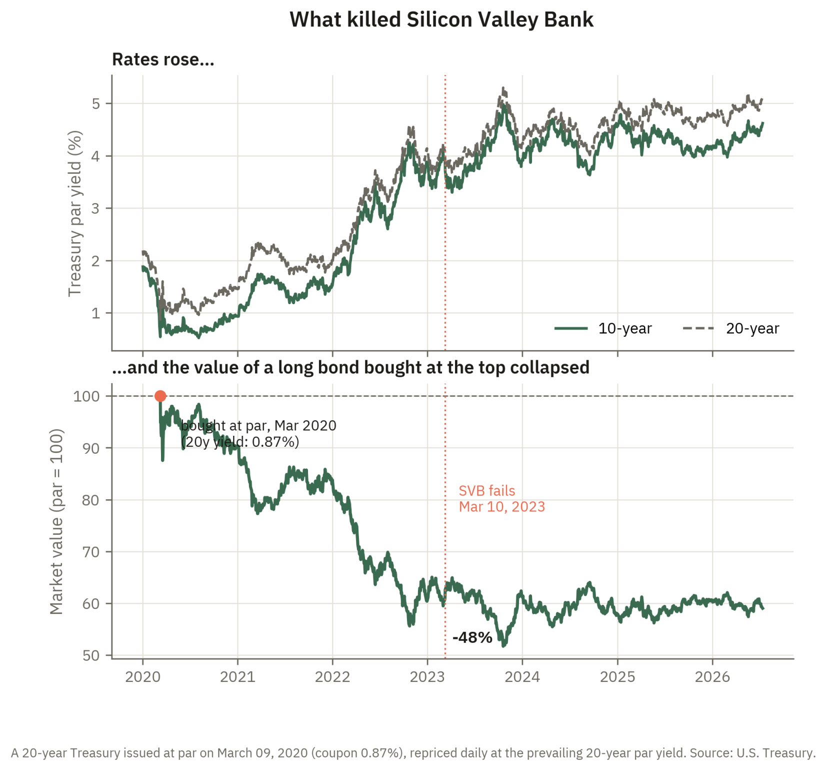

Figure 11.2: A 20-year Treasury bought at par at the March 2020 low in yields, repriced daily at the prevailing 20-year par yield using the pricing formula of this chapter. The bond lost nearly half its market value by October 2023 without any risk of default. Source: U.S. Department of the Treasury.

arising from a change in yield to the payment obligation.

Example. Consider a company that has agreed to pay an annuity of T annual payments with value M to an individual. If the current yield (discount rate) for such a payment stream is y, then the value of such an annuity is

$$V = \frac{M}{y}\left(1 - \frac{1}{(1+y)^T}\right)$$

The duration of this payment obligation is

$$D = \frac{1}{P}\sum_{t=1}^{T}\frac{tM}{(1+y)^t}$$

The company wants to fund the annuity using a zero coupon bond and they want to simultaneously insulate themselves from changes in yield. To do so, they must choose the face value of the debt (F) that they will purchase and then choose the maturity (τ) (i.e. choose the duration of the bond). The portfolio of obligations that the company will have after immunizing will be the fixed annuity payments and a single payment from the bond. The value of the portfolio of payment obligations is

$$V = -\frac{M}{y}\left(1 - \frac{1}{(1+y)^T}\right) + \frac{F}{(1+y)^\tau}$$

This company will choose F and τ such that $V = 0$ and $\partial V/\partial y = 0$.

These two conditions pin down F and τ. Write the value of the obligation as $P_{obl} = \frac{M}{y}\left(1 - (1+y)^{-T}\right)$ and the value of the zero as $Z = F(1+y)^{-\tau}$, so that $V = -P_{obl} + Z$. The first condition, $V = 0$, requires $Z = P_{obl}$: the zero must be purchased so that its present value equals the present value of the obligation.

For the second condition, differentiate. The zero satisfies $\frac{\partial Z}{\partial(1+y)} =$

$-\tau F(1+y)^{-\tau-1} = -\dfrac{\tau}{1+y} Z$, so its modified duration is $\tau/(1+y)$ and its Macaulay duration is exactly its maturity τ. The obligation satisfies $\frac{\partial P_{obl}}{\partial(1+y)} = -\dfrac{D_{obl}}{1+y} P_{obl}$, where D_{obl} is its Macaulay duration. Setting the derivative of V to zero,

$$\frac{\partial V}{\partial(1+y)} = \frac{D_{obl}}{1+y} P_{obl} - \frac{\tau}{1+y} Z = 0.$$

Because $V = 0$ already forces $Z = P_{obl}$, the common factor $P_{obl}/(1+y)$ cancels and we are left with $D_{obl} = \tau$. Duration-matching therefore forces the zero's maturity τ to equal the Macaulay duration of the obligation.

Example. Consider the previous example of a bond that matures in three years, makes annual coupon payments of 10% and currently sells at par ($1000). It was shown that the duration of the bond is 2.736. Consider the company choosing to meet the obligations of the bond by selling a zero coupon bond. So that the payment of the zero coupon bond has the same duration as the payment of the obligation. In order to hedge the obligation (which has a present value of $1000) the company purchases the zero in a quantity such that its present value is exactly $1000. The duration of the zero that they purchase should be $\tau = 2.736$. Thus, F solves

$$\$1000 = \frac{F}{1.10^{2.736}}$$

which implies that $F = \$1297.927$. Let's check that this works. Suppose that yields increase by a very small amount. What is the approximate change in the value of your portfolio?

$$\frac{\partial V}{\partial y} \frac{1+y}{V} = -D + D = -2.736 + 2.736 = 0$$

Now suppose that yields on both the zero and the payment obligation increase by .0001 (1 basis point). What is the new value of the bond that you issued? It is given by

$$V_B = \frac{100}{1.1001}\left(1 - \frac{1}{(1.1001)^3}\right) + \frac{1000}{1.1001^3} = \$999.751358$$

The value of the zero that you own is

$$V_Z = \frac{1297.927}{1.1001^3} = 999.751039$$

so the change in the value of your portfolio from a 1 basis point change in yields is $-999.751358 + 999.751039 = -\$0.000319$. So your portfolio is largely immunized from small changes in yields.

Notice however that duration is an (first order) approximation of the change in the value of your portfolio as yields change. This approximation gets worse as the changes in yields get higher. To see this consider the following.

Example cont'd. Now suppose that instead of a 1 basis point change in yields you have a 50 basis point change in yields. Then the value of your portfolio becomes

$$\begin{aligned} V &= -\frac{100}{1.105}\left(1 - \frac{1}{(1.105)^3}\right) + \frac{1000}{1.105^3} + \frac{1297.927}{1.105^3} \\ &= -987.67438 + 987.668207 \\ &= -.0061723 \end{aligned}$$

What if there were a 200 basis point change in yields?

$$\begin{aligned} V &= -\frac{100}{1.12}\left(1 - \frac{1}{(1.12)^3}\right) + \frac{1000}{1.12^3} + \frac{1297.927}{1.12^3} \\ &= -951.96337 + 951.896559 \\ &= -0.066811 \end{aligned}$$

So the approximation gets worse as changes in yield get larger. Why is this so?

11.5 Convexity

Recall that the change in the value of the bond portfolio was approximately the modified duration multiplied by the change in yield in the same sense that a finite change in the argument of a function leads to a change in the value of the function that is approximately the derivative of the function multiplied by the change in the argument. We can also do a second order Taylor series approximation of the price of the bond as a function of yield. To find the second derivative, differentiate the first derivative $\frac{\partial P}{\partial(1+y)} = -\sum_{t=1}^{T} t\, CF_t(1+y)^{-t-1}$ once more with respect to $(1+y)$. Applying the power rule to $(1+y)^{-t-1}$ gives $-(t+1)(1+y)^{-t-2}$, so each term's two negative signs combine to a positive:

$$\frac{\partial^2 P}{\partial(1+y)^2} = -\sum_{t=1}^{T} t\, CF_t \cdot (-(t+1))\,(1+y)^{-t-2} = \sum_{t=1}^{T} \frac{t(t+1)CF_t}{(1+y)^{t+2}}.$$

Because y and $1+y$ differ by a constant, $\partial/\partial y = \partial/\partial(1+y)$, and the second derivative of the price of a bond as a function of yield is

$$\frac{\partial^2 P}{\partial y^2} = \sum_{t=1}^{T} \frac{t(t+1)CF_t}{(1+y)^{t+2}}$$

which is positive, confirming that the price-yield curve is convex.

Notice that bond price as a function of yield is convex for standard bonds. As such, we have the following definition.

Convexity. The second derivative of the bond price with respect to yield expressed as a fraction of the bond price.

$$Convexity(P(y)) = \frac{1}{P(1+y)^2} \sum_{t=1}^{T} \frac{t(t+1)CF_t}{(1+y)^t}$$

In general, convexity is seen as a desirable property. Notice that since bonds are convex in yields, if yields go up the decrease in price that occurs will be much smaller than the increase in price that would occur if yields go down.

Example. Suppose that you have a payment obligation of $b a year for each of the next T years. You are concerned that the value of this obligation will increase (i.e. you are concerned that yields will decrease.) You want to provide a hedge against this occurence by purchasing assets with the money that you receive from selling the obligation. The present value of the payment obligation is

$$V = \sum_{t=1}^{T} \frac{b}{(1+y)^t}$$

and the modified duration is

$$D = \frac{b}{P(1+y)} \sum_{t=1}^{T} \frac{1}{(1+y)^t}$$

You want to purchase two different zero coupon bonds (of possibly different maturities) that have the same present value, modified duration and convexity when taken together as a portfolio. Notice that the convexity of a zero coupon bond is given by

$$\begin{aligned} Convexity\left(\frac{M}{(1+y)^T}\right) &= \frac{1}{P}\frac{T(T+1)M}{(1+y)^{T+1}} \\ &= \frac{(1+y)^T}{M}\frac{T(T+1)M}{(1+y)^{T+1}} \\ &= \frac{T(T+1)}{(1+y)^2} \end{aligned}$$

The (Macaulay) duration of your obligation is given by D and the convexity is given by C. Let p_1 and p_2 denote the present values (the dollar amounts) invested in the two zeros, with weights $w_i = p_i/V$, so that

$p_1 + p_2 = V$ and $w_1 + w_2 = 1$. Because each zero has Macaulay duration equal to its maturity τ_i and convexity $\tau_i(\tau_i + 1)/(1 + y)^2$, and the duration and convexity of a portfolio are the present-value-weighted averages of the components', you must select p_1, p_2, τ_1, τ_2 such that

$$\begin{aligned} V &= \frac{M_1}{(1+y)^{\tau_1}} + \frac{M_2}{(1+y)^{\tau_2}} = p_1 + p_2 \\ D &= w_1\tau_1 + w_2\tau_2 \\ C(1+y)^2 &= w_1\,\tau_1(\tau_1+1) + w_2\,\tau_2(\tau_2+1) \end{aligned}$$

where $M_i = p_i(1+y)^{\tau_i}$ is the face value of zero i. Note that matching the *Macaulay* duration D is equivalent to matching the modified duration $D^* = D/(1+y)$, since both the obligation and the hedge portfolio are discounted at the same yield, so the common factor $1/(1+y)$ cancels from both sides.

Example. Consider the previous example in which the obligation is the three-year 10% par bond ($1000 face, annual coupons of $100, and the $1000 principal repaid at maturity), currently selling at par. Its duration was 2.736. The convexity of this bond is

$$\begin{aligned} Convexity &= \frac{1}{1000(1.1)^2} \\ &\quad \times \left(\frac{1(1+1)100}{1.1^1} + \frac{2(2+1)100}{1.1^2} + \frac{3(3+1)1100}{(1.1)^3} \right) \\ &= \frac{10595.04}{1210} = 8.7562 \end{aligned}$$

(The final cash flow is $1100 — the last coupon plus the $1000 principal — not $100; including the principal is what makes the third term dominate.) To keep the algebra clean, suppose we split the hedge equally between the two zeros, so $w_1 = w_2 = \frac{1}{2}$ (each zero has present value $500). The second and third matching conditions from the general example then become

$$\begin{aligned} 2.736 &= \tfrac{1}{2}\tau_1 + \tfrac{1}{2}\tau_2 \quad \Longrightarrow \quad \tau_1 + \tau_2 = 5.472, \text{ and} \\ 8.7562\,(1.1^2) &= \tfrac{1}{2}\tau_1(\tau_1+1) + \tfrac{1}{2}\tau_2(\tau_2+1) \\ &\Longrightarrow \quad \tau_1^2 + \tau_2^2 + \tau_1 + \tau_2 = 21.190. \end{aligned}$$

Substituting $\tau_1 + \tau_2 = 5.472$ into the second equation gives $\tau_1^2 + \tau_2^2 = 21.190 - 5.472 = 15.718$. Combined with the identity $(\tau_1 + \tau_2)^2 = \tau_1^2 + \tau_2^2 + 2\tau_1\tau_2$, this yields

$$2\tau_1\tau_2 = 5.472^2 - 15.718 = 29.943 - 15.718 = 14.225, \qquad \tau_1\tau_2 = 7.113.$$

So τ_1 and τ_2 are the two roots of $z^2 - 5.472\,z + 7.113 = 0$:

$$\tau = \frac{5.472 \pm \sqrt{5.472^2 - 4(7.113)}}{2} = \frac{5.472 \pm \sqrt{1.505}}{2} = \frac{5.472 \pm 1.227}{2},$$

giving $\tau_1 = 2.12$ years and $\tau_2 = 3.35$ years. Purchasing \$500 of a 2.12-year zero and \$500 of a 3.35-year zero matches the obligation's present value, duration, and convexity, so the hedge remains accurate even for sizeable yield moves.

11.6 Application: A pension manager immunizing liabilities

A manager of a defined-benefit pension fund has promised a stream of payments to retirees stretching decades into the future, and must invest today's assets so those promises are met whatever happens to interest rates. Duration and convexity are the instruments that make this possible. By measuring the duration of both the liabilities and the bond portfolio, the manager can match them so that a rise or fall in rates moves the value of assets and obligations together, immunizing the plan against the very shocks that would otherwise open a funding gap. The same arithmetic that quantifies how much a long bond falls when yields rise — the mechanism behind well-publicized bank failures — is, in the manager's hands, the tool that keeps a pension solvent.

Further listening

The Indicator from Planet Money (NPR) — "How Silicon Valley Bank became the largest bank failure since 2008": how an unhedged duration mismatch — the exact risk this chapter manages — brought down a major bank.

11.7 Homework problems

11.7.1 Conceptual

FIP-C1. Two bonds have the same maturity and yield to maturity y, but bond A pays a 3% annual coupon while bond B pays a 9% annual coupon. Using property 2 of duration, explain why bond A has the larger Macaulay duration D, and state which of the two bonds will lose a larger percentage of its value when yields rise by a small amount Δy.

FIP-C2. The chapter interprets Macaulay duration as the "center of gravity" of a bond's payment stream — the time-weighted average of the present values of its cash flows. Using the formula $D = \frac{1}{P}\sum_{t=1}^{T}\frac{t\,CF_t}{(1+y)^t}$, explain why raising the yield y (holding the cash flows fixed) *lowers* the Macaulay duration. Which cash flows lose the most weight in the average when y rises, and why does that pull the center of gravity toward the present?

FIP-C3. A colleague argues that a 30-year zero coupon bond and a 30-year coupon bond selling at par are equally risky because "they mature at the same time." Using the definition of modified duration D^* as a measure of interest-rate sensitivity, and the chapter's result that a zero's Macaulay duration equals its maturity T, explain why the two bonds do not have the same sensitivity to a change in yield. Which one would you expect to fall more in price for a given increase in y?

FIP-C4. The chapter states that "long-maturity, low-coupon bonds are far more sensitive to rate changes than short-maturity, high-coupon bonds." Using the relation $\frac{\Delta P}{P} \approx -D^{*}\Delta y$ together with the qualitative properties of duration, explain why both the *maturity* effect and the *coupon* effect push in the same direction here, and why D^{*} is the single number that lets a manager compare and aggregate this sensitivity across an entire portfolio.

FIP-C5. State property 2 of duration ("duration decreases in the coupon rate") and explain the mechanism in terms of *when, on average, the bondholder receives value*. A perpetuity paying a fixed coupon forever and a zero coupon bond have very different duration behavior as the coupon rate changes — which of these two has a duration that does not depend on maturity at all, and how does that connect to property 2?

FIP-C6. Property 3 says duration "often increases with the maturity of bonds," but the chapter warns this "may not be true for some bonds selling at a very deep discount." Explain intuitively why lengthening maturity usually raises duration (in terms of spreading payments further into the future), and describe qualitatively why a deep-discount bond can break the pattern.

FIP-C7. A pension fund holds long-maturity Treasury bonds as assets and owes a stream of retiree payments as liabilities. Explain what it means to say the fund's assets and liabilities are "duration-matched," and describe what happens to the fund's net position after a small parallel rise in yields if (a) asset duration equals liability duration, versus (b) asset duration is much shorter than liability duration. Connect part (b) to the Silicon Valley Bank episode described in the chapter.

FIP-C8. In the immunization example the company chooses the face value F and maturity τ of a zero to satisfy $V = 0$ and $\partial V/\partial y = 0$. Explain in words what each of these two conditions guarantees, and show why together they force the zero's maturity to equal the Macaulay duration of the obligation ($\tau = D_{obl}$). Why is it the *Macaulay* duration of the obligation, and not its maturity T, that determines τ?

FIP-C9. The chapter shows that a duration-matched portfolio still shows a small residual loss for a 50-basis-point move and a larger one for a 200-basis-point move. Explain, in terms of the second derivative $\partial^2 P/\partial y^2$ and the curvature of the price-yield relationship, why a duration-only measure understates the true price for large yield moves. Why is the sign of the approximation error the same whether yields rise or fall?

FIP-C10. A bond fund manager says, "Given a choice between two bonds with identical duration, I will always take the one with higher convexity, and I would even pay a slightly higher price for it." Using the chapter's statement that a more convex bond "gains more when yields fall and loses less when yields rise," explain why convexity is a desirable property and why it commands a price premium.

FIP-C11. In the chapter's convexity-hedging example, the manager hedges a coupon obligation by buying *two* zero coupon bonds rather than one. Explain why one zero is not enough: how many conditions (present value, duration, convexity) must the hedge satisfy, and how many free choices does a single zero versus two zeros provide? Relate your count to the fact that in the worked example the two required conditions on maturities were $\tau_1 + \tau_2 = 5.472$ and a second equation in $\tau_1(\tau_1 + 1) + \tau_2(\tau_2 + 1)$.

FIP-C12. The barbell example matches present value, duration, *and* convexity using present-value weights $w_i = p_i/V$. Explain why the duration and convexity of the two-bond portfolio are the present-value-weighted averages of the individual zeros' durations and convexities (rather than, say, face-value-weighted averages). Why does matching Macaulay duration D automatically match modified duration $D^* = D/(1 + y)$ here, so that only one duration condition is needed?

11.7.2 Quantitative

FIP-Q1. A bond makes annual coupon payments of \$60 (a 6% coupon on \$1000 face) and matures in 3 years. Its YTM is $y = 0.06$, so it sells at par

($P = 1000$). Compute its Macaulay duration D using $D = \frac{1}{P}\sum_{t=1}^{T}\frac{t\,CF_t}{(1+y)^t}$.

FIP-Q2. A bond makes annual coupon payments of \$40 (a 4% coupon on \$1000 face) and matures in 2 years, so its cash flows are \$40 at $t = 1$ and \$1040 at $t = 2$. Its YTM is $y = 0.04$, so it sells at par ($P = 1000$). Compute its Macaulay duration $D = \frac{1}{P}\sum_{t=1}^{T}\frac{t\,CF_t}{(1+y)^t}$.

FIP-Q3. A zero coupon bond has face value $M = 5000$, maturity $T = 8$ years, and YTM $y = 0.05$. Using the chapter's result that a zero's Macaulay duration equals its maturity, state D, and then compute its modified duration $D^* = D/(1 + y)$.

FIP-Q4. A coupon bond has Macaulay duration $D = 2.833$ and YTM $y = 0.06$. Compute its modified duration $D^* = D/(1+y)$, and state the economic meaning of the number: for a 1-percentage-point rise in yield, approximately what percentage of its value does the bond lose?

FIP-Q5. A bond portfolio has price $P = 40{,}000$ and modified duration $D^* = 7.2$. Using $\frac{\Delta P}{P} = -D^*\Delta y$, compute the approximate percentage change and the approximate dollar change in the portfolio's value if yields rise by 40 basis points ($\Delta y = 0.0040$).

FIP-Q6. A bond portfolio has price $P = 25{,}000$ and modified duration $D^* = 6.5$. Using $\frac{\Delta P}{P} = -D^*\Delta y$, compute the approximate percentage change and the approximate dollar change in the portfolio's value if yields *fall* by 25 basis points ($\Delta y = -0.0025$).

FIP-Q7. A liability has present value $V = 250{,}000$ and modified duration $D^* = 11$. A manager wishes to immunize it with a single zero coupon bond of the same present value. Using $\frac{\Delta P}{P} = -D^*\Delta y$, what must the zero's modified duration be for the net position to be first-order immune to small yield changes, and (with $y = 0.04$) what is the required *maturity* τ of the zero (recall a zero's $D^* = \tau/(1 + y)$)?

FIP-Q8. A company owes an annuity of $M = 200$ per year for $T = 4$ years, discounted at $y = 0.05$. Its present value is $V = \frac{M}{y}\left(1 - \frac{1}{(1+y)^T}\right)$. Compute V. Then it funds the annuity with a zero coupon bond whose present value equals V and whose maturity is $\tau = 2.5$. Compute the required face value F

from $V = \frac{F}{(1+y)^{\tau}}$.

FIP-Q9. A 2-year bond pays an annual coupon of \$50 (5% on \$1000 face) and matures at \$1050 in year 2. Its YTM is $y = 0.05$ so $P = 1000$. Compute its convexity using $Convexity = \frac{1}{P(1+y)^2} \sum_{t=1}^{T} \frac{t(t+1)CF_t}{(1+y)^t}$.

FIP-Q10. A zero coupon bond has maturity $T = 3$ years and YTM $y = 0.05$. Using the chapter's closed form for the convexity of a zero, $Convexity = \frac{T(T+1)}{(1+y)^2}$, compute its convexity. (Confirm you get the same answer as evaluating the general convexity formula on a single cash flow at $t = 3$.)

FIP-Q11. Using the bond from FIP-Q9 ($P = 1000$, $D^* = D/(1 + y)$ with D its Macaulay duration, and convexity C from FIP-Q9), estimate the change in price for a large yield increase of $\Delta y = 0.02$ using the second-order approximation $\Delta P \approx -D^* P\, \Delta y + \frac{1}{2} CP(\Delta y)^2$. (Take $D = 1.9524$ for this bond.)

FIP-Q12. For the same bond as FIP-Q11 ($P = 1000$, $D = 1.9524$, $y = 0.05$, convexity $C = 5.2694$), estimate the change in price for a large yield *decrease* of $\Delta y = -0.03$ using $\Delta P \approx -D^* P\, \Delta y + \frac{1}{2} CP(\Delta y)^2$. Comment on how the convexity term affects the estimate relative to the duration-only prediction.

11.7.3 Data exercises

These problems use the live-data figures in this chapter.

FIP-D1. Figure 11.1 shows a 10-year Treasury carrying the par coupon 4.62% (semiannual payments, \$1,000 face), so it is priced at par with modified duration $D^* \approx 7.9$. (a) Use the duration approximation to estimate the percentage price change if the yield rises by 100 basis points to 5.62%. (b) Compute the exact new price from the pricing equation and the exact percentage change. (c) The two answers differ; state which is more negative, and connect the sign of the difference to the shaded region in the figure.

FIP-D2. In Figure 11.2, a 20-year Treasury issued at par in March 2020 with a 0.87% coupon had modified duration of roughly 17 at issue. Between March 2020 and October 2023 the 20-year yield rose by about 4.2 percentage points. (a) What loss does the first-order approximation $\Delta P/P \approx -D^* \Delta y$

predict? (b) The figure shows the realized loss was about −48%. Give two reasons the linear prediction overstates the loss for a move this large. (c) Explain why this loss required no default by the U.S. Treasury.

Part IV

Derivatives

Chapter 12

Forwards and Futures

 # -- coding: utf-8 --

12.1 Introduction

Three questions sit at the heart of the study of forwards and futures. First, what is the precise relationship between the current spot price of an asset and the price agreed upon today for delivery at a future date — and why must that relationship hold by arbitrage? Second, how do futures contracts differ from forward contracts in their institutional structure, and what practical consequences follow from those differences for hedgers? Third, how does the futures price compare to the expected future spot price, and what does that gap reveal about how risk is allocated between hedgers and speculators? These questions are fundamental to financial markets because forwards and futures are among the most widely traded instruments for managing price risk. Commodity producers hedge against price declines; currency traders lock in exchange rates; portfolio managers use equity index futures to adjust market exposure rapidly without trading the underlying securities. The theoretical foundations of these instruments underpin an enormous volume of global financial activity.

The first question is answered by a straightforward arbitrage argument. Owning a forward contract that delivers one unit of an asset in the future is economically equivalent to borrowing money today, buying the asset in the spot market, and holding it until delivery. Because these two strategies produce the same outcome, they must have the same cost, which implies that the forward price must equal the spot price compounded at the risk-free rate over the contract's life. Any deviation from this cost-of-carry relationship creates a riskless profit opportunity that arbitrageurs will immediately exploit, restoring the equilibrium. This principle is derived formally in the chapter and illustrated with a currency forward example.

The second question concerns the institutional differences between forward and futures contracts and their practical implications. Futures contracts are standardized and traded on organized exchanges, which solves the counterparty credit risk problem inherent in over-the-counter forward contracts but introduces the need for margin accounts and daily marking to market. When a futures position moves against a trader, losses are collected from the margin account each day rather than accumulated until expiration; this daily settlement can create cash flow demands that differ from the economics of the underlying hedge. The chapter works through margin mechanics in detail, using a corn futures example from the CBOT, and explains basis risk — the residual price uncertainty that arises when the hedging contract does not perfectly match the timing or specification of the underlying exposure.

The third question moves from no-arbitrage pricing to an equilibrium view of the futures price. The expectations hypothesis holds that the futures price is simply the market's best forecast of the future spot price. Normal backwardation and contango offer richer explanations grounded in the asymmetric hedging demands of producers and consumers: if natural sellers dominate, speculators must be compensated with a price discount to take the opposite side, so the futures price lies below the expected spot price. Modern portfolio theory provides a further refinement, linking the gap between the futures price and the expected spot price to the asset's systematic risk, in a way that is fully consistent with the CAPM.

The chapter opens by defining forward contracts and deriving the spot-forward price relationship through the arbitrage argument. It then turns to futures contracts, explaining exchange-based trading, margin requirements, marking to market, and basis risk through a series of concrete examples. The chapter closes by surveying the three classical theories of futures pricing — the expectations hypothesis, normal backwardation, and the modern portfolio theory perspective — providing both the technical tools and the economic intuition needed to analyze the pricing and strategic use of these instruments.

In the news

Futures prices move on tomorrow's expected supply, not just today's. CNBC reported Brent crude swinging around $80 a barrel in mid-2026 as U.S.–Iran talks faltered and traders reassessed the risk of disruption at the Strait of Hormuz — after an earlier conflict had briefly pushed prices above $120. Anyone exposed to those swings, from airlines to producers, manages them with the forward and futures contracts, and the cost-of-carry pricing, developed in this chapter. Read it at CNBC.

12.2 Forward contracts

A forward contract is an agreement to buy or sell a certain asset at a certain future time for a certain price. A *spot* market is a market where an asset is sold for immediate delivery. Forward contracts are sold in over-the-counter markets. The investor with the long position agrees to buy the asset in the future, while the investor who has the short position agrees to sell the asset in the future. Notice that there are two financial instruments working here. The first is the asset that will be bought or sold in the future. The second is the forward contract, or the agreement to buy or sell that asset in the future. Thus, one can speak of the price of the asset (usually called the spot price if speaking of the current price of the asset) or the price of the forward contract.

Example. As of 21 March 2011 the Euro spot rate was \$1.4208. The 3 months forward rate for Euros was \$1.4187. Forward rates can be used to hedge currency risk. To see this, consider a U.S. bank that knows that in 3 months it will be receiving a payment of EUR 1.5M. The bank faces currency risk because it knows that it will want to exchange the euros that it receives for dollars as soon as it receives them. Thus, if the spot rate is low in three months then the dollar value of the payment will be low and oppositely if the spot rate is high. In order to hedge this risk, the bank could enter into a forward contract today. It would agree to sell EUR 1.5M in three months. The current forward rate indicates that a counterparty would agree to pay $1.4187 \cdot 1,500,000 = \$2,128,050.0$ in three months to receive that EUR 1.5M. The bank is short a forward contract on the euro.

The payoff to a forward contract depends on the spot price of the underlying asset at the date of maturity of the contract. For a trader that is long the forward contract with delivery price K, the payoff of the contract at maturity is $S_T - K$ where S_T represents the spot price of the underlying asset at time T (maturity). If the spot price S_T is larger than the delivery price agreed upon in the contract K, then at maturity the long trader owns an agreement to purchase the asset for less than its current price. Thus the payoff to owning the contract is positive. If at maturity the spot price S_T is less than the agreed upon delivery price K then the long trader is required to purchase the asset at a price that is higher than she would have to pay if she were purchasing the asset in the spot market. The payoff to being short the contract is $K - S_T$ and analogous statements can be made.

The spot price of an asset and the forward price of the asset are related. To see why, consider what it means to be long a forward contract that matures in 1 year and will deliver Y units of the asset for a price of F per unit. This means that in 1 year the trader will need to pay FY and take receipt of Y units of the asset. An alternative way to have Y units (and pay for them) in one year is for the investor to borrow some amount of money today at interest rate r to be paid back in one year and use it to buy Y units of the asset today. Thus, the investor could borrow \$$SY$ today and pay it back in one year (i.e. pay back $SY(1+r)$). These strategies are equivalent since both give you possession of the asset in one year in exchange for some amount of money. Thus, the payoff to them must be the same. In other words $(1+r)SY = FY$ or $(1+r)S = F$.

To see why, assume that $(1+r)S > F$. Then the investor could sell short the asset today and receive S to be invested at the interest rate r. At the same time the investor would go long the forward contract. In one year, the investor now has $\$(1+r)S$ out of which she pays F to receive the asset that is then used to close out the short position. The profit made is $(1+r)S - F$ and it is risk free. On the other hand, if $(1+r)S < F$ then the trader can borrow S to buy one unit of the asset and takes a short position in the forward contract. In one year the investor delivers the unit of the asset to fulfill the short position on the contract and from the proceeds of the short contract (F) pays off the loan ($(1+r)S$) that was taken out. The risk-free proceeds are $F - S(1+r)$. Since both of these strategies are risk-free, they should provide the same rate of return in equilibrium. Hence $(1+r)S = F$.

The reason that the forward price is higher than the spot price is that purchasing the asset today requires using money that would otherwise be used for something else. In the forward contract the payment does not have to be made until the contract matures.

Example. Suppose that the current one-year, continuously compounded interest rate is r. The spot price for one Euro is \$1.4208 and the three-month forward price is \$1.4187. What is the implied interest rate r? Using continuous compounding we have that the relationship between the spot and future prices is $Se^{0.25r} = F$ (the horizon is $T = 0.25$ years). Solving, take logs of $e^{0.25r} = F/S$ to get $0.25\,r = \ln(F/S)$, so

$$r = \frac{1}{0.25}\ln\left(\frac{F}{S}\right) = 4\ln\left(\frac{1.4187}{1.4208}\right) = 4\,(-0.0014791) = -0.0059165.$$

Because the forward price is *below* the spot price ($F < S$), the log is negative and the implied rate is **negative**: about –0.59%. (The forward selling at a discount to spot signals a slightly negative annualized dollar interest rate over the quarter.)

12.3 Types of traders

Traders can generally (although not perfectly) be categorized as hedgers or speculators. Hedgers wish to reduce the risk inherent in their portfolios by taking forward and futures positions that offset their asset positions. Speculators generally have no asset position and wish to take a view (i.e. speculate) on changes in either the future/forward price, or the relative price of the futures and spot markets.

12.4 Futures contracts

Futures contracts are fundamentally similar to forward contracts. A futues contract is an agreement to buy or sell an asset at a future date for a specified price. There are some important differences between futures and forwards however. The first is that futures contracts are generally standardized to the point that they can be traded on an exchange.

Example. The futures contract for corn trades on the Chicago Board of Trade and is a contract for 5000 bushels of number 2 yellow corn. The party that is short the futures contract (that is has agreed to deliver the corn) may also substitute number 1 yellow corn for an increased price to the long position or number 3 yellow corn for a discount. The CBOT establishes contracts that expire in December, March, May, July and September on the 15th of the month. The last day of trading is the last business day before the 15th and delivery must be made by the second business day following the last trading date of the delivery month. The exchange also stipulates that for each contract, those in a hedging position must have $1500 in margin on reserve in their margin account while those in a speculative position must have $2025 in their margin account. Currently (as of 13 November 2008) the futures price for March 09 delivery is $388\frac{6}{32}$ cents per bushel or \$19,409.375. The contract also must specify where delivery will occur.

12.4.1 Convergence of futures and spot prices

The first thing to note about futures prices is that they must converge to the spot price on the date of delivery. To see why this must be the case, suppose that the price of delivery of March delivery of on 13 March 2009 (the last day of trading of the contract) is something other than the spot price for the purchase of corn. If the futures price is higher, then an investor would sell the futures contract agreeing to deliver corn and buy the corn to be delivered in the spot market for less than the futures price. This profit would be risk free and nearly immediate. Similarly if the futures price is lower than the spot price on the last day of trading then a trader could go long the futures contract agreeing to take delivery of the corn and sell the corn short in the spot market. The delivery from the futures contract would fulfill the obligation of the short selling in the spot market and the difference in prices would be risk-free profit for the trader.

12.4.2 Closing out positions

Most futures contracts are closed out before the end of trading. That is, most of the corn traded on the CBOT is never delivered. This is done by entering into the opposite trade with the market before the close of trading. If a trader is short 15 futures contracts then she must purchase 15 contracts to close her position. Notice that since a futures contract is an agreement to sell or buy in the future, going short in the futures markets is easy and does not require finding some of the asset in the market to borrow before shorting. We will see that the fact that the futures price converges to the spot price means that the futures contract still provides a hedge against price movements even if delivery is not taken.

Looking at futures price data[1] notice the column `OpenInt` is the measure

[1] Satchell, Stephen, and Alan Scowcroft. "A demystification of the Black-Litterman model: Managing quantitative and traditional portfolio construction." In Forecasting expected returns

of *one side* of the market. That is, it is the number of short positions, or equivalently long positions. If you look at rice today (13 March 2008) it can be noticed that the open interest is rather small since the November contract is about to stop trading.

12.4.3 Settlement and Margins

Futures contracts are settled *daily* based on the market value of that futures contract. This process is called *marking to market*.

Example. Consider the March 2009 corn contract discussed earlier. The current value of one contract is \$19,409.375. The margin required for a hedger is \$1,500 per contract. Consider a hedger that is long one contract and whose margin account has \$2000 in it. Suppose that tomorrow the price per bushel of corn for March delivery goes up to 390 cents per bushel. Now being long a March corn contract has value $3.90 \cdot 5000 = 19,500$. Thus, the value of one contract has increased by 19,500 - 19,409.375 = 90.625. At the end of the day then, the clearinghouse/exchange deducts \$90.625 from the margin account of each trader that is short the March contract and adds \$90.625 to the margin account of each trader that is long the March contract. So the balance of our trader's margin account is now \$2090.625. If the maintenance margin is \$1500, then when the price/bushel of the futures contract solves $19409.375 - 5000 \cdot \frac{p}{100} = 500$ then the investor will have to meet a margin call. This occurs when the price is $378\frac{6}{32}$ cents.

As mentioned previously, even if the position is closed before the close of trading, the futures contract provides a hedge for a hedging investor. Consider the following example.

Example. Consider an oil producer that has just agreed to sell 1 million barrels of crude oil. It is now 15 May. The price of the sale will be the market price on 15 August. Thus, if the price of oil increases

in the financial markets, pp. 39-53. Academic Press, 2007.

by \$0.01, then the increase in the proceeds of this sale increase by $1,000,000 \cdot 0.01 = 10,000$. Likewise, for every penny that the price of oil decreases the revenue from the sale decreases by \$10,000. Suppose that the spot price of crude oil on 15 May is \$60 per barrel and the crude oil futures price for August delivery (NYMEX) is \$59/barrel. Each oil futures contract on NYMEX is for 1000 barrels, so to hedge this position the producer will go short 1000 oil contracts. Closing out the contract near 15 August will imply that the producer locks in an approximate price of \$59/barrel. To see this, suppose that the spot price on 15 August is \$56. Then the sale of oil under the original agreement brings in \$56M. Since the futures contract expires near 15 August, the price of the futures contract will be approximately \$56/barrel. Therefore, for each contract the company has gained $3 \cdot 1000 = \$3000$. Multiplying this by the number of contracts shows that the gains from the futures position are $3000 * 1000 = 3,000,000$. So the net proceeds from the futures position and the sale of oil is $\$56M + \$3M = \$59M$. On the other hand, suppose that the spot price of oil on 15 August is \$63. In this case, the sale of the oil brings revenue of \$63M while the futures contracts show a loss of $4 \cdot 1000 \cdot 1000 = 4,000,000$. Thus, the total proceeds from the oil sale and the futures contracts is $\$63M - \$4M = \$59M$.

12.4.4 Basis risk

Many times when using futures to hedge there does not exist a contract that expires at exactly the same time as the asset that will be delivered. As such, one may have to hedge using an imperfect instrument. This leads to *basis risk*. For example, if a farmer is going to deliver wheat in April 2009, then she may use the May 2009 contract to lock in the price of the wheat for delivery. However, since May comes after April, it is likely that the spot price of wheat in April will not be equal to the futures price of wheat for May delivery. The difference between these is the basis. In particular, if T is the time at which the farmer will make delivery and $S(t)$ is the spot price with $F_\tau(t)$ the futures price for delivery at time τ when the time is t then

$$Basis(T) = S(T) - F_\tau(T)$$

12.5 Pricing futures contracts

Consider first the forward price. By a cost-of-carry (no-arbitrage) argument, buying the asset today at the spot price S with money borrowed at the continuously compounded rate r and holding it to delivery costs Se^{rt} at time t; since this replicates the forward's delivery, the forward price must be $\tilde{F} = Se^{rt}$, for otherwise one could borrow-and-carry (or short-and-lend) for a riskless profit.

When the interest rate is constant from the time that the futures contract is purchased until it expires, the futures price equals this forward price, $F = \tilde{F} = Se^{rt}$. The reason is a *rollover* argument. A futures position is marked to market daily, so it throws off (or absorbs) cash each day rather than only at maturity. But with a known, constant rate r those daily settlement flows can be borrowed or invested at exactly that same rate, and by holding a slightly scaled ("tailed") number of contracts — $e^{-r(t-s)}$ contracts on day s — the reinvested daily gains accumulate to precisely the single terminal payoff of the forward. The two contracts therefore deliver identical cash at time t and must carry the same price, Se^{rt}. However, there are differences between these in the case when interest rates are not constant. This arises because futures contracts are marked to market, so the proceeds from movements in the value of the contract accrue every day, as opposed to when the contract matures. Thus, the present value of the contract must discount immediate payments less than the payments at maturity are discounted. For example, consider a contract for which the value of a long position in the contract increases when the interest rate increases. Thus, the futures contract provides income to the long position exactly when this investor can invest those payments for a high rate. As such, the long position becomes relatively more desirable than the short position and the price of the futures contract will increase above the equivalent forward contract. On the other hand, if the value of being long the contract tends to decrease when the interest rate increases then the futures contract will be less desirable than the forward contract and the price will go down.

12.6 Futures prices and expected spot prices

There are three traditional hypotheses of the relationship between the futures price and the expected future spot price. These are:

Expectations hypothesis. That $ES_T = F$.

Normal Backwardation. For many contracts there are natural hedgers who want to reduce their risk. As such, speculators have to be rewarded to hold the risk that the hedgers are shedding. Thus it should be that $ES_T > F$.

Contango. The opposite of normal backwardation. The natural hedgers want to take a long position so the short side of the market must be rewarded to hold that risk. Thus $ES_T < F$.

Modern portfolio theory. Investors will enter the market if the rate of return is commensurate with the risk involved. Thus, if k is the discount rate required in order to hold the asset we have that

$$S = e^{-kT} ES_T$$

from the spot-futures relationship (assuming no correlation between interest rates and mark-to-market)

$$S = e^{-rT} F.$$

Both expressions equal the same spot price S, so we can eliminate S by setting them equal:

$$e^{-rT} F = e^{-kT} ES_T.$$

Multiplying both sides by e^{rT} isolates F,

$$F = e^{rT}e^{-kT}ES_T = e^{(r-k)T}ES_T.$$

It can be seen from this that if the variability of the spot price S_T is really high (which implies that k is high) then the price of the futures contract will be less than the expected spot price. In general this will happen for all positive β stocks. Thus, in general the long position in the futures contract will cost less than the expected future spot price.

12.7 Application: An airline hedging fuel

The treasurer of an airline knows that a sudden jump in the price of crude oil can erase a quarter's profit, and must decide how to protect the company against a spike it cannot control. Forward and futures contracts are the answer, and the cost-of-carry relationship developed in this chapter is what lets the treasurer use them intelligently. By locking in a price for fuel months ahead, the airline converts an uncertain future cost into a known one; understanding how futures prices relate to the spot price through storage, financing, and convenience yields tells the treasurer whether the hedge is fairly priced and how the futures curve will behave as delivery approaches. When geopolitical news sends oil swinging, the difference between a hedged and an unhedged airline is the difference this chapter's tools are built to manage.

> Further listening
>
> *Planet Money* (NPR) — "Oil #2: The Price of Oil": how the price of oil is really set in the futures market, where traders buy and sell promises to deliver months or years ahead.

12.8 Homework problems

12.8.1 Conceptual

FF-C1. A gold-mining company will produce and sell 10,000 ounces of gold in six months and worries that the price could fall before then. A trader at a hedge fund, holding no gold, believes prices will rise and wants to profit from that view. Explain how each party would use a gold forward or futures contract, identifying which side (long or short) each takes and why. Which party is the hedger and which is the speculator?

FF-C2. A wheat farmer expecting to harvest and sell wheat in three months hedges by going short wheat futures. Suppose that, contrary to fears, the spot price of wheat rises sharply by delivery. Explain what happens to the farmer's revenue from selling the physical wheat versus the payoff on the short futures position, and why the farmer is not made worse off by having hedged even though prices rose. What has the farmer actually "locked in," and what has the farmer given up?

FF-C3. Explain the no-arbitrage (cost-of-carry) argument that pins down the forward price. Describe the two portfolios that must have the same payoff — a long forward, versus borrowing to buy and carry the asset — and show how their equivalence forces $F = (1 + r)S$. Then explain what riskless trade an arbitrageur would execute if instead $(1 + r)S > F$.

FF-C4. The chapter states that the forward price exceeds the spot price for a non-dividend asset because "purchasing the asset today requires using money that would otherwise be used for something else." Explain this financing (cost-of-carry) intuition, and describe how the relationship would change if holding the asset instead threw off income or a convenience yield k, giving $F = Se^{(r-k)T}$.

FF-C5. Compare a forward contract negotiated over the counter with an exchange-traded futures contract on the same underlying asset with the same maturity. Explain how the futures contract addresses counterparty credit risk that the forward does not, and describe the mechanism (standardization, margin accounts, and daily marking to market) the exchange uses to do so.

FF-C6. Two traders each hold a long futures position on corn. Over a week the corn futures price falls steadily. Explain, using the marking-to-market mechanism, what happens in each trader's margin account each day, when a margin call is triggered, and why this daily cash-flow pattern differs from the economics of an otherwise-identical forward contract that settles only at maturity.

FF-C7. A jet-fuel-consuming airline cannot find a jet-fuel futures contract with enough liquidity, so it hedges using crude-oil futures whose delivery date is one month after it will actually buy fuel. Explain what basis risk is in this setting and identify the two distinct sources of basis risk in the airline's hedge (the mismatch in asset and the mismatch in timing). Why does basis risk mean the hedge is imperfect even though the position is closed out?

FF-C8. A farmer hedging a June wheat delivery uses the July futures contract. Explain, using the definition $Basis(T) = S(T) - F_\tau(T)$, why the basis is generally not zero at the time the farmer closes the position, but why it is nonetheless smaller and more predictable than the outright price risk the farmer would face with no hedge at all.

FF-C9. Consider a commodity for which producers (natural sellers) overwhelmingly dominate the demand for hedging, so they crowd the short side of the futures market. Using the theory of normal backwardation, explain why speculators must be offered a futures price below the expected future spot price ($ES_T > F$) to induce them to take the long side, and what this price discount represents economically.

FF-C10. Contrast the expectations hypothesis ($ES_T = F$) with contango. Describe a market — in terms of which side (producers or consumers) dominates the natural demand for hedging — in which contango arises, and explain why in that case the short side of the market must be rewarded so that $ES_T < F$.

FF-C11. Under the modern portfolio theory view, the relationship $F = e^{(r-k)T} ES_T$ ties the gap between the futures price and the expected spot price to the discount rate k required to hold the asset. Explain what happens to the futures price relative to the expected spot price for a high-β (positive systematic risk) commodity, and contrast this with what the pure expectations hypothesis ($ES_T = F$) would predict. Why are the two views inconsistent for a positive-β asset?

FF-C12. Derive the modern-portfolio-theory relation $F = e^{(r-k)T} ES_T$ from the two pricing statements $S = e^{-kT} ES_T$ and $S = e^{-rT} F$. Explain the economic content of each of the two starting equations, and interpret why a negative-β asset (with $k < r$) ends up with a futures price *above* its expected spot price.

12.8.2 Quantitative

FF-Q1. The spot price of one ounce of silver is $S = \$30$ and the annual simple risk-free interest rate is $r = 0.04$. Using the cost-of-carry relationship $F = (1 + r)S$, compute the fair one-year forward price F per ounce. Then suppose the one-year forward is instead quoted in the market at $F = \$32$. Describe the arbitrage trade and compute the risk-free profit per ounce.

FF-Q2. A stock index has spot value $S = \$2{,}000$ and the continuously compounded risk-free rate is $r = 0.05$. Using $F = Se^{rt}$, find the fair forward price for delivery in $t = 0.5$ years. If the actual six-month forward price is \$2,080, is the forward overpriced or underpriced relative to cost-of-carry, and by how much?

FF-Q3. Gold has spot price $S = \$1{,}800$ per ounce and the continuously compounded risk-free rate is $r = 0.05$. Using $F = Se^{rt}$, compute the fair six-month ($t = 0.5$) forward price. Separately, a currency has spot rate $S = \$1.4208$ and a three-month ($T = 0.25$) forward rate $F = \$1.4187$; using the inverted relationship $r = \frac{1}{T} \ln(F/S)$, compute the implied continuously compounded interest rate and state its sign, explaining why $F < S$ produces that sign.

FF-Q4. A trader is long one forward contract on one barrel of oil with delivery price $K = \$59$. Compute the payoff $S_T - K$ to the long position if the spot price at maturity is (a) $S_T = \$56$ and (b) $S_T = \$63$. Then compute the corresponding payoff $K - S_T$ to the short position in each case, and confirm that the long and short payoffs sum to zero.

FF-Q5. An oil producer will sell 1,000,000 barrels at the 15 August spot price and hedges by going short 1,000 NYMEX crude contracts (1,000 barrels each) at a futures price of \$59/barrel. Suppose the 15 August spot price turns out to be S_T = \$52/barrel and the futures price converges to the spot price. Compute (a) the revenue from selling the physical oil, (b) the total gain or loss on the short futures position using the per-barrel payoff $K - S_T$, and (c) the combined net proceeds, confirming the hedge locks in approximately \$59/barrel.

FF-Q6. A farmer will deliver wheat at time T and hedges with a futures contract for a later delivery date τ. At time T the spot price is $S(T)$ = \$6.20 per bushel and the futures price for τ delivery is $F_\tau(T)$ = \$6.35 per bushel. Using $Basis(T) = S(T) - F_\tau(T)$, compute the basis. State whether the basis is positive or negative and, if the farmer is short the futures to hedge a sale, whether this basis movement helped or hurt relative to a zero basis.

FF-Q7. On 13 March the spot price of corn is $S(T) = 385$ cents per bushel, while the futures price for the same-month delivery, one day before the contract stops trading, is $F_\tau(T) = 383$ cents per bushel. Compute $Basis(T)$ in cents. Explain, given that convergence forces basis toward zero at expiration, the arbitrage a trader could attempt to exploit this 2-cent gap, and compute the gross profit per 5,000-bushel contract.

FF-Q8. A commodity has expected spot price at maturity ES_T = \$100. The continuously compounded risk-free rate is $r = 0.03$, the discount rate required to hold the asset is $k = 0.08$, and the horizon is $T = 1$ year. Using the modern portfolio theory relationship $F = e^{(r-k)T} ES_T$, compute the futures price F. Is F above or below ES_T, and is this consistent with normal backwardation or contango?

FF-Q9. Suppose instead that a commodity behaves like a negative-β asset, so the required discount rate is $k = 0.01$ while $r = 0.04$, with ES_T = \$50 and $T = 2$ years. Using $F = e^{(r-k)T} ES_T$, compute the futures price F. State whether F exceeds or falls short of ES_T and which of the three classical hypotheses this outcome matches.

FF-Q10. A March corn futures contract (5,000 bushels) is currently valued at

\$19,409.375, i.e. a futures price of $388\frac{6}{32}$ cents per bushel. A hedger is long one contract with \$2,000 in a margin account and a maintenance margin of \$1,500. Suppose the next day the futures price rises to 390 cents per bushel. Compute (a) the new contract value, (b) the daily marking-to-market gain credited to the long margin account, and (c) the new margin balance. Then compute (d) the futures price (in cents per bushel) at which the account would fall to the \$1,500 maintenance level and trigger a margin call.

Chapter 13

Options

13.1 Introduction

Four questions define the study of options. First, what factors determine the price of an option, and what qualitative predictions can be made before any formal model is specified? Second, how should an option be priced — that is, what is the fair value of the right, without obligation, to buy or sell an asset at a fixed price on or before a future date? Third, how can the relative prices of puts and calls on the same underlying asset be constrained by no-arbitrage, and what does that constraint imply about early exercise? Fourth, how can options be combined with one another to engineer specific payoff profiles, and how are the resulting positions managed dynamically? These questions matter profoundly for financial markets. Options allow investors and corporations to hedge specific risks with surgical precision, to speculate on volatility independently of the direction of asset prices, and to tailor their exposure to future outcomes in ways that cannot be achieved with forwards, futures, or the underlying asset alone. The solution to the pricing question, provided by Black, Scholes, and Merton in the early 1970s, effectively founded the modern era of quantitative finance.

The first question establishes a qualitative framework before any formula is derived. A call option increases in value when the underlying stock price rises (more likely to expire in the money), when the strike price falls (a lower hurdle), when time to expiration increases (more opportunity for the stock to move favorably), when volatility rises (greater dispersion of outcomes, and only the favorable tail matters to the option holder), and when the risk-free rate rises (reducing the present value of the fixed payment owed upon exercise). These five factors — stock price, strike, time, volatility, and rate — are the inputs to every option pricing model and understanding their effects qualitatively provides a vital intuition check on any formula.

The second question is answered by the replication argument. If the payoff of an option can be reproduced exactly by a portfolio of the underlying stock and a risk-free bond, then no-arbitrage forces the option price to equal the cost of that portfolio. The binomial model makes this argument transparent in a discrete-time, two-state setting: the hedge ratio — the number of shares needed to offset the option's risk — can be computed directly from the range of possible payoffs and prices, and the resulting option price requires no assumption about investor risk preferences or the probability of an up move. The Black-Scholes formula extends this replication logic to a continuous-time setting with lognormal stock prices, yielding a closed-form expression for the call price and revealing that implied volatility — the only unobservable input — is what option markets are ultimately pricing.

The third question leads to put-call parity: a constraint relating the prices of European puts and calls with the same strike and expiration. The derivation is a straightforward portfolio comparison, but its implications are far-reaching. It immediately implies that an American call on a non-dividend-paying stock should never be exercised early — selling the option is always worth at least as much as exercising it — so American and European calls on such stocks have the same price. For puts, this logic fails, and early exercise can be optimal.

The fourth question concerns the vast space of strategies that options make possible. Even a small number of options can be combined into bull spreads, bear spreads, butterfly spreads, straddles, and strangles to express views on direction, on the magnitude of price movement, or on the range within which prices will stay. Dynamic management of these positions requires understanding the Greeks — delta, gamma, theta, vega, and rho — which measure the option's sensitivity to each of its inputs and are the primary tools of risk management for any options portfolio.

The chapter begins by cataloguing the factors affecting option prices and deriving put-call parity with its early-exercise implications. A survey of option strategies follows. It then introduces the binomial model to build intuition for replication and hedge ratios, before presenting the Black-Scholes formula and working through its interpretation. The chapter concludes with the Greeks, deriving delta, gamma, vega, and rho from the Black-Scholes formula and explaining delta hedging and why higher-order sensitivities matter for managing options positions over time.

> **ℹ In the news**
>
> The options market this chapter prices has exploded in size and speed. IFR reported that zero-days-to-expiry ("0DTE") contracts have become the *dominant* force in the S&P 500 options market — now around half of all index-options volume, much of it traded by individuals. Every one of those contracts is priced by the same Black–Scholes logic and hedged using the Greeks introduced here. Read it at IFR.

13.2 Option contracts

An options contract is an agreement that gives the long position the right, but not the obligation to purchase or sell a particular asset by a certain date in the future for a particular price. The price at which the option may be bought or sold is called the *strike price* or *exercise price* and the date specified in the contract is the *maturity date* or *expiration date*. Options can be either *European* which means that they must be exercised on a given day, or *American* which means that they may be exercised at any point before the expiration date.

Call options. A call option gives the long position the right to buy an underlying asset for the strike price. The seller or short position on the option is said to "write" the call. If the price of the call option is c, the price of the stock at expiration is S_T and the strike price is K then the total payoff to the option is $\max\{S_T - K, 0\} - c$. The payoff to the short position will be $c - \max\{S_T - K, 0\}$. (Draw)

Put options. Put options give the long position the right to sell a particular asset at a given strike price. If the price of the put is p then the payoff to the put will be $\max\{K - S_T, 0\} - p$. The payoff to the short position will be $p - \max\{K - S_T, 0\}$. (Draw)

Options can be written on many underlying assets including: stocks, market indices, futures contracts, interest rate derivatives, etc. For most of our discussion we will consider options whose underlying asset is a stock. For equity options the contract is usually for 100 shares of the underlying stock. Thus, if the price of a call option on stock XYZ is \$5.00 per share then the initial investment for one options contract is \$500.

13.3 Stock Options

There are a few factors that affect the price of equity options. These include

1. The current stock price S_0. Holding all other factors constant (especially the strike price of the option) an increase in S_0 will increase the price of any particular call option and decrease the price of a put option.
2. The strike price K. Holding all else constant, increasing K will lead calls to be less likely to be in the money and cause puts to be more likely to be in the money. Thus the price of a call decreases in K while the price of the put increases in K.
3. Time to expiration. For American options, both put and call options weakly increase in value with the time to expiration. This happens because they each have more chance to be in the money as the expiration date increases. With European options it is not so simple.
4. Volatility. Both call and put options (American and European) increase

in the volatility of the underlying stock. As the stock gets more volatile, it has more likelihood of being in the money.

5. Risk-free rate. Holding the stock price constant, an increase in the risk-free rate will increase the value of calls and decrease the value of puts.

13.4 Put-Call Parity

There is a relationship between the price p of a European put and the price c of a European call. To see this, consider the following two portfolios.

1. Holding one European call option with strike price K and cash in the amount of Ke^{-rT}.
2. Hold one share of the stock and one European put option with strike price K.

Consider the payoff to portfolio 1. If $S_T < K$ then the total value of the portfolio is K (the value of the cash at time T). If $S_T > K$ then the value of the portfolio is $S_T - K + K = S_T$. As such, the value of portfolio 1 is $\max(S_T, K)$.

Now consider the value of portfolio 2. If $S_T < K$ then the payoff to the put is $K - S_T$ and the value of the stock is S_T so the total value of the portfolio is K. On the other hand, if $S_T > K$ then the value of the put is zero and the value of the stock is S_T. Thus the value of portfolio 2 is also $\max(S_T, K)$ and the payoffs to each portfolio are the same. Since their payoffs are the same, they must have the same price. This implies that

$$c + Ke^{-rT} = p + S_0.$$

That is, the price of a call plus the cash position must be equal to the price of the put and the current stock price.

Notice first that $c \geq 0$ and $p \geq 0$ since puts and calls may always be left unexercised and as such will not impose any liability on the holder. Rearranging the put-call parity relationship then shows that

$$c = p + S_0 - Ke^{-rT}$$

If we call C the price of an American call we note that $C \geq c$. Since $p \geq 0$ this implies that

$$C \geq c \geq S_0 - Ke^{-rT} > S_0 - K$$

Notice that the right hand side of this inequality is the value that would be received if the American option were to be exercised today. The left hand side is the value that would be obtained if the American call were sold. Since the left hand side is always larger than the right hand side, we see that it is never optimal to exercise an American call option. If a person wants to close out her position, she should just sell the option.

Since it is never optimal to exercise an American call early, the early exercise option must have no value. Hence it should be that $C = c$. For put options this is not the case. If a firm on which you hold a put goes bankrupt, there is no possibility that the stock price will fall any lower (it is already 0) so the option should be exercised so that the money can be invested somewhere else. Hence $P > p$ (the price of an American put is higher than a European put).

13.5 Option Strategies

Options provide an investor a wide range of opportunities for taking a view on stock prices. What follows are some possible investment strategies using options.

Spread. A spread involves taking a position in 2 or more options of the same type.

Bull spread. One way to construct a bull spread is to buy a call option with a strike price K_1 and selling a call option with a strike price $K_2 > K_1$. The payoff to the first option is $\max(S_T - K_1, 0) - c_1$ while the payoff to the second is $c_2 - \max(S_T - K_2, 0)$. Therefore, the payoff to the total position is $\max(S_T - K_1, 0) - \max(S_T - K_2, 0) - c_1 + c_2$. Notice that since $K_2 > K_1$, $c_2 < c_1$.

Bear spread. A bear spread is a bet on stock prices decreasing. One way to construct one is by buying a put with one strike price and writing a put with a lower strike price. In this case the payoff to the option that is purchased is $\max(K_1 - S_T, 0) - p_1$ while the payoff to the option that is written is $p_2 - \max(K_2 - S_T, 0)$ so the total payoff is $p_2 - p_1 + \max(K_1 - S_T, 0) - \max(K_2 - S_T, 0)$.

Butterfly spread. A butterfly spread allows the holder to bet that prices won't vary much. It can be constructed by buying two call options with differing strike prices and selling a call option with a strike price in between these two.

Calendar spread.

Straddle. Allows the investor to bet on prices having significant movement. Implemented by buying a put and a call with the same strike price.

Strangle. Similar to a straddle except the call has a strike price that is higher than the put. Thus, prices must move even more than under a straddle for the strangle to be profitable.

These strategies are not textbook artifacts; every one of them can be priced from a public quote screen in seconds. Figure 13.1 draws the profit diagrams for a straddle and a bull spread on SPY, the S&P 500 ETF, using actual quoted premiums. The prices themselves are instructive. The at-the-money straddle costs about 3.6% of the index level, which means the market is charging exactly that much for the right to profit from a large move in *either* direction over roughly five weeks — buy it, and the index must move more than 3.6% for you to come out ahead. The bull spread shows the other signature property of option strategies: by giving up the upside beyond the higher strike, the investor cuts the cost of the position by roughly a dollar for every dollar of maximum payoff surrendered, capping both the loss and the gain at amounts known in advance.

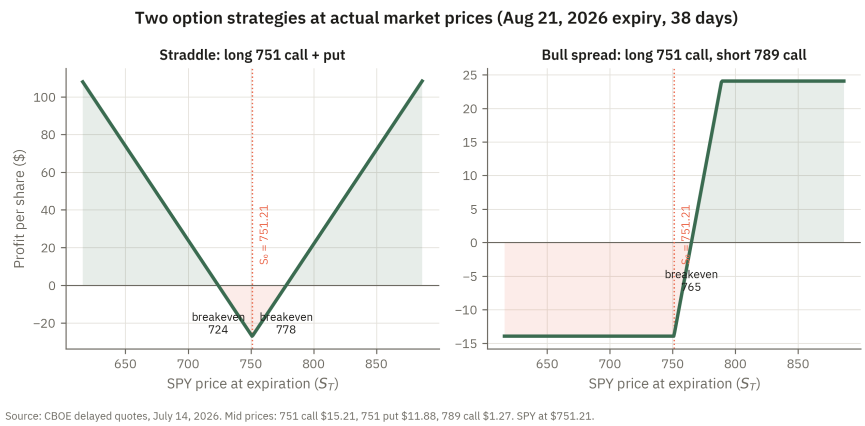

Figure 13.1: Profit at expiration for a straddle and a bull spread on SPY, constructed from quoted mid prices (July 2026, expiry about five weeks out). Breakeven prices and the current index level are marked. Source: CBOE delayed quotes.

Interestingly, it can be shown that if there were options at every possible strike price then any possible payoff profile could be constructed.

13.6 Binomial Option Pricing Models

The binomial model is a somewhat flexible model for pricing options contracts. As an introduction, consider a stock that currently sells for \$50. In one month, there is a 50% chance that the price of the stock will have increased by 10% to \$55 and a 50% chance that the price will have decreased by 10% to \$45. That is, there is a 50% chance that the price in 1 month will be $u50$ and a 50% chance that in 1 month the price will be $d50$ where $u = 1.1$ and $d = 0.9$. Consider a call option that expires in 1 month and has a strike price of \$52. Suppose that the monthly interest rate is 0.01 (1%). If one holds the option in one month, the payoff will either be 0 (if the price is \$45) or \$3 if the price is \$55. How much should one be willing to pay to purchase the call option? Let's use C to represent the price of the call option today. So purchasing the call option today implies paying C today and in one month getting \$3.000 if the price is \$55 and zero otherwise. Another way to get the same payoff in one month is to borrow $\frac{45}{1.01} = 44.554$ today at 1% interest to be paid back in one month and to purchase one share of the stock. Why exactly $\frac{45}{1.01}$? The loan is chosen to be the present value of the *down-state* stock price. In one month the loan requires repayment of $44.554 \times 1.01 = 45$, so in the down state the \$45 from selling the share exactly repays the loan and the levered position is worth $45 - 45 = 0$ — matching the option, which also pays 0 there. Setting the loan to the present value of the down payoff is precisely what zeroes out the down state and leaves a payoff only in the up state. The total cash outlay for doing this is 50 - 44.554 = 5.446. Doing so means that if the price in one month is \$45, then one may sell the share of stock to pay back the loan and have a payoff of 0. If the price is \$55 then the total payoff is \$55 - \$45 = \$10. This payoff is $\frac{10}{3}$ the payoff of owning the option in every state (price going up or price going down). That is, owning 3.33 options contracts will give you exactly the payoff that this portfolio gives you. Holding the one share of stock and taking out the loan costs 5.446, so the price of the options contract should satisfy

$$3.33C = 5.446$$

which implies that $C = 1.635$.

Another way to look at this is to note that the payoff to writing 3.33 calls and one share of stock is perfectly hedged. That is, its payoff doesn't depend on the realization of the stock price. If the price of the stock is \$55 then the stock pays \$55 and the obligation from the calls is $\$3 \cdot 3.33 = \$10$, so the total payoff is \$45. If the price is \$45 then the obligation from the options is 0 and the stock is worth \$45. So writing 3.33 options and owning the share of stock offers a riskfree payoff. Thus the current price of such a payoff should be $\frac{45}{1.01} = 44.554$. This is the same as buying the stock (\$50) and gaining the income from selling the puts ($3.33 \cdot 1.635 = 5.446$) which gives a total cost of $50 - 5.446 = 44.554$.

Deriving this price depends on the ability to buy stock and borrow cash in such a way that perfectly offsets the payoff to the option. The number of shares of stock per option needed to replicate the option portfolio is called the hedge ratio. The hedge ratio for this example is $\frac{1}{3.33...} = .3$. Notice that in the binomial model the hedge ratio is the ratio of the range of possible option payoffs to the range of possible stock prices. This is $\frac{3-0}{55-45} = 0.3$.

13.7 Black-Scholes Option Pricing

The Black-Scholes formula gives an explicit formula for the price of an option under very specific assumptions. These assumptions include

1. The stock pays no dividends.
2. The price path of the stock moves continuously and the joint distribution between prices at any two points in time is normally distributed, with constant mean and variance.
3. Trading can be conducted continuously.

Under these assumptions the price of a European call option is given by the formula below. (It is derived by extending the binomial replication argument above to continuous time: a continuously rebalanced portfolio of stock and cash replicates the option, and requiring that this self-financing hedge earn the risk-free rate leads to the Black-Scholes partial differential equation, whose solution is the formula quoted here.)

$$\begin{aligned} C_0 &= S_0 N(d_1) - Ke^{-rT} N(d_2) \\ d_1 &= \frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} \\ d_2 &= d_1 - \sigma\sqrt{T} \end{aligned}$$

where C_0 is the current call option value, S_0 is the current stock price, $N(d)$ is the probability that a random draw from a standard normal distribution will be less than d. K is the exercise price, r is the risk free rate, T is the time to expiration and σ is the standard deviation of the stock.

This formula makes intuitive sense. If both of the N terms are close to one, then payoff is nearly certain so the value of the call is $S_0 - Ke^{-rT}$ which is exactly what it should be given that you have a nearly certain obligation to buy the stock at K and the current value of that stock is S_0. If both of the N terms are really low (close to zero) then the value of the call option will be worth nothing.

Example. Consider a stock that has current price $S_0 = \$50$, with a one-month risk-free rate of 0.01 and a monthly standard deviation of $\sigma = 0.08$ (time is measured in months, so $T = 1$). What is the price of an option with strike price \$53? Plugging into the Black-Scholes formula we get that

$$\begin{aligned}
d_1 &= \frac{\ln\left(\frac{50}{53}\right) + \left(0.01 + \frac{0.08^2}{2}\right)1}{0.08\sqrt{1}} = -0.5634 \\
d_2 &= d_1 - 0.08 = -0.6434 \\
N(d_1) &= 0.28659 \\
N(d_2) &= 0.25999 \\
C &= 50(0.28659) - 53e^{-0.01}(0.25999) = \$0.687
\end{aligned}$$

Now suppose that the strike price is 50, so the option is at the money. Then the price of the call would be

$$\begin{aligned}
d_1 &= \frac{\ln\left(\frac{50}{50}\right) + \left(0.01 + \frac{0.08^2}{2}\right)1}{0.08\sqrt{1}} = 0.165 \\
d_2 &= d_1 - 0.08 = 0.085 \\
N(d_1) &= 0.56553 \\
N(d_2) &= 0.53387 \\
C &= 50 \cdot 0.56553 - 50e^{-0.01}0.53387 = \$1.8485
\end{aligned}$$

Moving the strike just 6% out of the money cuts the call's value by nearly two-thirds — an option's value is concentrated near the money.

Puts can be priced by using Black-Scholes and put-call parity. Notice that the only unobservable quantity in the formula is volatility. Hence, by observing option prices one may backout the implied volatility of the stock. This is the principle behind the CBOE Vix index (volatility index.)

13.8 First-order analytics: The Greeks

The partial derivative of the Black-Scholes formula with respect to particular variables is of practical and theoretical interest. These derivatives are often referred to by greek letters and are often collectively called "the Greeks."

Δ The delta (Δ) of an option is the change in the price of an option that occurs from an underlying change in the price of the asset on which the option is based. Thus, for a call option on a stock, the Δ is defined as

$$\Delta = \frac{\partial c}{\partial S}$$

For a call option that meets the assumptions of the Black-Scholes formula, this derivative is

$$\Delta = N(d_1)$$

For a put that meets the Black-Scholes formula assumptions this is

$$\Delta = N(d_1) - 1$$

This is not obvious, since both d_1 and d_2 are functions of S. Differentiating $C_0 = S_0 N(d_1) - Ke^{-rT} N(d_2)$ with the product and chain rules, and using $\frac{\partial d_1}{\partial S} = \frac{\partial d_2}{\partial S} = \frac{1}{S\sigma\sqrt{T}}$ (they differ only by the constant $\sigma\sqrt{T}$), gives

$$\Delta = N(d_1) + \frac{1}{S\sigma\sqrt{T}}\left[SN'(d_1) - Ke^{-rT}N'(d_2) \right].$$

The bracketed term vanishes because of the identity

$$SN'(d_1) = Ke^{-rT}N'(d_2),$$

which follows from $N'(d) = \frac{1}{\sqrt{2\pi}}e^{-d^2/2}$ and the fact that $\frac{1}{2}(d_1^2 - d_2^2) = \ln(S/K) + rT$, so that $N'(d_1)/N'(d_2) = (K/S)e^{-rT}$. The two product-rule terms therefore cancel and $\Delta = N(d_1)$. This same identity $SN'(d_1) = Ke^{-rT}N'(d_2)$ is what makes the other Greeks simplify below.

Δ gives you the number of call options necessary to hedge a long position in a particular stock (at least to a first order approximation). Notice that if a call option is deeply in the money then $\Delta \approx 1$. If it is deeply out of the money then $\Delta \approx 0$.

Example (Δ hedging). Consider an investor that has purchased 10 calls on the S&P 500. Each contract is for 100 shares of an S&P ETF. Suppose that the current price of the ETFs is \$50 and that $\Delta = 0.4$. Suppose that the price of the call is \$5.00. The investor could sell short $0.4 \cdot 1000 = 400$ shares of the ETF to hedge this call position. To see this, note that if the price of the stock decreases by \$1.00 (i.e. the new price is \$49), then the value of the stock position is now $400 \cdot 1 = 400$. On the other hand, the calls that are held have gone down in price by the amount $0.4 \cdot 1.00$ (= Δ times the dollar decrease in the price of the stock) so the price has now become \$4.60 per share which implies that the value of the total position is $4.60 \cdot 1000 = 4,600$. Thus, the changes in the stock portfolio offset the changes in the value of the call.

Notice that the Δ of the option changes as the stock price changes. That is, once the price moves, in order to remain Δ-neutral the stock position must be changed.

Θ. Θ is the rate of change of the value of the portfolio with respect to the passage of time with all else remaining constant. This is often quoted in days.

Γ. Γ is the rate of change of Δ with respect to the stock price. Hence it is the second derivative of the option price with respect to the stock price. Since $\Delta = N(d_1)$, differentiate once more: $\Gamma = \frac{\partial}{\partial S}N(d_1) = N'(d_1)\frac{\partial d_1}{\partial S}$, and with $\frac{\partial d_1}{\partial S} = \frac{1}{S\sigma\sqrt{T}}$ this gives

$$\Gamma = \frac{N'(d_1)}{S_0\sigma\sqrt{T}}$$

$\mathcal{V}$ **(vega).** $\mathcal{V}$ is the derivative of the value of an option with respect to volatility. Differentiating $C_0 = S_0N(d_1) - Ke^{-rT}N(d_2)$ in σ gives $\mathcal{V} = S_0N'(d_1)\frac{\partial d_1}{\partial \sigma} - Ke^{-rT}N'(d_2)\frac{\partial d_2}{\partial \sigma}$. Applying the identity $S_0N'(d_1) = Ke^{-rT}N'(d_2)$ collapses this to $S_0N'(d_1)\left(\frac{\partial d_1}{\partial \sigma} - \frac{\partial d_2}{\partial \sigma}\right)$, and since $d_1 - d_2 = \sigma\sqrt{T}$ has derivative $\sqrt{T}$, it is given by (for the Black-Scholes model).

$$\mathcal{V} = S_0\sqrt{T}N'(d_1)$$

ρ**.** The derivative of the value of the option with respect to changes in the interest rate. Differentiating in r produces three terms: $S_0N'(d_1)\frac{\partial d_1}{\partial r}$ from the first piece, and $+TKe^{-rT}N(d_2) - Ke^{-rT}N'(d_2)\frac{\partial d_2}{\partial r}$ from the second (the $+TKe^{-rT}N(d_2)$ coming from differentiating the discount factor e^{-rT}). Because $\frac{\partial d_1}{\partial r} = \frac{\partial d_2}{\partial r} = \frac{\sqrt{T}}{\sigma}$ and $S_0N'(d_1) = Ke^{-rT}N'(d_2)$, the two N' terms cancel, leaving only the discount-factor term. For Black-Scholes calls this is

$$\rho = KTe^{-rT}N(d_2)$$

for puts this is

$$\rho = -KTe^{-rT}N(-d_2)$$

13.9 Application: A manager buying protection

A portfolio manager running a technology-heavy fund worries about a sharp drop through earnings season but does not want to sell her positions and incur taxes and transaction costs. Buying put options on a market index lets her insure the portfolio instead, and the pricing tools of this chapter tell her what that insurance costs and how it will behave. The Black–Scholes formula translates the index level, the strike, the time to expiry, and — crucially — volatility into a price, while the Greeks tell her how the value of her hedge will respond as the market moves, as time passes, and as volatility itself changes. With volatility historically low, she can see that protection is inexpensive; understanding why, and how the hedge's delta and gamma will evolve, is what lets her buy the right amount rather than guessing.

> Further listening
>
> *Odd Lots* (Bloomberg) — "What the Dramatic Boom in Zero-Day Options Means for Stocks": how same-day options work, why volume has exploded, and how dealer hedging can move the market.

13.10 Homework problems

13.10.1 Conceptual

OPT-C1. Two traders each commit \$4 to a bet on stock XYZ, currently trading at $S_0 = \$40$. One buys a call with strike $K = \$42$ for a \$4 premium; the other uses the \$4 as margin to control 100 shares outright. The stock then falls to \$30 at expiration. Using the distinction between a *right* and an *obligation*, explain why the call holder's loss is capped at the \$4 premium while the leveraged stock holder can lose much more. What feature of the payoff $\max\{S_T - K, 0\} - c$ produces this floor?

OPT-C2. Explain why a call and a put with the same strike behave as mirror images with respect to the underlying price and the strike: an increase in

S_0 raises the call but lowers the put, and an increase in K lowers the call but raises the put. Then explain why volatility is the one factor that raises *both*. Frame each answer around the chapter's language of an option becoming "more likely to be in the money."

OPT-C3. A colleague claims that a call and a put on the same stock must respond to the risk-free rate r in the same direction, "since r is just a discount rate." Using the roles r plays in the payoffs — the present value of the fixed strike owed on exercise — explain why an increase in r *raises* a call but *lowers* a put. Connect your answer to the $-Ke^{-rT}$ term that appears in both put-call parity and the Black-Scholes call formula.

OPT-C4. A European call and a European put on the same non-dividend-paying stock share strike K and expiration T. The call trades at \$6, the put at \$1, the stock at $S_0 = \$50$, and $Ke^{-rT} = \$46$. Show that the two put-call-parity portfolios are *not* currently equal in price, and explain in words the riskless trade an arbitrageur would put on: which portfolio she buys, which she sells, and why the position locks in a profit with no exposure at T.

OPT-C5. Using put-call parity, explain why it is never optimal to exercise an American call on a non-dividend-paying stock early, but why early exercise *can* be optimal for an American put. Frame your answer around the inequality $C \geq c \geq S_0 - Ke^{-rT} > S_0 - K$ (comparing the value of selling the option to the value of exercising it) and the bankruptcy example from the chapter.

OPT-C6. Explain the economic meaning of put-call parity, $c + Ke^{-rT} = p + S_0$, as a statement that two *different* portfolios deliver the *same* payoff $\max(S_T, K)$ at expiration. Describe the payoff of each of the two chapter portfolios in the states $S_T < K$ and $S_T > K$, and explain why identical payoffs force identical prices today — the argument that makes parity a no-arbitrage identity rather than an empirical approximation.

OPT-C7. In the binomial example, the chapter replicates a call by borrowing cash and buying stock, then notes that "writing 3.33 calls and one share of stock is perfectly hedged." Explain what "perfectly hedged" means — why the combined position pays the same amount in both the up and the down state — and explain why this forces the option to a single price that does *not* depend on the 50% probability of an up move.

OPT-C8. Two stocks share the same current price, strike, rate, and time to expiration, but stock A can move to \$55 or \$45 next month while stock B can move to \$60 or \$40. Using the chapter's definition of the hedge ratio as "the ratio of the range of possible option payoffs to the range of possible stock prices," explain how the wider spread of stock B changes the replicating portfolio and the option's value, and connect this to the general claim that more volatile stocks have more valuable options.

OPT-C9. The chapter says that in Black-Scholes "the only unobservable quantity in the formula is volatility." Explain what *implied volatility* is, how it is extracted from an observed market price of a call, and why this is the principle behind the CBOE VIX index. What would it mean if two options on the same stock, with different strikes, implied different volatilities?

OPT-C10. Interpret the two limiting cases in the chapter: when both $N(d_1)$ and $N(d_2)$ are close to 1, the call is worth about $S_0 - Ke^{-rT}$, and when both are close to 0 the call is worth nearly nothing. Explain, in economic terms, why each limit is the value it should be, referring to the near-certain (or near-impossible) obligation to buy the stock at K and to the interpretation of $N(d_2)$ as an exercise probability.

OPT-C11. A trader believes a biotech stock will make a large move on an upcoming FDA decision but has no view on the direction. Explain why a straddle (buy a put and a call at the same strike) expresses this view, and explain how a strangle differs and why it requires an even larger price move to be profitable. What is the trader implicitly buying a view on, and how does this contrast with a directional bet like a bull spread?

OPT-C12. Compare a bull spread (buy a call at K_1, write a call at $K_2 > K_1$) to simply buying the call at K_1 alone. Explain what the investor gives up and what she gains by adding the written call, referencing the chapter's observation that $c_2 < c_1$. Why might an investor with a *moderately* bullish view prefer the spread over the outright call?

OPT-C13. The delta of the option in the Δ-hedging example is 0.4, and the chapter notes that "once the price moves, in order to remain Δ-neutral the stock position must be changed." Explain why delta is not constant, name the Greek that measures how fast delta changes, and explain why an option that is deeply in the money has $\Delta \approx 1$ while one deeply out of the money has $\Delta \approx 0$.

OPT-C14. A market maker is short many short-dated (0DTE) call options and delta-hedges continuously. Explain what gamma ($\Gamma = N'(d_1)/(S\sigma\sqrt{T})$) and vega ($\mathcal{V} = S_0\sqrt{T}N'(d_1)$) each measure. Describe the tension the market maker faces: why large gamma forces frequent, costly rehedging as the underlying moves, and why a spike in volatility hurts her through vega even if the underlying price does not move.

13.10.2 Quantitative

OPT-Q1. A call has strike $K = \$45$ and premium $c = \$3$; a put on the same stock has strike $K = \$45$ and premium $p = \$2$. Using the payoffs $\max\{S_T - K, 0\} - c$ and $\max\{K - S_T, 0\} - p$, compute the payoff to the long call and the long put at each of $S_T = \$38$, $\$45$, and $\$52$. State the breakeven stock price for each position.

OPT-Q2. An investor forms a bull spread by buying a call at $K_1 = \$50$ for $c_1 = \$6$ and writing a call at $K_2 = \$55$ for $c_2 = \$3$. Write the total payoff $\max(S_T - K_1, 0) - \max(S_T - K_2, 0) - c_1 + c_2$ and evaluate it at $S_T = \$48$, 53, and $\$60$. What are the maximum profit and the maximum loss of the spread?

OPT-Q3. A European call and put share strike $K = \$100$ and expiration $T = 0.5$ years. The stock trades at $S_0 = \$98$, the continuously compounded rate is $r = 0.04$, and the call trades at $c = \$7.50$. Use put-call parity $c + Ke^{-rT} = p + S_0$ to solve for the fair price of the put p.

OPT-Q4. With $S_0 = \$60$, $K = \$62$, $r = 0.03$, $T = 1$, a call trades at $c = \$5$ and a put at $p = \$5.50$. Compute both sides of $c + Ke^{-rT}$ and $p + S_0$. Do they match? If not, state the size of the mispricing and which side is overpriced relative to parity.

OPT-Q5. A stock sells for \$50. In one month it will be worth $u \cdot 50$ or $d \cdot 50$

with $u = 1.1$ and $d = 0.9$. A call has strike $K = \$52$ and the monthly rate is $r = 0.01$. Using the chapter's method, (a) compute the option payoffs in the up and down states, (b) compute the hedge ratio as the range of option payoffs divided by the range of stock prices, and (c) find the call price C. Confirm your C against the chapter's answer.

OPT-Q6. Reprice the option in OPT-Q5 but with strike $K = \$48$. Give the payoffs in the up (\$55) and down (\$45) states, the hedge ratio, and the replicated call price C. (Hint: build the riskless "write n calls, hold one share" portfolio, discount its guaranteed payoff at 1%, and back out C.)

OPT-Q7. Use the Black-Scholes formula with $S_0 = \$100$, $K = \$100$, $r = 0.05$, $T = 0.5$ years, and $\sigma = 0.20$. Compute d_1 and d_2, look up $N(d_1) \approx 0.5977$ and $N(d_2) \approx 0.5422$, and compute $C_0 = S_0 N(d_1) - Ke^{-rT} N(d_2)$. (These are standard, self-consistent inputs with $\sigma\sqrt{T} \approx 0.141$.)

OPT-Q8. A stock has $S_0 = \$100$, $K = \$105$, $r = 0.05$, $T = 0.25$ years, and $\sigma = 0.30$. Compute d_1 and d_2, find $N(d_1) \approx 0.4337$ and $N(d_2) \approx 0.3756$, and compute the Black-Scholes call price C_0. Then use put-call parity to find the corresponding put price.

OPT-Q9. For the stock in OPT-Q8 ($S_0 = \$100$, $K = \$105$, $r = 0.05$, $T = 0.25$, $\sigma = 0.30$), you are given $N'(d_1) = \frac{1}{\sqrt{2\pi}} e^{-d_1^2/2} \approx 0.3934$. Compute the option's gamma $\Gamma = N'(d_1)/(S_0 \sigma \sqrt{T})$ and vega $\mathcal{V} = S_0 \sqrt{T}\, N'(d_1)$. Interpret vega: approximately how much does the call price change if σ rises by one percentage point (from 0.30 to 0.31)?

OPT-Q10. An investor holds 20 call contracts (each on 100 shares) on an ETF priced at \$50, with $\Delta = N(d_1) = 0.55$. (a) How many shares must be shorted to make the position delta-neutral? (b) If the ETF falls by \$1, approximate the change in the value of the call position and confirm it is offset by the short stock position. (c) If instead the calls were deep in the money with $\Delta \approx 1$, how would the required hedge change?

OPT-Q11. A stock has a Black-Scholes call delta of $\Delta = N(d_1) = 0.62$. (a) State the call delta and, using $\Delta_{\text{put}} = N(d_1) - 1$, the put delta. (b) If the stock rises by \$2, use these deltas to approximate the dollar change in the value of one long call and one long put. (c) Explain why the put delta is negative and why $\Delta_{\text{call}} - \Delta_{\text{put}} = 1$, connecting this to differentiating put-call parity in S_0.

OPT-Q12. In the Δ-hedging example an investor holds 10 calls (each on 100 shares) on an ETF priced at $50 with $\Delta = 0.4$ and call price $5.00. (a) How many shares must be shorted to delta-hedge? (b) If the ETF falls to $49, compute the new value of the short stock position and the new value of the call position (using the delta approximation), and verify the two changes offset. (c) Explain why this hedge must be adjusted after the price moves.

13.10.3 Data exercises

These problems use the live-data figure in this chapter.

OPT-D1. In Figure 13.1, SPY trades at $S_0 = 751.21$ and the five-week 751-strike straddle costs $c + p = 15.21 + 11.88 = \$27.09$. (a) Compute the two breakeven prices at expiration and express the required move as a percentage of S_0. (b) Compute the profit per share if SPY finishes at 700. (c) In one sentence, interpret the straddle's cost as the market's price of volatility over the five weeks.

OPT-D2. The bull spread in Figure 13.1 buys the 751 call at $15.21 and writes the 789 call at $1.27. (a) Compute the net cost, the maximum profit, and the breakeven price. (b) Compute the profit if SPY finishes at 800 and at 740. (c) The chapter's definition of a bull spread requires $c_2 < c_1$; explain economically why the 789 call must be cheaper than the 751 call.

Part V

Advanced market models

Chapter 14

The Black-Litterman model and active portfolio management

14.1 Introduction

Three questions motivate this chapter. First, why does standard mean-variance optimization so often produce extreme and unstable portfolio weights in practice, and what structural remedy does the Black-Litterman model offer? Second, how should an investor who holds both a market-derived prior over expected returns and her own subjective views about specific securities or portfolios combine those two sources of information in a formally coherent way? Third, what is the relationship between the investor's confidence in a view and the magnitude of the resulting deviation from the market portfolio, and how can that relationship be made precise and controllable? These questions sit at the intersection of Bayesian statistics, mean-variance optimization, and the practical challenge of translating investment research into disciplined portfolio positions.

The first question matters because the failure mode of unconstrained mean-variance optimization is well documented and severe. Small perturbations in estimated expected returns — well within the range of estimation error — can produce wildly different optimal weights, including large short positions and extreme concentrations that no sensible investor would hold. The root cause is that the optimizer treats point estimates of expected returns as if they were known with certainty, fully exploiting any differences between them. The Black-Litterman model corrects this by treating expected returns themselves as random variables with a prior distribution, rather than as fixed parameters. The natural choice of prior is the set of expected returns that, when plugged into the mean-variance first-order conditions, produce the observed market capitalization weights as the optimal portfolio. This "reverse-engineered" equilibrium prior anchors the optimization and prevents it from over-fitting to noisy return estimates.

The second question — about combining the market prior with subjective views — is where Bayesian statistics enters. Each view takes the form of a belief about the expected return on a specific portfolio of assets, expressed as a linear combination of individual asset returns. The investor also specifies her uncertainty about each view through a variance parameter. Given these inputs, the standard formula for the conditional distribution of a jointly normal random vector delivers a closed-form posterior distribution over expected returns that blends the equilibrium prior and the investor's views in proportion to their relative precision. Assets about which the investor has no views are unaffected by the updating, ensuring that the model's output is conservative wherever research coverage is absent.

The third question — about the connection between conviction and position size — is answered by examining how the posterior expected returns feed back into the mean-variance first-order conditions. Portfolio weights that result from the Black-Litterman posterior deviate from market weights in a direction and by an amount that is mathematically tied to the precision of the investor's views: a view held with high confidence (low variance) pulls the weights more strongly toward the implied position, while a low-confidence view (high variance) produces only a modest tilt. This feature gives portfolio managers an interpretable, principled dial for scaling positions to match their actual degree of conviction. The chapter develops the full machinery — conditional normal distributions, equilibrium return extraction, Bayesian updating, and portfolio construction — and then applies it to a universe of U.S. sector exchange-traded funds to illustrate how the model functions with real data.

> **In the news**
>
> Black–Litterman was built for investors who effectively own the whole market and must decide how far to lean away from it. Norway's sovereign wealth fund — the world's largest, at roughly $2 trillion — earned a record return in 2025, driven by its huge stakes in technology names like Nvidia, Apple, and Microsoft. A fund that big starts from the market portfolio and tilts only where it has a view; turning those views into portfolio weights is exactly what this chapter formalizes. Read it at CNBC.

14.2 Conditional Normal Distributions

Consider a random vector

$$\begin{pmatrix} x \\ y \end{pmatrix} \sim N\left(\begin{pmatrix} \mu \\ Q \end{pmatrix}, \begin{pmatrix} \Sigma_x & \Sigma_{xy} \\ \Sigma_{xy} & \Sigma_y \end{pmatrix}\right)$$

In this setting, the conditional distribution of x given y is

$$x \mid y \sim N\left(\mu + \Sigma_{xy}\Sigma_y^{-1}(y - Q), \Sigma_x - \Sigma_{xy}\Sigma_y^{-1}\Sigma_{xy}\right)$$

14.3 Preferences

We start with traditional preferences given by the utility function

$$U = E(r_p) - \frac{1}{2}A\sigma_p^2$$

where r_p is the return on the portfolio, $E(r_p)$ is the expected return, σ_p^2 is the variance of the return, and A is the risk aversion coefficient. Given portfolio weights θ, the optimal portfolio weights are given by the first order condition

$$\mu = A\Sigma\theta$$

Assume that there are n assets in the market.

14.4 Non-consensus beliefs about expected returns

Suppose that instead of having fixed beliefs about the distribution of returns, the investor believes that the return vector is distributed

$$r \sim N(\mu, \Sigma)$$

but that the expected returns of the assets are given by the random process $\mu = \Pi + \epsilon^e$, where ϵ^e is distributed normally with zero mean and covariance $\tau\Sigma$.

Imagine that the investor has collected some information about stocks i and j and believes that taking an equal long-short position in i and j would lead to a portfolio that has mean q and standard deviation ω. Define the vector $p = (\cdots 1 \cdots - 1 \cdots)$ to represent a position of 1 in stock i and -1 in stock j. The expected return of this portfolio is $p\mu = q + \epsilon^v$ where ϵ^v is a random variable that is normally distributed with mean 0 and variance ω^2. Let's assume that ϵ^v is independent of ϵ^e. We interpret this view as conveying information on the expected means of asset payoffs so we try to calculate the value of $E[r|p\mu = q + \epsilon^v]$. This setup implies that the distribution of μ conditional on this view is

$$\mu|q+\epsilon^v \sim N\left(\left[(\tau\Sigma)^{-1}+p'\frac{1}{\omega^2}p\right]^{-1}\left[(\tau\Sigma)^{-1}\Pi+p'\frac{1}{\omega^2}q\right],\left[(\tau\Sigma)^{-1}+p'\frac{1}{\omega^2}p\right]^{-1}\right).$$

See the appendix of Satchell and Scowcroft (2007)[1] for a derivation of this result.

More generally, if the investor has m views that can be represented by the $m \times n$ matrix P. The expected return of the portfolio is $P\mu = Q + \epsilon^v$ where ϵ^v is a random variable that is normally distributed with mean 0 and diagonal covariance matrix Ω. Let's assume that ϵ^v is independent of ϵ^e. The distribution of μ conditional on these views is

$$\mu|Q{+}\epsilon^v \sim N\left(\left[(\tau\Sigma)^{-1}+P'\Omega^{-1}P\right]^{-1}\left[(\tau\Sigma)^{-1}\Pi+P'\Omega^{-1}Q\right],\left[(\tau\Sigma)^{-1}+P'\Omega^{-1}P\right]^{-1}\right).$$

14.5 Applying the Black-Litterman model

Let's collect data on returns to several industry sectors in the U.S. market. To do this, we consider several sector exchange traded funds.

..

XLB

XLC

XLE

XLF

XLI

XLK

[1] Satchell, Stephen, and Alan Scowcroft. "A demystification of the Black-Litterman model: Managing quantitative and traditional portfolio construction." In Forecasting expected returns in the financial markets, pp. 39-53. Academic Press, 2007.

XLP

XLRE

XLU

XLV

XLY

count

192.000

66.000

192.000

192.000

192.000

192.000

192.000

98.000

192.000

192.000

192.000

mean

0.008

0.008

0.006

0.007

0.009

0.013

0.008

0.007

0.006

0.009

0.012

std

0.062

0.061

0.081

0.068

0.058

0.055

0.037

0.051

0.043

0.042

0.058

min

-0.224

-0.141

-0.346

-0.262

-0.187

-0.161

-0.126

-0.152

-0.130

-0.124

-0.176

25%

-0.029

-0.028

-0.035

-0.027

-0.022

-0.016

-0.014

-0.023

-0.019

-0.018

-0.018

50%

0.009

0.012

0.014

0.018

0.011

0.018

0.012

0.010

0.012

0.011

0.013

75%

0.043

0.043

0.046

0.048

0.042

0.048

0.032

0.037

0.036

0.038

0.046

max

0.174

0.148

0.308

0.218

0.181

0.137

0.104

0.125

0.105

0.126

0.189

The covariance matrix of the returns is

TICKER

XLB

XLC

XLE

XLF

XLI

XLK

XLP

XLRE

XLU

XLV

XLY

XLB

0.004

0.003

0.003

0.003

0.003

0.003

0.001

0.002

0.001

0.002

0.003

XLC

0.003

0.004

0.003

0.003

0.003

0.003

0.001

0.003

0.001

0.002

0.004

XLE

0.003

0.003

0.006

0.003

0.003

0.002

0.001

0.002

0.001

0.002

0.003

XLF

0.003

0.003

0.003

0.005

0.003

0.003

0.002

0.002

0.001

0.002

0.003

XLI

0.003

0.003

0.003

0.003

0.003

0.003

0.002

0.002

0.001

0.002

0.003

XLK

0.003

0.003

0.002

0.003

0.003

0.003

0.001

0.002

0.001

0.002

0.003

XLP

0.001

0.001

0.001

0.002

0.002

0.001

0.001

0.001

0.001

0.001

0.001

XLRE

0.002

0.003

0.002

0.002

0.002

0.002

0.001

0.003

0.002

0.001

0.002

XLU

0.001

0.001

0.001

0.001

0.001

0.001

0.001

0.002

0.002

0.001

0.001

XLV

0.002

0.002

0.002

0.002

0.002

0.002

0.001

0.001

0.001

0.002

0.002

XLY

0.003

0.004

0.003

0.003

0.003

0.003

0.001

0.002

0.001

0.002

0.003

A covariance matrix printed as a table is hard to read; the same information is far more legible as the correlation heatmap in Figure 14.1, computed from the most recent eight years of monthly returns on the eleven sector ETFs. Two features matter for what follows. Every entry is positive — there is no true hedge among equity sectors, because all of them load on the market factor — but the entries range widely, from 0.29 between energy and technology to above 0.85 among the industrial-financial-materials cluster. This matrix is the Σ that every formula in this chapter consumes.

With Σ in hand, the reverse-optimization step $\Pi = A\Sigma\theta_{mkt}$ needs only the market weights, and those can be read off the actual composition of the S&P 500: aggregating the current SPY holdings by sector gives technology a weight of about 37%, financials 12%, and utilities barely 2%. Figure 14.2 shows the result next to the raw historical mean returns of the same sectors, and the contrast is the whole argument for the Black-Litterman prior in one picture. The equilibrium returns Π line up smoothly between about 3% and 9%, ordered by covariance with the market portfolio, exactly as the first-order condition requires. The sample means, by contrast, are scattered from 9% to 24% with no visible economic logic — technology's enormous mean reflects one extraordinary decade, not an expectation anyone should carry forward. Feeding those means into an unconstrained optimizer produces the extreme portfolios described in the introduction; starting from Π produces the market portfolio, which is why equilibrium returns, not sample means, are the model's point of departure.

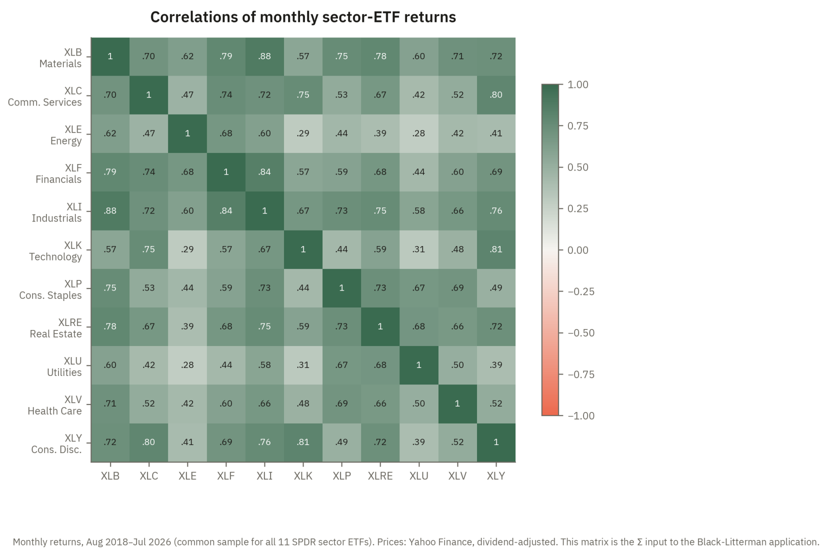

Figure 14.1: Correlations of monthly returns for the eleven SPDR sector ETFs over the most recent common sample. This is the Σ input to the Black-Litterman application, displayed as correlations. Source: Yahoo Finance.

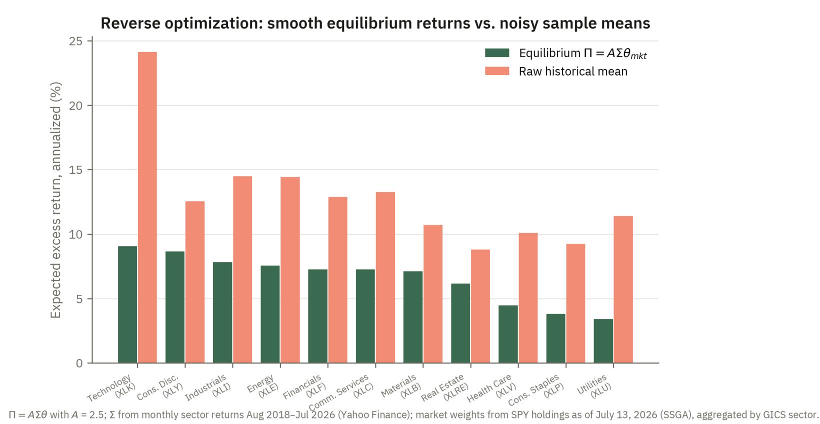

Figure 14.2: Reverse-optimized equilibrium expected returns $\Pi = A\Sigma\theta_{mkt}$ (with $A = 2.5$ and market weights from current SPY holdings aggregated by sector) next to raw historical mean returns. The equilibrium returns are smooth and economically ordered; the sample means are noise. Sources: Yahoo Finance; SSGA.

14.6 Application: A CIO sizing a view

The chief investment officer of a large pension or sovereign-wealth fund begins from an uncomfortable fact: simply owning the whole market means holding an enormous, concentrated bet on whatever the market happens to favor. The Black–Litterman framework is built for exactly her problem. It lets her start from the returns implied by market equilibrium — the portfolio she would hold if she had no special opinions — and then blend in her house views, such as a belief that one region or sector is overvalued, in proportion to how confident she is. The result is a set of portfolio tilts that are disciplined rather than arbitrary: large where conviction is high, small where it is weak, and always anchored to the market. For an investor managing hundreds of billions, this is the difference between expressing a view and accidentally betting the fund on it.

> Further listening
>
> *Odd Lots* (Bloomberg) — "Cullen Roche on the Art of Building a Perfect Portfolio": how to combine the market portfolio with your own views into a coherent allocation — the problem Black–Litterman formalizes.

14.7 Homework problems

14.7.1 Conceptual

BL-C1. An analyst runs an unconstrained mean-variance optimization on 11 sector ETFs using historical sample means as her estimates of μ. She reports that XLK receives a weight of +340% and XLE a weight of −210%. A colleague re-estimates the sample means using a data window that is only two months longer and now finds XLE receives +180% and XLK receives −90%. Using the chapter's discussion of the failure mode of unconstrained optimization, explain why such a small change in the input estimates can flip the signs and magnitudes of the optimal weights so dramatically. In your answer, refer to how the optimizer treats the point estimates of μ.

BL-C2. The chapter opens by asking why standard mean-variance optimization "so often produces extreme and unstable portfolio weights in practice." Explain the root cause the chapter identifies, and then explain the structural remedy Black-Litterman offers: how does treating expected returns as random variables with a prior distribution, rather than as fixed parameters, prevent the optimizer from over-fitting noisy return estimates?

BL-C3. The chapter builds its prior over expected returns by "reverse-engineering" the returns that make the observed market-cap weights optimal. Explain, in words, the logic of this reverse-optimization step: what equation is being solved, what is treated as known, and what is being backed out. Then explain why anchoring the prior to market weights is a sensible default choice for an investor who begins by effectively owning the whole market.

BL-C4. Two portfolio managers each build a mean-variance optimizer. Manager A feeds in raw historical sample means for μ. Manager B first computes the equilibrium prior $\Pi = A\Sigma\theta_{\text{mkt}}$ and uses that as her baseline, tilting away from it only where she has a view. Explain why Manager B's portfolios are likely to be more stable and more intuitively reasonable than Manager A's, even before either manager has expressed any subjective view. What structural feature of the reverse-optimization prior is doing the work here?

BL-C5. In the Black-Litterman setup the investor does not treat expected returns as fixed numbers; instead she writes $\mu = \Pi + \epsilon^e$ with $\epsilon^e \sim N(0, \tau\Sigma)$. Explain what economic content the parameter τ carries. What does it mean, in terms of the investor's confidence in the equilibrium prior, to choose a very small τ versus a large τ? Contrast this treatment of μ with the classical assumption that μ is a known constant.

BL-C6. Explain why the chapter scales the prior uncertainty of expected returns by $\tau\Sigma$ — that is, why the covariance of ϵ^e is taken to be proportional to the return covariance matrix Σ rather than, say, an unrelated diagonal matrix. What does this modeling choice say about how the investor believes uncertainty about mean returns is distributed across assets, and why is it convenient when the posterior formula is derived?

BL-C7. A manager holds a single view: that XLK will outperform XLE by some amount q. She has collected considerable research supporting this view and is highly confident. A second manager holds the identical directional view but describes it as "a hunch." Using the Bayesian updating logic of the chapter, explain how the two managers' posterior expected returns will differ, and identify precisely which input to the model encodes the difference between "highly confident" and "a hunch."

BL-C8. The chapter represents a set of m views through the matrix P, the vector Q, and the diagonal covariance Ω, writing $P\mu = Q + \epsilon^v$. Explain what a single row of P encodes, what the corresponding entry of Q specifies, and what the corresponding diagonal entry of Ω controls. Why is a view naturally written as a linear combination of asset returns rather than as a statement about one asset in isolation?

BL-C9. In the Black-Litterman model, assets about which the investor expresses no view are described as being "unaffected" (or only mildly affected) by the updating. Explain why the model has this conservative property, referring to the structure of the view matrix P. Contrast this with what happens in a naive optimizer where a single revised estimate of one asset's expected return can move the weights of many unrelated assets.

BL-C10. The posterior mean $\bar{\mu} = \left[(\tau\Sigma)^{-1} + P'\Omega^{-1}P\right]^{-1}\left[(\tau\Sigma)^{-1}\Pi + P'\Omega^{-1}Q\right]$ is described as a precision-weighted blend of the prior and the views. Explain, in terms of the matrices $(\tau\Sigma)^{-1}$ and $P'\Omega^{-1}P$, why the posterior mean is always a compromise between Π and the views, and explain what happens to the blend as $\Omega \to \infty$ (views become worthless) and as $\Omega \to 0$ (views become certain).

BL-C11. Explain the relationship the chapter draws between the confidence in a view and the size of the resulting tilt away from the market portfolio. Describe two extreme cases — a view held with near-certainty and a view held with almost no confidence — and state what happens to the posterior expected returns, and hence to the optimal weights, in each case.

BL-C12. A CIO says: "I like Black-Litterman because it lets me express opinions without betting the fund." Interpret this statement in light of the chapter's material. Explain how starting from the equilibrium prior and scaling tilts by conviction accomplishes exactly this, and describe what would go wrong if she instead simply set the expected returns of her favored sectors to large positive numbers and re-optimized.

14.7.2 Quantitative

BL-Q1. (Reverse optimization / implied equilibrium returns.) An investor has risk aversion $A = 3$. There are two assets with market-cap weights $\theta = (0.6, 0.4)'$ and covariance matrix

$$\Sigma = \begin{pmatrix} 0.04 & 0.01 \\ 0.01 & 0.02 \end{pmatrix}.$$

Using the first order condition $\mu = A\Sigma\theta$, compute the implied equilibrium expected return vector Π (i.e., set $\Pi = A\Sigma\theta$). Report both components.

BL-Q2. (Reverse optimization / implied equilibrium returns.) Using the same Σ and θ as in BL-Q1 but with risk aversion $A = 5$, recompute $\Pi = A\Sigma\theta$. Then verify, by plugging your Π back into $\mu = A\Sigma\theta$ and solving for θ, that the market weights $(0.6, 0.4)'$ are recovered as the optimal portfolio. Briefly state why they must be recovered.

BL-Q3. (Single-view specification p, q, ω.) Consider three assets ordered (XLB, XLK, XLE). An investor believes that a portfolio long XLK and short XLE by equal amounts will have an expected return of $q = 0.015$ per month, and she assigns this view a standard deviation of $\omega = 0.05$. Write down the view vector p, state the scalar q, and state the variance ω^2 of the view error ϵ^v. Then confirm that $p\mu$ picks out $\mu_{\text{XLK}} - \mu_{\text{XLE}}$ from the mean vector $\mu = (\mu_{\text{XLB}}, \mu_{\text{XLK}}, \mu_{\text{XLE}})'$.

BL-Q4. (General view specification P, Q, Ω.) An investor tracking four assets ordered (A, B, C, D) holds two views: (i) A will outperform B by 0.010; (ii) C will have an expected return of 0.008 in absolute terms. She assigns the views error standard deviations 0.04 and 0.06 respectively, and treats them as independent. Write down the 2×4 matrix P, the vector Q, and the 2×2

diagonal covariance matrix Ω.

BL-Q5. (Black-Litterman posterior mean, single view, scalar case.) To make the posterior mean formula hand-computable, take a one-asset world so all matrices are scalars. Let $\tau\Sigma = 0.02$, prior $\Pi = 0.006$, view $p = 1$, view value $q = 0.012$, and view variance $\omega^2 = 0.01$. Using

$$\bar{\mu} = \left[(\tau\Sigma)^{-1} + p' \tfrac{1}{\omega^2} p\right]^{-1} \left[(\tau\Sigma)^{-1}\Pi + p' \tfrac{1}{\omega^2} q\right],$$

compute the posterior expected return $\bar{\mu}$. Show that it lies between the prior 0.006 and the view 0.012, and state which one it is closer to and why.

BL-Q6. (Black-Litterman posterior mean and confidence.) Using all the same numbers as in BL-Q5 ($\tau\Sigma = 0.02$, $\Pi = 0.006$, $p = 1$, $q = 0.012$), recompute the posterior mean $\bar{\mu}$ for two different confidence levels: (a) a very confident view with $\omega^2 = 0.001$, and (b) a very unconfident view with $\omega^2 = 0.2$. Report both values of $\bar{\mu}$ and explain how they illustrate the chapter's point that higher confidence (lower ω^2) pulls the posterior more strongly toward the view.

BL-Q7. (Posterior variance, scalar case.) Using the same scalar inputs as in BL-Q5 ($\tau\Sigma = 0.02$, $p = 1$, $\omega^2 = 0.01$), compute the posterior variance of μ,

$$\left[(\tau\Sigma)^{-1} + p' \tfrac{1}{\omega^2} p\right]^{-1}.$$

Compare it to the prior variance $\tau\Sigma = 0.02$ and to the view variance $\omega^2 = 0.01$, and explain why the posterior variance must be smaller than both. Then recompute it for the very confident case $\omega^2 = 0.001$ and comment on what happens.

BL-Q8. (Role of τ in the posterior weighting.) In the scalar posterior mean of BL-Q5, the relative weight placed on the prior versus the view is governed by $(\tau\Sigma)^{-1}$ against $1/\omega^2$. Fix $\Sigma = 0.02$ (so with $\tau = 1$, $\tau\Sigma = 0.02$), $\Pi = 0.006$, $q = 0.012$, and $\omega^2 = 0.01$. Compute the posterior mean $\bar{\mu}$ for $\tau = 0.1$ and for $\tau = 1.0$, and explain the direction in which shrinking τ moves the posterior, in terms of confidence in the equilibrium prior.

BL-Q9. (From posterior returns to a tilt.) Continue from BL-Q1, where $A = 3$, $\Sigma = \begin{pmatrix} 0.04 & 0.01 \\ 0.01 & 0.02 \end{pmatrix}$, and market weights $\theta_{\text{mkt}} = (0.6, 0.4)'$ give equilibrium returns Π. Suppose Bayesian updating raises the posterior expected return of asset 1 above its equilibrium value while leaving asset 2 unchanged, giving posterior mean $\bar{\mu} = \Pi + (0.006, 0)'$. Using the FOC $\bar{\mu} = A\Sigma\theta$, solve for the new optimal weights $\theta = \frac{1}{A}\Sigma^{-1}\bar{\mu}$ and report the tilt $\theta - \theta_{\text{mkt}}$. Comment on why raising only asset 1's expected return nonetheless changes both weights.

BL-Q10. (From posterior returns to a tilt, negative view.) Using the same $A = 3$, Σ, and $\theta_{\text{mkt}} = (0.6, 0.4)'$ as in BL-Q9, suppose instead that updating lowers asset 2's posterior expected return while leaving asset 1 unchanged, giving $\bar{\mu} = \Pi + (0, -0.007)'$. Solve for the new optimal weights $\theta = \frac{1}{A}\Sigma^{-1}\bar{\mu}$ and report the tilt $\theta - \theta_{\text{mkt}}$. Explain the sign of each component of the tilt in light of the positive covariance between the two assets.

14.7.3 Data exercises

These problems use the live-data figures in this chapter.

BL-D1. A two-sector version of Figure 14.2: let $A = 2.5$, market weights $\theta_{mkt} = (0.7, 0.3)'$, and $\Sigma = \begin{pmatrix} 0.04 & 0.01 \\ 0.01 & 0.02 \end{pmatrix}$ (annualized). (a) Compute the implied equilibrium excess returns $\Pi = A\Sigma\theta_{mkt}$. (b) Explain why the first sector's Π is higher, and relate this to the figure, where technology (weight 37%, high covariance with the market) has the highest equilibrium return and utilities (weight 2%, low covariance) the lowest.

BL-D2. In Figure 14.2 the raw historical mean for technology is about 24% per year, while its equilibrium return Π is about 9%. (a) Describe what an unconstrained mean-variance optimizer would do if fed the historical means, and why (cite the correlations in Figure 14.1 in your answer). (b) Explain why the Black-Litterman model instead anchors on Π and lets the investor's *views* — not sample means — move the portfolio away from market weights.

Chapter 15

Credit Risk Models

15.1 Introduction

Two questions define the study of credit risk models in this chapter. First, how can the probability that a borrower defaults — and the timing of that default — be modelled in a way that is mathematically tractable even though default times do not follow a normal distribution? Second, how does correlation in default probabilities across borrowers affect the pricing and risk of structured credit products, such as the tranched securities that pool many individual loans? These questions are of fundamental importance to financial markets. Credit risk is the defining feature of corporate bonds, loans, and mortgage-backed securities, and the failure to model it correctly — particularly the second question about correlation — was a central cause of the 2007–2009 global financial crisis. Understanding these models is therefore both an intellectual necessity and a lesson drawn from one of the most consequential episodes in modern financial history.

The first question motivates the Gaussian copula approach to modelling default times. Default times are non-negative and typically right-skewed random variables; they cannot be directly plugged into standard multivariate normal models. The Gaussian copula resolves this by applying a transformation: each entity's default time is first mapped through its own marginal cumulative distribution function, producing a uniform random variable, and then through the inverse of the standard normal CDF, producing a standard normal variable that preserves the original marginal default probability. This transformation makes it possible to impose a correlation structure using familiar normal distribution tools. The key structural assumption is that each entity's transformed default indicator can be decomposed into a common systematic factor — shared by all borrowers — and an idiosyncratic component specific to that borrower, with the parameter a_i controlling the relative weight of each. Conditional on the common factor, defaults become independent, which makes computation of joint default probabilities tractable.

The second question is explored through a stylized two-loan example that builds direct intuition for the effects of correlation on tranche values. When the two loans default independently, the senior tranche — which collects the first payment regardless of which borrower makes it — defaults only if both loans fail simultaneously, an event with probability ϕ^2. As positive correlation is introduced through a common factor, this probability rises: because defaults tend to cluster together in bad states of the world, the protection that diversification provides to the senior tranche evaporates. The junior tranche, which collects only when both loans perform, correspondingly increases in expected value as correlation rises. This result captures in miniature the essential fragility of AAA-rated tranches of mortgage-backed securities: when a common economic shock can drive many borrowers into default simultaneously, the senior position is far less safe than its rating under an independence assumption would suggest.

The chapter begins by developing the Gaussian copula model formally, deriving the conditional default probability given the common factor and discussing the role of the loading parameter in governing default correlation. It then turns to the two-loan tranche example, computing expected payoffs under both independence and correlated defaults and showing analytically how correlation redistributes value between the senior and junior tranches. Together, these two parts illuminate both the power and the limitations of factor-based credit models.

> **In the news**
>
> Pricing the risk that a borrower fails to repay is the business of a fast-growing — and increasingly scrutinized — corner of finance. Global Finance reported that the roughly $2 trillion private-credit market is reaching a "crossroads," with defaults rising and regulators warning about thin oversight even as spreads remain tight. Estimating a borrower's probability of default and the loss it would cause, and reading those probabilities out of market spreads, is precisely what the credit models in this chapter do. Read it at Global Finance.

15.2 The Gaussian Copula Model of Time to Default

Let t be the random variable denoting the time to default of a particular entity (mortgage, bond payment, etc.). In general t will not be normally distributed. Suppose that the CDF of t is given by $F(t)$. So $F(t_0)$ is the probability that the entity defaults up to the time t_0. Let $N(x)$ be the CDF of the standard normal distribution (i.e. the normal distribution with mean 0 and variance 1). One can transform the non-normal random variable t into a normally distributed random variable x. To do so, let $x(t)$ be the solution to the equation

$$N(x(t)) = F(t)$$

Note that by definition, the probability that $t \leq t_0$ is equal to the probability that $x \leq x(t_0)$. Thus, by letting $x(t) = N^{-1}(F(t))$ we have transformed t into a normally distributed random variable.

Consider now a set of entities indexed by $\{1, \ldots, n\}$ with times to default given by $\{t_1, t_2, t_3, \ldots, t_n\}$. We will assume that the transformation of these times to default can be modelled by the function

$$x_i = a_i Y + \sqrt{1 - a_i^2}\epsilon_i$$

where $-1 < a_i < 1$. The variables ϵ_i represent the individual factors that affect default so it is assumed that $cov(\epsilon_i, \epsilon_j) = 0$ for all i and j. Y is a common factor affecting default amongst all entities. It is also assumed that Y and ϵ_i have independent standard normal distributions for all i. Note that

$$cov(x_i, x_j) = Ex_i x_j = a_i a_j$$

because of the independent standard normal assumption.

If the probability that entity i will default by time T is $F_i(T)$, then this occurs when $x_i \leq N^{-1}(F_i(T))$. From our model of x_i, this will happen when

$$a_i Y + \sqrt{1 - a_i^2}\epsilon_i < N^{-1}(F_i(T))$$

or rearranging, when

$$\epsilon_i \leq \frac{N^{-1}(F_i(T)) - a_i Y}{\sqrt{1 - a_i^2}}.$$

Therefore, if we knew the value of the factor Y, then we would know the probability of default which is

$$F_i(T|Y) = N\left(\frac{N^{-1}(F_i(T)) - a_i Y}{\sqrt{1 - a_i^2}}\right)$$

If we assume a common loading $a_i = \rho$ for all i and a common marginal default probability $F_i(T) = F(T)$, then this becomes

$$F(T|Y) = N\left(\frac{N^{-1}(F(T)) - \rho Y}{\sqrt{1-\rho^2}}\right).$$

This is the key result of the model: conditional on the common factor Y, every entity defaults with this same probability $F(T|Y)$ and — because the idiosyncratic shocks ϵ_i are mutually independent — the defaults are *conditionally independent*. The joint probability that a group of entities all default by T is therefore the product of their conditional probabilities, integrated over the distribution of Y:

$$\Pr(\text{all default}) = \int_{-\infty}^{\infty} \left[F(T|Y)\right]^k n(Y)\, dY,$$

where k is the number of entities and $n(\cdot)$ is the standard normal density. Correlation enters entirely through the shared factor Y: a bad draw of Y raises $F(T|Y)$ for every entity at once, which is what makes many defaults cluster in bad states.

15.3 A simple example

Consider a one year loan that will make a payment of M if not in default and 0 under default. The probability of the loan defaulting is ϕ. This asset has an expected value of $(1-\phi)M$ and a variance of $\phi(1-\phi)M^2$. For simplicity, suppose that the covariance of this asset with the market is 0. Thus, the CAPM says that the price of this asset would satisfy

$$E\left(\frac{M-P}{P}\right) - r_f = 0$$

which implies that its price is given by

$$P = \frac{EM}{1+r_f} = \frac{(1-\phi)M}{1+r_f}$$

Now assume that there are two such loans, each identical and each having independent probabilities of defaulting. The likelihood that both will default is ϕ^2, while the likelihood that one of these loans defaults is $2\phi(1-\phi)$ while the probability that none will default is $(1-\phi)^2$.

Consider constructing a security that will receive the first mortgage payment, no matter who it comes from. Such a security will default only if both of the loans default. As such, it has an expected payoff of $(1-\phi^2)M$, a variance of $\phi^2(1-\phi^2)M^2$ and a market price of

$$P = \frac{EM}{1+r_f} = \frac{(1-\phi^2)M}{1+r_f}$$

A second security will receive the second payoff regardless of from whom it comes. Such a security will payoff as long as both mortgages don't default. This security has expected payoff $(1-\phi)^2M$, variance $(1-\phi)^2(1-(1-\phi)^2)M^2$ and market price

$$P = \frac{(1-\phi)^2M}{1+r_f}$$

Now suppose that the probabilities of each of these loans defaulting are not independent. Specifically, suppose that the probability of each loan defaulting is given by $\phi_i = ay + \phi$ where y is a random, common factor that affects default probabilities for all of the loans. Notice that if a is 0 then this model becomes the old model. Suppose that conditional on a particular y, the default probabilities are still independent, so the way that dependence is introduced into the model is through the random factor y. Assume that $y \sim N(0, \sigma^2)$, and, to keep the results below in the same units as the independent case, adopt the normalization $\sigma^2 = \phi^2$ (the common factor has standard deviation equal to the baseline default probability). The two facts we need about y are $E_y[y] = 0$ and $E_y[y^2] = \sigma^2$.

A security based on the first loan payment received will default only if both loans default. Conditional on y the two defaults are independent, so that conditional probability is $\phi_i^2 = (ay + \phi)^2$, and the payoff probability is $1 - (ay + \phi)^2$. Taking the expectation over y, and being careful to keep the $a^2 y^2$ term,

$$\begin{aligned} E_y\left[1 - (ay + \phi)^2\right] &= 1 - E_y\left[a^2 y^2 + 2a\phi\, y + \phi^2\right] \\ &= 1 - a^2 E_y[y^2] - 2a\phi\, E_y[y] - \phi^2 \\ &= 1 - a^2\sigma^2 - \phi^2 = 1 - \phi^2 - a^2\phi^2, \end{aligned}$$

where the last step uses $E_y[y] = 0$, $E_y[(ay)^2] = a^2 E_y[y^2] = a^2\sigma^2$, and the normalization $\sigma^2 = \phi^2$. (The earlier version dropped the $a^2 y^2$ contribution; it is precisely this term that produces the $a^2\phi^2$ correction.) This implies a new price of

$$P = \frac{(1 - \phi^2 - a^2\phi^2)M}{1 + r_f}$$

Thus, when correlation increases, the relatively risk-free asset becomes less valuable.

The second asset pays off only when both assets do not default. Conditional on y the probability that neither defaults is $(1-\phi_i)^2 = (1-ay-\phi)^2$. Expanding fully — again keeping the $a^2 y^2$ term — and taking the expectation,

$$\begin{aligned} E_y\left[(1 - \phi - ay)^2\right] &= E_y\left[(1 - \phi)^2 - 2(1 - \phi)a\, y + a^2 y^2\right] \\ &= (1 - \phi)^2 - 2(1 - \phi)a\, E_y[y] + a^2 E_y[y^2] \\ &= (1 - \phi)^2 + a^2\sigma^2 = (1 - \phi)^2 + a^2\phi^2, \end{aligned}$$

using $E_y[y] = 0$, $E_y[y^2] = \sigma^2$, and the normalization $\sigma^2 = \phi^2$. Note that the expected value of the second asset increases. It now has value given by

$$P = \frac{[(1 - \phi)^2 + a^2\phi^2]M}{1 + r_f}$$

Comparing the two tranches, the common factor moves value from the senior security to the junior one. The senior tranche loses $a^2\phi^2 M/(1+r_f)$ and the junior tranche gains exactly the same amount, so the two effects offset: the combined value of the two tranches is still $\frac{[2(1-\phi)]M}{1+r_f}$, equal to the value of simply holding both loans. Correlation does not create or destroy value; it *redistributes* it. The intuition is that positive correlation makes joint outcomes more likely — both loans defaulting together, or both surviving together — which thins out the "one defaults, one survives" middle case. That hurts the senior tranche, whose safety relied on it being unlikely that *both* borrowers fail at once, and helps the junior tranche, whose payoff relied on it being likely that *both* survive.

This is the essential lesson of the chapter. A senior ("AAA") tranche rated on the assumption of independent defaults ($a = 0$, default probability ϕ^2) is materially riskier once a common factor is present, because its default probability rises to $\phi^2 + a^2\phi^2$. In the language of the 2007–2009 crisis, the common factor y is a nationwide shock to house prices: when it turns bad, many mortgages default together, and the diversification that supposedly protected the senior tranche evaporates precisely when it is needed most.

15.4 Application: An analyst underwriting a loan

An analyst at a private-credit fund must decide whether to extend a loan to a mid-sized company and what interest rate would compensate for the risk of not being repaid. The models in this chapter are the analyst's underwriting toolkit. By estimating the borrower's probability of default, the loss given default, and the exposure at default, the analyst can compute the expected loss the loan carries and judge whether the offered yield is adequate. Structural models tie the chance of default to the firm's assets and leverage, while reduced-form models read default intensity from market prices; together they let the analyst translate a credit spread into an implied default probability and ask whether the market — currently pricing very little default risk into a fast-growing, opaque asset class — is paying enough for the danger being taken on.

Further listening

Odd Lots (Bloomberg) — "What's Actually Going On With Private Credit": inside the $2-trillion private-credit boom and the rising default risk these credit models are built to price.

15.5 Homework problems

15.5.1 Conceptual

CR-C1. A junior analyst argues that because a borrower's time to default t is non-negative and right-skewed, "credit risk simply cannot be handled with normal-distribution machinery." Explain how the transformation $x(t) = N^{-1}(F(t))$ addresses this objection. In your answer, state precisely what is preserved by the transformation and why the probability that $t \le t_0$ equals the probability that $x \le x(t_0)$.

CR-C2. A modeler applies the transformation $x(t) = N^{-1}(F(t))$ to two entities whose marginal default distributions $F_1(t)$ and $F_2(t)$ are very different — one a sub-prime mortgage that defaults quickly, the other an investment-grade bond that defaults slowly. Explain why, after transformation, both x_1 and x_2 are standard normal even though t_1 and t_2 are not identically distributed, and explain what this common normal scale buys the modeler when he wants to impose correlation across the two entities.

CR-C3. In the factor model $x_i = a_i Y + \sqrt{1 - a_i^2}\,\epsilon_i$, describe the economic meaning of the common factor Y and the idiosyncratic term ϵ_i. Contrast a scenario in which every a_i is close to 1 with one in which every a_i is close to 0, explaining what each implies for how defaults cluster, and give a real-world example of an economic force that would be captured by Y.

CR-C4. Explain why the decomposition $x_i = a_i Y + \sqrt{1 - a_i^2}\,\epsilon_i$ is written with the coefficient $\sqrt{1 - a_i^2}$ on the idiosyncratic term rather than an arbitrary constant. Show that this choice keeps each x_i standard normal, and explain how it makes a_i interpretable directly as the correlation of x_i with the common factor Y. Why is preserving the marginal distribution of x_i essential to the copula construction?

CR-C5. The chapter shows $F_i(T|Y) = N\left(\frac{N^{-1}(F_i(T))-a_iY}{\sqrt{1-a_i^2}}\right)$. Explain why conditioning on the common factor Y makes the defaults of different entities independent of one another, and why this conditional-independence property is what makes joint default probabilities computationally tractable. What is lost if one instead tries to model the joint default distribution directly?

CR-C6. Interpret the conditional default probability $F(T|Y) = N\left(\frac{N^{-1}(F(T))-\rho Y}{\sqrt{1-\rho^2}}\right)$ as Y moves. Explain why a low (unfavorable) draw of Y raises the conditional default probability for *every* entity at once, and describe the limiting behavior as $\rho \to 0$ and as $\rho \to 1$. Why does this single expression capture the entire correlation structure of the model?

CR-C7. In the two-loan example, the senior security (first payment received) has default probability ϕ^2 under independence but $\phi^2 + a^2\phi^2$ once the common factor is introduced, while the junior security's survival probability rises from $(1-\phi)^2$ to $(1-\phi)^2 + a^2\phi^2$. Explain in words why positive correlation makes the senior tranche riskier while making the junior tranche more valuable, and relate this to why diversification "evaporates" for the senior position in bad states of the world.

CR-C8. The chapter argues that when the common factor is introduced, the senior tranche loses exactly $a^2\phi^2M/(1+r_f)$ in value and the junior tranche gains exactly the same amount, so that the combined value of the two tranches is unchanged. Explain why correlation *redistributes* value rather than creating or destroying it, and why this conservation is what one should expect given that the two tranches together simply reproduce holding both loans outright.

CR-C9. Explain how the two-loan example illuminates the 2007–2009 financial crisis. In particular, discuss why a AAA rating assigned to a senior mortgage-backed tranche under an independence assumption ($a = 0$) could be dangerously misleading, and what feature of the housing market corresponds to the common factor y in the model.

CR-C10. A ratings analyst insists that because each individual mortgage in a pool has only a small default probability ϕ, the senior tranche built on that pool must be nearly riskless. Using the result that the senior tranche's default probability is $\phi^2 + a^2\phi^2$ rather than ϕ^2, explain the flaw in this reasoning and identify the single modeling assumption whose misjudgment does the most damage to the senior tranche's safety.

15.5.2 Quantitative

CR-Q1. An entity has a constant hazard (default intensity) of $\lambda = 0.03$ per year, so its default-time CDF is $F(t) = 1 - e^{-\lambda t}$. (a) Compute the probability the entity defaults within $T = 5$ years, $F(5)$. (b) Using N^{-1}, find the transformed threshold $x(5) = N^{-1}(F(5))$. Report both numbers.

CR-Q2. An entity's default-time CDF is $F(t) = 1 - e^{-\lambda t}$ with $\lambda = 0.05$. (a) Find the time-to-default value t^* that transforms to the standard-normal threshold $x = 0$, i.e. solve $N^{-1}(F(t^*)) = 0$. (b) Verify by computing $F(t^*)$ and confirming it equals $N(0)$. Report t^* and $F(t^*)$.

CR-Q3. Using the conditional default probability $F_i(T|Y) = N\left(\frac{N^{-1}(F_i(T)) - a_i Y}{\sqrt{1-a_i^2}}\right)$, take $F_i(T) = 0.05$ and loading $a_i = 0.5$. (a) Compute $F_i(T|Y)$ when the common factor takes a "good" value $Y = +1$. (b) Compute it when the common factor takes a "bad" value $Y = -1$. Comment on which state produces more defaults.

CR-Q4. Using $F(T|Y) = N\left(\frac{N^{-1}(F(T)) - \rho Y}{\sqrt{1-\rho^2}}\right)$, take an unconditional default probability $F(T) = 0.10$ and a bad common-factor draw $Y = -1.5$. Compute the conditional default probability for (a) $\rho = 0.2$ and (b) $\rho = 0.7$. Report both and explain why the higher loading produces a larger conditional default probability in the bad state.

CR-Q5. The pairwise correlation of the default drivers is $cov(x_i, x_j) = a_i a_j$. (a) If two entities have loadings $a_i = 0.6$ and $a_j = 0.8$, compute $cov(x_i, x_j)$. (b) State what single common loading a (assumed equal across both entities) would be required to produce the same pairwise correlation. Report both.

CR-Q6. A pool has three entities with loadings $a_1 = 0.5$, $a_2 = 0.5$, $a_3 = 0.9$ on the common factor Y. (a) Compute the pairwise default-driver correlation $cov(x_i, x_j) = a_i a_j$ for each of the three pairs $(1,2)$, $(1,3)$, $(2,3)$. (b) Which pair is most correlated, and explain in one sentence why the entity with the largest loading dominates the correlation structure. Report all three correlations.

CR-Q7. Two identical one-year loans each pay $M = \$1{,}000$ if not in default and 0 otherwise, with default probability $\phi = 0.10$ and risk-free rate $r_f = 0.04$. Assuming independence, compute the market price of (a) the senior security (receives the first payment, defaults only if both default) and (b) the junior security (receives the second payment, pays only if neither defaults). Use $P_{\text{senior}} = \frac{(1-\phi^2)M}{1+r_f}$ and $P_{\text{junior}} = \frac{(1-\phi)^2 M}{1+r_f}$.

CR-Q8. Now introduce the common factor with $\phi_i = ay + \phi$, keeping $\phi = 0.10$, $M = \$1{,}000$, $r_f = 0.04$, and loading $a = 0.20$. Using the correlated-default results $P_{\text{senior}} = \frac{(1-\phi^2-a^2\phi^2)M}{1+r_f}$ and $P_{\text{junior}} = \frac{[(1-\phi)^2+a^2\phi^2]M}{1+r_f}$, compute both prices and report the change in each relative to the independent-default prices from CR-Q7.

CR-Q9. For the correlated two-loan model with $\phi = 0.10$, $M = \$1{,}000$, $r_f = 0.04$, and loading $a = 0.30$: (a) compute the dollar amount $\frac{a^2\phi^2 M}{1+r_f}$ that is transferred from the senior tranche to the junior tranche. (b) Confirm value conservation by showing that $P_{\text{senior}} + P_{\text{junior}}$ equals $\frac{2(1-\phi)M}{1+r_f}$, the value of holding both loans outright. Report the transfer amount and the combined value.

CR-Q10. Under the common-factor model the senior tranche's default probability is $\phi^2 + a^2\phi^2$. With $\phi = 0.10$: (a) compute the senior tranche's default probability under independence ($a = 0$) and under a loading of $a = 0.5$. (b) Compute the percentage increase in the senior tranche's default probability caused by moving from $a = 0$ to $a = 0.5$. Report both probabilities and the percentage increase.

Chapter 16

Market Makers

16.1 Introduction

Three questions structure this chapter. First, why does a bid-ask spread exist at all — why will a dealer simultaneously quote a lower price to buy and a higher price to sell the same asset? Second, how does an informed trader's strategic demand choice interact with competitive market makers' pricing rules to determine how efficiently private information is incorporated into prices? Third, if prices do reveal private information, what incentive remains for anyone to acquire that information in the first place? These questions sit at the heart of market microstructure, and the three canonical models examined here — Glosten-Milgrom, Kyle, and Grossman-Stiglitz — each isolates a different facet of the same underlying tension between informed and uninformed trading.

The first question matters because the bid-ask spread is the most visible cost of transacting in financial markets. A spread is not the result of dealer greed or market power alone; it arises even among perfectly competitive market makers as a rational response to the risk of trading against someone who knows more than they do. When an investor places a buy order, a competitive market maker cannot tell whether that investor is acting on a private signal that the asset is undervalued or merely has liquidity needs. Setting the ask equal to the expected value of the asset conditional on receiving a buy order — higher than the unconditional expectation — is the only way to avoid expected losses. The Glosten-Milgrom model formalizes this logic, showing that the spread is determined by the proportion of informed traders in the market and the magnitude of the information asymmetry, and that it shrinks as competition among market makers intensifies or as underlying asset volatility falls.

The second question takes the analysis a step further by asking how an informed trader should optimally exploit her informational advantage when she knows the market maker will adjust prices against her. The Kyle model addresses this: the insider trades strategically, spreading her demand across time to camouflage it within the noise generated by uninformed liquidity traders. In equilibrium the market maker sets a linear pricing rule whose slope — the Kyle lambda — measures market depth, the price impact of each additional unit of order flow. A higher fraction of noise trading relative to fundamental uncertainty produces a deeper market and a lower price impact, but never eliminates it entirely.

The third question strikes at the foundations of price informativeness. If prices were to perfectly aggregate and reveal all private information, the value of acquiring that information would vanish and no one would bother to collect it — yet without anyone collecting it, prices could not be informative. The Grossman-Stiglitz model resolves this paradox by showing that an equilibrium with fully revealing prices cannot exist when information acquisition is costly. Instead, prices are only partially informative: they reveal enough that the marginal informed trader is exactly indifferent between paying the cost of information and free-riding on the price signal. The chapter develops all three models in sequence, building from the simple binary-value setting of Glosten-Milgrom through the continuous Gaussian framework of Kyle to the competitive rational-expectations equilibrium of Grossman-Stiglitz.

In the news

When a retail trader places a "free" trade on an app, a market maker is on the other side, earning the bid–ask spread and paying the broker for the order. NBC News examined what lies "beneath the surface" of the 2021 GameStop frenzy — payment for order flow, and firms like Citadel Securities that handle a large share of retail volume. The spread those firms quote, and how it widens when they fear trading against someone better informed, is exactly the market-microstructure problem modeled in this chapter. Read it at NBC News.

16.2 The Glosten-Milgrom model of the spread

The Glosten-Milgrom model helps us to understand the motivation of liquidity providers when setting bid and ask prices.

Consider a set of identical, competitive market makers that do not know whether a stock will go up or down today. They must set prices at which they are willing to buy and sell shares of a particular stock. The price at which they will buy the stock is the bid (B) and the price at which they will sell the stock is the offer (or ask) (A).

Each market maker knows that the stock will be worth either V_0 or V_1 at the end of the day where $V_0 < V_1$. Prior to the opening of trading, that market maker believes that the probability that the stock will be worth V_0 is $P(V_0) = 1 - \pi$. Regardless of the future value of the stock, uninformed traders will arrive to both buy and sell the stock.

The market maker must decide where to set B and A in order to not lose money on average. If however, any market maker sets their spread wide enough to earn positive profits on average, competition will drive another market maker to come in and supply a better (narrower) bid/ask spread. The desire to not lose money, combined with competition means that in equilibrium, we will look for $B = E(V|\text{next order is a sell})$ and $A = E(V|\text{next order is a buy})$.

Suppose that amongst liquidity demanders, the fraction μ are informed, meaning they know the value of the signal that has been received. Fraction $1-\mu$ are uninformed. Each liquidity demander will demand exactly one share of the asset. The resolution of uncertainty is depicted in figure 1.

The resolution of uncertainty in the Glosten-Milgrom model

Thus, if an uninformed trader arrives at the market, the market maker will learn nothing about the value of the asset from trading with them. As such

$$E[V|\text{uninformed buy}] = E[V|\text{uninformed sell}] = E[V]$$

This implies that if traders were all uninformed, the spread on an asset would be zero. The bid $B = E[V|\text{sell order}] = E[V]$ while the ask is $A = E[V|\text{buy order}] = E[V]$.

However, an informed trader knows the true value of the asset. Therefore, if the market maker knew that the next trade to arrive is coming from an informed trader, then $A = E[V|\text{buy order}] = V_1$ and $B = E[V|\text{sell order}] = V_0$. The problem then for a market maker is that she doesn't know whether she is trading with an informed trader or an uninformed trader. As such, she must set the spread so that she doesn't lose money on average. That is, the money that she loses from trading with informed traders she recoups from trading with uninformed traders. We want to calculate $E[V|\text{buy}]$. To do that, we first calculate $P(\text{buy}) = \pi\mu+\pi(1-\mu)\frac{1}{2}+(1-\pi)(1-\mu)\frac{1}{2} = \pi\mu+\frac{1}{2}(1-\mu)$. (An informed trader buys only when the value is high, an event of probability $\pi\mu$; an uninformed trader buys with probability $\frac{1}{2}$ regardless of value, contributing $\frac{1}{2}(1-\mu)$.) Bayes' rule then gives the probability that a buyer is informed,

$$P(\text{In}|\text{buy}) = \frac{\pi\mu}{\pi\mu + \frac{1}{2}(1-\mu)}, \qquad P(\text{Un}|\text{buy}) = \frac{\frac{1}{2}(1-\mu)}{\pi\mu + \frac{1}{2}(1-\mu)},$$

and $E[V|\text{buy}, \text{In}] = V_1$ (an informed buyer knows the value is high), while $E[V|\text{buy}, \text{Un}] = E[V] = \pi V_1 + (1-\pi)V_0$. Then, to calculate the ask (similarly for the bid) we calculate

$$\begin{aligned}
A &= E[V|\text{buy order}] = P(In|\text{buy})E[V|\text{buy, In}] + P(\text{Un}|\text{buy})E[V|\text{buy, Un}] \\
&= \frac{1}{\pi\mu + \frac{1}{2}(1-\mu)}\left(\pi\mu V_1 + \frac{1-\mu}{2}\left[\pi V_1 + (1-\pi)V_0\right]\right) \\
&= \frac{\pi(1+\mu)V_1 + (1-\pi)(1-\mu)V_0}{1+(2\pi-1)\mu}
\end{aligned}$$

The last line multiplies the numerator and denominator of the previous line by 2: the denominator becomes $2\pi\mu + (1-\mu) = 1 + (2\pi-1)\mu$, and the numerator becomes $2\pi\mu V_1 + (1-\mu)[\pi V_1 + (1-\pi)V_0] = \pi(1+\mu)V_1 + (1-\pi)(1-\mu)V_0$.

Whereas the bid is

$$\begin{aligned}
B &= E[V|\text{sell order}] = P(\text{In}|\text{sell})E[V|\text{sell, In}] + P(Un|\text{sell})E[V|\text{sell, Un}] \\
&= \frac{1}{(1-\pi)\mu + \frac{1}{2}(1-\mu)}\left((1-\pi)\mu V_0 + \frac{1-\mu}{2}\left[\pi V_1 + (1-\pi)V_0\right]\right) \\
&= \frac{\pi(1-\mu)V_1 + (1-\pi)(1+\mu)V_0}{1+(1-2\pi)\mu}
\end{aligned}$$

From this we can compute the spread $A - B$. Note that the ask and bid have *different* denominators: writing $D_A = 1 + (2\pi-1)\mu$ and $D_B = 1 + (1-2\pi)\mu = 1 - (2\pi-1)\mu$, we have $A = N_A/D_A$ and $B = N_B/D_B$ with numerators $N_A = \pi(1+\mu)V_1 + (1-\pi)(1-\mu)V_0$ and $N_B = \pi(1-\mu)V_1 + (1-\pi)(1+\mu)V_0$. Combining over the common denominator $D_A D_B = 1 - (2\pi-1)^2\mu^2$,

$$A - B = \frac{N_A D_B - N_B D_A}{1-(2\pi-1)^2\mu^2}.$$

Expanding the numerator, the pieces symmetric in the two states cancel and one is left with $N_A D_B - N_B D_A = 4\pi(1-\pi)\mu(V_1 - V_0)$, so

$$A - B = \frac{4\pi(1-\pi)\mu(V_1 - V_0)}{1 - (2\pi - 1)^2\mu^2}.$$

If $\pi = 0.5$ the denominator is 1 and $2\pi - 1 = 0$, so this becomes $A - B = \mu(V_1 - V_0)$.

The Glosten-Milgrom model shows that there are several forces which narrow spreads and thus increase liquidity in limit order markets. These forces include

1. Competition between market makers
2. Decreases in the probability of informed trading, and
3. Decreases in the volatility of the underlying asset.

16.3 The Kyle Model of Orderbook Shape

The participants in the Kyle model are similar to those in the Glosten-Milgrom model. We have uninformed traders, informed traders (or insiders in Kyle's language) and market makers. Uninformed traders have random demand that has mean 0 and a known variance. Let the demand for uninformed traders be given by $u \sim N(0, \sigma_u^2)$. Informed traders observe the future value of the asset v and select their demand x as a function of this observed v and the price set by the market makers. Assume that $v \sim N(p_0, \Sigma_0)$. A competitive set of market makers establish an order book described as a price that is a function of the number of orders that they observe $y = x + u$. Since these market makers are competitive, they will in expectation earn zero profit. It is assumed that market makers cannot distinguish between informed and uninformed liquidity requests.

The expected profit to a market maker given the demand for liquidity y is $E\pi = E(v - p)(-y)$ (recall that the market maker must take the opposite position of the liquidity demanders). If expected profits are to be zero for every y, then in equilibrium $p = E[v|y]$. In order to find an equilibrium in this model, we proceed by making an assumption about the behavior of market makers, which at the end we verify to be true. Let us assume that market makers set the price according to a linear function of the demand that they observe. Specifically, let $P(y) = \mu + \lambda y$.

Now, we consider the problem of informed traders. The expected profit to informed traders is $E\pi = E[(v - p(y))x|v]$. In order to maximize this profit, informed traders solve (given the price function of market makers.)

$$max_x E[\pi] = \max_x E[(v - \mu - \lambda(x + u))x|v]$$

Since $Eu = 0$, the objective is $E[\pi] = (v - \mu - \lambda x)x = vx - \mu x - \lambda x^2$. Differentiating with respect to x and setting the derivative to zero gives the first order condition

$$\frac{\partial E[\pi]}{\partial x} = v - \mu - 2\lambda x = 0.$$

Solving for x gives $x = \frac{v-\mu}{2\lambda} = -\frac{\mu}{2\lambda} + \frac{1}{2\lambda}v$. Notice if the market makers use a linear pricing rule, then the informed traders will set their demand as a linear function of the signal that they receive.

Now consider the problem of market makers. Given that the demand that they observe is $y = x + u$, they need to calculate $E[v|x + u]$. Note that v and y are related since v determines x which is a part of y. In fact, since y is a linear function of v, y and v are jointly normally distributed. The demand y is $y \sim N(-\frac{\mu}{2\lambda} + \frac{1}{\lambda}p_0, \frac{1}{\lambda^2}\Sigma_0)$. The covariance between y and v is

$$\begin{aligned}
E(y - Ey)(v - Ev) &= E(-\frac{\mu}{2\lambda} + \frac{1}{2\lambda}v + u + \frac{\mu}{2\lambda} + \frac{1}{2\lambda}p_0)(v - p_0) \\
&= E(\frac{1}{2\lambda}(v - p_0))(v - p_0) \\
&= \frac{1}{2\lambda}\Sigma_0
\end{aligned}$$

For two jointly normal random variables (a, b), $E[a|b] = Ea + \frac{cov(a,b)}{\sigma_b^2}(b - Eb)$. Applying this formula to y and v gives

$$E[v|y] = p_0 + \frac{\frac{1}{2\lambda}\Sigma_0}{\frac{1}{4\lambda^2}\Sigma_0 + \sigma_u^2}(y + \frac{\mu}{2\lambda} - \frac{1}{2\lambda}p_0).$$

Now, recall our original assumption that $P(y) = \mu + \lambda y$. Since $P(y) = E[v|y]$, we can compare coefficients and we notice that the coefficient on y implies that

$$\lambda = \frac{\frac{1}{2\lambda}\Sigma_0}{\frac{1}{4\lambda^2}\Sigma_0 + \sigma_u^2}$$

To solve, cross-multiply by the denominator:

$$\lambda\left(\frac{1}{4\lambda^2}\Sigma_0 + \sigma_u^2\right) = \frac{1}{2\lambda}\Sigma_0 \quad \Longrightarrow \quad \frac{1}{4\lambda}\Sigma_0 + \lambda\sigma_u^2 = \frac{1}{2\lambda}\Sigma_0.$$

Move the Σ_0 terms to one side, $\lambda\sigma_u^2 = \frac{1}{2\lambda}\Sigma_0 - \frac{1}{4\lambda}\Sigma_0 = \frac{1}{4\lambda}\Sigma_0$, and multiply through by λ:

$$\lambda^2\sigma_u^2 = \frac{\Sigma_0}{4} \quad \Longrightarrow \quad \lambda^2 = \frac{\Sigma_0}{4\sigma_u^2}.$$

Taking the positive root (depth must be positive) produces the result that

$$\lambda = \frac{\sqrt{\Sigma_0}}{2\sigma_u}$$

Furthermore, it can be seen by inspection that $\mu = p_0$ is a solution to that equation.

16.4 The Grossman-Stiglitz model of information and asset prices

In the Grossman-Stiglitz model, participants are similar to the Kyle model, except that they are risk averse (and have CARA utility) and markets are competitive. Consider a fraction a large number of traders. Fraction λ of them are given a signal that takes the form $s = v + \epsilon$ where v is the true future value of the asset and $\epsilon \sim N(0, \sigma_\epsilon^2)$ is the noise term. The future value of the asset $v \sim N(\mu_0, \sigma_v^2)$. The risk-free asset has a net return of 0 and a normalized price of 1. Noise traders take the form of a noisy supply $Z \sim N(0, \sigma_z^2)$. We assume that Z, v and ϵ are all independent of each other.

First, we consider the problem of the informed agents. As was shown in class, when traders have utility of the form $U(w) = -e^{-\gamma w}$, and w is distributed normally, we can simplify their utility problem to become

$$\max_\theta \theta(E[v|I] - P) - \frac{\gamma}{2}\sigma^2(v|I)$$

where I represents the information that a trader has. This implies a demand function of

$$\theta(I) = \frac{E[v|I] - P}{\gamma\sigma^2(v|I)}$$

Using the demand functions for both informed and uninformed traders, equilibrium requires that

$$\lambda\frac{E[v|s] - P}{\gamma\sigma^2(v|s)} + (1-\lambda)\frac{E[v|P] - P}{\gamma\sigma^2(v|P)} = Z$$

To isolate P, write the two demand weights as $A_s = \frac{\lambda}{\gamma\sigma^2(v|s)}$ and $A_p = \frac{1-\lambda}{\gamma\sigma^2(v|P)}$. The market-clearing condition is $A_s(E[v|s]-P)+A_p(E[v|P]-P) = Z$. Distribute and collect the terms in P:

$$A_s E[v|s] + A_p E[v|P] - (A_s + A_p)P = Z.$$

Moving the P term to one side and dividing by $A_s + A_p$ gives

$$P = \frac{\lambda \frac{E[v|s]}{\gamma\sigma^2(v|s)} + (1-\lambda)\frac{E[v|P]}{\gamma\sigma^2(v|P)}}{\frac{\lambda}{\gamma\sigma^2(v|s)} + \frac{1-\lambda}{\gamma\sigma^2(v|P)}} - \frac{1}{\frac{\lambda}{\gamma\sigma^2(v|s)} + \frac{1-\lambda}{\gamma\sigma^2(v|P)}} Z,$$

which is exactly the conjectured form $P = a + bs - dZ$ once the conditional expectations below are substituted.

By the properties of updating for normal distributions, $E[v|s] = \mu + \frac{\sigma_v^2}{\sigma_v^2+\sigma_\epsilon^2} s$ and given our assumption about the price function $E[v|P] = \mu + \frac{b\sigma_v^2}{b^2(\sigma_\epsilon^2)+d^2\sigma_z^2}\left(P - a - b\mu\right)$.

Thus, the previous equation becomes $KP = \lambda(\mu + Ms) + (1-\lambda)(\mu + N(P - a - b\mu))) - Z$ where

$$K = \frac{\lambda}{\gamma\sigma^2(v|s)} + \frac{1-\lambda}{\gamma\sigma^2(v|P)},$$

$$M = \frac{\sigma_v^2}{\sigma_v^2 + \sigma_\epsilon^2}$$

and

$$N = \frac{b\sigma_v^2}{b^2(\sigma_\epsilon^2) + d^2\sigma_z^2}.$$

We notice that this has a linear form and hence can be solved for a, b and d. Concretely, substituting $E[v|s] = \mu + Ms$ and $E[v|P] = \mu + N(P - a - b\mu)$ into $KP = \lambda(\mu + Ms) + (1-\lambda)(\mu + N(P - a - b\mu)) - Z$ and equating the coefficients of 1, s, and Z on both sides of $P = a + bs - dZ$ yields three equations in a, b, d. Because N itself depends on b and d (see its definition above), this is a fixed-point system rather than a simple linear solve.

The thing to get out of this solution is that the price depends on the signal s but, because it is contaminated by the noisy supply Z, does not allow someone who is only observing the price to completely infer the signal. This partial revelation is exactly what preserves a positive return to becoming informed, resolving the Grossman-Stiglitz paradox.

16.5 Application: A trading desk pricing the spread

Picture the trader at a market-making firm responsible for quoting bid and ask prices in the minutes before a Federal Reserve announcement. Every quote exposes the desk to the possibility that whoever trades against it knows something it does not, and the models in this chapter tell the trader exactly how to respond. The Glosten–Milgrom framework shows that the bid-ask spread is the desk's defense against informed traders: the greater the chance that an incoming order carries private information, the wider the quotes must be to avoid being systematically picked off. As the announcement nears and the risk of informed trading rises, the desk widens its spread; once the news is public and uncertainty resolves, it tightens again. The spread that a retail investor experiences as a small, almost invisible cost is, on the other side of the trade, the precise output of this adverse-selection calculation.

> Further listening
>
> *Planet Money* (NPR) — "Why Robinhood froze the buying of some stocks": how payment for order flow, market makers like Citadel Securities, and the clearinghouse sit behind every "free" trade — the microstructure this chapter models.

16.6 Homework problems

16.6.1 Conceptual

MM-C1. Two stocks trade on the same exchange with identical fundamental values, but for stock X the market maker believes fraction $\mu = 0.10$ of arriving traders are informed, while for stock Y she believes $\mu = 0.40$. Using the logic of the Glosten-Milgrom model, explain which stock will have the wider bid-ask spread and why. In your answer, describe the specific mechanism by which the presence of informed traders forces a competitive, zero-profit market maker to quote $A = E[V|\text{buy order}]$ above $E[V]$ and $B = E[V|\text{sell order}]$ below $E[V]$.

MM-C2. A commentator argues that bid-ask spreads exist purely because market makers have monopoly power and use it to gouge traders. Drawing on the Glosten-Milgrom model, explain why a positive spread can arise even among perfectly competitive market makers who earn zero expected profit. What would the spread be in this model if every arriving trader were uninformed ($\mu = 0$), and what does that imply about the commentator's claim?

MM-C3. Explain why, in the Glosten-Milgrom model, a competitive market maker sets the ask equal to $A = E[V|\text{buy order}]$ rather than the unconditional expectation $E[V]$. Walk through the role Bayes' rule plays in computing $P(\text{In}|\text{buy}) = \frac{\pi\mu}{\pi\mu + \frac{1}{2}(1-\mu)}$, and explain intuitively why observing a buy order is itself "bad news" for the market maker that pushes the quote above $E[V]$.

MM-C4. Using the spread formula $A - B = \frac{4\pi(1-\pi)\mu(V_1 - V_0)}{1-(2\pi-1)^2\mu^2}$, explain how the spread responds to (i) the gap $V_1 - V_0$ between the two possible asset values and (ii) the fraction μ of informed traders. Which of the three spread-narrowing forces listed in the chapter — competition, less informed trading, lower volatility — does each of these two channels correspond to?

MM-C5. In the Kyle model the insider does not simply buy as many shares as possible when she learns that v is high. Explain why the insider optimally restrains her demand, and describe the role that the noise traders' demand $u \sim N(0, \sigma_u^2)$ plays in making it profitable for her to trade at all. What would happen to the informativeness of order flow if the noise-trader variance σ_u^2

shrank toward zero?

MM-C6. The Kyle lambda $\lambda = \frac{\sqrt{\Sigma_0}}{2\sigma_u}$ is described as a measure of "market depth" or price impact. Consider two assets: asset A has high fundamental uncertainty Σ_0 and thin noise trading (low σ_u), while asset B has low Σ_0 and heavy noise trading (high σ_u). Explain which asset has the deeper market, what "deeper" means for a trader submitting a large order, and why market makers rationally choose a steeper pricing rule for asset A.

MM-C7. The insider's first-order condition in the Kyle model is $v - \mu - 2\lambda x = 0$, giving the optimal order $x = \frac{v-\mu}{2\lambda}$. Explain economically why the optimal order is proportional to the "surprise" $v - \mu$ in the insider's information and inversely proportional to 2λ. Why does a deeper market (smaller λ) induce the insider to trade more aggressively on the same signal, and how does the market maker's linear pricing rule remain self-consistent in equilibrium?

MM-C8. The Grossman-Stiglitz model is often summarized by the claim that "prices cannot be perfectly informative if information is costly." Explain the paradox this addresses: if the equilibrium price $P = a + bs - dZ$ perfectly revealed the signal s, why would no trader pay to become informed, and why does that in turn make it impossible for the price to be perfectly revealing? Describe the role of the noisy supply $Z \sim N(0, \sigma_z^2)$ in resolving the paradox.

MM-C9. Suppose a regulator succeeds in eliminating all noise trading from a market described by the Grossman-Stiglitz model (so $\sigma_z^2 \to 0$). Explain what happens to the ability of the price to reveal the informed traders' signal, and use this to explain why the fraction λ of traders who choose to acquire the costly signal would collapse. Why does a healthy market, somewhat paradoxically, require some traders who trade for reasons unrelated to fundamental value?

MM-C10. Both Glosten-Milgrom and Kyle produce a wedge between the price a trader pays and the market maker's unconditional expectation of value, yet the models attribute that wedge to different features of the trading environment. Compare the two: identify what plays the role of the "adverse-selection cost" in each, contrast the binary-value, single-share Glosten-Milgrom setting with the continuous Gaussian, strategic-quantity

Kyle setting, and explain why the market maker earns zero expected profit in both despite systematically losing to informed traders.

MM-C11. The Grossman-Stiglitz demand function $\theta(I) = \frac{E[v|I]-P}{\gamma\sigma^2(v|I)}$ shows how a CARA investor's position depends on her information I. Explain why informed traders (conditioning on the signal s) and uninformed traders (conditioning only on the price P) generally hold different positions, and how the term $\sigma^2(v|I)$ in the denominator encodes the value of information. Why does a smaller posterior variance make an informed trader willing to take a larger position?

MM-C12. Textbooks distinguish "information-based" theories of the bid-ask spread (like Glosten-Milgrom) from "inventory-based" theories. Explain the core mechanism of each: in an information model, what generates the spread, and in an inventory model, why would a dealer who accumulates a large long or short position adjust her quotes? Give one empirical prediction that could help distinguish which mechanism dominates for a given asset.

16.6.2 Quantitative

MM-Q1. A stock will be worth either $V_0 = 90$ or $V_1 = 110$ at the end of the day. The market maker's prior probability that the value is high is $\pi = 0.5$, and fraction $\mu = 0.3$ of arriving liquidity demanders are informed. Using the Glosten-Milgrom results, compute the ask A, the bid B, and the spread $A - B$. (Since $\pi = 0.5$ you may use the simplified spread formula $A - B = \mu(V_1 - V_0)$ and verify it against the full ask and bid expressions.)

MM-Q2. A stock will be worth $V_0 = 20$ or $V_1 = 80$. The market maker's prior on the high value is $\pi = 0.7$, and fraction $\mu = 0.5$ of arriving traders are informed. Compute $E[V]$ and the ask $A = \frac{\pi(1+\mu)V_1+(1-\pi)(1-\mu)V_0}{1+(2\pi-1)\mu}$. Verify that $A > E[V]$ and explain in one sentence why the ask lies above the unconditional expectation.

MM-Q3. A stock will be worth $V_0 = 40$ or $V_1 = 60$. The market maker's prior on the high value is $\pi = 0.6$, and fraction $\mu = 0.25$ of traders are informed. Compute the ask $A = \frac{\pi(1+\mu)V_1+(1-\pi)(1-\mu)V_0}{1+(2\pi-1)\mu}$, the bid $B = \frac{\pi(1-\mu)V_1+(1-\pi)(1+\mu)V_0}{1+(1-2\pi)\mu}$, and the spread $A - B$. Then confirm your spread against the closed form $A - B = \frac{4\pi(1-\pi)\mu(V_1-V_0)}{1-(2\pi-1)^2\mu^2}$.

MM-Q4. A stock will be worth $V_0 = 100$ or $V_1 = 140$, with $\pi = 0.5$ and informed fraction $\mu = 0.4$. Compute the spread using the simplified formula $A - B = \mu(V_1 - V_0)$, then find A and B individually (recall that when $\pi = 0.5$ the spread is symmetric about $E[V]$). Finally, state what the spread would be if the informed fraction rose to $\mu = 0.8$, holding everything else fixed.

MM-Q5. In a Kyle-model market the fundamental value has variance $\Sigma_0 = 16$ and the noise-trader demand has variance $\sigma_u^2 = 4$ (so $\sigma_u = 2$). Compute the equilibrium price-impact coefficient $\lambda = \frac{\sqrt{\Sigma_0}}{2\sigma_u}$. Then, if total order flow observed by the market maker is $y = 3$ and the prior mean is $p_0 = 100$ (so $\mu = p_0$), compute the price $P(y) = \mu + \lambda y$ that the market maker sets.

MM-Q6. In a Kyle market the fundamental variance is $\Sigma_0 = 36$ and the noise-trader variance is $\sigma_u^2 = 9$ (so $\sigma_u = 3$). (a) Compute λ. (b) Now suppose noise trading intensifies so that $\sigma_u^2 = 36$ (so $\sigma_u = 6$), with Σ_0 unchanged; recompute λ. (c) State whether the market became deeper or shallower and explain, in one sentence, what this means for the price impact of a large order.

MM-Q7. Consider a Kyle market with prior mean $p_0 = 50$, fundamental variance $\Sigma_0 = 25$, and noise variance $\sigma_u^2 = 100$. (a) Compute λ. (b) An insider observes $v = 60$. Using the insider's optimal demand $x = \frac{v-\mu}{2\lambda}$ with $\mu = p_0$, compute her demand x. (c) Now suppose the noise variance quadruples to $\sigma_u^2 = 400$ while Σ_0 is unchanged; recompute λ and, for the same signal $v = 60$, recompute x, commenting on whether the insider trades more or less aggressively.

MM-Q8. A Kyle market has prior mean $p_0 = 30$, fundamental variance $\Sigma_0 = 64$, and noise variance $\sigma_u^2 = 16$ (so $\sigma_u = 4$). (a) Compute λ and set $\mu = p_0$. (b) An insider observes $v = 45$ and submits her optimal order $x = \frac{v-\mu}{2\lambda}$; compute x. (c) If the realized noise-trader demand is $u = -2.5$, compute the total order flow $y = x + u$ and the price $P(y) = \mu + \lambda y$ the market maker sets, and state whether the insider bought at a price below the true value $v = 45$.

MM-Q9. In a Grossman-Stiglitz setting the fundamental has variance $\sigma_v^2 = 9$ and the informed signal $s = v + \epsilon$ has noise variance $\sigma_\epsilon^2 = 3$. Using $E[v|s] = \mu + Ms$ with $M = \frac{\sigma_v^2}{\sigma_v^2+\sigma_\epsilon^2}$, compute the updating weight M. If the prior mean is $\mu = 20$ and an informed trader observes $s = 26$, compute her conditional expectation $E[v|s]$.

MM-Q10. In a Grossman-Stiglitz market the fundamental variance is $\sigma_v^2 = 16$ and the signal noise variance is $\sigma_\epsilon^2 = 16$. (a) Compute the updating weight $M = \frac{\sigma_v^2}{\sigma_v^2+\sigma_\epsilon^2}$. (b) With prior mean $\mu = 50$ and observed signal $s = 70$, compute $E[v|s] = \mu + Ms$. (c) State, in one sentence, what happens to M as the signal becomes very noisy ($\sigma_\epsilon^2 \to \infty$) and why that makes intuitive sense.

MM-Q11. An informed trader in the Grossman-Stiglitz model has CARA risk aversion $\gamma = 2$. Conditional on her signal she believes $E[v|s] = 25$ with conditional variance $\sigma^2(v|s) = 5$, and the market price is $P = 22$. Using the CARA demand $\theta(I) = \frac{E[v|I]-P}{\gamma\sigma^2(v|I)}$, compute her optimal share demand θ. Then state, in one sentence, how her demand would change if her posterior uncertainty $\sigma^2(v|s)$ doubled.

MM-Q12. Two Grossman-Stiglitz traders face the same price $P = 24$ and the same risk aversion $\gamma = 4$, and both hold the posterior belief $E[v|I] = 30$ with conditional variance $\sigma^2(v|I) = 2$. (a) Using $\theta(I) = \frac{E[v|I]-P}{\gamma\sigma^2(v|I)}$, compute each trader's demand θ. (b) A less risk-averse trader with $\gamma = 1$ shares the same beliefs; compute her demand. (c) State, in one sentence, the general relationship between risk aversion and the size of the position an investor is willing to take on a given expected gain.

Chapter 17

Solutions to homework problems

This chapter collects worked solutions to the homework problems posed at the end of each chapter. Conceptual answers are given as model paragraphs; quantitative answers show the intermediate steps and the final numeric or derived result. Each solution is labeled with the same problem identifier used in the corresponding chapter.

17.1 Markets

MKT-C1. Equity represents *ownership*: a shareholder owns a fractional stake in the firm itself. Fixed income represents *credit*: a bondholder has lent money and holds a promise of repayment on a fixed schedule. The ordering of payment follows directly. Creditors are paid first — coupons and principal are contractual obligations that must be met before any distribution to shareholders — and the equityholder receives only what remains after employees, suppliers, and debtholders have been satisfied. That is what *residual claimant* means, and it applies to the shareholder alone. This ordering, not any difference in the underlying business, is the fundamental reason equity earns a higher average return than the same firm's debt: the same stream of business cash flows is being divided, and the equityholder has agreed to absorb the variation in it. The bondholder's claim is senior and largely insulated from ordinary fluctuations; the shareholder's claim is junior and absorbs them in full. Bearing that variation is a service, and the higher average return is its price.

MKT-C2. *Notional* value is the face amount on which a derivative's payoffs are computed. *Gross market value* is the cost of replacing every outstanding contract at current market prices — that is, what the claims created by those contracts are actually worth today. The swap example makes the gap concrete: an interest-rate swap with $100 million of notional does not oblige either party to pay $100 million. It obliges them to exchange only the *difference* between two interest payments computed on that amount, which is worth a small fraction of the notional and may be worth nothing at all if the two rates happen to coincide. Notional therefore measures the scale of promises written, not the value of claims created, and summing notionals double-counts offsetting positions besides. Gross market value belongs in the chart, because it is the only one of the two measured on the same basis as the market capitalization of stocks and the market value of bonds. Using notional would compare a face amount against market values and produce a category error.

MKT-C3. The attention mismatch has several plausible sources. Equity prices are quoted continuously, publicly, and per share, so they generate a constant stream of legible daily news, whereas most bonds trade infrequently and their price movements are conventionally reported as yields, which require interpretation and move in the opposite direction to price. Equity is also held directly by a broad retail public, while debt is concentrated in institutional hands — pension funds, insurers, banks, and central banks — that do not generate consumer news coverage. And equity returns are simply more volatile (Table 3.2), so there is more to report. Of the roughly $145 trillion in global debt securities, governments account for slightly more than half, about $75 trillion, with the remaining $70 trillion owed by corporations and securitization vehicles. The sovereign share has grown because both the 2008 financial crisis and the 2020 pandemic were met with large fiscal responses financed by heavy government issuance, and because the resulting debt stocks have been rolled over rather than retired.

MKT-C4. In a very good year the residual is large: after the fixed claims of creditors have been met, everything above them accrues to shareholders, and there is no ceiling on that amount. In a year in which the firm barely covers its obligations, the residual is near zero and shareholders receive nothing — and in bankruptcy they are wiped out entirely while creditors recover part of their claim. Residual status is thus the source of both features of the equity return distribution. It generates the higher average return because investors will not accept a junior, variable claim on the same cash flows as a senior, fixed one without compensation. And it generates the fat left tail because the residual is what absorbs bad outcomes in full: a shortfall that leaves bondholders untouched can eliminate the equity entirely. The occasional catastrophic month visible on the left of Figure 3.4 is the aggregate expression of this — losses concentrate in the junior claim.

MKT-C5. *Control* is the right, ordinarily attached to common shares in proportion to holdings, to vote for directors and on major corporate decisions; the shareholder is an owner and therefore has a say in governance. *Limited liability* means the shareholder's exposure is capped at what she paid: creditors of a failing firm may take the firm's assets but cannot pursue her personal wealth. Her downside is therefore bounded at –100% while her upside is unbounded, since the residual claim has no ceiling. The bondholder's payoff is the mirror image: the best outcome is repayment in full, exactly as promised and no more, while the worst is default and partial or total loss. So the bondholder's upside is capped and her downside is not. This is why *high-yield* corporate bond returns are strongly negatively skewed (–1.2 in Table 3.2) — a long left tail with no corresponding right tail. The asymmetry is muted for investment-grade credit (skewness +0.1), whose issuers sit far enough from the default boundary that the capped-upside effect rarely binds; there the price variation is dominated by interest rates rather than by credit, and the distribution is nearly symmetric. Equity returns are also negatively skewed, because crashes are sharper than rallies, but equity retains a genuine long right tail that bonds lack, since a successful firm's residual can grow without limit.

MKT-C6. Both facts are true because they measure different things: the *count* of listed firms and the *aggregate value* of listed firms are independent quantities. A market can contain half as many companies and still be worth several times more if the survivors have grown, and if the firms that disappeared were absorbed into them rather than liquidated. Three forces behind the decline in count: (i) waves of mergers and acquisitions, which fold two listings into one without destroying value; (ii) the growth of private capital — venture funds, growth equity, private credit — which lets companies raise very large sums without ever listing, so firms stay private longer and some never list at all; and (iii) the fixed costs and disclosure burdens of being a public company, which fall disproportionately on smaller firms and make listing unattractive at the bottom of the size distribution. A slower pace of new IPOs relative to delistings is the arithmetic through which these forces operate.

MKT-C7. An index summarizes the prices of many stocks in a single number so that its movement tracks the fortunes of a whole market rather than any one firm. *Capitalization-weighted* means each constituent enters in proportion to its total market value — shares outstanding times price — so a firm worth $2 trillion moves the index a thousand times as much as a firm worth $2 billion. Such an index of roughly five hundred large firms is a reasonable proxy for "the market" because those firms account for the bulk of total U.S. market value; the thousands of omitted small companies collectively represent a modest share of the wealth at stake. The same construction is also what the index systematically underweights: small companies. A cap-weighted index tells you almost nothing about how small firms performed, and a portfolio tracking it is far more concentrated in a handful of the largest names than the count of five hundred suggests.

MKT-C8. A return is a *scale-free* measure of performance: it expresses the gain per dollar invested, so it can be compared across a $5 stock and a $5,000 stock, or across a stock index and a bond index, without adjustment. A price change of $3 means something entirely different in those two cases; a return of 3% means the same thing in both. This is why the theory that follows — expected return, variance, covariance, and every pricing model built on them — is stated in returns. The level of an index is close to meaningless on its own because it depends entirely on the arbitrary constant chosen when the index was launched and on subsequent changes in its composition and divisor. That the S&P 500 stands at 7,000 rather than 700 or 70,000 conveys nothing about how expensive the market is or how well investors have done; only the *changes* in that level, expressed as returns, carry information.

MKT-C9. (i) The distribution is centered at a small positive number, averaging about 0.7% per month. Economically this is the risk premium: the compensation investors earn for bearing equity risk, and the reason for holding stocks at all. (ii) It is spread out, with a standard deviation of roughly 3.5% per month. This dispersion *is* the risk being compensated, and measuring it is the business of the chapters that follow. (iii) The fitted normal density describes the middle of the data reasonably well, which is why the normal is such a convenient modeling assumption throughout finance. (iv) The fit fails in the left tail: severe losses occur more often and run deeper than a normal allows, with a worst month near −20%. The failure in the left tail is what matters most to an investor because it is precisely the region in which a portfolio can be permanently impaired. Errors in the middle of the distribution average out over time; errors about the frequency of catastrophes do not, and a risk model calibrated on the well-fitting middle will understate exactly the outcomes that can end an investment program.

MKT-C10. The variation comes from *interest-rate risk*, not default risk. A note held for one month is sold at whatever yield prevails at the end of that month; if yields have risen, the fixed remaining payments are discounted more heavily and the note is worth less. Nothing about the government's willingness to pay has changed — only the rate at which its promises are discounted. The 10-year note is more volatile than the 2-year because a given change in yield is applied to a longer stream of payments, so more discounting periods are affected and the price responds more. *Duration* is the name for this sensitivity: the responsiveness of a bond's price to a change in its yield, which grows with the length of the promise. Hence the slogan: *default-free is not risk-free*. A Treasury guarantees that the promised payments will be made; it guarantees nothing about the price at which the security can be sold before those payments arrive, which is the only thing that matters to an investor who may not hold to maturity.

MKT-C11. A bond's payoff is asymmetric by construction. The best possible outcome is that every coupon and the principal are paid exactly as promised — there is no state of the world in which a performing bond pays *more* than its contract. The downside, by contrast, extends all the way to default and near-total loss. Aggregate this asymmetry over a market of borrowers and the return distribution inherits it: many months of small, similar gains as coupons accrue and spreads drift, punctuated by rare months in which defaults or a sharp repricing of default risk produce very large losses. That is negative skew with a long left tail, and it is far more pronounced for high yield than for investment grade because high-yield issuers are much closer to the boundary at which the downside becomes real. Hence "bond-like in good times, equity-like in bad times": in normal conditions high yield delivers steady coupon income like any bond, but in a downturn its losses are driven by the same business-cycle deterioration that hurts stocks. Table 3.3 corroborates this directly — high yield correlates 0.64 with equities but only 0.11 with the 10-year Treasury, so statistically it behaves far more like a stock than like a government bond.

MKT-C12. First comparison: the 10-year Treasury and investment-grade credit have nearly identical standard deviations (1.84 and 1.94 percent per month), yet they are exposed to different underlying forces — interest rates alone for the Treasury, interest rates *plus* default risk for the corporate bond. Two assets can be equally variable for entirely different reasons, and those reasons determine when the variation shows up. Second comparison: the 10-year Treasury and high yield have broadly similar standard deviations (1.84 versus 2.29), but the Treasury's distribution is roughly symmetric while high yield's has skewness of −1.2 and a worst month of −16.3% against the Treasury's −8.2%. One number cannot distinguish a symmetric distribution from one that concentrates its risk in rare catastrophes. Turning to Table 3.3, an asset's standard deviation in isolation is not the risk it contributes to a portfolio, because what matters in a portfolio is *when* an asset's bad months arrive relative to everything else held. An asset that falls when the rest of the portfolio is also falling adds far more risk than an equally volatile asset that falls at unrelated times. This is why the near-zero equity–Treasury correlation (about −0.1) makes the classic stock-and-bond portfolio work: the bad months of one are, on average, ordinary months for the other, so the combination is steadier than the volatility of the components alone would suggest.

MKT-Q1. (a) Applying $r_t = P_t/P_{t-1} - 1$:

$$r_{\text{Jan}} = \frac{4675}{4820} - 1 = -0.030083\,(-3.0083\%), \qquad r_{\text{Feb}} = \frac{4953}{4675} - 1 = 0.059465\,(5.9465\%).$$

(b) Directly over two months:

$$r_{0\to 2} = \frac{4953}{4820} - 1 = 0.027593\ (2.7593\%).$$

(c) The sum of the monthly returns is $-3.0083\% + 5.9465\% = 2.9382\%$, which exceeds the true two-month return of 2.7593% by about 18 basis points. Returns compound rather than add, because February's gain is earned on the smaller base left after January's loss. The correct relationship is multiplicative in *gross* returns:

$$1 + r_{0\to 2} = (1 + r_{\text{Jan}})(1 + r_{\text{Feb}}) = (0.969917)(1.059465) = 1.027593.$$

Expanding the product shows the discrepancy is the cross term $r_{\text{Jan}} r_{\text{Feb}} = -0.00179$, which is small for one-month returns but grows with both the horizon and the size of the returns involved.

MKT-Q2. (a) The total growth factor is $7000/17 \approx 411.8$: a dollar's worth of the index in 1950 became about \$412 by 2026. (b) Solving $17(1+g)^{76} = 7000$:

$$g = \left(\frac{7000}{17}\right)^{1/76} - 1 = (411.76)^{0.013158} - 1 \approx 0.0824\ (8.24\%\text{ per year}).$$

(c) Two reasons, and the second survives even on identical samples. First, the index level excludes *dividends*, while a total return of 8.7% includes them; the price index alone therefore understates what a shareholder actually earned by the dividend yield. Second, and more fundamentally, 8.24% here is a *geometric* (compound) average while the 8.7% in Table 3.1 is an *arithmetic* average of period returns. For any series with variation, the geometric mean is strictly below the arithmetic mean, and the gap widens with volatility — roughly half the variance, which for a 12% annual standard deviation is about 0.7 percentage points. Arithmetic averages answer "what did a typical single period return?"; geometric averages answer "what did wealth actually compound at?" Only the second is achievable over the long run.

MKT-Q3. Annualizing the monthly moments:

$$\bar{r}_{\text{ann}} = 12 \times 0.73\% = 8.76\%, \qquad \sigma_{\text{ann}} = \sqrt{12} \times 3.47\% = 3.4641 \times 3.47\% = 12.02\%,$$

reproducing the equity row of Table 3.1 (8.7% and 12.0%). The small discrepancy in the mean — 8.76% against the table's 8.7% — is an artifact of working from the rounded monthly figure: Table 3.1 was computed from the unrounded monthly mean, which is slightly below 0.73%. Carrying rounded inputs through a multiplication by 12 magnifies the rounding error twelvefold, which is worth remembering whenever you annualize from a published table. The standard deviation scales with $\sqrt{12}$ rather than 12 because *variances* add across independent periods, not standard deviations. If monthly returns are uncorrelated with one another, the variance of a 12-month sum is $12\sigma^2$, so its standard deviation is $\sqrt{12}\,\sigma$. Means, being linear, simply add. The practical consequence is important: as the horizon lengthens, expected return grows proportionally with time while risk grows only with its square root, so the ratio of return to risk *improves* with horizon — the formal core of the argument that long-horizon investors can bear more equity risk. Note the assumption doing the work: if monthly returns were positively autocorrelated, the $\sqrt{12}$ rule would understate annual volatility.

MKT-Q4. Annualizing as in MKT-Q3:

Asset	Ann. mean	Ann. std. dev.	Ratio
2-year Treasury	$12(0.42) = 5.04\%$	$\sqrt{12}(0.74) = 2.56\%$	1.97
High-yield corporate	$12(0.64) = 7.68\%$	$\sqrt{12}(2.29) = 7.93\%$	0.97
U.S. equities	$12(0.73) = 8.76\%$	$\sqrt{12}(3.47) = 12.02\%$	0.73

The 2-year Treasury has by far the highest ratio of return to risk, roughly twice high yield's and nearly three times equities'. The chapter's caution is that this is an artifact of the particular history sampled: the series begins in 1976, near the 1981 peak in U.S. interest rates, and then rides yields downward for four decades. Falling yields generate capital gains on bonds, so the sample captures a long, essentially non-repeatable tailwind. Averages computed over particular histories are evidence, not guarantees, and a short-Treasury allocation should not be expected to deliver twice the risk-adjusted return of high yield going forward.

MKT-Q5. (a) The standardized distance is

$$z = \frac{-10 - 0.73}{3.47} = -3.092.$$

A normal distribution assigns probability $\Phi(-3.092) \approx 0.00099$, or about one month in a thousand. Over $n = 917$ months this predicts $917 \times 0.00099 \approx 0.9$ months — that is, roughly one month worse than −10% in the entire 76-year sample. (b) For the actual worst month:

$$z = \frac{-20.4 - 0.73}{3.47} = -6.089,$$

with normal probability about 5.7×10^{-10} — one month in roughly 1.8 billion, or once every 150 million years. (c) The normal approximation fails catastrophically in the tail. A month that actually occurred once in 917 observations is one the model regards as effectively impossible on any timescale relevant to the universe, and even the milder –10% threshold has been breached more often than the predicted single occurrence. The distribution of equity returns has *fat tails*: extreme outcomes are both more frequent and more severe than the normal allows. Since the normal fits the middle of the data well, the danger is specific and insidious — a risk model calibrated on ordinary months will look accurate right up until the month that matters.

MKT-Q6. (a) The standardized worst month is

$$z = \frac{-16.3 - 0.64}{2.29} = -7.397,$$

confirming the chapter's "roughly seven standard deviations." (b) A normal distribution assigns a -7.4σ event probability about 6.9×10^{-14}, or once in roughly 14 trillion months. It was observed once in 478 months. The model understates the frequency of this outcome by something on the order of 10^{10} — ten billion-fold. (c) An investor sizing positions by standard deviation alone would have concluded that high yield was only modestly riskier than investment-grade credit (2.29% versus 1.94% per month, about 18% more volatile) and would have sized her high-yield position accordingly. That comparison is accurate for ordinary months and badly wrong for the months that determine survival: October 2008 delivered a loss of 16.3% against investment grade's worst of 7.5%, more than twice as deep. Standard deviation is a symmetric measure and therefore cannot see the asymmetry — the capped upside and open-ended downside — that is the defining economic feature of a credit-risky bond. The skewness (–1.2) and excess kurtosis (10.0) in Table 3.2 are precisely the warning that the single volatility number omits.

MKT-Q7. (a) The total is 126.7 + 75.5 + 69.6 + 17.6 = $289.4 trillion, and the shares are:

Class	Value ($T)	Share
Equities	126.7	43.8%
Sovereign debt	75.5	26.1%
Corporate debt	69.6	24.0%

Class	Value ($T)	Share
Derivatives (gross market value)	17.6	6.1%

Note that sovereign and corporate debt together are 50.1% of the total, slightly more than equities — the chapter's point that debt is the larger market. (b) Substituting the $700 trillion notional for the $17.6 trillion gross market value gives a total of 126.7 + 75.5 + 69.6 + 700 = $971.8 trillion, of which derivatives are 700/971.8 = 72.0%. (c) The same market moves from a 6% sliver to nearly three-quarters of the entire chart purely through a change of measurement convention. A reader shown the second version would conclude that stocks and bonds are a rounding error beside the derivatives market and that the financial system is overwhelmingly composed of derivative exposure. Both conclusions are artifacts. Notional value measures the face amounts referenced by contracts, not the value of the claims those contracts create, and it is not commensurable with the market values used for the other three slices. Version (a) is the honest chart.

MKT-Q8. (a) With $w = 0.5$, $\sigma_E = 3.47$, $\sigma_B = 1.84$, $\rho = -0.09$:

$$\sigma_P^2 = (0.25)(3.47)^2 + (0.25)(1.84)^2 + 2(0.5)(0.5)(-0.09)(3.47)(1.84) = 3.0102 + 0.8464 - 0.2873 = 3.5693,$$

so $\sigma_P = \sqrt{3.5693} = 1.889\%$ per month. (b) The weighted average of the two standard deviations is $0.5(3.47) + 0.5(1.84) = 2.655\%$. The portfolio's actual standard deviation of 1.889% falls short of this by 0.766 percentage points — nearly 29% lower. The shortfall is diversification. The weighted average would be the correct answer only if the two assets moved in lockstep ($\rho = 1$), so that every bad month for equities were also a bad month for Treasuries. Because $\rho = -0.09$, the two assets' fluctuations are essentially unrelated and partially offset one another within the portfolio, and the offsetting is what removes risk without removing expected return. (c) With $\rho = 0.9$:

$$\sigma_P^2 = 3.0102 + 0.8464 + 2(0.25)(0.9)(3.47)(1.84) = 3.0102 + 0.8464 + 2.8727 = 6.7293, \qquad \sigma_P = 2.594\%.$$

The diversification benefit collapses from 0.766 percentage points to 0.061

— a 92% reduction. As $\rho \to 1$ the benefit vanishes entirely and σ_P converges to the weighted average 2.655%: assets that always move together cannot hedge one another, and combining them merely averages their risks rather than cancelling any of it. Low correlation, not low volatility, is what makes diversification work.

17.2 Mean-Variance Portfolios

MV-C1. Because both investors use the *same* CAL, their complete portfolios lie on the single line $Er_C = r_f + \frac{\sigma_C}{\sigma_P}(Er_P - r_f)$; each investor simply picks a point on it. The point is pinned down by the standard deviation $\sigma_C = y\sigma_P$, so a larger σ_C means a larger risky fraction y. Ben, with $\sigma_C = 0.15 > 0.05 = \sigma_A$, holds more of the risky asset and sits farther up and to the right on the CAL, so Ana is the more risk-averse investor: for a mean-variance utility $U = Er_C - \frac{1}{2}A\sigma_C^2$, a higher A lowers the optimal y and hence σ_C. Crucially, the *composition* of the risky portion is identical for both — both hold the same risky portfolio P and differ only in how much of their wealth they put in it versus the risk-free asset. This is the separation property made visible: the CAL fixes the risky portfolio for everyone, and risk aversion only determines the position along the line.

MV-C2. The CAL slope $\frac{Er_P - r_f}{\sigma_P}$ — the Sharpe ratio — is determined entirely by the properties of the underlying risky portfolio and the risk-free rate, not by the broker. For the brokerage to genuinely offer a steeper line *using the same risky portfolio*, it would need a different (lower) r_f at which clients can borrow or lend, or it would have to change Er_P or σ_P — but by assumption the risky portfolio is the same, so Er_P and σ_P are fixed. An ordinary broker cannot make the line steeper at will because the slope reflects the market's actual reward-per-unit-of-risk tradeoff, which is a property of the assets, not a marketing choice. Economically, the slope is exactly that reward: the extra expected return the market grants per unit of standard deviation borne. A "steeper CAL" that is not backed by a better risky portfolio or a better risk-free rate is not a free lunch — it is either a repackaging or a claim of superior portfolio selection (a better tangency portfolio).

MV-C3. By $y = \frac{Er_P - r_f}{A\sigma_P^2}$, doubling the risk premium (with A and σ_P fixed) doubles the optimal y. Starting from $y = 1$, the investor should move toward $y = 2$ — i.e., borrow at the risk-free rate to lever up her position in the risky portfolio, moving up and to the right along the CAL past the point P. Both a very risk-averse and a nearly risk-neutral investor respond in the *same direction* (both increase y), because y is proportional to the risk premium for every $A > 0$. They differ only in *magnitude*: the near-risk-neutral investor (small A) has a large y that increases by a large absolute amount, while the very risk-averse investor (large A) has a small y that increases by a smaller absolute amount. The sign of the response is universal; the size scales inversely with risk aversion.

MV-C4. In $y = \frac{Er_P - r_f}{A\sigma_P^2}$, risk aversion A and variance σ_P^2 both sit in the denominator. A higher A means the investor penalizes each unit of portfolio variance more heavily in $U = Er_C - \frac{1}{2}A\sigma_C^2$, so she demands less risk exposure and cuts y. A lower σ_P^2 means each unit of y buys less risk (since $\sigma_C = y\sigma_P$), so for the same risk premium she can hold more of the risky asset, raising y. For two investors with the same Er_P, σ_P, r_f but $A_2 = 2A_1$, the more risk-averse one holds exactly half the risky fraction: $y_2 = y_1/2$, placing her lower and to the left on the CAL. But the difference in A changes only *how much* of the risky portfolio each holds; it does not change *which* risky portfolio — both hold the identical tangency portfolio, so composition is unaffected by risk aversion (separation).

MV-C5. Portfolio variance is $\sigma_P^2 = w_D^2\sigma_D^2 + w_E^2\sigma_E^2 + 2w_D w_E \sigma_D \sigma_E \rho_{DE}$; the cross term carries the correlation. Asset D has strongly negative correlation with E, so adding it drives the covariance term negative and pulls total portfolio variance down — often more than enough to offset D's slightly higher own variance. Asset F, with the same variance as E but *positive* correlation, adds a positive cross term and reduces the risk-reduction available. What matters for portfolio risk is not an asset's own variance in isolation but its covariance with the rest of the portfolio; a modestly volatile asset that moves opposite the rest is more valuable than a same-variance asset that moves with the rest. An asset like D, with negative correlation to the other holdings, is called a *hedge*.

MV-C6. Holding $w_D, w_E, \sigma_D, \sigma_E$ fixed, only the cross term $2w_D w_E \sigma_D \sigma_E \rho_{DE}$ varies with ρ_{DE}, and it is strictly increasing in ρ_{DE}. So portfolio variance is strictly *decreasing* in correlation: it is largest at $\rho_{DE} = 1$ and smallest at $\rho_{DE} = -1$. At $\rho_{DE} = 1$ the expression becomes the perfect square $(w_D\sigma_D + w_E\sigma_E)^2$, so $\sigma_P = w_D\sigma_D + w_E\sigma_E$ — exactly the weighted average of the individual standard deviations, meaning no risk reduction at all. For any $\rho_{DE} < 1$ the cross term is smaller, so σ_P is strictly *below* that weighted average. This reveals that the diversification benefit comes entirely from imperfect correlation: it is the fact that assets do not move in lockstep, not the mere act of holding several assets, that lowers risk below the average of the parts.

MV-C7. The number of holdings is irrelevant; what matters is how they co-move. Writing $\sigma_P^2 = \sum_j w_j\, Cov(r_j, r_P)$, total risk is the weight-averaged covariance of each holding with the whole portfolio. If 500 stocks are all strongly positively correlated — driven by common factors — then each $Cov(r_j, r_P)$ is large and positive, and the sum stays large despite the many names. Diversification requires *low* covariances, not a *high* count. This is exactly the Magnificent Seven point: once a handful of highly correlated technology names dominate the index, the "500-stock" fund behaves like a concentrated bet, because the covariances beneath it are large. Genuine diversification is a property of the covariance structure, not the number of tickers.

MV-C8. $Cov(r_j, r_P) = \sum_i w_i Cov(r_i, r_j)$ measures how asset j's return moves together with the return of the whole portfolio it sits in. It is the right notion of the risk asset j *contributes* because total portfolio variance decomposes exactly as $\sum_j w_j\, Cov(r_j, r_P) = Cov(r_P, r_P) = \sigma_P^2$: each asset's contribution to σ_P^2 is $w_j\, Cov(r_j, r_P)$, not $w_j^2\sigma_j^2$. An asset's own variance σ_j^2 overstates its marginal risk contribution if it moves opposite the rest of the portfolio, because part of that variance is offset by negative covariances with other holdings. An asset with low $Cov(r_j, r_P)$ contributes little to total risk; one with *negative* $Cov(r_j, r_P)$ actually *reduces* σ_P^2, which is why hedge assets are prized even when volatile on their own.

MV-C9. The minimum-variance frontier gives, for each target return $\bar{r}$, the lowest attainable variance. A portfolio strictly inside the frontier has more variance than some frontier portfolio at the same expected return (and less expected return than some frontier portfolio at the same variance). It is "dominated" because a mean-variance investor, who likes higher Er and dislikes higher σ, can be made strictly better off. Two distinct improvements are available by moving to the frontier: (1) hold expected return fixed and slide left to the frontier, lowering variance for the same return; or (2) hold variance fixed and slide up to the frontier, raising expected return for the same risk. Because at least one of these strict improvements is always possible, no rational mean-variance investor holds an interior portfolio.

MV-C10. From $\sigma_P^2 = \frac{C\bar{r}^2 - 2A\bar{r} + B}{D}$, variance is a parabola in $\bar{r}$ opening upward with vertex at $\bar{r}^* = A/C$. Because a parabola is symmetric about its vertex, two target returns equidistant from $\bar{r}^*$ — one above, one below — yield the *same* variance but different expected returns. A mean-variance investor always prefers the one with the *higher* expected return, since more expected return at equal variance is strictly better. Hence only the branch with $\bar{r} \geq \bar{r}^* = A/C$ is ever chosen; portfolios below the vertex are dominated by their mirror images above it. The acceptable upper half of the frontier is called the *efficient frontier*.

MV-C11. The skeptic is right that the retiree and the 25-year-old hold different *complete portfolios* — but that is fully consistent with separation. Separation says the composition of the *risky* portion is the same for both: both hold the identical tangency (or minimum-variance-frontier) risky portfolio. Where they differ is in the *split* between that risky portfolio and the risk-free asset, i.e., the fraction $y = \frac{Er_P - r_f}{A\sigma_P^2}$. The cautious retiree (high A) holds a small y — mostly risk-free — while the aggressive youth (low A) holds a large y, possibly levered. The difference shows up in the risk-free/risky mix; it does *not* show up in the internal makeup of the risky holding, which is common to all investors who agree on the inputs.

MV-C12. Because every frontier portfolio can be written $w(\bar{r}) = g + \bar{r}\,h$ with g, h fixed vectors, the entire frontier is a one-parameter straight line in weight space indexed by $\bar{r}$. Pick any two distinct frontier portfolios, say $w(\bar{r}_1)$ and $w(\bar{r}_2)$; any other frontier portfolio $w(\bar{r})$ is an affine combination of these two (choosing the mixing coefficient so the target returns line up), because affine combinations of points on a line stay on the line. Thus two fixed funds span the whole frontier — the two-fund separation result. When a risk-free asset is added, the efficient set collapses to the single straight CAL from r_f to the tangency portfolio; the only risky fund anyone needs is that tangency portfolio, so the "two funds" become the risk-free asset plus one common risky portfolio, identical across investors.

MV-C13. The weights $w^* = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}'\Sigma^{-1}\mathbf{1}}$ solve $\min w'\Sigma w$ s.t. $w'\mathbf{1} = 1$ with *no* return constraint imposed — the objective is pure variance and the only side condition is that weights sum to one. Since neither the objective (Σ) nor the constraint ($\mathbf{1}$) involves μ, expected returns never enter, and the answer depends solely on the covariance structure. Intuitively, if you only want the least-risky portfolio and do not care what return you earn, expected returns are simply irrelevant to the problem. A practitioner might deliberately hold w^* when expected-return estimates are too unreliable to trust: covariances are estimated far more precisely than means, so a portfolio that ignores μ entirely can be more robust, accepting lower expected return in exchange for stability and minimal risk.

MV-C14. The weights are proportional to $\Sigma^{-1}\mathbf{1}$. Loosely, Σ^{-1} "downweights" directions of high variance and high covariance: assets that are volatile or strongly co-move with others receive smaller entries in $\Sigma^{-1}\mathbf{1}$, while low-variance, low-covariance assets receive larger ones, because putting weight on the latter reduces total variance most effectively. (In the diagonal, uncorrelated case this is exact: weights are proportional to $1/\sigma_i^2$.) In practice, sample expected returns are extremely noisy, and portfolio rules that target a particular $\bar{r}$ amplify that noise, producing unstable, extreme weights. Because w^* uses only Σ — which is estimated much more accurately — it sidesteps the worst estimation errors, giving stable, well-behaved weights. That robustness is why some practitioners favor the global minimum-variance portfolio even though it makes no attempt to earn a high expected return.

MV-Q1. (a) The CAL is $Er_C = r_f + \frac{\sigma_C}{\sigma_P}(Er_P - r_f) = 0.03 + \frac{0.11-0.03}{0.20}\,\sigma_C = 0.03 + 0.4\,\sigma_C$. (b) The Sharpe ratio is the slope, $\frac{Er_P - r_f}{\sigma_P} = \frac{0.08}{0.20} = 0.4$. (c) At $\sigma_C = 0.08$: $Er_C = 0.03 + 0.4(0.08) = 0.03 + 0.032 = 0.062$, i.e. 6.2%.

MV-Q2. (a) Solve $Er_C = r_f + \frac{\sigma_C}{\sigma_P}(Er_P - r_f)$ for σ_C: $0.06 = 0.02 + \frac{\sigma_C}{0.25}(0.10 - 0.02) = 0.02 + 0.32\,\sigma_C$, so $\sigma_C = \frac{0.04}{0.32} = 0.125$, i.e. 12.5%. (b) Since $\sigma_C = y\sigma_P$, $y = \frac{\sigma_C}{\sigma_P} = \frac{0.125}{0.25} = 0.5$. She holds half her wealth in the risky portfolio.

MV-Q3. (a) $y = \frac{Er_P - r_f}{A\sigma_P^2} = \frac{0.12-0.04}{4\times(0.20)^2} = \frac{0.08}{4\times 0.04} = \frac{0.08}{0.16} = 0.5$. (b) $Er_C = r_f + y(Er_P - r_f) = 0.04 + 0.5(0.08) = 0.08$, i.e. 8%; $\sigma_C = y\sigma_P = 0.5 \times 0.20 = 0.10$, i.e. 10%. (c) With $A = 2$: $y = \frac{0.08}{2\times 0.04} = \frac{0.08}{0.08} = 1.0$. Halving risk aversion doubles the risky fraction (from 0.5 to 1.0); the less risk-averse investor holds the entire portfolio in the risky asset, sitting farther out on the CAL.

MV-Q4. (a) From $y = \frac{Er_P - r_f}{A\sigma_P^2}$: $0.5 = \frac{Er_P - r_f}{3\times(0.20)^2} = \frac{Er_P - r_f}{3\times 0.04} = \frac{Er_P - r_f}{0.12}$, so $Er_P - r_f = 0.5 \times 0.12 = 0.06$, a risk premium of 6%. (b) With $\sigma_P = 0.25$: $y = \frac{0.06}{3\times(0.25)^2} = \frac{0.06}{3\times 0.0625} = \frac{0.06}{0.1875} = 0.32$. The optimal risky fraction falls from 0.5 to 0.32 because higher portfolio variance makes each unit of y riskier, so the investor reduces her exposure.

MV-Q5. $\sigma_P^2 = (0.4)^2(0.30)^2 + (0.6)^2(0.20)^2 + 2(0.4)(0.6)(0.30)(0.20)(0.25)$. Term by term: $0.16 \times 0.09 = 0.0144$; $0.36 \times 0.04 = 0.0144$; cross term $= 2 \times 0.24 \times 0.06 \times 0.25 = 2 \times 0.0036 = 0.0072$. So $\sigma_P^2 = 0.0144 + 0.0144 + 0.0072 = 0.0360$, and $\sigma_P = \sqrt{0.036} = 0.1897 \approx 0.19$. The weighted average of the standard deviations is $0.4(0.30) + 0.6(0.20) = 0.12 + 0.12 = 0.24$. The portfolio's $\sigma_P \approx 0.19$ is well below 0.24; the gap (≈ 0.05) is the diversification benefit, arising because $\rho_{DE} = 0.25 < 1$.

MV-Q6. With $\rho_{DE} = -1$, $\sigma_P = 0$ requires $w_D\sigma_D - w_E\sigma_E = 0$, i.e. $w_D(0.40) = w_E(0.10)$, so $w_D = 0.25\,w_E$. Combined with $w_D + w_E = 1$: $0.25w_E + w_E = 1 \Rightarrow 1.25\,w_E = 1 \Rightarrow w_E = 0.8$, $w_D = 0.2$. Verification: $w_D\sigma_D - w_E\sigma_E = 0.2(0.40) - 0.8(0.10) = 0.08 - 0.08 = 0$, so $\sigma_P^2 = 0^2 = 0$. The portfolio is risk-free.

MV-Q7. (a) $D = BC - A^2 = (0.30)(20) - (2.0)^2 = 6.0 - 4.0 = 2.0$. (b) $\sigma_P^2 = \frac{C\bar{r}^2 - 2A\bar{r} + B}{D} = \frac{20(0.15)^2 - 2(2.0)(0.15) + 0.30}{2.0} = \frac{20(0.0225) - 0.60 + 0.30}{2.0} = \frac{0.45 - 0.60 + 0.30}{2.0} = \frac{0.15}{2.0} = 0.075$. (c) $\bar{r}^* = A/C = 2.0/20 = 0.10$; $\sigma^2_{min} = 1/C = 1/20 = 0.05$.

MV-Q8. (a) $D = BC - A^2 = (0.20)(8) - (1.2)^2 = 1.6 - 1.44 = 0.16$. (b) At $\bar{r} = 0.10$: $\sigma_P^2 = \frac{8(0.10)^2 - 2(1.2)(0.10) + 0.20}{0.16} = \frac{0.08 - 0.24 + 0.20}{0.16} = \frac{0.04}{0.16} = 0.25$. At $\bar{r} = 0.20$: $\sigma_P^2 = \frac{8(0.20)^2 - 2(1.2)(0.20) + 0.20}{0.16} = \frac{0.32 - 0.48 + 0.20}{0.16} = \frac{0.04}{0.16} = 0.25$. (c) Both give 0.25 because $\bar{r}^* = A/C = 1.2/8 = 0.15$ is exactly midway, so 0.10 and 0.20 are symmetric about the vertex and yield equal variance. At $\bar{r}^* = 0.15$: $\sigma_P^2 = \frac{8(0.0225) - 2(1.2)(0.15) + 0.20}{0.16} = \frac{0.18 - 0.36 + 0.20}{0.16} = \frac{0.02}{0.16} = 0.125 = 1/8 = 1/C$, confirming the minimum equals $1/C$.

MV-Q9. $\Sigma^{-1} = \frac{1}{\det \Sigma} \begin{pmatrix} 0.09 & -0.01 \\ -0.01 & 0.04 \end{pmatrix}$ with $\det \Sigma = (0.04)(0.09) - (0.01)^2 = 0.0036 - 0.0001 = 0.0035$. Then $\Sigma^{-1}\mathbf{1} = \frac{1}{0.0035} \begin{pmatrix} 0.09 - 0.01 \\ -0.01 + 0.04 \end{pmatrix} = \frac{1}{0.0035} \begin{pmatrix} 0.08 \\ 0.03 \end{pmatrix}$. The sum of these entries is $\mathbf{1}'\Sigma^{-1}\mathbf{1} = \frac{0.08+0.03}{0.0035} = \frac{0.11}{0.0035} = 31.43 = C$. Normalizing, $w^* = \frac{1}{0.11} \begin{pmatrix} 0.08 \\ 0.03 \end{pmatrix} = \begin{pmatrix} 0.727 \\ 0.273 \end{pmatrix}$. Minimum variance $= 1/C = 0.0035/0.11 = 0.0318$. Asset 1 gets the larger weight (≈ 0.73) because it has the lower variance (0.04 vs. 0.09); the min-variance portfolio tilts toward the less volatile asset.

MV-Q10. For a diagonal Σ, $\Sigma^{-1} = \begin{pmatrix} 1/0.01 & 0 \\ 0 & 1/0.04 \end{pmatrix} = \begin{pmatrix} 100 & 0 \\ 0 & 25 \end{pmatrix}$. Then $\Sigma^{-1}\mathbf{1} = \begin{pmatrix} 100 \\ 25 \end{pmatrix}$ and $\mathbf{1}'\Sigma^{-1}\mathbf{1} = 100 + 25 = 125 = C$. So $w^* = \frac{1}{125} \begin{pmatrix} 100 \\ 25 \end{pmatrix} = \begin{pmatrix} 0.8 \\ 0.2 \end{pmatrix}$, and the minimum variance is $1/C = 1/125 = 0.008$. The weights $0.8 : 0.2$ are in ratio $100 : 25 = (1/0.01) : (1/0.04)$, i.e. inversely proportional to the variances. This makes sense: with no correlation, the only way to cut risk is to load up on the lower-variance asset, so the safer asset (variance 0.01) gets four times the weight of the riskier one (variance 0.04).

MV-D1. (a) Mean: $0.5(14.3\%) + 0.5(1.95\%) = 8.13\%$. Variance: $0.25(0.149)^2 + 0.25(0.070)^2 + 2(0.5)(0.5)(0.149)(0.070)(-0.03) = 0.005550 + 0.001225 - 0.000156 = 0.006619$, so $\sigma_P = 8.14\%$. (b) With $\rho = +1$: $\sigma_P = 0.5(14.9\%) + 0.5(7.0\%) = 10.95\%$. Diversification cuts the 50/50 portfolio's risk by about 2.8 percentage points — the horizontal gap between the chord and the curve in the figure.

MV-D2. Numerator: $\sigma_E^2 - \sigma_D\sigma_E\rho = 0.0049 - (0.149)(0.070)(-0.03) = 0.0049 + 0.000313 = 0.005213$. Denominator: $0.022201 + 0.0049 + 2(0.000313) = 0.027740$. So $w_D^* = 0.1879 \approx 19\%$ in stocks, matching the figure. The minimum-variance portfolio holds stocks because, with $\rho \approx 0$, a small stock allocation adds almost no covariance risk while its own variance enters with weight w_D^2 — a second-order cost — so the first small holdings of the volatile asset actually *reduce* total risk relative to holding bonds alone.

17.3 The Capital Asset Pricing Model

CAPM-C1. Because both investors have homogeneous expectations and are mean-variance optimizers, they face identical inputs (μ, Σ, r_f) and solve the same optimization problem. The first-order condition gives risky weights proportional to $\Sigma^{-1}(\mu - r_f\mathbf{1})$ for every investor; the only thing that differs across them is the scalar $1/A_i$ that multiplies this common vector. Normalizing to unit weight, both investors hold the identical tangency portfolio w_T, so the *composition* of the risky part is the same. Each investor expresses her risk preference solely through the fraction y of wealth she allocates to that risky portfolio versus the risk-free asset: the cautious retiree chooses a small y (much in the risk-free asset), while the aggressive professional chooses a large y (possibly borrowing, $y > 1$). The risk-free asset is the "dial" that scales overall risk without changing the mix of risky holdings — this is the separation property.

CAPM-C2. Every investor holds the same risky portfolio with the same weights on each asset. Since all assets are publicly traded and must be held by someone, aggregating across all investors means the common weights must equal the value weights of the assets in the market as a whole. Concretely, if every investor puts, say, 20% of her risky wealth in asset 1, then across all investors 20% of aggregate risky wealth is in asset 1, so asset 1 constitutes 20% of the market's value. Hence the common risky portfolio must be the value-weighted market portfolio. If the common risky portfolio placed zero weight on some traded asset, then no one would hold it, yet the asset exists and must be held — a contradiction of market clearing. Its price would have to fall until its expected return rose enough for investors to want to hold it, restoring positive weight.

CAPM-C3. With homogeneous expectations, all investors agree on (μ, Σ) and therefore compute the same tangency portfolio, which market clearing forces to be the market portfolio. If a group of investors instead believes technology stocks have a higher variance (or covariances) than everyone else, that group solves a *different* mean-variance problem: their Σ differs, so their $\Sigma^{-1}(\mu - r_f\mathbf{1})$ differs, and they choose a different-composition risky portfolio — typically underweighting tech relative to the rest of the market. Because investors now hold different risky portfolios, there is no single portfolio that is simultaneously tangency-optimal for everyone. The aggregate of these differing portfolios is still the market (all assets are held), but the market portfolio is no longer the tangency portfolio for any given investor, and the clean single-beta CAPM pricing relation breaks down.

CAPM-C4. Mean-variance optimization (Assumption 5) means every investor cares only about the mean and variance of her portfolio return, so her demand is fully summarized by the first-order condition $w_i^* = \frac{1}{A_i}\Sigma^{-1}(\mu - r_f\mathbf{1})$; the *form* of the solution is common to all. Homogeneous expectations (Assumption 6) means the inputs μ and Σ in that solution are literally the same numbers for everyone, so the direction $\Sigma^{-1}(\mu - r_f\mathbf{1})$ — and hence the composition of the risky portfolio — is identical across investors. Market clearing then forces this common tangency portfolio to be the market portfolio. If investors instead disagreed about expected returns (different μ_i), they would compute different directions $\Sigma^{-1}(\mu_i - r_f\mathbf{1})$ and hold different-composition risky portfolios; no single portfolio would be tangency-optimal for all, and the identification of the market portfolio with the tangency portfolio would fail.

CAPM-C5. In equilibrium the expected excess return on any asset is $Er_i - r_f = \beta_i(Er_M - r_f)$ with $\beta_i = Cov(r_i, r_M)/\sigma_M^2$. An asset's *standalone* variance σ_i^2 can be decomposed as $\sigma_i^2 = \beta_i^2\sigma_M^2 + \sigma_{\epsilon_i}^2$; the idiosyncratic piece $\sigma_{\epsilon_i}^2$ can be diversified away for free and earns no reward, so only beta is priced. Asset A, with $\beta_A = 1.4$, contributes more to the variance of the market portfolio — the only portfolio anyone holds — than asset B with $\beta_B = 0.3$, so A must offer the higher expected return to be held willingly. The fact that A and B have equal σ_i is irrelevant: the CAPM prices the *systematic* contribution to aggregate risk (beta), not total variance, because a diversified investor bears only the systematic part.

CAPM-C6. Beta, $\beta_i = Cov(r_i, r_M)/\sigma_M^2$, measures how much an asset moves *with* the market and hence how much it adds to the variance of the market portfolio — the only portfolio a diversified investor holds. A stock can have very high standalone volatility σ_i yet low beta if most of that volatility is idiosyncratic (uncorrelated with the market). Because idiosyncratic risk diversifies away at no cost, the market pays nothing for it, so a high-σ_i but low-β_i stock does *not* command a large premium. If the stock had a beta of exactly zero, its covariance with the market is zero, it adds nothing to market variance, and in equilibrium it earns only the risk-free rate — despite its large standalone volatility. The advisor's error is confusing total variance with systematic (priced) risk.

CAPM-C7. The equilibrium condition $Er_M - r_f = \bar{A}\,\sigma_M^2$ says the market risk premium is the product of the average (wealth-weighted harmonic mean) risk aversion $\bar{A}$ and market variance σ_M^2. Holding σ_M^2 fixed, if investors collectively become more risk-averse during a panic — each A_i rises, so $\bar{A}$ rises — the equilibrium premium $\bar{A}\,\sigma_M^2$ rises. Intuitively, more risk-averse investors demand a larger reward per unit of market variance to be induced to hold the same aggregate risky supply; since supply is fixed, prices fall and expected returns (the premium) rise until markets clear. The mechanism is entirely on the risk-aversion side, so no change in σ_M^2 is required.

CAPM-C8. Each investor i places dollars $W_i y_i$ into the risky market portfolio and $W_i(1-y_i)$ into the risk-free asset. Because any borrowing by one investor (negative risk-free holding) must be matched by lending from another, the risk-free asset is in zero net supply: $\sum_i W_i(1 - y_i) = 0$. This is equivalent to $\sum_i W_i y_i = W$, i.e. the wealth-weighted average of y equals one, $\sum_i (W_i/W) y_i =$

1. Equivalently, all aggregate wealth must be held in real (risky) assets because that is the only net supply that exists. Substituting the individual demands $y_i = (Er_M - r_f)/(A_i\sigma_M^2)$ into this clearing condition and solving is exactly what pins down the market risk premium $Er_M - r_f = \bar{A}\sigma_M^2$.

CAPM-C9. Writing $r_i = \alpha_i + \beta_i r_M + \epsilon_i$ and taking variances gives $\sigma_i^2 = \beta_i^2\sigma_M^2 + \sigma_{\epsilon_i}^2$. The *systematic* term $\beta_i^2\sigma_M^2$ is the risk that comes from the asset's exposure to economy-wide movements — it is shared with the whole market and cannot be escaped by diversification. The *idiosyncratic* term $\sigma_{\epsilon_i}^2$ is firm-specific: it reflects events affecting this asset alone and is uncorrelated with the market. A concrete example that would appear in ϵ_i but not in r_M is a company-specific event such as a product recall, a failed drug trial, a lawsuit, or a surprise CEO resignation at that one firm — such news moves that stock without moving the broad market. By contrast, a change in interest rates or a recession moves r_M and is systematic.

CAPM-C10. The residual ϵ_i is defined as the part of r_i left over after projecting on the market: it is the regression residual, and by construction of a regression it is uncorrelated with the regressor, $Cov(\epsilon_i, r_M) = 0$, with $E\epsilon_i = 0$. When we take the variance of $r_i = \alpha_i + \beta_i r_M + \epsilon_i$, the cross term is $2\beta_i Cov(r_M, \epsilon_i)$, which is exactly zero by that construction. That is why the variance splits cleanly into $\sigma_i^2 = \beta_i^2\sigma_M^2 + \sigma_{\epsilon_i}^2$ with no cross term. The first term, $\beta_i^2\sigma_M^2$, captures the asset's exposure to economy-wide (market) movements; the second, $\sigma_{\epsilon_i}^2$, captures firm-specific events uncorrelated with the market.

CAPM-C11. For an equally weighted portfolio of n uncorrelated-residual assets, the idiosyncratic variance is $Var\left(\frac{1}{n}\sum\epsilon_i\right) = \frac{1}{n^2}\sum\sigma_{\epsilon_i}^2 \le \bar{\sigma}_\epsilon^2/n$, which shrinks toward zero as n grows. With only three stocks, $n = 3$ is small, so the firm-specific portion of the variance is large and the portfolio swings on idiosyncratic news. Adding many more uncorrelated stocks drives $\bar{\sigma}_\epsilon^2/n \to 0$ at essentially no cost (no expected-return sacrifice, since idiosyncratic risk is unpriced). What remains no matter how many stocks she adds is the systematic variance $\bar{\beta}^2\sigma_M^2$, the market-risk portion, which does not diversify away because it is shared across all assets.

CAPM-C12. In $r_P = \bar{\alpha} + \bar{\beta}\, r_M + \frac{1}{n}\sum\epsilon_i$, the systematic loading is the *average* beta

$\bar{\beta} = \frac{1}{n}\sum \beta_i$, which stays of order one as n grows, so the systematic variance $\bar{\beta}^2\sigma_M^2$ does not shrink. The idiosyncratic term is an average of uncorrelated shocks, whose variance falls like $1/n$. The fundamental difference is correlation: idiosyncratic shocks are uncorrelated across assets, so averaging them causes cancellation (the law-of-large-numbers effect), whereas the market factor r_M hits *every* asset in the same direction and cannot cancel. Systematic risk is common risk that everyone bears together; there is no free lunch because it cannot be netted out by combining assets. Only idiosyncratic risk is "free" to eliminate.

CAPM-C13. Suppose an asset offered a positive premium as compensation for its idiosyncratic risk $\sigma_{\epsilon_i}^2$. Investors would recognize that this firm-specific risk can be diversified away at no cost by holding the asset inside a large well-diversified portfolio. They would rush to buy the asset to capture the "free" extra return without bearing extra undiversifiable risk. This buying pressure raises the asset's price and lowers its expected return, and it continues until the idiosyncratic premium is competed down to zero. In equilibrium, then, only systematic risk — the part that survives in every diversified portfolio and cannot be escaped — is compensated. This is precisely why the Security Market Line prices assets by beta rather than by total variance.

CAPM-C14. By the Security Market Line, $Er_i - r_f = \beta_i(Er_M - r_f)$. With $\beta_i = 0$, the asset earns only the risk-free rate, $Er_i = r_f$, regardless of how large its standalone standard deviation is. The reason is the decomposition $\sigma_i^2 = \beta_i^2\sigma_M^2 + \sigma_{\epsilon_i}^2$: with $\beta_i = 0$ the systematic part vanishes and *all* of the stock's large variance is idiosyncratic, which diversifies away for free inside the market portfolio and hence earns no premium. A naive investor who prices the stock by its total volatility would demand a large risk premium and expect a high return; the CAPM says she is wrong, because the market pays only for the systematic (beta) contribution, which here is zero.

CAPM-Q1. Apply the Security Market Line $Er_i - r_f = \beta_i(Er_M - r_f)$. The market risk premium is $Er_M - r_f = 0.09 - 0.02 = 0.07$. Then $Er_i - r_f = 1.3 \times 0.07 = 0.091$, so $Er_i = 0.02 + 0.091 = 0.111$. The asset's equilibrium expected return is $Er_i = 0.111$ (11.1%).

CAPM-Q2. With premium $Er_M - r_f = 0.06$: for asset A, $Er_A = r_f + \beta_A(Er_M - r_f) = 0.03 + 0.5 \times 0.06 = 0.03 + 0.03 = 0.06$ (6%). For asset B, $Er_B = 0.03 + 1.6 \times 0.06 = 0.03 + 0.096 = 0.126$ (12.6%). The difference is $Er_B - Er_A = 0.126 - 0.06 = 0.066$ (6.6 percentage points), equal to $(\beta_B - \beta_A)(Er_M - r_f) = 1.1 \times 0.06$.

CAPM-Q3. First, $\beta_i = Cov(r_i, r_M)/\sigma_M^2 = 0.018/0.036 = 0.5$. Then apply the SML: $Er_i - r_f = \beta_i(Er_M - r_f) = 0.5 \times 0.055 = 0.0275$, so $Er_i = 0.025 + 0.0275 = 0.0525$. The stock's expected return is $Er_i = 0.0525$ (5.25%).

CAPM-Q4. The covariance is $Cov(r_i, r_M) = \rho_{iM}\,\sigma_i\,\sigma_M = 0.4 \times 0.30 \times 0.20 = 0.024$. The market variance is $\sigma_M^2 = (0.20)^2 = 0.04$. Therefore $\beta_i = Cov(r_i, r_M)/\sigma_M^2 = 0.024/0.04 = 0.6$. The stock's beta is $\beta_i = 0.6$.

CAPM-Q5. Using $Er_M - r_f = \bar{A}\,\sigma_M^2 = 2.5 \times 0.04 = 0.10$, the equilibrium market risk premium is 0.10 (10%). With $r_f = 0.03$, the expected market return is $Er_M = r_f + 0.10 = 0.03 + 0.10 = 0.13$ (13%).

CAPM-Q6. Solve $Er_M - r_f = \bar{A}\,\sigma_M^2$ for $\bar{A}$: $\bar{A} = (Er_M - r_f)/\sigma_M^2 = 0.06/0.05 = 1.2$. The implied average degree of risk aversion is $\bar{A} = 1.2$.

CAPM-Q7. Using the demand rule $y = (Er_M - r_f)/(A\sigma_M^2) = 0.07/(3 \times 0.04) = 0.07/0.12 \approx 0.583$. Since $y < 1$, she places about 58.3% of wealth in the risky market portfolio and the remaining 41.7% in the risk-free asset, so she is a net *lender* at the risk-free rate.

CAPM-Q8. Set $y = 1$ in $y = (Er_M - r_f)/(A\sigma_M^2)$: $1 = 0.05/(A \times 0.025)$, so $A = 0.05/0.025 = 2$. An investor with $A = 2$ holds exactly the market portfolio and nothing in the risk-free asset. Since $Er_M - r_f = \bar{A}\sigma_M^2$, an investor whose A equals the market's average risk aversion $\bar{A}$ chooses $y = 1$; here $\bar{A} = (Er_M - r_f)/\sigma_M^2 = 0.05/0.025 = 2 = A$, confirming that this investor's risk aversion is exactly the equilibrium average.

CAPM-Q9. Systematic variance $= \beta_i^2\sigma_M^2 = (1.2)^2 \times 0.05 = 1.44 \times 0.05 = 0.072$. Total variance $\sigma_i^2 = \beta_i^2\sigma_M^2 + \sigma_{\epsilon_i}^2 = 0.072 + 0.02 = 0.092$. Total standard deviation $\sigma_i = \sqrt{0.092} \approx 0.303$. The systematic fraction is $0.072/0.092 \approx 0.783$, so about 78.3% of the total variance is systematic.

CAPM-Q10. Systematic variance = $\beta_i^2\sigma_M^2 = (0.8)^2 \times 0.045 = 0.64 \times 0.045 = 0.0288$. From $\sigma_i^2 = \beta_i^2\sigma_M^2 + \sigma_{\epsilon_i}^2$, the idiosyncratic variance is $\sigma_{\epsilon_i}^2 = \sigma_i^2 - \beta_i^2\sigma_M^2 = 0.10 - 0.0288 = 0.0712$. The systematic share of total variance is $0.0288/0.10 = 0.288$, i.e. about 28.8%.

CAPM-D1. (a) Systematic variance = $(2.15)^2(4.4)^2 = 4.6225 \times 19.36 \approx 89.5$ ($\%^2$ monthly). (b) Total variance = $89.5/0.45 \approx 198.9$, so $\sigma_i \approx 14.1\%$ monthly; idiosyncratic variance = $198.9 - 89.5 = 109.4$, so $\sigma_\epsilon \approx 10.5\%$ monthly. (c) Only the systematic share — the R^2 of 45% of variance — commands compensation; the majority of NVIDIA's risk (55%) is diversifiable and therefore unpriced.

CAPM-D2. (a) NVIDIA: $4\% + 2.15 \times 6\% = 16.9\%$. Coca-Cola: $4\% + 0.37 \times 6\% = 6.2\%$. (b) First, sampling error: with roughly five years of monthly data and residual volatility above 10% per month, the standard error of $\hat{\alpha}$ is enormous, so even a large point estimate may not be statistically distinguishable from zero. Second, the joint-hypothesis problem: a measured alpha is evidence against the *pair* (CAPM, market efficiency); it may reflect an incomplete risk model — an omitted factor on which NVIDIA loads heavily — rather than a genuine free lunch.

17.4 Portfolio performance evaluation

PERF-C1. She should choose Fund B, the fund with the higher Sharpe ratio, even though it has the lower average return. Because she is holding all of her risky wealth in one fund and then mixing it with the risk-free asset, the set of expected-return/risk combinations available to her is the capital allocation line through the fund, a straight line rising from r_f with slope equal to the fund's Sharpe ratio. A higher Sharpe ratio means a steeper CAL, and a steeper CAL delivers more expected return at *every* level of total risk she might choose to bear. So Fund B dominates Fund A: at any risk level she likes, the CAL through B lies above the CAL through A. She is not stuck with Fund B's own expected return, however. By levering up — placing more than 100% of her wealth in Fund B (borrowing at r_f to do so) — she moves out along the steeper CAL and can reach an expected return as high as, or higher than, Fund A's, but at a *lower* standard deviation than Fund A would impose for that same expected return. The Sharpe ratio, not the raw average return, is what ranks the two funds because it is the slope of the opportunity set she actually faces.

PERF-C2. The first analyst, who used total standard deviation σ_P, has applied the Sharpe ratio correctly: the Sharpe ratio is by definition excess return divided by total risk, $S_P = (Er_P - r_f)/\sigma_P$, and it is the appropriate measure precisely when the fund is (or would be) held as the investor's entire risky position, so that the investor bears all of the fund's volatility. The second analyst, by dividing excess return by systematic risk (beta) instead, has not computed a Sharpe ratio at all — he has effectively computed the **Treynor measure**, $T_P = (Er_P - r_f)/\beta_P$. That measure is the right one, but only under a different assumption about how the fund is held: when the fund is one component of a large, well-diversified portfolio, its idiosyncratic risk is diversified away and only its beta contributes to the risk the investor ultimately bears. The two analysts have not made an arithmetic error against a common standard; they have implicitly assumed different roles for the fund, and each measure is correct for its own role.

PERF-C3. With forty sleeves already in place, any single sleeve is only a small component of a large, well-diversified whole, and its idiosyncratic risk is diversified away by the other thirty-nine holdings. What the new sleeve actually adds to the fund's risk is therefore not its total volatility but its covariance with everything else — summarized by its beta. The Sharpe ratio charges the sleeve for its *total* standard deviation, including diversifiable risk the pension fund will never actually bear, so it is the wrong yardstick here; it would penalize a sleeve for idiosyncratic swings that vanish in the aggregate. The Treynor measure, $T_P = (Er_P - r_f)/\beta_P$, charges the sleeve only for its systematic risk and so measures reward per unit of the risk the sleeve genuinely contributes. The feature of the CAPM that justifies this is the lesson that in equilibrium only systematic (non-diversifiable) risk is priced: idiosyncratic risk earns no premium because it can be diversified away at no cost, so a component held inside a diversified portfolio should be rewarded — and evaluated — only for the market risk it brings.

PERF-C4. Equal Treynor measures with $\sigma_X \gg \sigma_Y$ tells you the two portfolios differ substantially in idiosyncratic risk. Recall that total variance decomposes as $\sigma_P^2 = \beta_P^2\sigma_M^2 + \sigma^2(\varepsilon_P)$. Since the Treynor measures are equal, the portfolios earn the same excess return per unit of beta; if they also have similar betas (as their equal Treynor and equal reward-to-beta pricing suggest), their systematic-risk components $\beta_P^2\sigma_M^2$ are comparable, so the much larger total variance of X must come from a much larger residual variance $\sigma^2(\varepsilon_X)$. Portfolio X therefore carries far more diversifiable risk than Y. Yet an investor holding either as one small piece of a well-diversified whole would be nearly indifferent between them: that extra idiosyncratic risk in X is exactly the kind of risk that gets diversified away inside the larger portfolio, so it does not affect the risk the investor ultimately bears. What matters for a diversified holder is the systematic contribution and the reward per unit of it — and on that dimension, captured by the Treynor measure, the two portfolios are identical.

PERF-C5. "Abnormal" means the fund's expected return lies 2% *above* the Security Market Line: the CAPM assigns a fund of beta β_P the fair return $r_f + \beta_P(Er_M - r_f)$, and Jensen's alpha is the vertical gap between the return actually earned and that fair return. A positive alpha is return the single-factor CAPM cannot explain by market exposure alone. Two distinct reasons it is not, by itself, convincing evidence of skill: (1) **Statistical noise.** An estimated alpha is the intercept of a regression run on a finite sample and carries a standard error; a positive point estimate can easily arise from luck. Whether it is distinguishable from zero is a question about its t-statistic, which is roughly $IR_P\sqrt{T}$ and typically requires many years of data to clear the significance threshold. (2) **Model misspecification.** A positive alpha may reflect exposure to a priced source of risk that the one-factor CAPM omits (a value, size, or momentum tilt, say) rather than any genuine informational edge. What looks like skill relative to a single-factor benchmark can be ordinary compensation for a risk the benchmark fails to account for.

PERF-C6. Fund G, the lower-beta fund, almost certainly has the larger

Jensen's alpha. Alpha is the raw return minus the CAPM-required return $r_f + \beta_P(Er_M - r_f)$. Both funds earned the same raw return of 11%, but Fund H's higher beta of 1.4 entitles it to a *higher* CAPM-required return than Fund G's beta of 0.8 (assuming a positive market risk premium). Subtracting a larger required return from the same raw return leaves Fund H with the smaller — possibly negative — alpha, while Fund G, held to a lower bar, keeps more of its return as abnormal. The general lesson is that raw return can mask poor risk-adjusted performance: a fund can post an impressive headline return simply by taking more systematic risk, and once you charge it for the return that risk was *supposed* to earn, the apparent outperformance can shrink or vanish. Two funds with identical raw returns are not equally good if one reached that return by running a much higher beta.

PERF-C7. The high-conviction active fund has the larger tracking error. Tracking error is the volatility of the active return $r_P - r_b$, and its square decomposes as $TE_P^2 = \sigma_P^2 + \sigma_b^2 - 2\rho_{Pb}\sigma_P\sigma_b$. A pure index fund is built to replicate the S&P 500, so its return moves almost in lockstep with the benchmark: ρ_{Pb} is near 1 and $\sigma_P \approx \sigma_b$, which drives the expression toward zero and gives a tracking error near zero. The high-conviction active fund takes large positions away from the index, lowering ρ_{Pb} and often raising σ_P relative to σ_b; both effects enlarge the right-hand side, so its tracking error is large. But a large tracking error is neither good nor bad news on its own, because tracking error is a measure of *risk*, not of performance: it says only how far the manager is willing to stray from the benchmark, not whether straying paid off. A manager can run a large tracking error and still underperform; tracking error becomes informative only as the denominator of the information ratio, which pairs it with active return.

PERF-C8. The statement is wrong because tracking error measures only *risk*, not performance. Two funds with the same tracking error have taken the same amount of active (benchmark-relative) risk, but that says nothing about what they *earned* for taking it. The information ratio, $IR_P = \overline{r_P - r_b}/TE_P$, is the measure of performance: it divides the active return — how much the fund actually beat or trailed its benchmark — by the tracking error. Two funds with equal tracking error are distinguished by their numerators: the one with the higher average active return has the higher information ratio and is the better active manager, while a fund with the same tracking error but zero or negative active return is worse despite the identical risk. To rank the two funds one needs their active returns $\overline{r_P - r_b}$; with those in hand, the information ratio orders them by reward per unit of active risk.

PERF-C9. The parallel is exact once the right analogues are identified. The Sharpe ratio measures the excess return over the *risk-free rate* per unit of *total* risk, $S_P = (Er_P - r_f)/\sigma_P$. The information ratio replaces both pieces with their active-management counterparts: the numerator becomes the excess return over the *benchmark* — the active return $Er_P - Er_b$ — and the denominator becomes the *tracking error*, the volatility of that active return, which is the total risk of the active bet. So where the Sharpe ratio measures reward per unit of risk for total investing, the information ratio measures reward per unit of risk for the active-management overlay, treating the benchmark rather than cash as the zero point. The information ratio, not Jensen's alpha, is the natural reward-to-risk measure for a benchmarked manager because alpha is an *absolute* number of return units that says nothing about how much risk was taken to earn it: two managers with the same alpha are not equally good if one incurred far more active risk. Dividing by tracking error converts alpha into a reward-per-risk ratio, exactly the standard by which a benchmarked manager should be judged.

PERF-C10. The two definitions — active return over tracking error, $\overline{r_P - r_b}/TE_P$, and alpha over residual risk, $\alpha_P/\sigma(\varepsilon_P)$ — coincide in the special case where the benchmark *is* the market portfolio and the manager's only systematic exposure is to that market. In that case the active return is exactly the alpha plus residual noise, the tracking error is exactly the residual volatility, and the two ratios are the same object written two ways. They diverge whenever the benchmark differs from the market, or the portfolio has systematic exposures beyond the single market factor: then the active return $r_P - r_b$ contains a component due to the portfolio's beta differing from the benchmark's (a systematic tilt), so tracking error includes systematic as well as residual risk, and it will generally differ from $\sigma(\varepsilon_P)$. It matters to state which definition is in use because the two can give materially different numbers for the same fund (as the chapter's worked example shows, 0.50 versus about 0.32); a reported information ratio is not interpretable without knowing whether its denominator is benchmark-relative tracking error or regression residual risk.

PERF-C11. From $t(\hat{\alpha}_P) \approx IR_P\sqrt{T}$, an information ratio of 0.5 produces a t-statistic of 2 — the conventional threshold for statistical significance — only when $0.5\sqrt{T} = 2$, i.e. $\sqrt{T} = 4$, i.e. $T = 16$ years. So even a respectable, *sustained* information ratio of 0.5 requires sixteen years of data before the manager's alpha is distinguishable from luck at the usual standard. This is what makes the information ratio a far more demanding standard than a single year's outperformance: one good year, or even a handful, produces a t-statistic far below 2 and is statistically indistinguishable from a lucky streak. Because the t-statistic grows only with the *square root* of the sample length, establishing skill with confidence takes a long, consistent track record; a headline number from one year carries almost no evidentiary weight about whether genuine skill is present.

PERF-C12. The relationship rests on two ingredients. First, the t-statistic for testing whether the true alpha is zero is, as always, the estimate divided by its standard error, $t(\hat{\alpha}_P) = \hat{\alpha}_P/se(\hat{\alpha}_P)$. Second, the standard error of the regression intercept is approximately the residual volatility divided by the square root of the number of observations, $se(\hat{\alpha}_P) \approx \sigma(\varepsilon_P)/\sqrt{T}$ — more data shrinks the sampling uncertainty in the estimated intercept at the usual $1/\sqrt{T}$ rate. Substituting the second into the first, $t(\hat{\alpha}_P) \approx \hat{\alpha}_P/[\sigma(\varepsilon_P)/\sqrt{T}] = [\hat{\alpha}_P/\sigma(\varepsilon_P)]\sqrt{T} = IR_P\sqrt{T}$, since $IR_P = \alpha_P/\sigma(\varepsilon_P)$ is the information ratio in its residual-risk form. The $\sqrt{T}$ factor is why two managers with the *same* information ratio can have very different t-statistics: the one with the longer track record has accumulated more evidence and so a larger t. It also pins down what a manager can and cannot do to raise her t-statistic: she can raise her information ratio (more skill relative to residual risk) or accumulate more years of data, but she cannot make a short record convincing — the $\sqrt{T}$ term guarantees that a genuinely skilled manager evaluated over too few periods will still fail to clear the significance threshold.

PERF-Q1. Apply the Sharpe ratio, excess return over total risk, to each. For the fund (ex post form): $\hat{S}_P = (\bar{r}_P - \bar{r}_f)/s_P = (10 - 2.5)/15 = 7.5/15 = 0.5$. For the market: $S_M = (8 - 2.5)/16 = 5.5/16 \approx 0.344$. Since $0.5 > 0.344$, the fund delivered more reward per unit of total risk than the market.

PERF-Q2. (a) The Sharpe ratio is $S_P = (Er_P - r_f)/\sigma_P = (14 - 4)/25 = 10/25 = 0.4$. (b) On the capital allocation line, a fraction y in Fund Z gives expected return $r_f + y(Er_P - r_f)$. Setting this to 11%: $11 = 4 + y(14 - 4) = 4 + 10y$, so $10y = 7$ and $y = 0.7$. She places 70% of her wealth in Fund Z (and 30% in the risk-free asset). (c) The standard deviation of the combined position is $y\sigma_P = 0.7 \times 25 = 17.5\%$. (Check: slope times risk gives excess return, $0.4 \times 17.5 = 7 = 11 - 4$, consistent.)

PERF-Q3. Treynor measure is excess return over beta. Portfolio: $T_P = (Er_P - r_f)/\beta_P = (13 - 3)/1.5 = 10/1.5 \approx 6.67\%$. Market: $T_M = (Er_M - r_f)/1 = (9 - 3)/1 = 6.0\%$. Since $T_P = 6.67\% > T_M = 6.0\%$, the portfolio earns more excess return per unit of systematic risk than the market, which is equivalent to saying it plots **above** the Security Market Line.

PERF-Q4. $T_D = (Er_D - r_f)/\beta_D = (10 - 3)/0.9 = 7/0.9 \approx 7.78\%$. $T_E = (Er_E - r_f)/\beta_E = (15 - 3)/1.6 = 12/1.6 = 7.5\%$. Since $T_D \approx 7.78\% > T_E = 7.5\%$, portfolio **D** is the better performer per unit of systematic risk, even though E has the higher raw return: E's larger excess return is more than accounted for by its larger beta.

PERF-Q5. CAPM-predicted return: $r_f + \beta_P(Er_M - r_f) = 2 + 0.7(10 - 2) = 2 + 0.7 \times 8 = 2 + 5.6 = 7.6\%$. Jensen's alpha: $\alpha_P = Er_P - 7.6 = 11 - 7.6 = 3.4\%$. The alpha is positive, so the fund earned 3.4% more than its market exposure alone would justify — it plots above the Security Market Line, putative (though not conclusive) evidence of skill.

PERF-Q6. Alpha is zero when the raw return equals the CAPM-required return: $Er_P = r_f + \beta_P(Er_M - r_f)$, i.e. $12 = 3 + \beta_P(8 - 3) = 3 + 5\beta_P$. Solving, $5\beta_P = 9$, so $\beta_P = 1.8$. A beta *above* 1.8 raises the CAPM-required return above the fund's 12% raw return, so the required return exceeds the realized return and the fund's alpha becomes **negative**.

PERF-Q7. The information ratio is active return over tracking error: $IR_P = \overline{r_P - r_b}/TE_P = 3/6 = 0.5$. Interpretation: the fund earned half a percentage point of active return for each percentage point of active (benchmark-relative) risk it took. Equivalently, its reward-to-risk ratio for active management is 0.5 — a respectable but not exceptional figure, and one that (by $t \approx IR\sqrt{T}$) would take roughly sixteen years to establish as statistically significant.

PERF-Q8. First compute the tracking error from the decomposition: $TE_P^2 = \sigma_P^2 + \sigma_b^2 - 2\rho_{Pb}\sigma_P\sigma_b = 18^2 + 15^2 - 2(0.9)(18)(15) = 324 + 225 - 486 = 63$. So $TE_P = \sqrt{63} \approx 7.94\%$. The average active return is $\bar{r}_P - \bar{r}_b = 10.5 - 8 = 2.5\%$. The information ratio is therefore $IR_P = 2.5/7.94 \approx 0.315$.

PERF-Q9. The information ratio is $IR_P = 3/6 = 0.5$. Using $t(\hat{\alpha}_P) \approx IR_P\sqrt{T}$ and requiring $t = 2$: $2 = 0.5\sqrt{T}$, so $\sqrt{T} = 4$ and $T = 16$. The manager needs about **sixteen years** of data for her alpha to reach a t-statistic of 2, the conventional significance threshold.

PERF-Q10. Information ratio (residual-risk form): $IR_P = \alpha_P/\sigma(\varepsilon_P) = 1.5/6 = 0.25$. Approximate t-statistic: $t(\hat{\alpha}_P) \approx IR_P\sqrt{T} = 0.25 \times \sqrt{25} = 0.25 \times 5 = 1.25$. Since $1.25 < 2$, the manager's alpha is **not** statistically distinguishable from zero at the conventional threshold, despite twenty-five years of data — her information ratio is simply too low for that sample length to establish skill.

PERF-Q11. Residual-risk definition: $IR_P = \alpha_P/\sigma(\varepsilon_P) = 2.4/8 = 0.30$. Benchmark-difference definition: $IR_P = \overline{r_P - r_b}/TE_P = 2.0/5 = 0.40$. They differ because the two definitions measure active risk differently: the residual standard deviation $\sigma(\varepsilon_P)$ from the market-model regression is not the same quantity as the tracking error TE_P relative to the benchmark (they coincide only when the benchmark is the market and the portfolio's sole systematic exposure is to it), so dividing by different denominators yields different ratios.

PERF-Q12. From the identity $\sigma_P^2 = \beta_P^2\sigma_M^2 + \sigma^2(\varepsilon_P)$, solve for residual variance: $\sigma^2(\varepsilon_P) = \sigma_P^2 - \beta_P^2\sigma_M^2 = 22^2 - 1.1^2 \times 15^2 = 484 - 1.21 \times 225 = 484 - 272.25 = 211.75$. So $\sigma(\varepsilon_P) = \sqrt{211.75} \approx 14.55\%$. The residual-risk information ratio is then $\alpha_P/\sigma(\varepsilon_P) = 1.9/14.55 \approx 0.131$.

PERF-D1. (a) On raw excess return the fund wins: 6.8% > 6.5%. (b) Sharpe ratios: fund = 0.068/0.22 = 0.309; index = 0.065/0.17 = 0.382. The index delivered more reward per unit of risk. (c) The fund "outperformed" only by taking proportionally more risk; putting funds and benchmark on an equal-risk footing converts many raw-return winners into risk-adjusted losers, which is why the risk-adjusted bars are uniformly higher.

PERF-D2. If risk adjustment worsens measured performance, active funds must on average run *higher* total volatility than the benchmark — otherwise adjusting for risk would flatter them. The gap between 90% and 99% is direct evidence that the average active large-cap fund is riskier than the index it is judged against, so its raw returns embed a risk premium rather than skill. This is the chapter's central point: return means nothing until divided by, or benchmarked against, the risk taken to earn it.

17.5 Rational expectations and the efficient markets hypothesis

EMH-C1. The chart-based strategy relies only on the history of prices and trading volume — exactly the information set covered by the *weak form* of the EMH. If weak-form efficiency holds, all information in past prices and volume is already impounded in the current price, so patterns in that history cannot be exploited for expected returns beyond compensation for risk; this is precisely why weak-form efficiency rules out profitable technical analysis. The fund trading on newly released accounting ratios uses publicly available (but non-price) information, so ruling it out requires *semi-strong* efficiency, which asserts that all public information — earnings, filings, ratios — is already priced. Semi-strong efficiency is a stronger assumption because its information set strictly contains the weak-form set (past prices are a subset of all public information), so it makes a broader claim about what prices already reflect and rules out both technical *and* fundamental/factor analysis.

EMH-C2. The three forms differ by the information set they assume is already reflected in prices. Weak form: the history of market prices (and volume). Semi-strong form: all publicly available information — the price history plus earnings, filings, news, and other public data. Strong form: all information relevant to the asset's payoffs, including private/insider information. The forms are nested because each information set contains the previous one: private-plus-public information (strong) includes all public information (semi-strong), which includes past prices (weak). Hence if prices reflect the larger set they necessarily reflect the smaller subset, so strong implies semi-strong implies weak. An example of information in the semi-strong set but not the weak set is a company's just-released quarterly earnings report: it is public but is not contained in the past price history.

EMH-C3. The observation bears on the *strong form* of the EMH, because insider (private) information is exactly what the strong form asserts is already impounded in prices. A profitable insider trade shows only that non-public information had value, which contradicts strong-form efficiency; it is not evidence against the weak or semi-strong forms, since those make claims only about past-price and public information and are silent on the value of insider information. The strong form is generally the hardest to defend empirically precisely because there is abundant evidence — and legal cases — that insiders can and do profit from private information; it is implausible that prices already reflect everything a CEO or insider knows before it becomes public.

EMH-C4. Technical analysis uses only past prices, so *weak-form* efficiency is enough to render it unprofitable in expectation: if past-price information is already priced, chart patterns carry no exploitable signal. Fundamental and factor analysis use public non-price data (earnings, balance-sheet items, characteristics), so ruling them out requires *semi-strong* efficiency, under which all public information is already impounded. Insider trading uses private information, so only *strong-form* efficiency rules it out. If the strong form holds, then prices already reflect literally all payoff-relevant information, so no analysis of any kind — including insider information — can identify an asset expected to outperform its risk-adjusted benchmark.

In that world there is nothing to be gained from active security selection, so every investor should simply hold the market portfolio and pursue a passive strategy.

EMH-C5. Two reasons the chapter gives: (1) *Volatility masks the signal.* Market returns are so volatile that a strategy raising returns by only 5% is very hard to distinguish statistically from noise over any realistic sample; a much larger edge (say 20%) would be detectable, but an edge that large would itself imply the market is grossly inefficient. (2) *The finite-sample "dart-throwing monkey" problem.* We observe only finite histories, and out of many managers some will beat a benchmark for 5, 10, or 20 years purely by luck — like a monkey throwing darts who happens to throw in the right direction. Ten years of outperformance therefore does not establish a genuine repeatable strategy. More convincing evidence would include a long out-of-sample track record, performance persistence across many independent periods and assets, a clear economic mechanism, and results robust to the benchmark used (addressing the joint-hypothesis problem) — ideally with statistical significance that survives correction for the number of strategies searched.

EMH-C6. If a trader has a genuinely profitable strategy, she has every incentive to keep it secret and trade on it herself until it stops working, only revealing it afterward. This means the strategies most damaging to the EMH are precisely the ones researchers cannot observe while they are effective, biasing the observable record toward strategies that have already decayed. On detectability: because market returns have high variance, a 5% improvement in expected return is small relative to the year-to-year swings in realized returns, so it is easily lost in the noise and would require a very long sample to confirm statistically. A 20% improvement is large enough to stand out against that volatility and would be observable — but the ability to earn an extra 20% reliably would itself mean the market was leaving enormous, easily exploited profits on the table, i.e., that it was highly inefficient. So the very edges we could detect are the ones an efficient market should not permit.

EMH-C7. The conclusion does not follow because biases at the *individual* level need not translate into biases in *market prices*. If rational arbitrageurs can profit from trading against biased investors, they will buy underpriced and sell overpriced assets, pushing prices back toward fundamental value and eliminating the mispricing. For individual biases to move equilibrium prices, arbitrage must be *limited* — the arbitrageurs must be unable or unwilling to trade away the mispricing. Two concrete limits: (1) transaction costs and bid-ask spreads that make small mispricings unprofitable to correct; and (2) the risk that the mispricing widens before it converges (so a correctly identified bet can lose money in the interim), possibly combined with financing/short-sale constraints or capital withdrawals that force the arbitrageur to unwind at a loss. When such limits bind, biased traders can affect prices.

EMH-C8. *Overconfidence* is overestimating the precision of one's own beliefs or forecasts; it can lead an investor to take a larger position in a particular stock than a correctly calibrated assessment of the odds would justify, and to trade too much — behavior inconsistent with expected-utility maximization under accurate beliefs. *Representativeness bias* (the "law of small numbers") is treating small samples as more informative than they are; it leads investors to extrapolate recent trends, underestimating how noisy those trends are. On mutual-fund survival: if many investors suffer representativeness bias and chase recently strong (extrapolated) performance, they allocate capital to funds that had good recent runs regardless of whether the managers are skilled. Because biased investors keep supplying capital to such funds, even funds run by otherwise irrational managers can attract flows and survive rather than being competed out of existence — the biased beliefs of the investors sustain them.

EMH-C9. With frictionless arbitrage, any gap between price and fundamental value is a riskless (or nearly riskless) profit opportunity: arbitrageurs would immediately buy the underpriced asset and sell the overpriced one in unlimited size, and their trading would move prices back to fundamentals before any lasting mispricing could persist. So biased traders would have no durable effect. The risk that a mispricing worsens before it corrects deters arbitrage in a way transaction costs do not: even an arbitrageur who has correctly identified that an asset is overpriced can suffer mark-to-market losses if the overpricing grows further in the short run. If she faces a finite horizon, margin calls, or investors who withdraw capital after interim losses, she may be forced to close the position at the worst time. This "noise-trader" or convergence risk means arbitrage capital is limited even when the arbitrageur is right, leaving room for biases to persist in prices.

EMH-C10. (Any three of the six.) *All investors are price takers:* fails when a large institution unwinding a position, an activist, or a large order in a thin market moves the price by its own trading, disturbing the competitive equilibrium. *One identical holding period:* fails because a young saver, a retiree, and a quarterly-judged trading desk optimize over very different horizons, and long-horizon investors may want to hedge shifts in the opportunity set that a one-period model cannot see. *All investments are publicly traded:* fails because human capital, private businesses, and closely held real estate are large, non-tradable, market-correlated assets. *No transaction costs or taxes:* fails because commissions, spreads, and differing tax situations drive wedges so investors need not agree on prices. *All investors are mean-variance optimizers:* fails because investors care about downside risk, skewness, and ruin. *Homogeneous expectations:* fails because investors visibly disagree, and the sheer volume of trade is hard to reconcile with universal agreement. Any one failure undermines the sharp conclusion because the security-market-line result is derived *from* these assumptions holding jointly; if even one breaks, the derived equilibrium in which every asset must lie exactly on the SML no longer follows, and assets can plausibly earn returns off the line.

EMH-C11. Non-traded wealth includes human capital and future labor income, privately held businesses, closely held real estate, and the returns to education — none of which trades in public markets. Because these assets are large and correlated with the market yet cannot be freely bought, sold, diversified, or hedged, the "market portfolio" of publicly traded assets that the CAPM prices is not the true portfolio of aggregate wealth; a large slice of real wealth is simply missing from it. This matters for an investor whose largest asset is her own future labor income: her overall risk-return position is dominated by an asset the CAPM does not describe, so the CAPM's prescription (hold the traded market portfolio) can be badly wrong for her. For instance, if her human capital already behaves like a risky, market-correlated asset, she may rationally want to tilt her *traded* holdings away from assets correlated with her earnings, something the standard CAPM cannot capture.

EMH-C12. Alpha is the return an asset or portfolio earns in excess of what the benchmark model — here the CAPM — says it should earn given its risk. Relative to the CAPM the benchmark return is the security market line, $r_f + \beta_i(Er_M - r_f)$, so $\alpha_i = Er_i - [\, r_f + \beta_i(Er_M - r_f)\,]$. Geometrically, alpha is the vertical distance between the asset's expected return and the security market line: a positive alpha means the asset plots *above* the line (more return than its beta entitles it to), and a negative alpha means it plots *below*. If both the CAPM and the EMH hold, the CAPM makes the SML the correct description of equilibrium expected returns and the EMH ensures prices actually equal those fair values, so every asset lies exactly on the line and $\alpha_i = 0$ for everything.

EMH-C13. The joint-hypothesis problem is that any test of alpha is simultaneously a test of the asset-pricing model used to define the benchmark *and* of market efficiency, so a nonzero alpha cannot by itself say which of the two has failed. For a significant positive estimated alpha the chapter offers three distinct explanations: (1) *bad benchmark* — the CAPM is the wrong model (an assumption fails, e.g., non-traded human capital or heterogeneous beliefs), so the asset earns fair compensation for a risk the CAPM omits and only *appears* to have alpha; (2) *genuine mispricing* — the CAPM is right but the EMH fails, so the price departs from fundamental value and the alpha is a real, temporarily available return; and (3) *finite-sample chance* — the true alpha is zero and the estimate is nonzero purely by luck in a limited sample of returns. A single nonzero estimate cannot discriminate among these because all three produce the same number; you need additional evidence (out-of-sample tests, alternative benchmarks, larger samples) to separate them.

EMH-C14. The CAPM says that unless one of its six assumptions is violated, no opportunity can offer returns higher than its risk warrants — every legitimate opportunity sits on the security market line. So a “high return, low risk” pitch leaves only two possibilities: the claim is false, or an assumption is genuinely being violated. The checklist walks the assumptions one at a time: Is the return really compensation for illiquidity/non-tradability (assumption 3, e.g., a private business or real estate)? Is there a genuine tax advantage specific to you (assumption 4)? Do you actually possess private information others lack (strong-form EMH and assumption 6)? Is the “high return” hiding a fat downside or tail risk (assumption 5)? Could a large player be moving the price (assumption 1)? If you can name the assumption the opportunity relies on, that names the specific risk you are being paid to bear. If, after going through the whole list, you cannot identify any assumption that is plausibly violated, the CAPM’s verdict is that the promised returns are almost certainly illusory, mismeasured because risk is being ignored, or outright fraudulent.

EMH-C15. Affinity fraud is fraud that spreads through tight-knit groups — families, religious congregations, ethnic or professional communities — in which shared identity substitutes for due diligence. The chapter argues that a trusted-community source is a reason to work through the checklist *more* carefully, not to skip it, because trust in the *messenger* does exactly the work the CAPM says the *numbers* cannot: it makes an off-the-security-market-line promise “feel safe” even though nothing about the returns has been justified. The empirical record (e.g., recruitment concentrated in affinity groups in the Fortune Hi-Tech Marketing scheme) shows these frauds exploit precisely that trust, often turning victims into recruiters. So the right response is to run the assumptions regardless of who is pitching, and to treat a familiar, trusted source as raising the stakes of skipping analysis rather than lowering the need for it.

EMH-C16. The EMH predicts this pattern because if prices already impound available information, consistently identifying mispriced securities is extremely hard, so the fees active managers charge become a near-certain drag on returns; the average active fund should therefore underperform a cheap index after fees, and rational savers should shift toward low-cost index funds — which is what the flows show. However, persistent active underperformance is not decisive proof of efficiency. The same evidence is also consistent with a world where some mispricings exist but are too small, too costly, or too risky to exploit reliably (limits to arbitrage), where genuine skill exists but is rare and hard to identify ex ante among many luck-driven track records (the dart-throwing-monkey problem), or where the benchmark used to judge "underperformance" is itself the wrong model (the joint-hypothesis problem). Fee drag alone can explain average underperformance without the market being fully efficient.

EMH-Q1. (a) The security-market-line (CAPM) required return is $r_f + \beta_i(Er_M - r_f) = 0.03 + 1.2\,(0.09 - 0.03) = 0.03 + 1.2(0.06) = 0.03 + 0.072 = 0.102 = 10.2\%$.

(b) $\alpha_i = Er_i - [\, r_f + \beta_i(Er_M - r_f)\,] = 0.12 - 0.102 = 0.018 = 1.8\%$.

(c) Since the alpha is positive, the stock plots *above* the security market line — the analyst forecasts 1.8% more than its beta entitles it to. By the joint-hypothesis problem, this nonzero alpha is not self-interpreting: it could mean the CAPM is the wrong benchmark (a "bad benchmark," so the stock is fairly compensating some risk the CAPM omits), it could reflect a genuine mispricing (the CAPM holds but the EMH fails), or it could be an artifact of a finite sample / an inaccurate forecast. The estimate alone cannot say which.

EMH-Q2. (a) CAPM required return is $r_f + \beta(Er_M - r_f)$ with $r_f = 0.02$ and market premium 0.05.

Portfolio A: required $= 0.02 + 0.8(0.05) = 0.02 + 0.04 = 0.06 = 6\%$; $\alpha_A = 0.07 - 0.06 = 0.01 = 1\%$.

Portfolio B: required $= 0.02 + 1.5(0.05) = 0.02 + 0.075 = 0.095 = 9.5\%$; $\alpha_B = 0.09 - 0.095 = -0.005 = -0.5\%$.

(b) Portfolio A has the larger alpha (+1% versus –0.5%). Portfolio B has the higher *raw* expected return (9% versus 7%) but the *lower* alpha — in fact a negative one. This shows that raw expected return is a misleading basis for comparison: B earns more only because it takes more systematic risk (higher beta), and once you adjust for that risk it actually falls short of its CAPM benchmark, whereas A exceeds its benchmark. Risk-adjusted return (alpha), not raw return, is the correct comparison.

(c) If the true population alphas are both zero, the nonzero estimates would be explained by finite-sample chance — the estimated alpha is nonzero purely by luck in a limited sample of returns.

EMH-D1. (a) The wealth ratio is $(1.07/1.08)^t$: after 1 year 0.9907; after 5 years 0.9546; after 20 years 0.830. (b) A one-year gap of less than 1% is easily masked by luck, so a substantial minority of funds beat the index in any single year. But the fee drag compounds geometrically while idiosyncratic luck averages away, so the fraction able to stay ahead shrinks as the horizon lengthens — exactly the rising staircase in the figure.

EMH-D2. The one-year small-cap number does not refute the EMH. (i) Small-cap prices are the most plausible place for informational inefficiencies — less analyst coverage and higher trading costs — so some short-horizon edge is consistent with near-efficiency elsewhere. (ii) In one-year samples luck dominates: with hundreds of funds, many will beat a benchmark by chance. (iii) The fifteen-year figure shows that whatever edge exists does not survive compounding fees; roughly nine in ten small-cap funds still trail their benchmark, which is precisely the EMH's long-run prediction after costs.

17.6 Interest

INT-C1. The time value of money says most people are not indifferent between \$1000 today and \$1000 in a year: a dollar today can be invested at the risk-free rate r and will grow to $\$1000(1+r) > \$1000$ by next year, so the future payment is worth strictly less. By the arbitrage principle, the present value of the delayed \$1000 is exactly the amount you would need to set aside today to reproduce it — namely $\$1000/(1+r)$, which is below \$1000 whenever $r > 0$. Taking the payment today therefore leaves you strictly wealthier (you can always wait and lend), so a certain payment today dominates the same certain payment later.

INT-C2. Asset A (paid in 3 years) has the larger present value: discounting compounds with the horizon, so $PV_T = (1+r)^{-T}x$ shrinks as T grows, and 3 years of discounting removes less value than 8 years. As r rises, the discount factor $(1+r)^{-T}$ falls for both assets, so *both* present values decline; because the longer-dated asset is discounted over more periods, its value falls proportionally faster, so the *dollar* gap between A and B narrows toward zero at high r (both values head to zero). Only in the limit $r \to 0$ do the two present values coincide at \$5000, since with no discounting timing is irrelevant.

INT-C3. The value $V = M/r$ falls as r rises because a higher interest rate means a smaller present-day deposit is needed to throw off the same perpetual payment (equivalently, each future dollar is discounted more heavily). Even a small M can be worth a large lump sum because the payments continue *forever*: dividing by a small r magnifies the sum. The replication/arbitrage argument guarantees exactness because M/r is precisely the principal that, invested at rate r, earns interest $r \cdot (M/r) = M$ each period; paying out exactly the interest M leaves the principal intact to earn M again next period, perpetually reproducing the stream without depleting capital. Any other price would let someone arbitrage the difference.

INT-C4. Both routes price the *same* cash flows — a payment of M at each date $1, \ldots, T$ — so by the law of one price they must carry the same value; if they differed, one could buy the cheap version and sell the dear one for a riskless profit. The arbitrage route makes this concrete: buying a perpetuity (M/r today) and selling the rights to all payments after date T (worth M/r at date T, hence $\frac{M/r}{(1+r)^T}$ today) leaves exactly the first T payments, so the annuity is worth $M/r - \frac{M/r}{(1+r)^T} = M\frac{1}{r}(1-(1+r)^{-T})$, matching the geometric-series result. As $T \to \infty$ the term $(1+r)^{-T} \to 0$ and the annuity value approaches M/r — obvious from the picture, since selling payments infinitely far in the future removes something of zero present value, leaving the whole perpetuity.

INT-C5. For a one-year horizon with interest credited a single time at year-end, simple and compound interest give the identical balance $x(1+r)$: there has been no prior interest for compounding to act upon, so the student is right in that narrow case. The student is wrong once interest is credited more than once (e.g. compounded n times, giving $x(1 + r/n)^n > x(1 + r)$ for $n > 1$) or the money is left for several years: compound interest earns interest on previously credited interest, so after t years compound growth $x(1+r)^t$ pulls ahead of simple growth $x(1+rt)$, and the gap widens with both the frequency and the horizon.

INT-C6. Under compound interest the first year's interest xr is itself left in the account and earns interest in the second year, so the two-year balance is $x(1+r)^2 = x(1+2r+r^2)$. Under simple interest only the original principal earns, giving $x(1 + 2r)$. The extra term compounding adds is xr^2 — the interest earned *on the first year's interest* xr. Because each extra year adds further "interest on interest," the gap between the compound balance $x(1 + r)^t$ and the simple balance $x(1 + rt)$ widens as the horizon t lengthens.

INT-C7. Both CDs quote the same *annual rate* $r = 6\%$, but the annual rate ignores compounding within the year. The monthly-compounding CD credits its interest twelve times, and each credited amount itself earns interest for the rest of the year, so it accumulates more than the semiannual CD, which compounds only twice. The *effective annual rate* is $r^* = (1 + r/n)^n - 1$, the actual one-year percentage increase in the balance; with $n = 12$ it exceeds the $n = 2$ value. Because r^* folds in the compounding frequency while the quoted rate r does not, r^* is the correct apples-to-apples figure for comparing the two CDs.

INT-C8. Raising n credits interest more often, and each earlier credit earns interest for the remainder of the year, so $r^* = (1 + r/n)^n - 1$ rises with n. It does not grow without bound because the extra "interest-on-interest" from each additional split is progressively smaller; the sequence $(1 + r/n)^n$ converges. The chapter shows via L'Hôpital's rule that $\lim_{n\to\infty}(1+r/n)^n = e^r$, so the limiting gross annual return is e^r and the limiting effective rate is $e^r - 1$. The natural exponential appears as the ceiling because continuous compounding is the mathematical limit of ever-finer compounding, and e^r is exactly the value that self-consistent instantaneous growth at rate r

produces.

INT-C9. The expression $(1+r/n)^n$ has the variable n in both the base and the exponent, which makes the limit hard to attack directly; taking the logarithm turns it into $n\log(1+r/n)$, a product, and then a quotient $\frac{\log(1+r/n)}{1/n}$ that ordinary limit tools can handle. As $n\to\infty$ the numerator $\log(1+r/n)\to\log 1=0$ and the denominator $1/n\to 0$, so the quotient has the indeterminate form $0/0$, exactly the setting in which L'Hôpital's rule applies; differentiating top and bottom yields the limit r. Exponentiating back gives e^r. The balance formula $B_t = xe^{rt}$ then says a continuously compounded deposit grows exponentially and smoothly through time, with the instantaneous growth rate equal to r at every moment.

INT-C10. The *nominal* rate is the rate the account literally pays in dollars; the *real* rate, by the Fisher relation, is roughly the nominal rate minus expected inflation. When the Fed's nominal rate (~3.6%) equals inflation (~3.6%), the real return is about zero: the saver's dollar balance grows at 3.6% per year, but prices rise just as fast, so the basket of goods those dollars can buy is unchanged (and shrinks if inflation exceeds the rate). The compounding machinery operates on the *nominal* rate because that is the rate at which actual dollars in the account accumulate; inflation is a separate erosion of what those dollars are worth, applied after the nominal growth is computed.

INT-C11. She should discount the payment stream at a rate consistent with its nominal terms and then judge the result in real terms. The stream of monthly payments is an annuity (a fixed-life level payment), so its present value is the annuity value of this chapter; a lifelong-but-uncertain-horizon pension behaves closer to a perpetuity-like stream. The key Fisher point is that a *fixed nominal* annuity pays the same dollar amount every month regardless of prices, so inflation steadily erodes its purchasing power — the real value of each payment shrinks over time. A payment *indexed to prices* rises with inflation, preserving real value, so it does not suffer this erosion. Thus the lump-sum-versus-annuity choice hinges on both the discount rate used and whether the annuity is nominal or inflation-indexed.

INT-C12. With a positive interest rate, a dollar received at maturity is worth less than a dollar today, and by $PV_T = (1 + r)^{-T} x$ the bill's price is the face value discounted back over its life: $(1 + r)^{-T} < 1$ when $r > 0$, so the price sits strictly below face value. This discount is precisely the investor's return — the gap between purchase price and the face value paid at maturity. If the six-month rate r suddenly rose, the discount factor $(1 + r)^{-T}$ would fall, so the bill's price would drop: higher required return means a lower price today for the same fixed future payment.

INT-Q1. Discount each payment and sum, with $r = 0.08$:

$$PV = \frac{400}{1.08} + \frac{600}{1.08^2} + \frac{1000}{1.08^3}.$$

Term by term: $400/1.08 = 370.37$; $600/1.1664 = 514.40$; $1000/1.259712 = 793.83$. Summing gives $PV \approx \$1678.61$.

INT-Q2. Single payment, $x = 2500$. At $T = 4$, $r = 0.07$:

$$PV = 2500(1.07)^{-4} = 2500/1.310796 \approx \$1907.24.$$

At $T = 9$, $r = 0.07$ held fixed:

$$PV = 2500(1.07)^{-9} = 2500/1.838459 \approx \$1359.83.$$

The longer horizon lowers the present value.

INT-Q3. Perpetuity value $V = M/r$ with $M = 750$. At $r = 0.05$:

$$V = 750/0.05 = \$15{,}000.$$

At $r = 0.04$:

$$V = 750/0.04 = \$18{,}750.$$

Lowering the interest rate raises the perpetuity's value — value moves inversely with r, since each future payment is discounted less heavily.

INT-Q4. Annual rate $r = 0.08$, so periodic (quarterly) rate $r/4 = 0.02$ and quarterly coupon $rM/4 = 0.08 \cdot 1000/4 = \$20$. The perpetuity of quarterly coupons, valued immediately *after* a payment, is

$$P = \frac{rM/4}{r/4} = \frac{20}{0.02} = \$1000.$$

Immediately *before* a coupon, the buyer also collects the imminent \$20 payment, so the price is $1000 + 20 = \$1020$.

INT-Q5. Annuity factor with $M = 1500$, $r = 0.06$, $T = 15$:

$$V = 1500 \cdot \frac{1}{0.06}\left(1 - \frac{1}{1.06^{15}}\right).$$

Here $1.06^{15} = 2.396558$, so $1/1.06^{15} = 0.417265$ and $1 - 0.417265 = 0.582735$. Then $V = 1500 \cdot (1/0.06) \cdot 0.582735 = 25000 \cdot 0.582735 \approx \$14{,}568.37$.

INT-Q6. Annuity with $M = 800$, $r = 0.09$, $T = 25$:

$$V = 800 \cdot \frac{1}{0.09}\left(1 - \frac{1}{1.09^{25}}\right).$$

Here $1.09^{25} = 8.623081$, so $1/1.09^{25} = 0.115968$ and $1 - 0.115968 = 0.884032$. Then $V = 800 \cdot (1/0.09) \cdot 0.884032 = 8888.89 \cdot 0.884032 \approx \$7858.06$. Check: the corresponding perpetuity is $M/r = 800/0.09 = \$8888.89$, and indeed $7858.06 < 8888.89$, as expected since the annuity omits all payments after year 25.

INT-Q7. Compounded balance with $x = 5000$, $r = 0.10$, $n = 4$, $t = 8$:

$$B_t = 5000\left(1 + \frac{0.10}{4}\right)^{4\cdot 8} = 5000(1.025)^{32}.$$

Since $(1.025)^{32} = 2.203757$, $B_t = 5000 \cdot 2.203757 \approx \$11{,}018.78$.

INT-Q8. Effective annual rate with $r = 0.18$. Monthly ($n = 12$):

$$r^* = \left(1 + \frac{0.18}{12}\right)^{12} - 1 = (1.015)^{12} - 1 = 1.195618 - 1 \approx 0.19562\ (19.56\%).$$

Daily ($n = 365$):

$$r^* = \left(1 + \frac{0.18}{365}\right)^{365} - 1 \approx 1.197164 - 1 \approx 0.19716\ (19.72\%).$$

More frequent compounding raises the effective rate slightly.

INT-Q9. Continuous compounding with $x = 3000$, $r = 0.05$, $t = 4$:

$$B_t = 3000\, e^{0.05\cdot 4} = 3000\, e^{0.2} = 3000 \cdot 1.221403 \approx \$3664.21.$$

Effective annual rate: $r^* = e^{0.05} - 1 = 1.051271 - 1 \approx 0.051271$ (5.13%).

INT-Q10. Continuous compounding with $x = 10{,}000$, $r = 0.03$, $t = 10$:

$$B_t = 10000\, e^{0.03\cdot 10} = 10000\, e^{0.3} = 10000 \cdot 1.349859 \approx \$13{,}498.59.$$

Effective annual rate: $r^* = e^{0.03} - 1 = 1.030455 - 1 \approx 0.030455$ (3.05%).

INT-Q11. Apply $PV_T = (1+r)^{-T}(1000)$ with $r = 0.10$:

$$PV_1 = \frac{1000}{1.10} \approx \$909.09, \qquad PV_2 = \frac{1000}{1.10^2} \approx \$826.45, \qquad PV_7 = \frac{1000}{1.10^7} \approx \$513.16.$$

Between year 1 and year 7 the present value falls by $(909.09 - 513.16)/909.09 \approx 43.6\%$. The decline is not seven times the one-year discount because discounting is *multiplicative*, not additive: each additional year multiplies the value by another factor of $1/1.10$, so the cumulative discount is 1.10^{-7} rather than 7×0.10. Geometric decay falls more slowly than the naive linear extrapolation would suggest — seven times a 9.1% one-year haircut would be 63.6%, far more than the actual 43.6%.

INT-Q12. (a) A zero-coupon bill is a single cash flow of F one period (six months) away, so by the single-payment present value formula

$$P = \frac{F}{1+r}.$$

(b) At $r = 0.021$: $P = 1000/1.021 \approx \$979.43$. The bill trades at a discount of \$20.57 to face value, and that discount *is* the investor's entire return. (c) Rolling the six-month rate over twice gives a gross annual return of $(1.021)^2 = 1.042441$, an effective annual rate of about 4.24% — slightly more than double 2.1% because of compounding within the year. (d) At $r = 0.025$: $P = 1000/1.025 \approx \$975.61$. The price *falls* by about \$3.82. Bond prices move inversely to interest rates: the promised payment is fixed, so a higher discount rate can only reduce what it is worth today.

INT-Q13. (a) Continuous compounding with $x = 1000$, $r = 0.05$, $t = 1$:

$$B = 1000\,e^{0.05} = 1000 \cdot 1.051271 \approx \$1051.27.$$

The effective annual rate is $r^* = e^{0.05} - 1 \approx 0.051271$ (5.1271%). (b) Using $B = x\left(1 + \frac{r}{n}\right)^n$ for one year:

- Semiannually ($n = 2$): $1000(1 + 0.05/2)^2 \approx \$1050.63$, $r^* \approx 5.0625\%$.
- Monthly ($n = 12$): $1000(1 + 0.05/12)^{12} \approx \$1051.16$, $r^* \approx 5.1162\%$.
- Daily ($n = 365$): $1000(1 + 0.05/365)^{365} \approx \$1051.27$, $r^* \approx 5.1267\%$.

(c) The ordering is semiannual < monthly < daily < continuous, exactly as the monotonicity of $(1 + r/n)^n$ in n requires. The total gain available from moving all the way from semiannual to continuous compounding is $1051.2711 - 1050.625 = \$0.65$; the move from semiannual to monthly alone captures $1051.1619 - 1050.625 = \$0.54$, or about 83% of it. Daily compounding is within a twentieth of a cent of the continuous limit. The practical lesson is that the continuous case is analytically convenient rather than economically dramatic: beyond monthly compounding, frequency scarcely matters.

INT-Q14. Each phone costs an immediate payment plus a 24-month annuity of data fees. With annuity factor $A(r) = \frac{1}{r}\left(1 - (1 + r)^{-24}\right)$:

(a) At $r = 0.004074$, $A = 22.8198$:

$$PV(\text{original}) = 399 + 20(22.8198) = 399 + 456.40 = \$855.40,$$

$$PV(\text{later model}) = 199 + 30(22.8198) = 199 + 684.59 = \$883.59.$$

The original phone is cheaper by \$28.19 despite its higher sticker price — the \$200 saved up front is more than undone by \$10 per month for two years.

(b) At $r = 0.01$, $A = 21.2434$: $PV(\text{original}) = 399 + 424.87 = \$823.87$ and $PV(\text{later}) = 199 + 637.30 = \$836.30$; the original is still cheaper, but only by \$12.43. At $r = 0.02$, $A = 18.9139$: $PV(\text{original}) = 399 + 378.28 = \$777.28$ and $PV(\text{later}) = 199 + 567.42 = \$766.42$; now the later model is cheaper by \$10.86.

(c) The ranking reverses somewhere between a 1% and a 2% monthly rate. The intuition is that a higher discount rate shrinks the present value of *future* payments but leaves the *immediate* outlay untouched. The later model's disadvantage lies entirely in the future (an extra \$10 per month), while its advantage (\$200 less today) is undiscounted. Raising r therefore erodes the disadvantage without touching the advantage, and once r is high enough the up-front saving wins. This is the general principle that impatience — a high discount rate — favors back-loading costs and front-loading benefits.

17.7 Fixed Income Yields

YLD-C1. The yield to maturity y is defined as the single discount rate that equates the present value of a bond's promised cash flows to its observed price, i.e. the solution to $P = \sum_{t=1}^{T} \frac{b}{(1+y)^t} + \frac{M}{(1+y)^T}$. The colleague's statement is correct only under a specific condition: the yield to maturity is the return earned by holding to maturity *provided every coupon can be reinvested at the rate y until maturity*. It is a promised or "if-all-goes-as-assumed" return, not a guaranteed one. If reinvestment rates differ from y over the bond's life, the realized compound return will differ from the quoted yield. So the statement is correct as a definition of the internal rate of return implied by today's price, but incorrect as an unconditional forecast of the return the investor will actually earn.

YLD-C2. A single discount rate is used because the yield to maturity is defined as the one number y that makes the discounted sum of *all* promised payments equal the price. It is the internal rate of return of the bond's cash-flow stream: rather than assigning a distinct spot rate to each maturity, it compresses the whole term structure into one summary rate specific to that bond. This is what lets markets quote value and compare instruments in a single figure — two bonds with different coupons and maturities can be ranked by their yields even though their cash-flow patterns differ. The cost of this convenience is that y is bond-specific and embeds the reinvestment assumption; it is not a set of maturity-by-maturity discount rates.

YLD-C3. Write the per-period coupon as $b = cM$, where c is the coupon rate, and evaluate the pricing equation at the candidate yield $y = c$. The coupon stream is an annuity, so using the annuity factor $\frac{1}{c}\left(1 - (1+c)^{-T}\right)$, $P = cM \cdot \frac{1}{c}\left(1 - \frac{1}{(1+c)^T}\right) + \frac{M}{(1+c)^T} = M\left(1 - \frac{1}{(1+c)^T}\right) + \frac{M}{(1+c)^T}$. The two occurrences of $M(1+c)^{-T}$ cancel, leaving $P = M$. So discounting at the coupon rate prices the bond exactly at par. Because the pricing equation has a unique yield solution for any given price, the converse follows: if $P = M$ then y must equal c.

YLD-C4. If a bond sells above par, buyers are paying more than the face value they will get back, so the promised payments must be discounted at a rate *below* the coupon rate to make their present value large enough to reach that high price; hence YTM $< c$. If a bond sells below par, the promised payments must be discounted at a rate *above* the coupon rate to shrink their present value down to that low price; hence YTM $> c$. Both rules follow from the definition of y as the discount rate equating the present value of promised payments to P: raising y lowers present value and raising the price requires lowering y, so price and yield move inversely around the par/coupon-rate pivot.

YLD-C5. The realized compound return will be *below* the quoted yield to maturity. The yield to maturity implicitly assumes that every coupon received is reinvested at the original yield until the bond matures. When market rates fall below that original yield, the coupons are reinvested at lower rates, so the terminal value of the reinvested coupon stream is smaller than the yield-to-maturity calculation presumed. Solving $M(1+r)^T$ = final cash then produces a realized r less than y. The violated assumption is precisely the constant-reinvestment-at-y assumption embedded in the yield to maturity.

YLD-C6. The objection misunderstands what the reinvestment assumption governs. The assumption matters whenever the return is measured as a *compound* return over the holding horizon: even coupons that are spent still have a value, and comparing the yield to maturity to a realized compound return implicitly asks what those coupons could have earned. An investor who spends every coupon and holds to maturity does earn the coupon cash flows, but relative to the quoted yield he forgoes the interest-on-interest that the yield to maturity credited him with. When rates rise after purchase, an investor who *reinvests* gains because coupons compound at higher rates; the investor who spends them captures no such gain, so his realized experience diverges from the yield to maturity exactly through the reinvestment channel — even though, for him, that divergence shows up as forgone reinvestment income rather than lower cash in hand.

YLD-C7. The holding period return is $R = \frac{P_1+b-P_0}{P_0}$, which splits into a coupon component $\frac{b}{P_0}$ and a capital-gain component $\frac{P_1-P_0}{P_0}$. The coupon component is fixed and positive over the period. When market interest rates rise sharply just after purchase, the bond must be discounted at a higher yield, so its resale price P_1 falls below the purchase price P_0, making the capital-gain component negative. If that capital loss is large enough to exceed the coupon income, the total return turns negative. So it is the capital-gain (price) component — driven by the inverse price–yield relationship — that can convert a positive coupon return into a negative total return.

YLD-C8. A rate increase raises the discount rate applied to all remaining cash flows, so the market price of the bond falls immediately. For the investor who plans to sell after one year, this is a direct loss: the price at which she can sell has dropped, so she realizes a capital loss on the sale. For the investor holding to maturity, the price drop is irrelevant to the terminal payoff — he still collects every coupon and the face value M — but the higher rates actually help him, because the coupons he receives can now be *reinvested* at the higher prevailing rates, raising his realized compound return. The same rate move thus hurts the early seller through the price channel and helps the buy-and-hold investor through the reinvestment channel.

YLD-C9. In $P = \sum_{t=1}^{T} \frac{b}{(1+y)^t} + \frac{M}{(1+y)^T}$, every term has $(1+y)$ in the denominator, so raising y shrinks each discounted payment and lowers the total present value P; lowering y raises P. Price and yield therefore move in opposite directions by construction. The news item illustrates the same mechanism from the holder's side: a hot inflation report pushed the 10-year Treasury yield up to about 4.46%, and because the promised coupon and face payments are fixed, the only way the market can offer a higher yield on those fixed payments is by paying a lower price for them. Investors already holding the bond therefore saw its price — and hence its market value — fall as yields rose.

YLD-C10. For a given rise in the market yield, the 20-year bond's price falls by more in percentage terms than the 2-year bond's. In the pricing equation, a longer maturity means more distant cash flows and a larger power of $(1 + y)$ in the denominators; distant payments are far more sensitive to a change in the discount rate because the discount factor $(1 + y)^{-t}$ changes proportionally faster as t grows. The 20-year bond has a large fraction of its value in payments many periods away, so when y rises, those far-off terms shrink dramatically and the whole price drops sharply. The 2-year bond's value is concentrated in near-term payments that are barely affected by the higher discount rate, so its price moves little. Longer maturity therefore means greater price sensitivity to yield changes.

YLD-C11. The current yield is annual coupon income divided by price, b/P; it measures only the cash coupon return relative to the amount invested. The yield to maturity is the single discount rate solving the full pricing equation, and it incorporates *all* sources of return — coupon income *and* the capital gain or loss realized as the price converges to face value at maturity, along with the timing of every payment. The current yield omits this capital-gain/loss and timing dimension. A scenario in which the higher current yield delivers the lower yield to maturity: a premium bond (priced above par) has a high coupon and thus a high current yield, but because it will decline toward face value by maturity, that capital loss drags its yield to maturity below the current yield; a discount bond with a lower coupon (lower current yield) enjoys a capital gain to par that lifts its yield to maturity. Comparing the two by current yield alone can reverse their true ranking.

YLD-C12. For a bond with semiannual coupons and periodic yield y, the bond equivalent yield annualizes simply by doubling, $2y$, ignoring compounding of the mid-year coupon. The effective annual yield compounds the periodic yield over the two half-years, $(1+y)^2 - 1$, thereby crediting interest earned on the first semiannual coupon during the second half-year. Because $(1+y)^2 - 1 = 2y + y^2 \geq 2y$ for any $y \geq 0$, the effective annual yield is always at least as large as the bond equivalent yield; the extra term y^2 is exactly the interest-on-interest that simple annualization ignores. The two coincide only when $y = 0$ (no compounding to accumulate).

YLD-Q1. The bond sells at par: $P = M = \$1000$. By the par-bond result, a bond priced at par has yield to maturity equal to its coupon rate. The coupon rate here is $b/M = 70/1000 = 0.07$, so $y = 0.07$. No numerical root-finding is needed — one can verify directly that $\sum_{t=1}^{3} \frac{70}{1.07^t} + \frac{1000}{1.07^3} = 1000$. Result: $y = 0.07$ (7%).

YLD-Q2. Substitute $y = 0.07$: $\frac{60}{1.07} + \frac{60}{1.07^2} + \frac{1000}{1.07^2} = 56.075 + 52.406 + 873.439 = 981.92$, which equals the observed price $P = \$981.92$. So $y = 0.07$ is confirmed. Since $P = \$981.92 < M = \$1000$, the bond trades *below par*, and consistent with the chapter's rule the yield to maturity (7%) exceeds the coupon rate $b/M = 60/1000 = 6\%$.

YLD-Q3. Coupons of \$80 are received at $t = 1, 2, 3$; reinvested to maturity ($t = 3$) at 10%: the $t = 1$ coupon grows to $80(1.10)^2 = 96.80$, the $t = 2$ coupon to $80(1.10) = 88.00$, and the $t = 3$ coupon stays 80.00. Adding the face value \$1000: final cash = 96.80 + 88.00 + 80.00 + 1000 = \$1264.80. Then $1000(1+r)^3 = 1264.80 \Rightarrow r = (1.2648)^{1/3} - 1 = 0.0815$, i.e. about 8.15%. Because reinvestment occurred at 10% > 8%, the realized compound return exceeds the quoted yield of 8%.

YLD-Q4. The first coupon (\$50, received at $t = 1$) is reinvested at 3% to

$t = 2$: 50(1.03) = 51.50. The second coupon ($50) is received at $t = 2$, plus the face value $1000. Final cash = 51.50 + 50 + 1000 = $1101.50. Then $1000(1 + r)^2 = 1101.50 \Rightarrow r = (1.1015)^{1/2} - 1 = 0.0495$, about 4.95%. Because the coupon was reinvested at 3% < 5%, the realized compound return (4.95%) falls short of the quoted yield to maturity (5%).

YLD-Q5. Purchase price at periodic yield $y = 0.04$, 20 periods: $P_0 = \frac{6}{0.04}\left(1 - \frac{1}{1.04^{20}}\right) + \frac{100}{1.04^{20}} = 150(1 - 0.456387) + 45.639 = 81.542 + 45.639 =$ \$127.181. Six months later, 19 periods remain: $P_1 = \frac{6}{0.04}\left(1 - \frac{1}{1.04^{19}}\right) + \frac{100}{1.04^{19}} = 150(1 - 0.474642) + 47.464 = 78.804 + 47.464 =$ \$126.268. Holding period return over the first six months: $R = \frac{P_1 + b - P_0}{P_0} = \frac{126.268 + 6 - 127.181}{127.181} = \frac{5.087}{127.181} = 0.040$, i.e. 4% — exactly the new periodic market rate, as expected.

YLD-Q6. $R = \frac{P_1 + b - P_0}{P_0} = \frac{955 + 40 - 980}{980} = \frac{15}{980} = 0.0153$, about 1.53%. Coupon component: $\frac{b}{P_0} = \frac{40}{980} = 0.0408$ (4.08%). Capital-gain component: $\frac{P_1 - P_0}{P_0} = \frac{955 - 980}{980} = -0.0255$ (–2.55%). The two sum to the total: 0.0408 – 0.0255 = 0.0153. The capital loss partly offsets the coupon income.

YLD-Q7. The periodic yield is $y = 0.04$. The bond equivalent yield (simple annualization) is $2y = 2(0.04) = 0.08$, or 8%. The effective annual yield is $(1 + y)^2 - 1 = (1.04)^2 - 1 = 1.0816 - 1 = 0.0816$, or 8.16%. The effective annual yield exceeds the bond equivalent yield by 16 basis points, reflecting compounding of the first semiannual coupon.

YLD-Q8. The periodic yield is $y = 0.035$. Bond equivalent yield: $2y = 2(0.035) = 0.07$, or 7%. Effective annual yield: $(1 + y)^2 - 1 = (1.035)^2 - 1 = 1.071225 - 1 = 0.071225$, or about 7.1225%. The effective annual yield exceeds the bond equivalent yield by 0.071225 – 0.07 = 0.001225, i.e. about 12.25 basis points (the interest-on-interest term $y^2 = 0.001225$).

YLD-Q9. Current yield = $\frac{b}{P} = \frac{65}{928.57} = 0.0700$, i.e. 7%. The coupon rate is $b/M = 65/1000 = 6.5\%$. Since $P = \$928.57 < M = \$1000$, the bond sells *below par*, so by the chapter's rule the yield to maturity is *above* the coupon rate of 6.5%.

YLD-Q10. Current yield = $\frac{b}{P} = \frac{70}{1050} = 0.0667$, i.e. 6.67%. Coupon rate $= b/M = 70/1000 = 0.07$, i.e. 7%. Since $P = \$1050 > M = \$1000$, the bond trades *above par*, so the yield to maturity is *below* the coupon rate. For a premium bond the ordering from highest to lowest is: coupon rate (7%) > current yield (6.67%) > yield to maturity (below 6.67%). The coupon rate tops the list because it is measured against face value; the current yield is lower because the price paid exceeds face; and the yield to maturity is lowest because it further subtracts the capital loss the holder will incur as the premium price declines to par at maturity.

17.8 The Yield Curve

YC-C1. When the yield curve is not flat, each cash flow of the note must be discounted at the spot rate whose maturity matches the timing of that cash flow, not at a single rate. The stripping-and-reconstitution argument shows why: the coupon due in one year is, by itself, a one-year zero and must sell at the one-year spot rate y_1; the coupon-plus-principal due in two years is a two-year zero and must sell at the two-year spot rate y_2. If the note were instead priced by discounting both flows at y_2, its price would differ from the sum of the stripped pieces, and one could strip (or reconstitute) the note through the Treasury to capture a riskless profit. The correct pricing rule is therefore $P = \frac{b}{1+y_1} + \frac{b+M}{(1+y_2)^2}$: coupon b discounted at y_1, and the final coupon-plus-face $b + M$ discounted at y_2. Discounting everything at y_2 is correct only in the special case of a flat curve, $y_1 = y_2$.

YC-C2. If the coupon bond and a portfolio of zeros with identical cash flows traded at different prices, the arbitrageur buys the cheaper and sells the dearer *today*. Concretely, if the coupon bond is cheaper than its stripped pieces, buy the bond, ask the Treasury to strip it into its component zeros, and sell those zeros for more than you paid — a riskless profit realized immediately. If the bond is more expensive than the pieces, do the reverse: buy the individual zeros, reconstitute them into the coupon bond through the Treasury, and sell the reconstituted bond. Because the cash flows are identical by construction, the trade carries no risk and requires no waiting for the market to "correct." The mere availability of this trade forces the coupon bond's price to equal the sum of the values of its stripped cash flows:

any gap would be seized instantly by arbitrageurs, so no gap can persist. This is the law of one price, and it holds independently of investor optimism or pessimism because it rests on cash-flow identity, not on beliefs.

YC-C3. The statement confuses a spot rate with a forward rate. The three-year spot rate y_3 is the yield to maturity, quoted today, on a zero-coupon bond that matures in three years; it summarizes the average rate an investor locks in over the entire three-year horizon starting now. It is not a forecast of any single future short rate. The forward rate f_3 is the future one-year rate, for the interval from year 2 to year 3, that is implied by today's curve; it is what the market's spot rates say the year-2-to-year-3 rate must be to rule out arbitrage. The two are linked by $1 + f_3 = \frac{(1+y_3)^3}{(1+y_2)^2}$: the three-year spot rate embeds the forward rates for each year, so f_3 is extracted from y_3 and y_2 rather than being read off y_3 directly. Only in a world of certainty does the forward rate f_3 equal the short rate r_3 that will actually prevail three years out.

YC-C4. The spot rate y_n is the yield to maturity observed *today* on an n-period zero, so it is observable now. The forward rate f_n is computed *today* from the observed spot curve via $1 + f_n = \frac{(1+y_n)^n}{(1+y_{n-1})^{n-1}}$, so it too is observable now — it is a deterministic function of today's data. The short rate r_n is the one-period rate that will actually prevail over interval n; except for r_1 (today's short rate, equal to y_1), the future short rates $r_2, r_3, \ldots$ are random variables not known until they arrive. Under certainty there is no randomness, so the no-arbitrage forces the realized short rate to equal the implied forward rate, $r_n = f_n$. Under uncertainty they generally differ, and the chapter's theories concern the sign of $f_n - Er_n$. Finally, y_n is always a geometric average because compounding over n periods gives $(1 + y_n)^n = (1 + r_1)(1 + r_2)\cdots(1 + r_n)$ (under certainty) or the analogous product of one-plus-forward-rates, so $1 + y_n = [(1 + r_1)\cdots(1 + r_n)]^{1/n}$.

YC-C5. The investor buys x/M units of the two-year zero today and simultaneously shorts enough one-year zeros to pay for them, so the net cash outlay today is zero. Wait — for a loan *received* at time 1 and repaid at time 2, the strategy is: short (sell) x/M one-year zeros today, which raises cash now, and use it to buy two-year zeros; equivalently, following the chapter's general construction, short enough two-year zeros to fund a purchase of one-year zeros so that x arrives at time 1 and $(1 + f_2)x$ is owed at time 2. Either way, the one-year zeros deliver exactly x at time 1 (the loan proceeds), and the short position in two-year zeros must be closed at time 2 by paying their face value, which totals $(1 + f_2)x$. Because every quantity is fixed today from the observed spot curve, there is no uncertainty. The effective borrowing rate is exactly f_2 — not y_1 (which covers only year 1) or y_2 (which covers the full two years) — because the trade isolates the single year-1-to-year-2 interval, and f_2 is precisely the rate the current curve implies for that interval.

YC-C6. No explicit forward contract is needed because a complete set of zeros lets the borrower *manufacture* the forward loan out of spot instruments. The construction shorts $(1 + f_2)\,x/M$ two-year zeros today and uses the proceeds to buy x/M one-year zeros today; the two legs are sized so that their time-0 cash flows exactly cancel, so nothing net changes hands today. At time 1, the one-year zeros mature and pay the borrower x — this is the loan being "received." Between time 0 and time 1 the two-year zero short position simply sits open, generating no intermediate cash flow. At time 2, the borrower must buy back the two-year zeros to close the short, paying their face value $(1+f_2)x$ — this is the loan being "repaid." Thus cash flows out only at time 2 and in only at time 1, exactly like a one-period loan running from year 1 to year 2, and the synthetic contract replicates a forward loan at rate f_2 without any counterparty signing a forward agreement.

YC-C7. With only long-horizon (two-year) investors, everyone can lock in a certain two-year payoff by holding the two-year zero; the roll-over strategy (one-year zero, then reinvest at the random r_2) exposes them to interest-rate risk they do not want. Since investors are risk-averse, they will hold the risky roll-over strategy only if it offers a *higher expected* payoff than the safe two-year zero. Formally, indifference requires $EU((1 + y_1)(1 + r_2)) = U((1 + y_2)^2)$, and Jensen's inequality (for the strictly concave U) gives $U(E[(1+y_1)(1+r_2)]) > EU((1+y_1)(1+r_2)) = U((1+y_2)^2)$. Since U is increasing, $E[(1 + y_1)(1 + r_2)] > (1 + y_2)^2 = (1 + y_1)(1 + f_2)$, hence $E(1 + r_2) > 1 + f_2$, i.e. $f_2 < Er_2$. The forward rate lies below the expected future short rate because long-horizon investors accept a lower implied rate on the safe long bond in exchange for avoiding reinvestment risk.

YC-C8. Under the expectations hypothesis $f_n = Er_n$, so the forward rates embedded in today's curve are exactly the market's forecasts of future short rates. Because the two-year (and longer) spot rate is a geometric average of the sequence of forward rates, $(1 + y_2)^2 = (1 + y_1)(1 + f_2)$, the curve slopes upward ($y_2 > y_1$) precisely when $f_2 > y_1$, i.e. when the market expects the future short rate Er_2 to exceed today's short rate y_1. So an upward slope signals expected rate increases. A downward-sloping (inverted) curve, $y_2 < y_1$, requires $f_2 < y_1$, which under the expectations hypothesis means $Er_2 < y_1$: the market expects future short rates to fall below today's short rate — the configuration often read as a recession signal.

YC-C9. Liquidity preference theory decomposes the forward rate as $f_2 = Er_2 + LP$ with $LP > 0$. Because the forward rate is what drives the slope — $(1 + y_2)^2 = (1 + y_1)(1 + f_2)$ — a positive LP can make $f_2 > y_1$, and hence $y_2 > y_1$, even when the market expects short rates to be flat ($Er_2 = y_1$). The economic force LP represents is a premium that risk-averse short-horizon investors demand as compensation for bearing the interest-rate risk of holding longer-term bonds. Contrast: under the expectations hypothesis an upward slope means the market forecasts *rising* short rates ($Er_2 > y_1$); under liquidity preference an upward slope can arise purely from the term premium LP, even with unchanged expected short rates. The two theories thus attribute the same observed slope to different causes — expectations versus risk compensation.

YC-C10. Short-horizon (one-year) investors would prefer to hold the safe one-year zero and earn y_1 with certainty. To hold the two-year zero for one year and sell it, they bear the risk that the resale price depends on the random future short rate r_2. Because they are risk-averse, they will accept this gamble only if its expected return exceeds the safe return, which forces $E\frac{(1+y_2)^2}{1+r_2} > \frac{(1+y_2)^2}{1+f_2}$; carrying the Jensen argument through shows this requires $f_2 > Er_2$, i.e. a positive liquidity premium $LP = f_2 - Er_2 > 0$. The liquidity preference theory posits that the market is dominated by these short-horizon investors — not the long-horizon investors of YC-C7 — so their demand for compensation dominates, pushing forward rates above expected future short rates and tilting the curve upward. It is the assumption that most real-world investors have short effective horizons (and are averse to interest-rate risk) that makes the short-investor case the empirically relevant one.

YC-C11. Under certainty the two strategies — (i) hold a one-year zero for one year, and (ii) buy a two-year zero and sell it after one year — deliver payoffs that are known in advance, and no-arbitrage forces their one-year returns to be equal. If they differed, an investor could short the lower-returning strategy and go long the higher-returning one for a riskless profit. After one year the two-year zero has become a one-year zero priced at $\frac{1000}{1+r_2}$, and substituting the no-arbitrage relation $1 + r_2 = \frac{(1+y_2)^2}{1+y_1}$ shows the resale value grows the purchase price by exactly a factor of $1 + y_1$. Therefore the *gross* one-year return is $1 + y_1$, and the *net* one-year holding-period return is $r = (1 + y_1) - 1 = y_1$. It is the net return that equals y_1; the gross return is $1 + y_1$.

YC-C12. The statement is wrong because it reports the *gross* return as if it were the holding-period (net) return. The gross return is the ratio of ending value to beginning value, $\frac{\text{resale price}}{\text{purchase price}}$, and under certainty this ratio equals $1 + y_1$. The *net* holding-period return is the gross return minus one, $\frac{\text{resale price}}{\text{purchase price}} - 1 = (1 + y_1) - 1 = y_1$. So the net one-year holding-period return is y_1, while $1 + y_1$ is the gross return. In the chapter's derivation the "−1" enters exactly at $r = \frac{(1+y_2)^2}{1+r_2} - 1$, i.e. when the ending-over-beginning price ratio is converted into a net return by subtracting the initial dollar. Saying the holding-period return is $1 + y_1$ double-counts the return of principal.

YC-C13. Under the expectations hypothesis, the forward rates embedded in the curve equal expected future short rates, so an inverted curve — long yields below short yields — means the market expects future short rates to *fall*. Since forward rates below today's short rate translate into $Er_n < y_1$, the market is forecasting rate cuts, which typically accompany an anticipated economic slowdown; this is why an inverted curve is historically a recession warning. A bank treasurer who borrows short and lends long should pay close attention: the profitability of that maturity mismatch depends on short funding rates staying low relative to the long assets, and an inversion signals both an expected reversal in rate direction and the heightened chance of a downturn that could impair the loan book and squeeze the funding spread. The configuration is precisely the term-structure signal the chapter frames as central to fixed-income risk management.

YC-C14. Borrowing short and lending long is typically profitable when the curve slopes upward because the bank pays the low short-term rate ($\approx y_1$) on its deposits while earning the higher long-term rate (y_n) on its loans, capturing the positive spread. In term-structure language, an upward slope means the forward rates — and hence the long spot rates that are their geometric average — exceed today's short rate, whether because the market expects rising short rates (expectations hypothesis) or because a positive term/liquidity premium LP is embedded (liquidity preference). The risk is that this spread is not locked in: the bank must keep rolling over its short-term funding at whatever short rate actually prevails, i.e. at the realized future short rates r_n, which are random. If realized short rates r_n turn out higher than the forward rates f_n priced into the long assets, the bank's funding cost rises above the yield on its fixed long-term loans and the spread can vanish or turn negative — the maturity-mismatch exposure that the forward-rate framework of this chapter lets the treasurer measure rather than ignore.

YC-Q1. No-arbitrage price with $b = 60$, $M = 1000$, $(y_1, y_2) = (0.030, 0.035)$:

$$P = \frac{60}{1.030} + \frac{60 + 1000}{1.035^2} = 58.2524 + \frac{1060}{1.071225} = 58.2524 + 989.5203 = 1047.773.$$

So the correct price is about \$1047.77. Discounting *both* flows at $y_2 = 0.035$ instead gives $\frac{60}{1.035} + \frac{1060}{1.035^2} = 57.9710 + 989.5203 = 1047.491$, about \$1047.49 — a different (incorrect) price because the year-1 coupon is over-discounted at $y_2 > y_1$. The two agree only if the curve were flat.

YC-Q2. (a) Spot rates from $P = \frac{1000}{(1+y_n)^n}$: y_1: $1 + y_1 = 1000/970.874 =$

$1.030000 \Rightarrow y_1 = 0.0300$. y_2: $(1+y_2)^2 = 1000/942.596 = 1.060897 \Rightarrow 1+y_2 = 1.030000 \Rightarrow y_2 = 0.0300\ldots$ check: $1.030000^2 = 1.0609$, and $1000/1.0609 = 942.60$, consistent, so $y_2 = 0.0300$. Recompute precisely: $1000/942.596 = 1.060897$; $\sqrt{1.060897} = 1.030$; $y_2 = 0.0300$. y_3: $(1+y_3)^3 = 1000/915.142 = 1.092727 \Rightarrow 1+y_3 = 1.092727^{1/3} = 1.030 \Rightarrow y_3 = 0.0300$. (Here the three prices happen to imply a flat 3.00% curve.) (b) Three-year note, face \$1000, annual coupon \$50, discount each flow at its own spot rate:

$$P = \frac{50}{1.030} + \frac{50}{1.030^2} + \frac{1050}{1.030^3} = 48.5437 + 47.1298 + 960.9787 = 1056.652.$$

Price is about \$1056.65. (Equivalently, using the observed zero prices per \$1000 par: $50 \cdot 0.970874 + 50 \cdot 0.942596 + 1050 \cdot 0.915142 = 48.5437 + 47.1298 + 960.899 = 1056.57$, matching up to rounding of y_3.)

YC-Q3. With $(y_1, y_2, y_3) = (0.030, 0.035, 0.038)$:

$$1 + f_2 = \frac{(1.035)^2}{1.030} = \frac{1.071225}{1.030} = 1.040024 \Rightarrow f_2 = 0.040024 \approx 4.00\%.$$

$$1 + f_3 = \frac{(1.038)^3}{(1.035)^2} = \frac{1.118386}{1.071225} = 1.044026 \Rightarrow f_3 = 0.044026 \approx 4.40\%.$$

YC-Q4. Spot rates from prices ($M = 1000$): $1 + y_1 = 1000/952.381 = 1.050000 \Rightarrow y_1 = 0.0500$. $(1+y_2)^2 = 1000/898.452 = 1.113023 \Rightarrow 1 + y_2 = 1.055000 \Rightarrow y_2 = 0.0550$. Forward rate:

$$1 + f_2 = \frac{(1.055)^2}{1.050} = \frac{1.113025}{1.050} = 1.060024 \Rightarrow f_2 = 0.060024 \approx 6.00\%.$$

YC-Q5. Under certainty $(1+y_3)^3 = (1+r_1)(1+r_2)(1+r_3)$:

$$(1+y_3)^3 = 1.020 \times 1.030 \times 1.040 = 1.092624.$$

$$1 + y_3 = 1.092624^{1/3} = 1.029968 \Rightarrow y_3 = 0.029968 \approx 3.00\%.$$

YC-Q6. Arbitrage relation $(1+y_2)^2 = (1+y_1)(1+r_2)$ with $y_2 = 0.05$, $y_1 = 0.04$:

$$1 + r_2 = \frac{(1.05)^2}{1.04} = \frac{1.1025}{1.04} = 1.060096 \Rightarrow r_2 = 0.060096 \approx 6.01\%.$$

YC-Q7. $(y_1, y_2) = (0.04, 0.05)$, face \$1000. (a) Purchase price of the two-year zero today: $P_0 = \frac{1000}{(1.05)^2} = \frac{1000}{1.1025} = 907.029$. (b) Short rate under certainty: $1 + r_2 = \frac{(1.05)^2}{1.04} = \frac{1.1025}{1.04} = 1.060096 \Rightarrow r_2 = 0.060096$. After one year the bond is a one-year zero, resale price $P_1 = \frac{1000}{1+r_2} = \frac{1000}{1.060096} = 943.310$. (c) Net one-year holding-period return: $r = \frac{P_1 - P_0}{P_0} = \frac{943.310 - 907.029}{907.029} = \frac{36.281}{907.029} = 0.04000 = y_1$. It equals $y_1 = 0.04$. ✓ (d) The corresponding gross return is $\frac{P_1}{P_0} = \frac{943.310}{907.029} = 1.04000 = 1 + y_1$.

YC-Q8. $(y_1, y_2) = (0.03, 0.045)$, face \$1000. Purchase price today: $P_0 = \frac{1000}{(1.045)^2} = \frac{1000}{1.092025} = 915.730$. Short rate: $1 + r_2 = \frac{(1.045)^2}{1.03} = \frac{1.092025}{1.03} = 1.060218 \Rightarrow r_2 = 0.060218$. Resale price after one year: $P_1 = \frac{1000}{1.060218} = 943.201$. Gross one-year holding-period return: $\frac{P_1}{P_0} = \frac{943.201}{915.730} = 1.03000 = 1 + y_1$. Net return: $1.03000 - 1 = 0.03000 = y_1$. ✓ Reporting $1 + y_1 = 1.03$ as "the holding-period return" would be an error because that is the *gross* return (ending value per dollar invested); the holding-period (net) return subtracts the returned principal, giving $y_1 = 0.03$.

YC-Q9. $(y_1, y_2) = (0.0154, 0.0174)$. Forward rate: $1 + f_2 = \frac{(1.0174)^2}{1.0154} = \frac{1.035118}{1.0154} = 1.019419 \Rightarrow f_2 = 0.019419 \approx 1.9419\%$. Under the expectations hypothesis $Er_2 = f_2 = 0.019419$. Under liquidity preference with $LP = 0.004$ and the same f_2: $Er_2 = f_2 - LP = 0.019419 - 0.004 = 0.015419 \approx 1.5419\%$. The expected future short rate is lower once the term premium is stripped out of the forward rate.

YC-Q10. $y_1 = 0.03$, $Er_2 = 0.05$. (a) Expectations hypothesis, $f_2 = Er_2 = 0.05$: $(1 + y_2)^2 = (1.03)(1.05) = 1.0815 \Rightarrow 1 + y_2 = 1.039952 \Rightarrow y_2 = 0.039952 \approx 4.00\%$. The curve slopes up ($y_2 > y_1$) because the market expects the short rate to rise. (b) With $LP = 0.01$, $f_2 = Er_2 + LP = 0.06$: $(1 + y_2)^2 = (1.03)(1.06) = 1.0918 \Rightarrow 1 + y_2 = 1.044892 \Rightarrow y_2 = 0.044892 \approx 4.49\%$. The liquidity premium raises the forward rate and hence the two-year spot rate, steepening the curve relative to the expectations-hypothesis case (4.49% vs 4.00%).

YC-Q11. $(y_1, y_2) = (0.020, 0.025)$, $M = 1000$, want \$2M at time 1. (a) Prices: one-year zero $P_1 = \frac{1000}{1.020} = 980.392$; two-year zero $P_2 = \frac{1000}{(1.025)^2} = \frac{1000}{1.050625} = 951.814$. (b) To receive \$2,000,000 at time 1, buy 2,000,000/1000 = 2000 one-year zeros, costing $2000 \times 980.392 = \$1{,}960{,}784$. (c) Fund this by shorting $q = 1{,}960{,}784/951.814 = 2060.264$ two-year zeros. (d) Repay at time 2 the face value of the shorted zeros: $qM = 2060.264 \times 1000 = \$2{,}060{,}264$. Verification: effective interest = $2{,}060{,}264 - 2{,}000{,}000 = \$60{,}264$ on \$2M, i.e. 60,264/2,000,000 = 0.030132. Forward rate $1 + f_2 = \frac{(1.025)^2}{1.020} = \frac{1.050625}{1.020} = 1.030025 \Rightarrow f_2 = 0.030025$, matching the effective rate (up to rounding). ✓

YC-Q12. $(y_1, y_2, y_3) = (0.030, 0.035, 0.040)$, $x = \$500{,}000$, $M = 1000$. Forward rate: $1 + f_3 = \frac{(1.040)^3}{(1.035)^2} = \frac{1.124864}{1.071225} = 1.050073 \Rightarrow f_3 = 0.050073 \approx 5.01\%$. Number of (time-3) zeros to short: $q = (1 + f_3)\frac{x}{M} = 1.050073 \times \frac{500{,}000}{1000} = 1.050073 \times 500 = 525.036$. Amount repaid at time 3: $(1 + f_3)x = 1.050073 \times 500{,}000 = \$525{,}036$ (equivalently $qM = 525.036 \times 1000 = \$525{,}036$). The interest cost is about \$25,036, the forward rate $f_3 \approx 5.01\%$ applied to the \$500,000 loan.

YC-D1. $f_2 = \dfrac{(1.0426)^2}{1.0412} - 1 = \dfrac{1.08702}{1.0412} - 1 \approx 0.0440$ (4.40%), and $f_3 = \dfrac{(1.0430)^3}{(1.0426)^2} - 1 = \dfrac{1.13463}{1.08702} - 1 \approx 0.0438$ (4.38%). Today's forwards sit within a few basis points of today's spot rates because the current curve is nearly flat beyond one year, whereas the steeply rising short end of the 2008 curve implied forwards well above spots — a market pricing in rising short rates.

YC-D2. Under the expectations hypothesis, forward rates equal expected future short rates, so an inverted curve implies the market expects short rates to *fall* substantially from 5.4%. Under liquidity preference, $f_n = Er_n + LP$ with $LP > 0$, so $Er_n = f_n - LP$ lies *below* the forward rate: the expected decline in short rates is even *larger* than the raw forwards suggest. An inverted curve is thus an especially strong signal of expected easing once a positive term premium is accounted for.

17.9 Fixed Income Portfolio Management

FIP-C1. Property 2 states that Macaulay duration decreases in the coupon rate. Bond A (3% coupon) delivers a larger fraction of its total present value at maturity, so the center of gravity of its cash-flow stream sits farther out in time; bond B (9% coupon) returns more value early, pulling its center of gravity toward the present. Hence $D_A > D_B$. Because modified duration $D^* = D/(1+y)$ inherits this ordering and $\frac{\Delta P}{P} \approx -D^*\Delta y$, the higher-duration bond A loses the larger percentage of its value when yields rise by a small Δy.

FIP-C2. In $D = \frac{1}{P}\sum_t \frac{t\,CF_t}{(1+y)^t}$, the weight attached to time t is the present value $CF_t/(1+y)^t$ divided by P. Raising y discounts distant cash flows more heavily than near ones, because $(1+y)^{-t}$ falls faster the larger t is. The far-future payments — which carry the largest time indices t and therefore the biggest terms in the numerator — lose the most weight in the average. This shifts the time-weighted average (the center of gravity) toward the present, so Macaulay duration falls as y rises.

FIP-C3. Modified duration measures interest-rate sensitivity: $\frac{\Delta P}{P} \approx -D^*\Delta y$. Although both 30-year bonds mature at the same time, the coupon bond returns substantial value before maturity, so its cash-flow center of gravity — its Macaulay duration — is well below 30, whereas the zero's Macaulay duration equals its maturity $T = 30$. The zero therefore has the far larger D^* and is much more sensitive, so it falls more in price for a given increase in y. “Same maturity” is not “same sensitivity.”

FIP-C4. With $\frac{\Delta P}{P} \approx -D^*\Delta y$, sensitivity is governed entirely by D^*. Longer maturity spreads cash flows further into the future, raising duration (property 3), while a lower coupon shifts more value toward the distant principal payment, also raising duration (property 2). For a long-maturity, low-coupon bond both effects push duration up, so it is far more rate-sensitive than a short-maturity, high-coupon bond, where both effects push duration down. Because D^* collapses the whole cash-flow timing into a single number, and portfolio duration is the value-weighted average of component durations, a manager can compare and aggregate this sensitivity across the entire portfolio with one statistic.

FIP-C5. Property 2: duration decreases in the coupon rate, because a higher coupon moves more of the bond's value forward in time, lowering the time-weighted average date at which the holder receives value. A fixed-coupon perpetuity has no maturity yet a duration of $(1 + y)/y$ that is independent of maturity (it has no maturity to depend on); a zero coupon bond has duration exactly equal to its maturity T and pays no coupon at all. It is the perpetuity whose duration does not depend on maturity. This connects to property 2 because as the coupon rate rises the bond behaves more like a stream of early payments — its center of gravity, and hence its duration, moves toward the present.

FIP-C6. Lengthening maturity usually adds cash flows further into the future and pushes the principal payment further out, both of which raise the time-weighted average date of receipt and hence duration. For a bond selling at a very deep discount, however, the price is dominated by the distant principal while the coupons are small relative to that principal's present value; adding still more maturity can change the balance of present values in a way that fails to increase — and can even slightly decrease — the time-weighted average, so duration need not rise monotonically with maturity.

FIP-C7. "Duration-matched" means the Macaulay (equivalently modified) duration of the asset portfolio equals that of the liability stream, so both sides of the balance sheet have the same first-order sensitivity to yield. (a) If asset duration equals liability duration, a small parallel rise in yields lowers the present value of assets and liabilities by approximately the same percentage, leaving the net position roughly unchanged — the fund is immunized. (b) If asset duration is much shorter than liability duration, a rise in yields shrinks the liabilities more than the assets, which happens to help; but the dangerous case is a rise when assets are *longer* than liabilities. Silicon Valley Bank held long-duration Treasuries against short-duration deposits: when rates rose in 2022 the assets fell far more in value than the liabilities, opening a large funding gap and forcing losses on sale.

FIP-C8. The condition $V = 0$ guarantees that the present value of the zero exactly funds the present value of the obligation today ($Z = P_{obl}$): the two sides start balanced. The condition $\partial V/\partial y = 0$ guarantees that a small yield change moves both sides equally, so the net position is first-order insensitive to yield. Writing $\partial P_{obl}/\partial(1+y) = -\frac{D_{obl}}{1+y}P_{obl}$ and the zero's $\partial Z/\partial(1+y) = -\frac{\tau}{1+y}Z$, the second condition becomes $\frac{D_{obl}}{1+y}P_{obl} - \frac{\tau}{1+y}Z = 0$; since $V = 0$ already forces $Z = P_{obl}$, the common factors cancel and $\tau = D_{obl}$. It is the Macaulay duration of the obligation, not its maturity T, because duration — not maturity — measures the yield sensitivity that must be matched, and the obligation pays value throughout its life rather than all at T.

FIP-C9. Bond price is convex in yield, so the price-yield curve lies above its tangent line. Duration is the slope of that tangent (a first-order, linear approximation). The positive second derivative $\partial^2 P/\partial y^2 = \sum_t \frac{t(t+1)CF_t}{(1+y)^{t+2}} > 0$ means the true price curves upward away from the tangent as Δy grows, so the linear estimate falls below the true price and understates it by more the larger the move. The sign of the error is the same whether yields rise or fall because convexity is a symmetric curvature term: the correction $+\frac{1}{2}CP(\Delta y)^2$ depends on $(\Delta y)^2$, which is positive regardless of the direction of the move.

FIP-C10. For two bonds of identical duration, convexity is a free upgrade: because the price-yield curve of the more convex bond bows out more, it gains more when yields fall and loses less when yields rise, so it weakly dominates the less convex bond for any yield move. A rational investor therefore always prefers higher convexity, and competition bids its price up — the manager is willing to pay a premium because the extra convexity delivers a favorable asymmetric payoff at no cost in duration.

FIP-C11. The hedge must satisfy three conditions: match present value, match duration, and match convexity. A single zero offers only two free choices (its face value F and its maturity τ), which can satisfy at most present value and duration — leaving convexity unmatched. Two zeros provide four free choices (p_1, p_2, τ_1, τ_2), enough to satisfy all three matching equations with a degree of freedom to spare. In the worked example, fixing equal present-value weights reduced the duration and convexity conditions to $\tau_1 + \tau_2 = 5.472$ and $\tau_1(\tau_1 + 1) + \tau_2(\tau_2 + 1) = 21.190$ — two equations in the two maturities, which a single zero could never satisfy simultaneously.

FIP-C12. The present value of a portfolio is the sum of the present values of its parts, and both duration and convexity are defined through derivatives of price divided by price. Differentiating $V = p_1 + p_2$ and dividing by V makes each component's contribution scale by its share of total value $w_i = p_i/V$; hence portfolio duration and convexity are the *present-value*-weighted (not face-value-weighted) averages of the components'. Matching Macaulay duration automatically matches modified duration because $D^* = D/(1 + y)$ and both the obligation and the hedge are discounted at the same yield y, so the common factor $1/(1 + y)$ cancels from both sides of the matching equation — only one duration condition is needed.

FIP-Q1. With cash flows $CF_1 = CF_2 = 60$ and $CF_3 = 1060$, $y = 0.06$, $P = 1000$:

$$D = \frac{1}{1000}\left(1 \cdot \frac{60}{1.06} + 2 \cdot \frac{60}{1.06^2} + 3 \cdot \frac{1060}{1.06^3}\right).$$

The terms are 56.60, 106.79, and 2669.99, summing to 2833.39. Dividing by 1000 gives $D = 2.8334$ years.

FIP-Q2. With $CF_1 = 40$, $CF_2 = 1040$, $y = 0.04$, $P = 1000$:

$$D = \frac{1}{1000}\left(1 \cdot \frac{40}{1.04} + 2 \cdot \frac{1040}{1.04^2}\right) = \frac{1}{1000}(38.46 + 1923.08) = \frac{1961.54}{1000} = 1.9615 \text{ years.}$$

FIP-Q3. A zero's Macaulay duration equals its maturity, so $D = T = 8$ years. Its modified duration is

$$D^* = \frac{D}{1 + y} = \frac{8}{1.05} = 7.619 \text{ years.}$$

(The face value $M = 5000$ does not affect duration.)

FIP-Q4. $D^* = \frac{D}{1+y} = \frac{2.833}{1.06} = 2.673$ years. Economic meaning: since $\frac{\Delta P}{P} \approx -D^*\Delta y$, a 1-percentage-point rise in yield ($\Delta y = 0.01$) lowers the bond's value by about $D^* \times 0.01 = 2.673\% \approx 2.67\%$.

FIP-Q5. Percentage change: $\frac{\Delta P}{P} = -D^*\Delta y = -7.2 \times 0.0040 = -0.0288 = -2.88\%$. Dollar change: $\Delta P = -0.0288 \times 40{,}000 = -\$1{,}152$. The portfolio loses about \$1,152.

FIP-Q6. Percentage change: $\frac{\Delta P}{P} = -D^*\Delta y = -6.5 \times (-0.0025) = +0.01625 = +1.625\%$. Dollar change: $\Delta P = 0.01625 \times 25{,}000 = +\$406.25$. A yield decline raises the portfolio value by about \$406.25.

FIP-Q7. For first-order immunity the zero's modified duration must equal the liability's, $D^* = 11$ years, so that both sides move together under $\frac{\Delta P}{P} \approx -D^*\Delta y$. A zero's modified duration is $D^* = \tau/(1+y)$, so

$$\tau = D^*(1+y) = 11 \times 1.04 = 11.44 \text{ years.}$$

(The zero must also have present value \$250,000 to satisfy $V = 0$.)

FIP-Q8. Present value: $V = \frac{200}{0.05}\left(1 - \frac{1}{1.05^4}\right) = 4000\,(1 - 0.822702) = 4000 \times 0.177298 = \$709.19$. Required face value: from $V = F/(1+y)^\tau$,

$$F = V(1+y)^\tau = 709.19 \times 1.05^{2.5} = 709.19 \times 1.12968 = \$801.19.$$

FIP-Q9. With $CF_1 = 50$, $CF_2 = 1050$, $y = 0.05$, $P = 1000$:

$$Convexity = \frac{1}{1000(1.05)^2}\left(\frac{1 \cdot 2 \cdot 50}{1.05^1} + \frac{2 \cdot 3 \cdot 1050}{1.05^2}\right).$$

The bracket is $\frac{100}{1.05} + \frac{6300}{1.1025} = 95.238 + 5714.286 = 5809.524$. Dividing by $1000 \times 1.1025 = 1102.5$ gives $C = 5.2694$.

FIP-Q10. Using the zero's closed form with $T = 3$, $y = 0.05$:

$$Convexity = \frac{T(T+1)}{(1+y)^2} = \frac{3 \cdot 4}{1.05^2} = \frac{12}{1.1025} = 10.884.$$

Check via the general formula on a single cash flow at $t = 3$ with $P = M/1.05^3$: the numerator $\frac{3 \cdot 4\, M}{1.05^3}$ divided by $P(1.05)^2 = \frac{M}{1.05^3}(1.05)^2$ leaves $\frac{12}{1.05^2} = 10.884$ — the same value.

FIP-Q11. $D^* = D/(1+y) = 1.9524/1.05 = 1.85943$. With $P = 1000$, $C = 5.2694$, $\Delta y = 0.02$:

$$\Delta P \approx -D^* P\,\Delta y + \tfrac{1}{2} C P (\Delta y)^2 = -1.85943(1000)(0.02) + \tfrac{1}{2}(5.2694)(1000)(0.0004).$$

This is $-37.1886 + 1.0539 = -\$36.13$. The convexity term adds back about \$1.05 relative to the duration-only estimate of $-\$37.19$.

FIP-Q12. $D^* = 1.9524/1.05 = 1.85943$. With $P = 1000$, $C = 5.2694$, $\Delta y = -0.03$:

$$\Delta P \approx -1.85943(1000)(-0.03) + \tfrac{1}{2}(5.2694)(1000)(0.0009) = +55.783 + 2.371 = +\$58.15.$$

Duration alone predicts a +\$55.78 gain; the convexity term (+\$2.37, always positive because it depends on $(\Delta y)^2$) raises the estimated gain. Convexity works in the investor's favor: it enlarges the gain when yields fall, just as (in FIP-Q11) it cushions the loss when yields rise.

FIP-D1. (a) $\Delta P/P \approx -7.9 \times 0.01 = -7.9\%$. (b) At $y = 5.62\%$ the price is $P = 23.10 \times \dfrac{1-(1.0281)^{-20}}{0.0281} + 1000(1.0281)^{-20} \approx 23.10 \times 15.144 + 574.46 \approx$ \$924.3, an exact change of −7.57%. (c) The duration estimate (−7.9%) is more negative than the exact change (−7.6%): the tangent line lies below the convex price curve, which is exactly the shaded convexity gap in the figure.

FIP-D2. (a) $\Delta P/P \approx -17 \times 0.042 = -71.4\%$. (b) First, convexity: for a move of four-plus percentage points the price curve bends far above its tangent, so the linear rule badly overstates the loss. Second, the bond aged: by late 2023 its remaining maturity (and hence duration) was materially shorter than at issue, and coupon accrual partly offset price decline. (c) Every cash flow will still be paid in full — the loss is purely a change in the *present value* of fixed future payments as the discount rate rose; duration risk requires no credit event.

17.10 Forwards and Futures

FF-C1. The gold-mining company is the *hedger*: it already has an asset position (10,000 ounces it will produce and sell) and wants to remove the price risk. It goes *short* a gold forward/futures contract — agreeing to deliver (sell) gold at a locked-in price K. If the spot price falls, the loss on the physical gold is offset by the gain $K - S_T$ on the short contract, so the company's realized sale price is fixed near K. The hedge-fund trader is the *speculator*: holding no gold, the trader has no underlying exposure to offset and simply wants to profit from a directional view. Believing prices will rise, the trader goes *long* a contract — agreeing to buy at K — earning $S_T - K > 0$ if the price rises as expected. The long trader takes on precisely the price risk the miner sheds.

FF-C2. The farmer is short futures at price K, so the short payoff is $K - S_T$. When the spot price rises sharply to a high S_T, the farmer sells the physical wheat in the spot market at that high price, earning *more* revenue than expected — but the short futures position loses $S_T - K$ (i.e. the payoff $K - S_T$ is negative). These two effects offset: the extra revenue on the physical sale is handed over on the futures, so the farmer's net proceeds are approximately the locked-in price K per bushel regardless of S_T. The farmer is not made worse off in absolute terms (net proceeds are what was targeted, near K); the hedge did exactly its job of removing uncertainty. What the farmer locked in is the delivery price K; what the farmer gave up is the *upside* — the windfall that an unhedged farmer would have kept when prices rose. Hedging trades away both downside and upside for certainty.

FF-C3. Being long a forward that delivers Y units in one year is economically equivalent to a second strategy: borrow SY today at rate r, buy Y units spot now, and hold them. Both strategies deliver Y units of the asset in one year in exchange for a cash outlay — the forward pays FY at delivery, while the borrow-and-carry strategy repays the loan $(1 + r)SY$. Because the two portfolios produce identical positions, no-arbitrage forces equal cost: $(1+r)SY = FY$, i.e. $F = (1+r)S$. If instead $(1+r)S > F$, an arbitrageur *short-sells the asset* today, receiving S and investing it at r, while simultaneously going *long the forward*. In one year the invested proceeds have grown to $(1 + r)S$; the trader pays F under the forward to receive the asset, uses it to close the short, and keeps the riskless profit $(1 + r)S - F > 0$. Such trading bids the forward price back up until $(1 + r)S = F$.

FF-C4. Buying the asset today ties up cash that could otherwise earn the risk-free rate (or fund another use); the forward defers payment to delivery. The forward price therefore embeds the financing cost of carrying the asset — this is the cost-of-carry intuition, giving $F = Se^{rT}$ (continuous compounding) so that $F > S$ when $r > 0$. If, however, holding the asset itself *generates* value while it is carried — a dividend, a lease/storage-adjusted income stream, or a convenience yield k (the benefit of physically holding a commodity) — then carrying the asset is cheaper by that yield. The net cost of carry falls to $r - k$, and the pricing relation becomes $F = Se^{(r-k)T}$. When $k > r$ the forward price falls *below* the spot price; when $k < r$ it remains above. The yield k thus directly offsets the financing cost in the exponent.

FF-C5. An over-the-counter forward is a private bilateral agreement, so each side bears *counterparty credit risk*: if the price moves far against one party, that party may default, and the winning party has no institutional protection. A futures contract removes this by being *standardized* (fixed quantity, grade, and delivery dates) so it can trade on an organized exchange, and by interposing the exchange/clearinghouse as counterparty to every trade. To keep default risk from accumulating, the exchange requires each trader to post *margin* and *marks the contract to market daily*: gains and losses are settled every day by transferring cash between the short and long margin accounts. Because losses are collected as they occur rather than allowed to build up until maturity, no party can accumulate a large unpaid obligation, and a trader whose margin falls below the maintenance level receives a margin call. Standardization plus daily settlement together convert the forward's open-ended credit exposure into a tightly controlled, continuously-settled position.

FF-C6. Both long traders are marked to market daily. Each day the corn futures price falls, the contract loses value, so the clearinghouse *deducts* that day's price decline (per bushel × 5,000 bushels per contract) from each long trader's margin account and credits it to the shorts. As the price keeps falling over the week, the long accounts are debited each day and the balances erode. A *margin call* is triggered when an account's balance falls to (or below) the maintenance margin level; the trader must then deposit additional funds to restore the account, or the position is closed out. This differs from an otherwise-identical forward, which involves *no interim cash flows at all*: a forward settles only once, at maturity, so the same cumulative loss would be realized as a single payment at delivery. The futures thus demands cash *as losses accrue* (creating interim liquidity/financing demands), whereas the forward defers the entire settlement to expiration.

FF-C7. Basis risk is the residual price uncertainty that remains because the hedging instrument does not perfectly match the exposure. In the airline's hedge there are two distinct mismatches. First, an *asset mismatch*: it hedges jet fuel using crude-oil futures, and the jet-fuel price and crude-oil price do not move one-for-one (refining spreads vary), so the spot price of fuel it must buy differs from the crude price the futures tracks. Second, a *timing mismatch*: the crude contract delivers one month after the airline actually buys fuel, so even for the same commodity the futures price at close-out need not equal the spot price on the airline's purchase date. Because of these gaps, the price the airline effectively pays after closing the futures position is the futures price plus the (uncertain) basis, not a perfectly fixed number. Closing out the position removes the outright price level risk but leaves the basis — the difference between the fuel spot and the crude futures — unhedged, so the hedge is imperfect.

FF-C8. By definition $Basis(T) = S(T) - F_\tau(T)$, where $S(T)$ is the June spot and $F_\tau(T)$ is the July futures price at the June close-out date. The basis is generally *not zero* because the July contract has not expired at time T: it still reflects an extra month of cost-of-carry (financing, storage, convenience yield) and its own supply/demand, so July futures and June spot differ. Convergence forces the basis toward zero only *at* the July contract's expiration, not in June. Nonetheless, the basis is far smaller and more predictable than the outright price: over a one-month horizon the *difference* between two closely related prices moves much less than either price level moves on its own. The farmer with no hedge faces the full swing in the spot price of wheat; the hedged farmer faces only the residual variation in the basis, which is a substantial reduction in risk even though it is not zero.

FF-C9. In such a market the natural hedgers are producers who are long the physical commodity and want to lock in a sale price, so they crowd onto the *short* side of the futures. For the market to clear, speculators must be induced to take the *long* side — bearing the price risk the producers shed. Speculators will only accept that risk if they expect to be compensated, which requires that the price they pay today be low enough to leave an expected profit: the futures price must sit *below* the expected future spot price, $F < ES_T$ (equivalently $ES_T > F$), the condition of normal backwardation. The long speculator then expects, on average, to buy at F and see the asset worth $ES_T > F$ at delivery, earning $ES_T - F$. Economically, this discount is a *risk premium* — the reward paid by hedging producers to speculators for absorbing commodity price risk.

FF-C10. The expectations hypothesis holds that the futures price is simply the market's unbiased forecast of the future spot, $ES_T = F$, with no risk premium built in either direction. Contango is the reverse of normal backwardation: it arises when the *consumers* of a commodity (natural buyers) dominate the demand for hedging. Wishing to lock in a purchase price, these hedgers crowd onto the *long* side of the futures market. To clear the market, speculators must be induced to take the *short* side and bear the risk the consumers shed. They will do so only if compensated, which requires selling futures at a price *above* the expected future spot, so that on average they gain $F - ES_T$. Hence in contango $ES_T < F$: the short side (the speculators) is rewarded for holding the risk the long-hedging consumers wanted to offload.

FF-C11. From $F = e^{(r-k)T} ES_T$, the sign of $F - ES_T$ is governed by $r - k$. A high-β commodity carries substantial systematic risk, so investors demand a high required discount rate k to hold it — in particular $k > r$. Then $r - k < 0$, the exponential factor $e^{(r-k)T} < 1$, and the futures price lies *below* the expected spot, $F < ES_T$ (normal backwardation). The pure expectations hypothesis, by contrast, predicts $F = ES_T$ exactly, with no discount. The two views are inconsistent for a positive-β asset precisely because expectations pricing ignores the risk premium: a positive-β asset commands $k > r$, which drives a wedge $F < ES_T$ that the expectations hypothesis assumes away. Only for a zero-risk-premium asset ($k = r$) do the two coincide.

FF-C12. The two starting equations are pricing statements for the same spot price S. First, $S = e^{-kT}ES_T$ discounts the *expected* future spot at the risk-adjusted rate k — the rate investors require to hold the (risky) asset, so it embeds the asset's risk premium. Second, $S = e^{-rT}F$ is the cost-of-carry relation $F = Se^{rT}$ rearranged, discounting the futures price at the *risk-free* rate (valid when interest rates are uncorrelated with the mark-to-market flows). Setting the two right-hand sides equal, $e^{-rT}F = e^{-kT}ES_T$, and multiplying both sides by e^{rT} isolates $F = e^{(r-k)T}ES_T$. For a negative-β asset, its returns move opposite the market, so it is *desirable* as a hedge and investors accept a low required return: $k < r$. Then $r - k > 0$, $e^{(r-k)T} > 1$, and $F > ES_T$ — the futures price exceeds the expected spot (a contango outcome), because the asset's negative systematic risk commands a *negative* risk premium.

FF-Q1. Cost-of-carry gives the fair forward $F = (1+r)S = (1.04)(30) = \$31.20$ per ounce. If the market instead quotes $F = \$32 > 31.20$, the forward is overpriced, so arbitrage the pattern $(1 + r)S < F$: *short the forward* (agree to deliver at 32) and *borrow* 30 *to buy one ounce spot now*. In one year, deliver the ounce to settle the short forward, receive \$32, and repay the loan $(1 + r)S = \$31.20$. Riskless profit $= F - (1 + r)S = 32 - 31.20 = \$0.80$ per ounce.

FF-Q2. Fair forward: $F = Se^{rt} = 2000\,e^{0.05\times0.5} = 2000\,e^{0.025} = 2000(1.025315) = \$2{,}050.63$. The actual six-month forward of \$2,080 exceeds this, so the forward is *overpriced* relative to cost-of-carry by $2080 - 2050.63 = \$29.37$.

FF-Q3. Gold forward: $F = Se^{rt} = 1800\,e^{0.05\times0.5} = 1800\,e^{0.025} = 1800(1.025315) = \$1{,}845.57$ per ounce. Currency implied rate: with

$S = 1.4208$, $F = 1.4187$, $T = 0.25$,

$$r = \frac{1}{T}\ln\left(\frac{F}{S}\right) = \frac{1}{0.25}\ln\left(\frac{1.4187}{1.4208}\right) = 4\ln(0.998522) = 4(-0.0014791) = -0.0059165 \approx -0.59\%.$$

The implied rate is **negative**. Because the forward is *below* the spot ($F < S$), the ratio $F/S < 1$, its natural log is negative, and so is r: the currency's forward trading at a discount to spot signals a slightly negative annualized implied dollar interest rate over the quarter.

FF-Q4. Long payoff $S_T - K$ with $K = 59$: (a) $56 - 59 = -\$3$; (b) $63 - 59 = +\$4$. Short payoff $K - S_T$: (a) $59 - 56 = +\$3$; (b) $59 - 63 = -\$4$. In each case the long and short payoffs sum to zero: $-3 + 3 = 0$ and $+4 + (-4) = 0$. A forward is a zero-sum contract — one party's gain is exactly the other's loss.

FF-Q5. (a) Physical revenue: $1{,}000{,}000 \times \$52 = \$52{,}000{,}000$. (b) Short futures payoff per barrel is $K - S_T = 59 - 52 = \$7$; over 1,000,000 barrels (1,000 contracts $\times$ 1,000 bbl) the gain is $7 \times 1{,}000{,}000 = \$7{,}000{,}000$. (c) Net proceeds $= \$52\text{M} + \$7\text{M} = \$59{,}000{,}000$, i.e. \$59 per barrel — the hedge locks in approximately the futures price of \$59/barrel regardless of the realized spot.

FF-Q6. $Basis(T) = S(T) - F_\tau(T) = 6.20 - 6.35 = -\$0.15$ per bushel. The basis is *negative* (spot below futures). A farmer who is *short* the futures to hedge a sale effectively receives $F_{\text{initial}} + Basis(T)$ at close-out (sell spot at $S(T)$, buy back the futures at $F_\tau(T)$); a negative basis means the spot came in below the futures level, which *hurt* the farmer relative to a zero basis — the realized proceeds are \$0.15/bushel lower than they would have been had the basis been zero.

FF-Q7. $Basis(T) = 385 - 383 = 2$ cents per bushel (spot above futures). Since convergence forces the basis to zero at expiration, a trader expecting the 2-cent gap to close can *buy the futures* (long, at 383) and *sell the corn short in the spot market* (at 385); as the futures price rises toward the spot at expiration, the trader closes both legs and captures the 2-cent convergence. Gross profit per 5,000-bushel contract $= 0.02 \times 5{,}000 = \$100$ (2 cents $\times$ 5,000 bushels).

FF-Q8. $F = e^{(r-k)T}ES_T = e^{(0.03-0.08)(1)}(100) = e^{-0.05}(100) = 0.951229 \times 100 =$ \$95.12. Since $k > r$, the exponent is negative and $F =$ \$95.12 $< ES_T =$ \$100: the futures price lies *below* the expected spot. This is consistent with *normal backwardation* ($ES_T > F$).

FF-Q9. $F = e^{(r-k)T}ES_T = e^{(0.04-0.01)(2)}(50) = e^{0.06}(50) = 1.061837 \times 50 =$ \$53.09. Here $k < r$ (negative-β asset), the exponent is positive, and $F =$ \$53.09 $> ES_T =$ \$50: the futures price *exceeds* the expected spot. This matches *contango* ($ES_T < F$).

FF-Q10. (a) New contract value at 390 cents: $3.90 \times 5{,}000 =$ \$19,500. (b) Marking-to-market gain to the long: $19{,}500 - 19{,}409.375 =$ \$90.625, credited to the long margin account. (c) New margin balance: $2{,}000 + 90.625 =$ \$2,090.625. (d) A margin call occurs when the balance falls to the \$1,500 maintenance level, a loss of $2{,}000 - 1{,}500 =$ \$500 from the starting \$2,000. That corresponds to the contract value falling by \$500, to $19{,}409.375 - 500 =$ \$18,909.375, i.e. price $= 18{,}909.375/5{,}000 =$ \$3.781875 per bushel $= 378.1875$ cents, or $378\frac{6}{32}$ cents per bushel.

17.11 Options

OPT-C1. The call is a *right*, not an obligation. If the stock falls to \$30, the strike-\$42 call is out of the money and the holder simply walks away, exercising nothing; the payoff is $\max\{30 - 42, 0\} - 4 = 0 - 4 = -4$. The premium c is the entire cost, and the $\max\{S_T - K, 0\}$ term can never go below zero, so it caps the intrinsic loss at that premium. The leveraged stock holder, by contrast, holds the underlying directly and bears every dollar of decline: 100 shares falling from \$40 to \$30 loses \$1{,}000, wiping out the \$4 margin many times over and generating a margin call. The floor in the option payoff comes precisely from the $\max\{\cdot, 0\}$ operator — the option converts an unlimited downside into a fixed, prepaid premium, which is what "right without obligation" means in cash terms.

OPT-C2. For the underlying price S_0: a higher S_0 makes a call more likely to finish above K (in the money) and less likely a put finishes below K, so the call rises and the put falls. For the strike K: a higher K is a higher hurdle for the call (less likely in the money, so the call falls) but a higher floor for the put (more likely in the money, so the put rises). These two factors are exact mirror images because a call profits when S_T exceeds K and a put profits when S_T falls below K — opposite tails of the same distribution. Volatility is different: greater dispersion of S_T fattens *both* tails, and since each option's payoff is truncated at zero on its unfavorable side, only the favorable tail matters to each holder. More volatility therefore raises the value of both the call and the put — the sole factor that moves them the same way.

OPT-C3. The strike is a *fixed* amount owed at exercise. A higher r lowers the present value Ke^{-rT} of that fixed payment. For the call holder, who will pay K to buy the stock, a smaller present value of the obligation makes the call more valuable — this is exactly the $-Ke^{-rT}$ term entering the Black-Scholes call price, which grows (less negative) as r rises. For the put holder, who will *receive* K by selling the stock, a higher r shrinks the present value of that receipt, lowering the put. Put-call parity $c - p = S_0 - Ke^{-rT}$ makes this crisp: raising r raises the right-hand side, so $c - p$ increases — the call gains and the put loses. The colleague's error is treating K as a cost to both parties; it is a payment *owed* by the call holder but *received* by the put holder, so discounting it helps one and hurts the other.

OPT-C4. Parity requires $c + Ke^{-rT} = p + S_0$. Portfolio 1 (call plus Ke^{-rT} cash) costs $c + Ke^{-rT} = 6 + 46 = \$52$. Portfolio 2 (stock plus put) costs $S_0 + p = 50 + 1 = \$51$. The two sides differ by \$1, so parity is violated and an arbitrage exists: portfolio 1 (the call-plus-cash side) is overpriced by \$1 relative to portfolio 2. The arbitrageur *sells* the expensive portfolio — writes the call and lends $Ke^{-rT} = \$46$ (collecting the \$6 call premium) — and *buys* the cheap one — buys the stock for \$50 and the put for \$1. The net cash today is $+6 + 1 - 50 = -\$43$ from the trades plus $-\$46$ lent… more simply, she collects \$52 from the short portfolio and pays \$51 for the long portfolio, netting +\$1 up front. At expiration both portfolios pay exactly $\max(S_T, K)$, so the long and short legs cancel dollar-for-dollar in every state, leaving zero exposure. The \$1 collected today is therefore a riskless profit.

OPT-C5. From parity, $c = p+S_0-Ke^{-rT}$, and since $p \geq 0$ we have $c \geq S_0-Ke^{-rT}$. An American call is worth at least its European counterpart, $C \geq c$, so $C \geq c \geq S_0 - Ke^{-rT} > S_0 - K$ (the last strict inequality because $Ke^{-rT} < K$ for $r > 0$). The right-hand quantity $S_0 - K$ is exactly what exercising the call today yields; the left-hand quantity C is what selling the option yields. Since selling always beats exercising, early exercise is never optimal and $C = c$. For a put the logic reverses: a put's value is bounded above by K, and once a firm goes bankrupt the stock price is 0 and can fall no further, so the payoff K is already maximal. Waiting only forgoes the interest that could be earned by collecting K now; exercising early to reinvest the proceeds can therefore be strictly optimal, which is why $P > p$.

OPT-C6. Put-call parity is the statement that two economically distinct portfolios have identical payoffs at expiration and must therefore cost the same today. Portfolio 1 is a call plus Ke^{-rT} in cash; if $S_T < K$ the call expires worthless and the cash grows to K, giving K; if $S_T > K$ the call pays $S_T - K$ and the cash gives K, totaling S_T. Portfolio 2 is one share plus a put; if $S_T < K$ the put pays $K - S_T$ and the share is worth S_T, giving K; if $S_T > K$ the put is worthless and the share is worth S_T. In both states each portfolio equals $\max(S_T, K)$. Because their future payoffs coincide state by state, no-arbitrage forces their present prices to coincide: $c + Ke^{-rT} = p + S_0$. This is an identity, not an approximation, because any price gap would let a trader buy the cheaper portfolio, sell the dearer one, and collect a certain profit while the payoffs cancel at T.

OPT-C7. "Perfectly hedged" means the combined position (write 3.33 calls, hold one share) delivers the *same* dollar payoff regardless of which state occurs. In the up state the share is worth \$55 and the written calls owe $3.33 \times \$3 = \$10$, netting \$45; in the down state the share is worth \$45 and the calls owe nothing, again \$45. Because the payoff is a certain \$45 in every state, the position is riskless and must be priced by discounting at the risk-free rate. This is what pins the option to one value: the replicating portfolio's cost is fixed by arbitrage, and the probability of an up move never enters. Two investors who disagree about that probability must still agree on the option price, because either could construct the same riskless hedge

— preferences and beliefs about the up-probability are irrelevant, which is the central insight of replication-based pricing.

OPT-C8. The hedge ratio is $(C_u - C_d)/(S_u - S_d)$, the range of option payoffs over the range of stock prices. For stock A the stock range is $55 - 45 = \$10$; for stock B it is $60 - 40 = \$20$. With the same strike, stock B's wider terminal spread produces a larger range of option payoffs in the favorable state while the unfavorable payoff stays floored at zero, so the *numerator* grows even as the denominator doubles. More importantly, the replicating portfolio for stock B must combine more borrowing and shares to span the wider payoff gap, and the cost of that portfolio — the option's value — is higher. Economically, the option holder captures only the favorable tail; a wider spread of possible terminal prices lengthens that favorable tail without adding downside (the payoff cannot go below zero), so the option on the more volatile stock B is worth more. This is the discrete-time version of the general result that option value increases in volatility.

OPT-C9. Volatility σ is the only input to Black-Scholes that cannot be read off a screen: S_0, K, r, and T are all observable, but the standard deviation of future returns is not. *Implied volatility* is the value of σ that, when plugged into the formula, reproduces the option's observed market price exactly. Because the call price is strictly increasing in σ, this inversion is unique: given a market price one solves $C^{\text{market}} = C^{\text{BS}}(\sigma)$ numerically for the implied σ. The CBOE VIX applies exactly this logic to a basket of S&P 500 options, aggregating their implied volatilities into a single index of the market's expected near-term volatility. If two options on the same stock with different strikes imply different volatilities, the model's assumption of a single constant σ is violated — the empirical pattern known as the volatility "smile" or "skew," signaling that the true return distribution has fatter tails or asymmetry that Black-Scholes does not capture.

OPT-C10. In the Black-Scholes call $C_0 = S_0N(d_1) - Ke^{-rT}N(d_2)$, the term $N(d_2)$ is the (risk-neutral) probability the option finishes in the money. When both $N(d_1)$ and $N(d_2)$ approach 1, exercise is virtually certain: the holder will almost surely pay K to acquire a share worth S_0 today, so the call is worth

$S_0 - Ke^{-rT}$ — the current stock price minus the present value of the near-certain payment. This is the correct value for an all-but-guaranteed forward purchase. When both terms approach 0, exercise is almost impossible: the stock will almost surely finish below K, the right to buy is never used, and the call is worth essentially nothing. Both limits confirm the formula respects the economics: value equals the stock's worth minus the discounted obligation, weighted by how likely that obligation is to be triggered.

OPT-C11. A straddle — a long call and a long put at the same strike — pays off whenever S_T moves far from K in *either* direction: the call captures large up moves, the put captures large down moves, and only a stock that stays near K leaves both nearly worthless. This is exactly a bet on the *magnitude* of the move, not its sign, which fits a trader who expects a big FDA-driven jump but does not know which way. A strangle uses a call struck above and a put struck below the current price, so both options start further out of the money and are cheaper — but the stock must travel past one of the two wider strikes before either pays, requiring a *larger* move to profit. In both cases the trader is implicitly buying a view on volatility (realized dispersion), in contrast to a bull spread, which is a directional bet that profits only if the stock rises.

OPT-C12. Buying the call at K_1 alone gives unlimited upside above K_1 for the cost c_1. Adding a written call at $K_2 > K_1$ brings in premium c_2 (with $c_2 < c_1$ because the higher-strike call is less likely to pay), lowering the net cost of the position to $c_1 - c_2$. What the investor *gives up* is all upside beyond K_2: above K_2 the written call's obligation grows one-for-one with the stock, capping the spread's payoff at $K_2 - K_1$. What she *gains* is the reduced entry cost and therefore a lower breakeven and smaller maximum loss. A moderately bullish investor — one who expects the stock to rise toward K_2 but not far beyond — prefers the spread because she is not sacrificing upside she expects to realize, and she pays less for the exposure she actually wants.

OPT-C13. Delta $\Delta = N(d_1)$ depends on d_1, which itself depends on the current stock price S; as S changes, d_1 and hence $N(d_1)$ change, so delta is not a constant. The Greek measuring the rate of change of delta with respect to the stock price is *gamma*, $\Gamma = N'(d_1)/(S\sigma\sqrt{T})$, the second derivative of the option price in S. When a call is deeply in the money, d_1 is large and positive, so $N(d_1) \approx 1$: the option moves almost dollar-for-dollar with the stock, behaving like a share. When it is deeply out of the money, d_1 is large and negative, so $N(d_1) \approx 0$: the option barely responds to stock moves because it is nearly certain to expire worthless. Because delta drifts as the stock moves, a delta-neutral hedge must be continually rebalanced — the practical reason gamma matters.

OPT-C14. Gamma $\Gamma = N'(d_1)/(S\sigma\sqrt{T})$ measures how fast delta changes as the underlying moves; vega $\mathcal{V} = S_0\sqrt{T}N'(d_1)$ measures how the option value changes with volatility. A short-dated (small T) at-the-money option has very large gamma — the $\sqrt{T}$ in the denominator blows up as expiry nears — so its delta swings sharply with even small price moves. A market maker short many such calls must therefore rehedge frequently and in the wrong direction (buying as the market rises, selling as it falls), incurring cumulative transaction costs; this is the cost of being short gamma. Separately, being short options means negative vega: if volatility spikes, the options she has written become more valuable and her position loses money *even if the underlying price does not move at all*. The tension is that the short-option position earns time decay but is punished by both realized moves (gamma) and rising implied volatility (vega).

OPT-Q1. Call payoffs $\max\{S_T - 45, 0\} - 3$: at \$38, $0 - 3 = -\$3$; at \$45, $0 - 3 = -\$3$; at \$52, $7 - 3 = +\$4$. Put payoffs $\max\{45 - S_T, 0\} - 2$: at \$38, $7 - 2 = +\$5$; at \$45, $0 - 2 = -\$2$; at \$52, $0 - 2 = -\$2$. Breakeven for the long call: $S_T = K + c = 45 + 3 = \$48$. Breakeven for the long put: $S_T = K - p = 45 - 2 = \$43$.

OPT-Q2. Payoff = $\max(S_T - 50, 0) - \max(S_T - 55, 0) - 6 + 3 = \max(S_T - 50, 0) - \max(S_T - 55, 0) - 3$. At $S_T = 48$: $0 - 0 - 3 = -\$3$. At $S_T = 53$: $3 - 0 - 3 = \$0$. At $S_T = 60$: $10 - 5 - 3 = +\$2$. Maximum profit occurs for $S_T \geq K_2$: $(K_2 - K_1) - (c_1 - c_2) = (55 - 50) - (6 - 3) = 5 - 3 = +\$2$. Maximum loss occurs for $S_T \leq K_1$: $-(c_1 - c_2) = -\$3$.

OPT-Q3. Put-call parity: $p = c + Ke^{-rT} - S_0$. Discount factor: $Ke^{-rT} =$

$100 \cdot e^{-0.04\times0.5} = 100 \cdot e^{-0.02} = 100 \cdot 0.98020 = 98.020$. Then $p = 7.50 + 98.020 - 98 = \$7.52$.

OPT-Q4. Left side: $c + Ke^{-rT} = 5 + 62e^{-0.03} = 5 + 62 \cdot 0.97045 = 5 + 60.168 = 65.168$. Right side: $p + S_0 = 5.50 + 60 = 65.50$. They do *not* match. The gap is $65.50 - 65.168 = \$0.332$. The right-hand portfolio (put plus stock) is overpriced by about \$0.33 relative to parity; equivalently, the put is roughly \$0.33 too expensive (or the call too cheap). An arbitrageur would sell the stock-plus-put portfolio and buy the call-plus-cash portfolio.

OPT-Q5. Up state $S_u = 1.1 \times 50 = \$55$, down state $S_d = 0.9 \times 50 = \$45$. (a) Payoffs: $C_u = \max\{55 - 52, 0\} = \$3$, $C_d = \max\{45 - 52, 0\} = \$0$. (b) Hedge ratio $= (C_u - C_d)/(S_u - S_d) = (3 - 0)/(55 - 45) = 3/10 = 0.3$. (c) Replicate: hold $h = 0.3$ shares and borrow the present value of the down-state stock value $h \cdot S_d = 0.3 \times 45 = 13.5$, i.e. borrow $13.5/1.01 = 13.366$. Cost today $C = hS_0 - 13.366 = 0.3 \times 50 - 13.366 = 15 - 13.366 = \$1.634$. This matches the chapter's answer $C = 1.635$ (rounding).

OPT-Q6. Up state \$55, down state \$45, strike \$48. Payoffs: $C_u = \max\{55 - 48, 0\} = \$7$, $C_d = \max\{45 - 48, 0\} = \$0$. Hedge ratio $= (7 - 0)/(55 - 45) = 7/10 = 0.7$. Replicate with $h = 0.7$ shares. The riskless "hold one share, write $1/h = 1.4286$ calls" portfolio pays 45 in each state (up: $55 - 1.4286 \times 7 = 55 - 10 = 45$; down: $45 - 0 = 45$), worth $45/1.01 = 44.554$ today. Equivalently, borrow $hS_d/1.01 = 0.7 \times 45/1.01 = 31.5/1.01 = 31.188$; then $C = hS_0 - 31.188 = 0.7 \times 50 - 31.188 = 35 - 31.188 = \$3.812$.

OPT-Q7. $\sigma\sqrt{T} = 0.20 \times \sqrt{0.5} = 0.20 \times 0.70711 = 0.14142$. $d_1 = \dfrac{\ln(100/100) + (0.05 + 0.02) \times 0.5}{0.14142} = \dfrac{0 + 0.035}{0.14142} = 0.2475$. $d_2 = d_1 - \sigma\sqrt{T} = 0.2475 - 0.14142 = 0.1061$. With $N(d_1) = 0.5977$ and $N(d_2) = 0.5422$: $Ke^{-rT} = 100e^{-0.025} = 97.531$. $C_0 = 100 \times 0.5977 - 97.531 \times 0.5422 = 59.77 - 52.882 = \$6.89$.

OPT-Q8. $\sigma\sqrt{T} = 0.30\times\sqrt{0.25} = 0.30\times0.5 = 0.15$. $d_1 = \dfrac{\ln(100/105) + (0.05 + 0.045) \times 0.25}{0.15} = \dfrac{-0.04879 + 0.023750}{0.15} = \dfrac{-0.025040}{0.15} = -0.1669$. $d_2 = -0.1669 - 0.15 = -0.3169$. With $N(d_1) = 0.4337$, $N(d_2) = 0.3756$: $Ke^{-rT} = 105e^{-0.0125} = 105 \times 0.98758 = 103.696$. $C_0 = 100\times0.4337 - 103.696\times0.3756 = 43.37 - 38.948 =$

\$4.42. Put by parity: $p = C_0 + Ke^{-rT} - S_0 = 4.42 + 103.696 - 100 = \$8.11$.

OPT-Q9. Using $S_0 = 100$, $\sigma = 0.30$, $T = 0.25$ so $\sigma\sqrt{T} = 0.15$, and $N'(d_1) = 0.3934$. Gamma: $\Gamma = \dfrac{N'(d_1)}{S_0\sigma\sqrt{T}} = \dfrac{0.3934}{100 \times 0.15} = \dfrac{0.3934}{15} = 0.02623$. Vega: $\mathcal{V} = S_0\sqrt{T}\,N'(d_1) = 100 \times 0.5 \times 0.3934 = 19.67$. Vega is the change in call price per one-unit (i.e. per 1.00, or 100 percentage points) change in σ; for a one-percentage-point rise ($\Delta\sigma = 0.01$) the call price changes by approximately $\mathcal{V} \times 0.01 = 19.67 \times 0.01 = \$0.197$, so about \$0.20 more expensive.

OPT-Q10. (a) Total shares controlled $= 20 \times 100 = 2000$. Delta-neutral short $= \Delta \times 2000 = 0.55 \times 2000 = 1100$ shares shorted. (b) If the ETF falls by \$1, the call position loses about $\Delta \times \$1 \times 2000 = 0.55 \times 2000 = \$1100$, while the 1100 short shares gain $1100 \times \$1 = \$1100$ — the two offset. (c) If $\Delta \approx 1$ (deep in the money), the hedge would require shorting $1 \times 2000 = 2000$ shares — the full share count, since the calls then move nearly one-for-one with the ETF and behave like the stock itself.

OPT-Q11. (a) Call delta $= N(d_1) = 0.62$; put delta $= N(d_1) - 1 = 0.62 - 1 = -0.38$. (b) For a \$2 rise: long call changes by $\Delta_{\text{call}} \times 2 = 0.62 \times 2 = +\$1.24$; long put changes by $\Delta_{\text{put}} \times 2 = -0.38 \times 2 = -\$0.76$. (c) The put delta is negative because a put *loses* value as the stock rises (it is a right to sell). Differentiating put-call parity $c - p = S_0 - Ke^{-rT}$ with respect to S_0 gives $\partial c/\partial S_0 - \partial p/\partial S_0 = 1$, i.e. $\Delta_{\text{call}} - \Delta_{\text{put}} = 1$; here $0.62 - (-0.38) = 1$, confirming the relationship.

OPT-Q12. (a) Total shares controlled $= 10 \times 100 = 1000$; delta-neutral short $= \Delta \times 1000 = 0.4 \times 1000 = 400$ shares. (b) If the ETF falls from \$50 to \$49, the 400 short shares gain $400 \times \$1 = \$400$. The call price falls by about $\Delta \times \$1 = 0.4 \times 1 = \$0.40$ per share, from \$5.00 to \$4.60, so the call position is now worth $4.60 \times 1000 = \$4600$, a loss of \$400 from the original \$5000. The \$400 stock gain offsets the \$400 call loss. (c) Delta is not constant: as the ETF price moves, $N(d_1)$ changes (gamma is nonzero), so the number of shares needed to stay delta-neutral changes, forcing the hedge to be rebalanced after the move.

OPT-D1. (a) Breakevens at 751 ± 27.09: S_T = 723.91 and S_T = 778.09, i.e. moves of about −3.6% and +3.6% from S_0 = 751.21. (b) At S_T = 700 the put finishes 51 in the money: profit = 51 – 27.09 = $23.91 per share. (c) The straddle's cost is what the market charges for exposure to *any* large move over five weeks — a direct market price of expected volatility.

OPT-D2. (a) Net cost = 15.21 – 1.27 = $13.94; maximum profit = (789 – 751) – 13.94 = $24.06; breakeven = 751 + 13.94 = 764.94. (b) At S_T = 800 both calls finish in the money and the payoff caps at the strike gap: profit = 38 – 13.94 = $24.06. At S_T = 740 both expire worthless: profit = −13.94. (c) The 789 call pays off only in states where SPY rises more than 5% in five weeks — a strict subset of the states in which the 751 call pays off, and with smaller payoffs in every common state, so it must cost less.

17.12 The Black-Litterman model and active portfolio management

BL-C1. The optimizer treats the point estimates of μ as if they were the true expected returns, known with certainty, and it fully exploits every difference between them. When two assets are highly correlated, the optimizer can construct a nearly offsetting long-short pair whose relative weight is enormous, because even a tiny estimated difference in their means looks like a "free" opportunity to earn return with little marginal variance. But that tiny estimated difference is well inside the range of sampling error. A two-month change in the estimation window shifts the sample means by amounts comparable to their own standard errors, and since the optimal weights depend on $\Sigma^{-1}\mu$ — where Σ^{-1} amplifies differences among correlated assets — the sign and magnitude of the large offsetting positions swing wildly. XLK and XLE flip because the optimizer was never estimating a stable quantity in the first place; it was leveraging noise.

BL-C2. The root cause the chapter identifies is that unconstrained mean-variance optimization treats point estimates of expected returns as if they were known with certainty and fully exploits any differences between them, so it over-fits estimation noise into extreme, unstable weights. Black-Litterman's remedy is to treat the expected returns themselves as random variables with a prior distribution rather than as fixed parameters. Anchoring the prior to the equilibrium returns Π (those that reproduce market-cap weights) and then updating with the investor's views in a Bayesian way means the posterior expected returns cannot drift far from the equilibrium anchor unless a view with real precision pushes them. Noise in a return estimate no longer translates directly into a large weight, because the model's output is a precision-weighted compromise that shrinks toward the market whenever the investor has no confident view to the contrary.

BL-C3. Reverse optimization starts from the mean-variance first order condition $\mu = A\Sigma\theta$. In forward optimization we treat μ, A, and Σ as known and solve for the optimal weights $\theta = \frac{1}{A}\Sigma^{-1}\mu$. In reverse optimization we instead treat the weights as known — we set them equal to the observed market-capitalization weights θ_{mkt} — together with A and Σ, and we back out the expected returns $\Pi = A\Sigma\theta_{\text{mkt}}$ that would make those weights optimal. The quantity being solved for is Π; the market weights and covariance are the knowns. Anchoring the prior to market weights is sensible because an investor who begins by effectively owning the whole market is, by revealed preference, already holding the portfolio that these implied returns rationalize; absent any special opinion, her best guess for expected returns is precisely the set consistent with holding the market.

BL-C4. Even before either manager expresses a view, Manager A's baseline is a vector of raw sample means, each carrying large sampling error, and the optimizer will lever up the noisy differences among them into extreme long-short positions. Manager B's baseline is $\Pi = A\Sigma\theta_{\text{mkt}}$, which by construction reproduces the market weights when run through the optimizer — a diversified, sensible, fully-invested portfolio with no extreme positions. The structural feature doing the work is that the reverse-optimization prior is internally consistent with a reasonable portfolio: it is the fixed point of the optimizer, so feeding it back in yields the market rather than a wild allocation. Manager B therefore starts from a stable, intuitive point and only departs from it deliberately, whereas Manager A starts from noise.

BL-C5. In the classical treatment μ is a known constant, fully trusted. Here $\mu = \Pi + \epsilon^e$ with $\epsilon^e \sim N(0, \tau\Sigma)$ makes the expected returns random, centered on the equilibrium prior Π but uncertain. The scalar τ scales the magnitude of that uncertainty: it measures how much the investor distrusts the equilibrium prior as a statement of the true means. A very small τ means ϵ^e has tiny variance, so the investor is highly confident that the true μ equals Π; the prior is nearly rigid and views must be very precise to move the posterior much. A large τ means the prior is loosely held, so even modestly confident views can pull the posterior substantially. Relative to the classical constant-μ assumption, this converts a single trusted point into a distribution whose spread the investor tunes through τ.

BL-C6. Scaling the prior uncertainty by $\tau\Sigma$ says that uncertainty about mean returns has the same correlation structure as returns themselves: assets whose returns are volatile and move together have expected returns that are correspondingly uncertain and jointly uncertain. This is a natural default — the same forces that make returns co-move make our estimates of their means co-move — and it means the investor needs to specify only the single scalar τ rather than an entire new covariance matrix. It is also convenient in the derivation: because both the prior covariance $\tau\Sigma$ and the return covariance Σ share the same structure, the conditional-normal algebra collapses cleanly, and the posterior precision $(\tau\Sigma)^{-1} + P'\Omega^{-1}P$ combines the equilibrium and view information without requiring any unrelated matrix to be estimated.

BL-C7. The two managers' posteriors differ only through the view variance — ω^2 in the single-view case, or the corresponding diagonal entry of Ω. The directional view p (long XLK, short XLE) and its target q are identical for both. The highly confident manager assigns a small ω^2, giving her view large precision $1/\omega^2$; in the posterior mean $\bar{\mu} = [(\tau\Sigma)^{-1} + p'\frac{1}{\omega^2}p]^{-1}[(\tau\Sigma)^{-1}\Pi + p'\frac{1}{\omega^2}q]$ this pulls the blend strongly toward q. The "hunch" manager assigns a large ω^2, low precision, so her posterior stays close to the equilibrium prior Π. The single input encoding the difference between conviction and hunch is

ω^2 (equivalently Ω), the variance of the view error ϵ^v.

BL-C8. A single row of P specifies the portfolio the view is about: its entries are the weights on each asset in the linear combination whose expected return the investor is expressing an opinion on (e.g. +1 on the asset believed to outperform, −1 on the asset it is believed to beat, zeros elsewhere for a relative view). The corresponding entry of Q gives the expected return the investor assigns to that portfolio. The corresponding diagonal entry of Ω is the variance ω^2 of the view's error ϵ^v — how uncertain the investor is about that particular view. A view is written as a linear combination because most investment opinions are naturally relative ("A will beat B") and because expressing views on portfolios lets the model translate an opinion on a spread into coherent implications for all correlated assets, rather than forcing every opinion to be an isolated absolute forecast.

BL-C9. The conservative property comes from the structure of P: a row that expresses no view about a given asset has a zero in that asset's column, so that asset never enters the view term $P'\Omega^{-1}Q$ nor the view precision $P'\Omega^{-1}P$ except through its correlation with assets that are in a view. Its posterior expected return therefore stays at (or very near) its equilibrium prior value Π, and moves only to the extent that Σ links it to an asset the investor has an opinion about. In a naive optimizer there is no such anchor: revising a single asset's estimated mean feeds directly into $\theta = \frac{1}{A}\Sigma^{-1}\mu$, and because Σ^{-1} is dense, that one revised number can swing the weights of many unrelated assets. Black-Litterman confines the disturbance to where the investor actually has information.

BL-C10. The posterior mean is a matrix-weighted average of the prior Π and the view target Q, with weights given by the two precision matrices $(\tau\Sigma)^{-1}$ (prior precision) and $P'\Omega^{-1}P$ (view precision). Writing $\bar{\mu} = W^{-1}[(\tau\Sigma)^{-1}\Pi + P'\Omega^{-1}Q]$ with $W = (\tau\Sigma)^{-1} + P'\Omega^{-1}P$, the coefficients on Π and on the view are each positive-semidefinite and sum (in the precision sense) to W, so $\bar{\mu}$ is always a convex-type compromise between the prior and the views — it can never lie outside the range they span. As $\Omega \to \infty$ the view precision $P'\Omega^{-1}P \to 0$: the views carry no information and $\bar{\mu} \to \Pi$, the pure equilibrium prior. As $\Omega \to 0$ the view precision explodes and, along the directions spanned by P, $\bar{\mu}$ is pulled all the way onto the views Q (the views are treated as certain), while directions orthogonal to P remain at the prior.

BL-C11. The chapter ties the size of the tilt directly to the precision of the view. Posterior expected returns deviate from equilibrium in a direction and by an amount governed by view precision, and those posterior returns feed back through the FOC $\theta = \frac{1}{A}\Sigma^{-1}\bar{\mu}$ into the weights. Near-certain view (very low ω^2): the posterior mean is pulled essentially onto the view value q, producing a large deviation from Π and hence a large tilt of the optimal weights toward the implied position. Almost-no-confidence view (very high ω^2): the view has negligible precision, the posterior mean stays essentially at Π, and the optimal weights barely move from the market portfolio. Confidence is thus an interpretable dial: conviction scales the tilt.

BL-C12. Starting from the equilibrium prior means the CIO's default portfolio is the market itself — she is not betting anything until she chooses to. Because the posterior is a precision-weighted blend, each view moves the weights only in proportion to the confidence she assigns it, so her opinions produce disciplined tilts: large where conviction is high, small where it is weak, and always anchored to the market. That is exactly "expressing an opinion without betting the fund." If instead she simply set her favored sectors' expected returns to large positive numbers and re-optimized, she would be back in the failure mode of BL-C1: the optimizer would treat those fabricated means as certain, lever them up through Σ^{-1}, and generate extreme, unstable, undiversified positions with no anchor to the market and no principled scaling by conviction.

BL-Q1. Compute $\Sigma\theta$ first. $\Sigma\theta = (0.04 \cdot 0.6 + 0.01 \cdot 0.4,\ 0.01 \cdot 0.6 + 0.02 \cdot 0.4)' = (0.024 + 0.004,\ 0.006 + 0.008)' = (0.028,\ 0.014)'$. Then $\Pi = A\Sigma\theta = 3 \cdot (0.028, 0.014)' = (0.084,\ 0.042)'$. So the implied equilibrium expected returns are $\Pi_1 = 0.084$ and $\Pi_2 = 0.042$ per period.

BL-Q2. With $A = 5$ and the same $\Sigma\theta = (0.028, 0.014)'$, $\Pi = 5 \cdot (0.028, 0.014)' = (0.140,\ 0.070)'$. To recover the weights, solve $\theta = \frac{1}{A}\Sigma^{-1}\Pi = \frac{1}{5}\Sigma^{-1}(0.140, 0.070)'$. But Π was defined as $5\Sigma\theta_{\text{mkt}}$, so

$\frac{1}{5}\Sigma^{-1}(5\Sigma\theta_{\text{mkt}}) = \theta_{\text{mkt}} = (0.6, 0.4)'$. The market weights are recovered exactly. They must be, because reverse optimization defines Π as precisely the return vector that makes θ_{mkt} satisfy the first order condition; forward-optimizing that same Π inverts the operation and returns the weights we started from.

BL-Q3. With assets ordered (XLB, XLK, XLE), a portfolio long XLK and short XLE by equal amounts has view vector $p = (0,\ 1,\ -1)$. The scalar is $q = 0.015$. The view error ϵ^v has variance $\omega^2 = 0.05^2 = 0.0025$. Applying p to the mean vector $\mu = (\mu_{\text{XLB}}, \mu_{\text{XLK}}, \mu_{\text{XLE}})'$ gives $p\mu = 0{\cdot}\mu_{\text{XLB}} + 1{\cdot}\mu_{\text{XLK}} - 1{\cdot}\mu_{\text{XLE}} = \mu_{\text{XLK}} - \mu_{\text{XLE}}$, exactly the expected outperformance of XLK over XLE that the view is about. The specification is thus $p\mu = q + \epsilon^v$ with $q = 0.015$ and $\text{Var}(\epsilon^v) = 0.0025$.

BL-Q4. Order the assets (A, B, C, D). View (i) is relative — A over B by 0.010 — so its row is $(1, -1, 0, 0)$ with target 0.010. View (ii) is absolute on C — expected return 0.008 — so its row is $(0, 0, 1, 0)$ with target 0.008. Hence

$$P = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0.010 \\ 0.008 \end{pmatrix}.$$

The error standard deviations are 0.04 and 0.06, and the views are independent, so

$$\Omega = \begin{pmatrix} 0.04^2 & 0 \\ 0 & 0.06^2 \end{pmatrix} = \begin{pmatrix} 0.0016 & 0 \\ 0 & 0.0036 \end{pmatrix}.$$

BL-Q5. In scalar form: prior precision $(\tau\Sigma)^{-1} = 1/0.02 = 50$; view precision $1/\omega^2 = 1/0.01 = 100$ (with $p = 1$). The posterior precision is $50 + 100 = 150$, so the posterior variance is $1/150$. The posterior mean is

$$\bar{\mu} = \frac{50 \cdot 0.006 + 100 \cdot 0.012}{150} = \frac{0.30 + 1.20}{150} = \frac{1.50}{150} = 0.010.$$

Since $0.006 < 0.010 < 0.012$, the posterior lies between the prior and the view. It is closer to the view (0.012) because the view carries more precision ($100 > 50$); the posterior is a precision-weighted average, so it leans toward whichever source is more confident.

BL-Q6. Keep $(\tau\Sigma)^{-1} = 50$, $\Pi = 0.006$, $q = 0.012$.

(a) $\omega^2 = 0.001 \Rightarrow 1/\omega^2 = 1000$. $\bar{\mu} = \dfrac{50 \cdot 0.006 + 1000 \cdot 0.012}{50 + 1000} = \dfrac{0.30 + 12.0}{1050} = \dfrac{12.3}{1050} \approx 0.01171$.

(b) $\omega^2 = 0.2 \Rightarrow 1/\omega^2 = 5$. $\bar{\mu} = \dfrac{50 \cdot 0.006 + 5 \cdot 0.012}{50 + 5} = \dfrac{0.30 + 0.06}{55} = \dfrac{0.36}{55} \approx 0.00655$.

In (a) the very confident view ($\omega^2 = 0.001$) has precision 1000, dwarfing the prior precision 50, so $\bar{\mu} \approx 0.0117$ sits almost on the view 0.012. In (b) the very unconfident view ($\omega^2 = 0.2$) has precision only 5, far below 50, so $\bar{\mu} \approx 0.0065$ barely leaves the prior 0.006. Lower ω^2 (higher confidence) pulls the posterior strongly toward the view, exactly as the chapter argues.

BL-Q7. Posterior variance $= [(\tau\Sigma)^{-1} + p' \frac{1}{\omega^2} p]^{-1} = [50 + 100]^{-1} = 1/150 \approx 0.00667$. Compare: prior variance $\tau\Sigma = 0.02$; view variance $\omega^2 = 0.01$. The posterior variance 0.00667 is smaller than both. It must be, because precisions add: the posterior precision $50 + 100 = 150$ exceeds either individual precision (50 or 100), and inverting a larger precision gives a smaller variance. Combining two independent noisy sources can only sharpen the estimate. For $\omega^2 = 0.001$ (precision 1000): posterior variance $= [50 + 1000]^{-1} = 1/1050 \approx 0.000952$, smaller still and now below even the view variance 0.001 — a highly confident view drives the posterior uncertainty down toward the view's own tiny variance.

BL-Q8. Fix $\Sigma = 0.02$, $\Pi = 0.006$, $q = 0.012$, $\omega^2 = 0.01$ (view precision $1/\omega^2 = 100$).

$\tau = 0.1$: $\tau\Sigma = 0.002$, prior precision $= 1/0.002 = 500$. $\bar{\mu} = \dfrac{500 \cdot 0.006 + 100 \cdot 0.012}{500 + 100} = \dfrac{3.0 + 1.2}{600} = \dfrac{4.2}{600} = 0.0070$.

$\tau = 1.0$: $\tau\Sigma = 0.02$, prior precision $= 50$. $\bar{\mu} = \dfrac{50 \cdot 0.006 + 100 \cdot 0.012}{150} = 0.010$ (as in BL-Q5).

Shrinking τ from 1.0 to 0.1 moves the posterior from 0.010 down toward the prior 0.006 (to 0.007). A smaller τ shrinks the prior variance $\tau\Sigma$, raising the prior precision, which means the investor is more confident in the equilibrium prior; the posterior therefore leans more heavily on Π and away from the view.

BL-Q9. Equilibrium returns $\Pi = (0.084, 0.042)'$ from BL-Q1, and $\bar{\mu} = \Pi + (0.006, 0)' = (0.090,\ 0.042)'$. Need $\theta = \frac{1}{A}\Sigma^{-1}\bar{\mu}$ with $A = 3$. First Σ^{-1}: $\det \Sigma = 0.04 \cdot 0.02 - 0.01^2 = 0.0008 - 0.0001 = 0.0007$, so

$$\Sigma^{-1} = \frac{1}{0.0007}\begin{pmatrix} 0.02 & -0.01 \\ -0.01 & 0.04 \end{pmatrix}.$$

Then $\Sigma^{-1}\bar{\mu}$: component 1 = $(0.02 \cdot 0.090 - 0.01 \cdot 0.042)/0.0007$ = $(0.0018 - 0.00042)/0.0007 = 0.00138/0.0007 = 1.9714$; component 2 = $(-0.01 \cdot 0.090 + 0.04 \cdot 0.042)/0.0007 = (-0.0009 + 0.00168)/0.0007 = 0.00078/0.0007 = 1.1143$. Divide by $A = 3$: $\theta = (0.6571,\ 0.3714)'$. Tilt $\theta - \theta_{\text{mkt}} = (0.6571 - 0.6,\ 0.3714 - 0.4)' = (+0.0571,\ -0.0286)'$. Raising only asset 1's expected return still changes both weights because Σ^{-1} is not diagonal: the assets are positively correlated, so overweighting asset 1 to exploit its higher return means the optimizer trims asset 2 to keep total portfolio risk in check.

BL-Q10. Now $\bar{\mu} = \Pi + (0, -0.007)' = (0.084,\ 0.035)'$, same $A = 3$ and Σ^{-1} as BL-Q9. $\Sigma^{-1}\bar{\mu}$: component 1 = $(0.02\cdot0.084-0.01\cdot0.035)/0.0007 = (0.00168-0.00035)/0.0007 = 0.00133/0.0007 = 1.9$; component 2 = $(-0.01 \cdot 0.084 + 0.04\cdot0.035)/0.0007 = (-0.00084+0.0014)/0.0007 = 0.00056/0.0007 = 0.8$. Divide by 3: $\theta = (0.6333,\ 0.2667)'$. Tilt $\theta - \theta_{\text{mkt}} = (+0.0333,\ -0.1333)'$. Asset 2's weight falls sharply because its posterior return was lowered — the investor wants to hold less of it. Asset 1's weight rises even though its own expected return is unchanged: because the two assets are positively correlated, cutting the position in asset 2 removes some exposure that the optimizer partly restores by adding to the still-attractive, positively-correlated asset 1, keeping the risk-return trade-off optimal.

BL-D1. (a) $\Sigma\theta = (0.04 \times 0.7 + 0.01 \times 0.3,\ \ 0.01 \times 0.7 + 0.02 \times 0.3)' = (0.031, 0.013)'$, so $\Pi = 2.5 \times (0.031, 0.013)' = (7.75\%, 3.25\%)'$. (b) The first sector has both the larger weight and the larger variance, so it contributes more to the market portfolio's risk; in equilibrium the market can only be willingly held if the assets that add the most covariance risk offer the most expected return. This is the figure's ordering: technology's huge weight and high covariance put it at the top, utilities' small weight and low covariance at the bottom.

BL-D2. (a) Fed a 24% expected return for technology alongside far lower means for highly correlated sectors, an unconstrained optimizer sees a near-arbitrage: it takes an enormous long position in technology financed by short positions in the sectors most correlated with it (the industrial-financial cluster, with correlations above 0.8 in Figure 14.1), producing the extreme weights described in the chapter's introduction. (b) Π is the unique return vector consistent with the market portfolio being optimal, so starting there guarantees a sensible default; deviations then arise only from views the investor actually holds, sized by stated confidence, rather than from statistical noise in historical means.

17.13 Credit Risk Models

CR-C1. The objection confuses the *shape* of the default-time distribution with the *tools* available to handle it. The transformation proceeds in two steps: first map t through its own CDF, $u = F(t)$, which is a probability-integral transform yielding a uniform-$(0, 1)$ variable; then map that uniform through the inverse standard-normal CDF, $x = N^{-1}(u)$, yielding a standard normal variable. Because F and N are (weakly) increasing, the event $\{t \leq t_0\}$ is identical to the event $\{x \leq x(t_0)\}$, so $\Pr(t \leq t_0) = \Pr(x \leq x(t_0))$: the transformation preserves the entity's marginal default probability exactly. Nothing about the borrower's economics is altered; we have only relabeled the outcome axis onto a normal scale so that normal-distribution machinery (correlation, factor models) can be brought to bear. The right-skewness of t is fully accommodated because it is baked into F, which is used to build x.

CR-C2. Each entity is transformed through *its own* marginal CDF: $x_1 = N^{-1}(F_1(t_1))$ and $x_2 = N^{-1}(F_2(t_2))$. The probability-integral transform guarantees that $F_i(t_i)$ is uniform on $(0, 1)$ regardless of the shape of F_i, and applying N^{-1} to a uniform variable produces a standard normal. Hence both x_1 and x_2 are standard normal even though the fast-defaulting subprime mortgage (t_1) and the slow-defaulting investment-grade bond (t_2) have wildly different, non-identical marginals. Putting both entities on a common standard-normal scale is exactly what lets the modeler impose correlation with familiar Gaussian tools — a single correlation parameter (or factor loading) now has a clean meaning as the correlation between two standard normals, and the joint behavior can be handled with the multivariate normal, without ever forcing $F_1 = F_2$.

CR-C3. Y is the *systematic* or common factor — a single economy-wide random driver shared by every borrower, representing forces like the business cycle, interest-rate environment, or nationwide house prices. ϵ_i is the *idiosyncratic* shock specific to entity i — firm-level or household-level fortunes independent across borrowers ($cov(\epsilon_i, \epsilon_j) = 0$). The loading a_i splits each borrower's default driver between these two sources. When every $a_i \to 1$, $x_i \approx Y$: all borrowers move together, defaults cluster heavily, and diversification fails because a bad Y sinks everyone at once. When every $a_i \to 0$, $x_i \approx \epsilon_i$: defaults are essentially independent and pool losses are highly diversifiable. A concrete real-world Y is a nationwide collapse in house prices (or a deep recession), which simultaneously pushes many mortgages toward default.

CR-C4. The coefficient $\sqrt{1 - a_i^2}$ is chosen so that the variance of x_i is exactly one. Since Y and ϵ_i are independent standard normals, $\mathrm{Var}(x_i) = a_i^2 \,\mathrm{Var}(Y) + (1 - a_i^2)\,\mathrm{Var}(\epsilon_i) = a_i^2 + (1 - a_i^2) = 1$, and the mean is zero; being a linear combination of independent normals, x_i is itself standard normal. With unit variance, the correlation of x_i with Y equals their covariance, $cov(x_i, Y) = a_i \,\mathrm{Var}(Y) = a_i$, so a_i *is* the correlation of the default driver with the common factor. Preserving the standard-normal marginal of x_i is essential because the copula construction required $x_i = N^{-1}(F_i(t_i))$ to be standard normal in the first place; if the factor model rescaled the variance, the threshold $N^{-1}(F_i(T))$ would no longer correspond to the entity's true default probability $F_i(T)$ and the marginal calibration would break.

CR-C5. Conditional on a fixed value of Y, each entity's default driver is $x_i = a_i Y + \sqrt{1 - a_i^2}\,\epsilon_i$, and once Y is held constant the only remaining randomness is the idiosyncratic ϵ_i, which are mutually independent across entities. Therefore, given Y, the default events are independent, and the joint probability that a group all default factors into the product of their individual conditional probabilities $F_i(T|Y)$. This is what makes the model tractable: to get an unconditional joint probability one integrates the *product* against the one-dimensional density of Y, a single integral, instead of confronting a high-dimensional dependent distribution. If one tried to model the joint default distribution directly, one would have to specify and estimate the full correlation structure among all entities simultaneously — an intractable, parameter-heavy object with no conditional-independence shortcut — losing exactly the computational and estimation simplicity the factor structure provides.

CR-C6. The expression says every entity's conditional default probability is a monotone function of the single scalar Y. Because Y enters with a minus sign, a low (unfavorable) draw of Y raises the numerator $N^{-1}(F(T)) - \rho Y$ and hence raises $F(T|Y)$ for *every* entity at once — this simultaneous shift is precisely the mechanism that makes defaults cluster in bad states. As $\rho \to 0$, the factor drops out and $F(T|Y) \to N(N^{-1}(F(T))) = F(T)$, the unconditional probability, so defaults are independent and the conditional probability no longer depends on Y. As $\rho \to 1$, the denominator $\sqrt{1-\rho^2} \to 0$ and the conditional probability collapses toward 0 or 1 depending on the sign of the numerator: defaults become perfectly comonotone, either everyone survives or everyone fails. This one expression captures the entire correlation structure because all cross-entity dependence in the model flows through the single shared factor Y; conditioning on it removes all dependence.

CR-C7. Positive correlation, introduced through the common factor, makes the two loans' outcomes move together: it makes the joint events "both default" and "both survive" more likely and thins out the middle "one defaults, one survives" case. The senior tranche (first payment received) defaults only when *both* borrowers fail; its safety rested on that joint failure being unlikely, so fattening the both-default event raises its default probability from ϕ^2 to $\phi^2 + a^2\phi^2$ and lowers its value. The junior tranche pays only when *neither* borrower defaults; fattening the both-survive event raises its survival probability from $(1-\phi)^2$ to $(1-\phi)^2 + a^2\phi^2$ and raises its value. Diversification "evaporates" for the senior position because in a bad common state (y low) many borrowers default at once — exactly when the senior tranche needed the loans to be independent, they instead all fail together.

CR-C8. The two tranches, taken together, receive the first payment plus the second payment — that is, whatever total the two loans pay — so their combined payoff is identical to simply holding both loans outright, whose value is $\frac{2(1-\phi)M}{1+r_f}$. This combined value does not depend on a at all: it is fixed by the two individual loans' expected payoffs. Correlation cannot change the sum, so any value the senior tranche loses must reappear in the junior tranche. Algebraically the senior falls by $a^2\phi^2 M/(1 + r_f)$ and the junior rises by the same $a^2\phi^2 M/(1 + r_f)$, and the terms cancel. Hence correlation *redistributes* value between the tranches rather than creating or destroying it — it reslices a pie whose total size is unchanged.

CR-C9. The two-loan example is a miniature of a mortgage-backed security: many individual loans pooled and carved into a safe senior tranche and a risky junior tranche. Rating the senior tranche under an independence assumption ($a = 0$) assigns it default probability ϕ^2, which for small ϕ is tiny — the basis for a AAA rating. But if a common factor is actually present, the senior tranche's true default probability is $\phi^2 + a^2\phi^2$, strictly larger. The dangerous error is that the rating ignored the correlation: in the crisis the common factor y was a nationwide shock to house prices, which drove many mortgages into default simultaneously. When that shock hit, the diversification the AAA rating assumed vanished, and "safe" senior tranches suffered losses far beyond what an independence-based rating implied.

CR-C10. The reasoning treats the senior tranche's default probability as if it were ϕ^2 — vanishingly small for small ϕ — which is correct only under the counterfactual assumption of independent defaults ($a = 0$). Once a common factor is present, the correct default probability is $\phi^2 + a^2\phi^2$, and the added $a^2\phi^2$ term is of the *same order* as ϕ^2 (it is a^2 times as large), so it can inflate the senior default probability substantially. The single assumption whose misjudgment does the most damage is the assumed default *correlation* — the loading a (equivalently ρ). Getting the marginal default probability ϕ roughly right is not enough; underestimating a makes the senior tranche look far safer than it is, because the senior tranche's entire risk sits in the joint-default event that correlation controls.

CR-Q1. (a) $F(5) = 1 - e^{-0.03\times5} = 1 - e^{-0.15} = 0.139292$. (b) $x(5) = N^{-1}(0.139292) = -1.083506$. So the entity defaults within 5 years with probability about 0.1393, and the corresponding normal threshold is about -1.0835.

CR-Q2. (a) We need $N^{-1}(F(t^*)) = 0$, i.e. $F(t^*) = N(0) = 0.5$. So $1 - e^{-0.05t^*} = 0.5 \Rightarrow e^{-0.05t^*} = 0.5 \Rightarrow t^* = \frac{\ln 2}{0.05} = 13.862944$ years. (b) Check: $F(13.862944) = 1 - e^{-0.05\times13.862944} = 1 - e^{-\ln 2} = 1 - 0.5 = 0.500000 = N(0)$. So $t^* \approx 13.863$ years and $F(t^*) = 0.5$, confirming the transform maps this default time to the normal threshold $x = 0$.

CR-Q3. With $F_i(T) = 0.05$, $N^{-1}(0.05) = -1.644854$, and $a_i = 0.5$ so $\sqrt{1-0.25} = 0.866025$. (a) $Y = +1$: $F_i(T|Y) = N\left(\frac{-1.644854-0.5}{0.866025}\right) = N(-2.4766) = 0.006631$. (b) $Y = -1$: $F_i(T|Y) = N\left(\frac{-1.644854+0.5}{0.866025}\right) = N(-1.3219) = 0.093090$. The "bad" state $Y = -1$ produces far more defaults (about 9.3% vs 0.66%), since a low common factor pushes every entity toward default.

CR-Q4. With $F(T) = 0.10$, $N^{-1}(0.10) = -1.281552$, and $Y = -1.5$. (a) $\rho = 0.2$: denominator $\sqrt{1-0.04} = 0.979796$; $F(T|Y) = N\left(\frac{-1.281552-0.2(-1.5)}{0.979796}\right) = N(-1.0018) = 0.158222$. (b) $\rho = 0.7$: denominator $\sqrt{1-0.49} = 0.714143$; $F(T|Y) = N\left(\frac{-1.281552-0.7(-1.5)}{0.714143}\right) = N(-0.3243) = 0.372879$. The higher loading gives a larger conditional default probability (37.3% vs 15.8%) in the bad state because a larger ρ ties the entity more tightly to the common factor, so an unfavorable Y raises its default probability more.

CR-Q5. (a) $cov(x_i, x_j) = a_i a_j = 0.6 \times 0.8 = 0.48$. (b) A common loading a with

$a^2 = 0.48$ gives the same pairwise correlation, so $a = \sqrt{0.48} = 0.692820$. Result: correlation 0.48; equivalent common loading ≈ 0.6928.

CR-Q6. (a) $cov(x_1, x_2) = 0.5 \times 0.5 = 0.25$; $cov(x_1, x_3) = 0.5 \times 0.9 = 0.45$; $cov(x_2, x_3) = 0.5 \times 0.9 = 0.45$. (b) The pairs $(1, 3)$ and $(2, 3)$ are most correlated, at 0.45. Entity 3's large loading $a_3 = 0.9$ dominates because pairwise correlation is the *product* of loadings, so any pair including the high-loading entity inherits a large correlation. Correlations: $0.25,\ 0.45,\ 0.45$.

CR-Q7. With $\phi = 0.10$, $M = \$1{,}000$, $r_f = 0.04$. (a) Senior: $P_{\text{senior}} = \frac{(1-\phi^2)M}{1+r_f} = \frac{(1-0.01)(1000)}{1.04} = \frac{990}{1.04} = \$951.92$. (b) Junior: $P_{\text{junior}} = \frac{(1-\phi)^2 M}{1+r_f} = \frac{(0.9)^2(1000)}{1.04} = \frac{810}{1.04} = \$778.85$.

CR-Q8. With $a = 0.20$, so $a^2\phi^2 = 0.04 \times 0.01 = 0.0004$. Senior: $P_{\text{senior}} = \frac{(1-0.01-0.0004)(1000)}{1.04} = \frac{989.6}{1.04} = \$951.54$. Junior: $P_{\text{junior}} = \frac{(0.81+0.0004)(1000)}{1.04} = \frac{810.4}{1.04} = \$779.23$. Changes relative to CR-Q7: senior falls by \$0.3846 (from \$951.92 to \$951.54); junior rises by \$0.3846 (from \$778.85 to \$779.23). The loss to the senior equals the gain to the junior.

CR-Q9. With $a = 0.30$, $a^2\phi^2 = 0.09 \times 0.01 = 0.0009$. (a) Transfer $= \frac{a^2\phi^2 M}{1+r_f} = \frac{0.0009\times1000}{1.04} = \frac{0.9}{1.04} = \$0.8654$ moves from senior to junior. (b) $P_{\text{senior}} = \frac{(0.99-0.0009)(1000)}{1.04}$ and $P_{\text{junior}} = \frac{(0.81+0.0009)(1000)}{1.04}$; summing, the ±0.0009 terms cancel: $P_{\text{senior}} + P_{\text{junior}} = \frac{(0.99+0.81)(1000)}{1.04} = \frac{1800}{1.04} = \$1{,}730.77$. Holding both loans outright: $\frac{2(1-\phi)M}{1+r_f} = \frac{2(0.9)(1000)}{1.04} = \frac{1800}{1.04} = \$1{,}730.77$. The two match, confirming value conservation. Transfer ≈ \$0.87; combined value \$1,730.77.

CR-Q10. With $\phi = 0.10$. (a) Under independence ($a = 0$): default probability $= \phi^2 = 0.01$. Under $a = 0.5$: $\phi^2 + a^2\phi^2 = 0.01 + 0.25 \times 0.01 = 0.0125$. (b) Percentage increase $= \frac{0.0125-0.01}{0.01} = \frac{a^2\phi^2}{\phi^2} = a^2 = 0.25 = 25\%$. So the senior tranche's default probability rises from 0.0100 to 0.0125, a 25% increase — a substantial jump driven entirely by the correlation loading.

17.14 Market Makers

MM-C1. Stock Y, with $\mu = 0.40$, has the wider spread. In the Glosten-Milgrom model a competitive market maker cannot tell an informed order from an uninformed one, so she must protect herself against the possibility that the trader on the other side knows the true value. When a buy order arrives, the posterior probability that it came from an informed trader (who buys only when $v = V_1$) rises with μ; the zero-profit condition then forces $A = E[V|\text{buy order}]$, which lies above $E[V]$ because a buy order is partly informative "good news." Symmetrically a sell order pushes $B = E[V|\text{sell order}]$ below $E[V]$. With more informed traders (stock Y), each order carries more information, the conditional expectations move farther from $E[V]$, and the spread $A - B$ widens. The money lost trading against informed traders is exactly recouped from the uninformed, so the market maker breaks even on average.

MM-C2. The spread does not require monopoly power; it is an adverse-selection cost. Even perfectly competitive market makers who earn zero expected profit must quote $A = E[V|\text{buy}] > E[V]$ and $B = E[V|\text{sell}] < E[V]$, because otherwise they would lose money on the informed trades and, being unable to distinguish informed from uninformed, could not stay solvent. If every arriving trader were uninformed ($\mu = 0$), then a buy or sell order conveys no information, $A = B = E[V]$, and the spread collapses to zero. This directly refutes the commentator: the spread is generated by information asymmetry, not market power, and it vanishes precisely when there is nothing to be adversely selected against.

MM-C3. A buy order is not a random event: informed traders buy only when the value is high, so seeing a buy order raises the market maker's assessment of the probability that $v = V_1$. Bayes' rule quantifies this. The unconditional probability of a buy is $P(\text{buy}) = \pi\mu + \frac{1}{2}(1 - \mu)$, and of that the informed contribute $\pi\mu$, giving $P(\text{In}|\text{buy}) = \frac{\pi\mu}{\pi\mu + \frac{1}{2}(1-\mu)}$. Because informed buyers are certain the value is high, the posterior expectation $E[V|\text{buy}]$ tilts upward relative to $E[V]$. Setting $A = E[V|\text{buy order}]$ rather than $E[V]$ means the market maker charges buyers for the "bad news to her" that a buy order represents; quoting the unconditional $E[V]$ would leave her expecting to lose money whenever the buyer turns out to be informed.

MM-C4. The spread $A - B = \frac{4\pi(1-\pi)\mu(V_1 - V_0)}{1-(2\pi-1)^2\mu^2}$ is (i) proportional to the value gap $V_1 - V_0$ and (ii) increasing in the informed fraction μ (the numerator rises linearly and the denominator falls as μ grows). Channel (i) corresponds to the chapter's third narrowing force — lower volatility of the underlying asset shrinks $V_1 - V_0$ and hence the spread. Channel (ii) corresponds to the second force — a decrease in the probability of informed trading. The first listed force, competition between market makers, is what drives the market maker to the zero-profit quotes $A = E[V|\text{buy}]$ and $B = E[V|\text{sell}]$ in the first place, without which the spread could be wider still.

MM-C5. If the insider bought without limit, her enormous order flow would reveal that v is high, the market maker's price $P(y) = \mu + \lambda y$ would jump, and she would end up buying at prices near the true value — eliminating her profit. She therefore restrains demand to the optimal $x = \frac{v-\mu}{2\lambda}$, trading enough to exploit her signal but little enough to keep the price impact from swallowing her gains. The noise traders' demand $u \sim N(0, \sigma_u^2)$ provides camouflage: because the market maker sees only total order flow $y = x + u$, the insider's trade is hidden inside random noise, so a large y could reflect either informed buying or a noise shock. If $\sigma_u^2 \to 0$ there is no camouflage, order flow becomes fully revealing of v, the price moves one-for-one against the insider, and her profit — and any incentive to trade on information — disappears.

MM-C6. From $\lambda = \frac{\sqrt{\Sigma_0}}{2\sigma_u}$, asset A (high Σ_0, low σ_u) has the larger λ and thus the *shallower* market; asset B (low Σ_0, high σ_u) has the smaller λ and the *deeper* market. "Deeper" means a trader can push through a large order with less price impact: each unit of order flow moves the price by only λ, so a small λ lets big orders execute close to the prior price. Market makers rationally choose a steeper pricing rule (larger λ) for asset A because its high fundamental uncertainty and thin noise trading make order flow very informative, so each unit of imbalance signals a large revision in value and must be priced aggressively to avoid losses to informed traders.

MM-C7. The optimal order $x = \frac{v-\mu}{2\lambda}$ is proportional to the surprise $v - \mu$ because the insider profits from the gap between the true value and the price the market maker would otherwise set; a bigger informational edge justifies a bigger position. It is inversely proportional to 2λ because λ measures how fast her own trading moves the price against her: the factor of 2 reflects that trading a marginal share both raises the price paid on that share and on all inframarginal shares. A deeper market (smaller λ) means less price impact per share, so she can exploit the same signal more aggressively before the price catches up. In equilibrium this linear demand feeds back into a normally distributed order flow, and matching the coefficient on y in $E[v|y]$ to λ pins down $\lambda = \frac{\sqrt{\Sigma_0}}{2\sigma_u}$, confirming the market maker's conjectured linear rule is self-consistent.

MM-C8. If the price $P = a + bs - dZ$ perfectly revealed s, an uninformed trader could read the signal straight off the price for free, so no one would pay the cost of becoming informed — the private return to information would be zero. But if no one acquires the signal, there is nothing for the price to reveal, so a fully revealing price cannot be sustained in equilibrium. The noisy supply $Z \sim N(0, \sigma_z^2)$ breaks this impossibility: because the price mixes the signal s together with the unobserved supply shock Z, observers cannot invert the price to recover s exactly. This partial revelation leaves a positive return to being informed, so a strictly interior fraction of traders is willing to pay for the signal, and equilibrium exists.

MM-C9. With $\sigma_z^2 \to 0$ the price becomes an exact, invertible function of the signal — there is no longer a noisy supply term to garble it, so the price fully reveals s. Uninformed traders can then infer the signal costlessly from the price, making the private value of paying for information zero. Rational traders would therefore stop acquiring the costly signal, so the fraction λ collapses toward zero. This is the Grossman-Stiglitz paradox in reverse: a healthy market needs some traders who trade for reasons unrelated to fundamental value (noise/liquidity traders), because their randomness is exactly what keeps prices from being perfectly revealing and thereby preserves the incentive for anyone to gather information at all.

MM-C10. In both models the transaction price sits away from the market maker's unconditional expectation of value, but the wedge is generated differently. In Glosten-Milgrom the wedge is the discrete bid-ask spread:

values are binary (V_0 or V_1), each trader demands exactly one share, and the adverse-selection cost is the chance that a single order comes from an informed trader. In Kyle values and quantities are continuous and Gaussian, the informed trader chooses a strategic *quantity* $x = \frac{v-\mu}{2\lambda}$, and the adverse-selection cost shows up as the price-impact slope λ applied to order flow. In both the market maker is competitive and earns zero expected profit: she systematically loses to informed traders but recovers exactly those losses from uninformed/noise traders, so the quotes (spread in one, λ in the other) are set precisely at the break-even level.

MM-C11. Informed and uninformed traders hold different positions because their conditioning information differs: the informed trader uses $E[v|s]$ and the tighter variance $\sigma^2(v|s)$, while the uninformed trader uses $E[v|P]$ and the wider $\sigma^2(v|P)$. Since the informed posterior mean typically diverges from the price more sharply and with more confidence, her demand $\theta = \frac{E[v|I]-P}{\gamma\sigma^2(v|I)}$ is generally larger in magnitude. The term $\sigma^2(v|I)$ in the denominator encodes the value of information: better information means a smaller posterior variance, which shrinks the denominator and lets the trader take a larger position on the same perceived mispricing. A risk-averse trader scales back positions she is uncertain about, so reducing that uncertainty is exactly what makes information valuable.

MM-C12. In an information-based theory such as Glosten-Milgrom, the spread compensates the dealer for adverse selection — the risk that the counterparty is better informed — so it is generated purely by information asymmetry and would exist even for a risk-neutral dealer with unlimited capital. In an inventory-based theory, the spread reflects the dealer's need to manage risk from holding unbalanced positions: a dealer who accumulates a large long position lowers both her bid and ask to encourage buyers and discourage further sellers, nudging her inventory back toward a target. One distinguishing empirical prediction: inventory models imply quote revisions that depend on the dealer's accumulated position and mean-revert as inventory normalizes, whereas information models predict quote changes that are permanent (prices ratchet in the direction of the informative order flow and do not revert).

MM-Q1. With $\pi = 0.5$, $E[V] = 0.5(110) + 0.5(90) = 100$. Ask numerator $\pi(1+\mu)V_1 + (1-\pi)(1-\mu)V_0 = 0.5(1.3)(110) + 0.5(0.7)(90) = 71.5 + 31.5 = 103$; denominator $1 + (2\pi - 1)\mu = 1 + 0 = 1$, so $A = 103$. Bid numerator $0.5(0.7)(110) + 0.5(1.3)(90) = 38.5 + 58.5 = 97$; denominator 1, so $B = 97$. Spread $A - B = 103 - 97 = 6$. The simplified formula gives $A - B = \mu(V_1 - V_0) = 0.3(20) = 6$, matching. **Result:** $A = 103$, $B = 97$, $A - B = 6$.

MM-Q2. $E[V] = 0.7(80) + 0.3(20) = 56 + 6 = 62$. Ask numerator $\pi(1+\mu)V_1 + (1-\pi)(1-\mu)V_0 = 0.7(1.5)(80) + 0.3(0.5)(20) = 84 + 3 = 87$; denominator $1 + (2\pi - 1)\mu = 1 + (0.4)(0.5) = 1.2$. So $A = 87/1.2 = 72.5$. Indeed $A = 72.5 > 62 = E[V]$: a buy order raises the posterior probability the value is high, so the market maker prices above the unconditional expectation to avoid losing to informed buyers. **Result:** $E[V] = 62$, $A = 72.5$.

MM-Q3. $E[V] = 0.6(60) + 0.4(40) = 52$ (for reference). Ask numerator $0.6(1.25)(60) + 0.4(0.75)(40) = 45 + 12 = 57$; denominator $1 + (2\pi - 1)\mu = 1 + (0.2)(0.25) = 1.05$; $A = 57/1.05 = 54.2857$. Bid numerator $0.6(0.75)(60) + 0.4(1.25)(40) = 27 + 20 = 47$; denominator $1 + (1 - 2\pi)\mu = 1 + (-0.2)(0.25) = 0.95$; $B = 47/0.95 = 49.4737$. Spread $A - B = 54.2857 - 49.4737 = 4.812$. Closed form: $\frac{4(0.6)(0.4)(0.25)(20)}{1-(0.2)^2(0.25)^2} = \frac{4.8}{1-0.0025} = \frac{4.8}{0.9975} = 4.812$, matching. **Result:** $A \approx 54.29$, $B \approx 49.47$, $A - B \approx 4.81$.

MM-Q4. With $\pi = 0.5$, spread $A - B = \mu(V_1 - V_0) = 0.4(140 - 100) = 0.4(40) = 16$. $E[V] = 0.5(140) + 0.5(100) = 120$; because $\pi = 0.5$ the spread is symmetric about $E[V]$, so $A = 120 + 8 = 128$ and $B = 120 - 8 = 112$. If μ rises to 0.8, the spread becomes $0.8(40) = 32$. **Result:** $A - B = 16$, $A = 128$, $B = 112$; at $\mu = 0.8$, $A - B = 32$.

MM-Q5. $\lambda = \frac{\sqrt{\Sigma_0}}{2\sigma_u} = \frac{\sqrt{16}}{2(2)} = \frac{4}{4} = 1$. With $\mu = p_0 = 100$ and $y = 3$: $P(y) = \mu + \lambda y = 100 + 1(3) = 103$. **Result:** $\lambda = 1$, $P(3) = 103$.

MM-Q6. (a) $\lambda = \frac{\sqrt{36}}{2(3)} = \frac{6}{6} = 1$. (b) With $\sigma_u^2 = 36$, $\sigma_u = 6$: $\lambda = \frac{\sqrt{36}}{2(6)} = \frac{6}{12} = 0.5$. (c) The market became deeper (smaller λ): with more noise trading to hide behind, each unit of order flow is less informative, so a large order moves the price by less — lower price impact. **Result:** $\lambda = 1$ then $\lambda = 0.5$; deeper.

MM-Q7. (a) $\lambda = \frac{\sqrt{25}}{2\sqrt{100}} = \frac{5}{2(10)} = \frac{5}{20} = 0.25$. (b) With $\mu = p_0 = 50$ and $v = 60$:

$x = \frac{v-\mu}{2\lambda} = \frac{60-50}{2(0.25)} = \frac{10}{0.5} = 20$. (c) With $\sigma_u^2 = 400$, $\sigma_u = 20$: $\lambda = \frac{5}{2(20)} = \frac{5}{40} = 0.125$; then $x = \frac{10}{2(0.125)} = \frac{10}{0.25} = 40$. The insider trades more aggressively because the deeper market (smaller λ) means less price impact per share. **Result:** $\lambda = 0.25$, $x = 20$; then $\lambda = 0.125$, $x = 40$.

MM-Q8. (a) $\lambda = \frac{\sqrt{64}}{2(4)} = \frac{8}{8} = 1$; $\mu = p_0 = 30$. (b) $x = \frac{v-\mu}{2\lambda} = \frac{45-30}{2(1)} = \frac{15}{2} = 7.5$. (c) $y = x + u = 7.5 + (-2.5) = 5$; $P(y) = \mu + \lambda y = 30 + 1(5) = 35$. The insider bought at 35, below the true value $v = 45$, so she earns a positive profit of $45 - 35 = 10$ per share. **Result:** $\lambda = 1$, $x = 7.5$, $y = 5$, $P = 35$ (below v).

MM-Q9. $M = \frac{\sigma_v^2}{\sigma_v^2+\sigma_\epsilon^2} = \frac{9}{9+3} = \frac{9}{12} = 0.75$. With $\mu = 20$ and $s = 26$: $E[v|s] = \mu + Ms = 20 + 0.75(26) = 20 + 19.5 = 39.5$. **Result:** $M = 0.75$, $E[v|s] = 39.5$.

MM-Q10. (a) $M = \frac{16}{16+16} = \frac{16}{32} = 0.5$. (b) $E[v|s] = \mu + Ms = 50 + 0.5(70) = 50 + 35 = 85$. (c) As $\sigma_\epsilon^2 \to \infty$, $M \to 0$: an extremely noisy signal carries almost no information, so the trader places essentially no weight on it and relies on the prior. **Result:** $M = 0.5$, $E[v|s] = 85$; $M \to 0$ as noise grows.

MM-Q11. $\theta = \frac{E[v|I]-P}{\gamma\sigma^2(v|I)} = \frac{25-22}{2(5)} = \frac{3}{10} = 0.3$. If posterior uncertainty $\sigma^2(v|s)$ doubled to 10, the denominator would double and her demand would halve to 0.15 — greater uncertainty makes her scale back her position on the same expected gain. **Result:** $\theta = 0.3$; doubling $\sigma^2(v|s)$ halves it to 0.15.

MM-Q12. (a) $\theta = \frac{E[v|I]-P}{\gamma\sigma^2(v|I)} = \frac{30-24}{4(2)} = \frac{6}{8} = 0.75$ for each trader. (b) With $\gamma = 1$: $\theta = \frac{30-24}{1(2)} = \frac{6}{2} = 3$. (c) Demand is inversely proportional to risk aversion: for a given expected gain and posterior variance, a less risk-averse investor takes a proportionally larger position. **Result:** $\theta = 0.75$ (at $\gamma = 4$), $\theta = 3$ (at $\gamma = 1$).